

MATHEMATICS

STUDENT'S BOOK

Senior 5

Version Edited in 2026

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Experimental version

FOREWORD

Dear students,

The Rwanda Basic Education Board (REB) is honoured to present this Mathematics textbook for Senior Five (S5) for Mathematics and Science Pathways. This textbook is designed to support the effective implementation of the Competence Based Curriculum (CBC). It promotes learner-centred teaching and learning practices that align with Rwanda's vision of empowering learners to reach their full potential, actively participate in society, and pursue meaningful careers. As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding learners to connect classroom knowledge with real-life experiences, thereby fostering both individual growth and national development.

This student book serves as a comprehensive resource for the teaching and learning of Mathematics at Senior Five (S5). Mathematics is an integral subject within the CBC, aimed at equipping learners with essential knowledge, skills, values, and attitudes necessary to become responsible and competent citizens. Through this subject, you are encouraged to develop problem-solving skills and contribute positively to your communities. In this regard, the only way to learn mathematics effectively is by doing mathematics through active problem-solving rather than passive reading. You are expected to work through problems, attempt proofs, and practise regularly and do not rely only on reviewing theorems or looking up answers.

In its pursuit of quality education, the Government of Rwanda emphasizes the importance of using this kind of teaching and learning materials aligned with the curriculum.

I sincerely thank everyone who contributed to the development and refinement of this textbook. Special recognition goes to the REB team that coordinated the entire process. We also thank all partners, teachers, lecturers, and individuals who were involved from the beginning to the end. All feedback was highly valued and will contribute to improving the learning outcomes of our students.



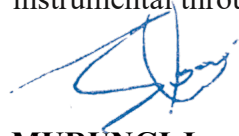
Dr. MBARUSHIMANA Nelson
Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Mathematics students' book for Senior Five (S5). This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff; subject experts; university lecturers and teachers for their valuable work in the writing of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.

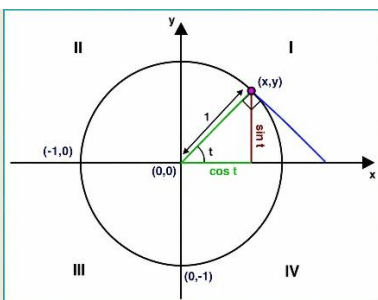


MURUNGI Joan
Head of CTLR Department

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UNIT 1: TRIGONOMETRIC FORMULAE, EQUATIONS AND INEQUALITIES

Key unit competence: Use trigonometric formulae, equations, and inequalities to solve mathematical and real-life problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Show how to use transformation formulas to simplify trigonometric expressions.
- Extend the concepts of trigonometric ratios and their properties to trigonometric equations.
- Analyze and discuss the solution of trigonometric inequalities.
- Explain the structure and solution methods for trigonometric equations and inequalities.
- Describe real-life applications of trigonometric formulas, including integration and harmonic motion in physics

Skills:

- Apply the transformation formulas to simple trigonometric expressions.
- Use trigonometric functions and equations to model and solve problems involving trigonometry concepts.
- Apply trigonometric functions and equations to perform operations in calculation, integration and solving problems including harmonic motion in physics.
- Solve any other problems related to trigonometry.

Attitudes and values:

- Develop precision and logical reasoning when manipulating trigonometric expressions.
- Appreciate the relevance of trigonometry in real-life situations, especially in physics.
- Show persistence and curiosity when exploring multiple solution methods.
- Appreciate the relationship between trigonometry and other subjects.

Introduction

Trigonometry studies relationship involving lengths and angles of a triangle. The techniques in trigonometry are used for finding relevance in navigation particularly satellite systems and astronomy, naval and aviation industries, oceanography, land surveying and in cartography (creation of maps). Now those are the scientific applications of the concepts in trigonometry, but most of the mathematics we study would seem (on the surface) to have little real-life application. Trigonometry is relevant in our day-to-day activities.



Introductory activity

A cooperative of coffee farmers in Rwanda has found that the monthly profit P (in thousands of FRW) from their sales fluctuates throughout the year due to seasonal demand.

The profit t months after January 1st is modelled by the trigonometric function: $P(t) = 80\sin(t)\cos(t) + 50$, where $0 \leq t \leq 12$ represents the months of the year.

The cooperative needs to identify the months during which their profit is above 70,000 FRW, [$P(t) > 70$] to decide when to invest in new processing equipment.

- Before solving the inequality, use the double-angle identity for sine to simplify $P(t)$.
- Find the reference angles (principal values) using the inverse sine function.
- Determine the intervals for $2t$ within the circle, then divide by 2 to find the months t .

1.1 Introduction of trigonometric function and their periods



Activity 1.1

Given the function $y = \sin x$.

Step 1: Represent this function on the graph.

Step 2: For a fixed value of x , locate and compare the value of $y = \sin x$ and $y = \sin(x + 2\pi)$.

What can you deduce from step 2?



Content summary

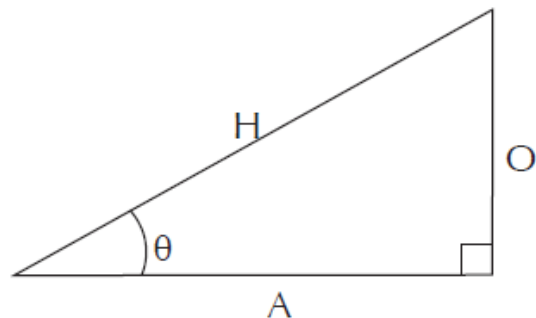
Trigonometric functions are functions that describe periodic phenomena and are defined using the coordinates of points on a unit circle. They relate the angles of a right-angled triangle to the **ratios** of its side lengths.

Basic trigonometric ratios:

These are **sine**, **cosine**, and **tangent**. The other common trigonometric ratios are: **secant**, **cosecant** and **cotangent**.

In the figure, the side H opposite the right angle is called the **hypotenuse**. Relative to the angle θ , the side O opposite the angle θ is called the **opposite side**. The remaining side A is called the **adjacent side**.

Trigonometric ratios provide relationships between the sides and angles of a right-angle triangle. The three most used ratios are sine, cosine and tangent.



Name	Formula
Sine	$\sin \theta = \frac{O}{H}$
Cosine	$\cos \theta = \frac{A}{H}$
Tangent	$\tan \theta = \frac{O}{A}$

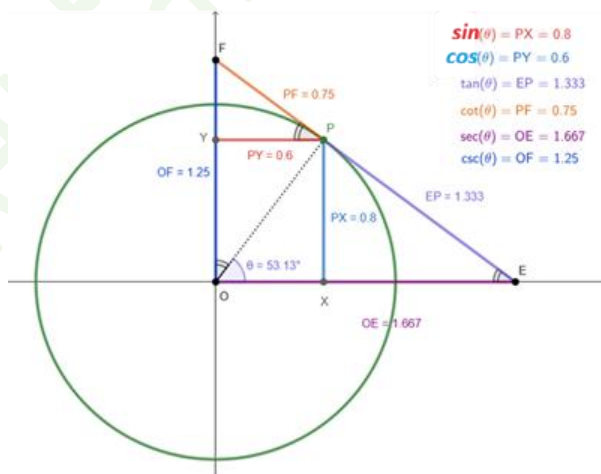
Note that $\tan \theta = \frac{\sin \theta}{\cos \theta}$. Other ratios are defined by using the above three:

Name	Formula
Cosecant	$\csc \theta = \frac{1}{\sin \theta}$
Secant	$\sec \theta = \frac{1}{\cos \theta}$
Cotangent	$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$

These six ratios define what are known as trigonometric **functions**.

If a line from the origin at an angle θ (measured counterclockwise from the positive x-axis) intersects the unit circle at a point (x, y) , then:

- $\cos \theta = x$
- $\sin \theta = y$
- $\tan \theta = \frac{y}{x}$



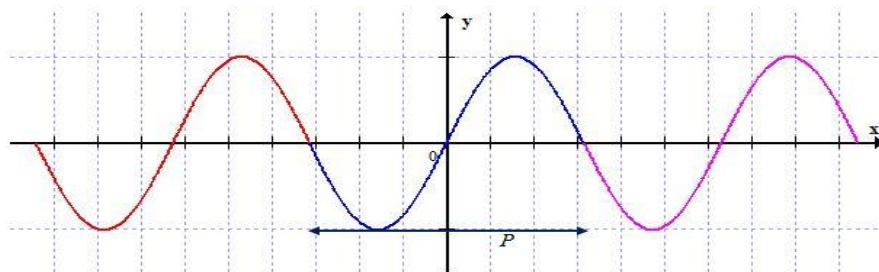
Period of a trigonometric function

A function f is called **periodic** if there is a positive number P such that $f(x + p) = f(x)$ whenever x and $x + p$ lie in the domain of f .

We call P a **period** of the function. The smallest positive period is called the **fundamental period** (also **primitive period**, **basic period**, or **prime period**) of function.

A function with period P will repeats on intervals of length P , and these intervals are referred to as **periods**.

Geometrically, a periodic function can be defined as a function whose graph exhibits translational symmetry. Specifically, a function is periodic with period P if its graph is invariant under translation in the x -direction by a distance of P .



The most important examples of periodic functions are the trigonometric functions.

Any function which is not periodic is called **aperiodic**.

Example 1:

The sine and cosine functions:

2π is the period since $\sin(x + 2\pi) = \sin x$ and $\cos(x + 2\pi) = \cos x$.

Also $4\pi, 6\pi, 8\pi, \dots$, are periods for sine and cosine functions since $\sin(x + 4\pi) = \sin x$, $\sin(x + 6\pi) = \sin x$, $\sin(x + 8\pi) = \sin x, \dots$ and $\cos(x + 4\pi) = \cos x$, $\cos(x + 6\pi) = \cos x$, $\cos(x + 8\pi) = \cos x, \dots$

The fundamental period of sine and cosine functions is 2π .

Example 2:

For $\sin 3x$ and $\cos 3x$ functions, the fundamental period is $\frac{2\pi}{3}$ since

$$\sin\left[3\left(x + \frac{2\pi}{3}\right)\right] = \sin(3x + 2\pi) = \sin(3x) \text{ and } \cos\left[3\left(x + \frac{2\pi}{3}\right)\right] = \cos(3x + 2\pi) = \cos(3x).$$

Tangent and cotangent functions:

π is a period since $\tan(x + \pi) = \tan x$ and $\cot(x + \pi) = \cot x$. Also $2\pi, 3\pi, 4\pi, \dots$ are periods, but π is the fundamental period.

For $\tan x$, we have $\tan(x + p) = \tan x \Leftrightarrow x + p = x + k\pi, k \in \mathbb{Z} \Leftrightarrow p = k\pi$.

Since we need the smallest positive period, we take. Thus, $p = \pi$.

Theorem 1:

If $a \neq 0$ and $b \neq 0$, then the functions $y = a \sin bx$ and $y = a \cos bx$ have fundamental period $\frac{2\pi}{|b|}$ and their graphs oscillate between $-a$ and a .

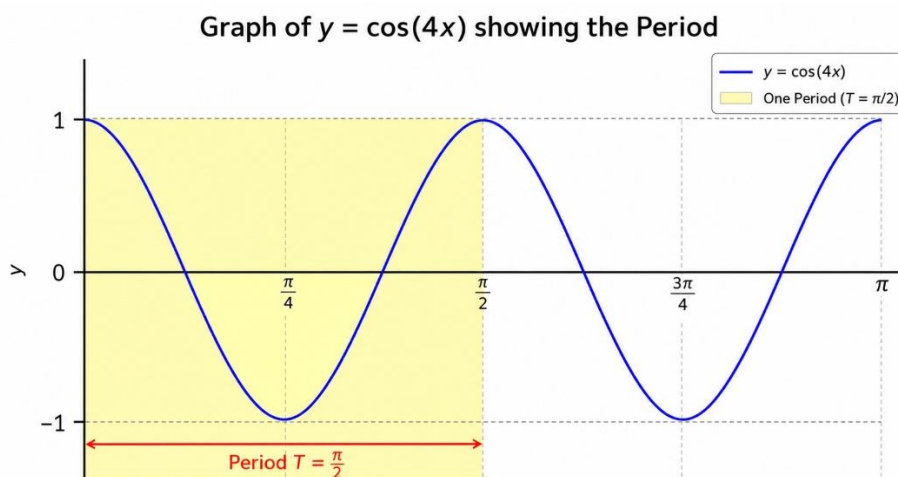
The number $|a|$ is called the **amplitude of the function**.

Example 3: The graph of the function $y = \cos(4x)$ is shown below.

For a cosine function of the form $y = \cos(kx)$, the **period** (T) is calculated using the formula:

$$T = \frac{2\pi}{k}, \text{ For } y = \cos(4x) \text{ where } k = 4: T = \frac{2\pi}{4} = \frac{\pi}{2}$$

This means the function completes one full cycle every $\frac{\pi}{2}$ units along the x-axis.



Example 4:

Find the fundamental period of $f(x) = 2 \sin 6x$, and $g(x) = 4 \cos 3x$.

For $f(x)$, we have $2 \sin 6(x + p) = 2 \sin 6x \Leftrightarrow 6x + 6p = 6x + 2k\pi, k \in \mathbb{Z} \Leftrightarrow 6p = 2k\pi$.

Since we need the smallest positive period, we take $k = 1$. Thus, $p = \frac{\pi}{3}$.

For $g(x)$, we have $4 \cos 3(x + p) = 4 \cos 3x \Leftrightarrow 3x + 3p = 3x + 2k\pi, k \in \mathbb{Z} \Leftrightarrow 3p = 2k\pi$.

Since we need the smallest positive period, we take $k = 1$. Thus, $p = \frac{2\pi}{3}$.

Period of the sum of functions

If we have two functions $f(x)$ and $g(x)$ with the same period, say p , we can throw them together any way we want.

Let $h(x) = f(x) + g(x)$, at any value a of x :

$$h(a) = f(a) + g(a) = c \Leftrightarrow h(a + p) = f(a + p) + g(a + p) = c \Rightarrow h(a) = h(a + p).$$

Thus, $h(a) = h(a + p)$.

This is different for functions that don't have the same fundamental period:

Let us say that we have two periodic functions $f(x)$ and $g(x)$ with period P and Q respectively:

$$f(x + p) = f(x) \text{ is true for all real } x, k \text{ integer.}$$

$$g(x + kQ) = g(x) \text{ is true for all real } x, k \text{ integer.}$$

Now, we cannot construct that nice $h(x)$ as we did before because we have different periods.

Consider the following case: If a function repeats every 2 units, then it will also repeat every 6 units. If we have one function with fundamental period 2, and another function with fundamental period 8, we have got no problem because 8 is a multiple of 2, and both functions will cycle every 8 units.

If we can patch up the periods to be the same. We know that if we combine them, we will get a function with the patched-up period.

What we must do is to find the **Lowest Common Multiple (LCM)** of two periods.

What can we do if one function has period 4 and another has period 5? We can see that in 20 units, both will cycle, so they are fine.

Example 4:

Find the fundamental period of the function $f(x) = \tan\left(\frac{x+1}{2}\right) \sin\left(\frac{2x+1}{5}\right)$.

$$\text{For } \tan\left(\frac{x+1}{2}\right), p_1 = 2\pi, \text{ for } \sin\left(\frac{2x+1}{5}\right), p_2 = 5\pi, LCM(2\pi, 5\pi) = 10\pi, \quad p = 10\pi.$$

Another important case is where the periods are fractions:

Suppose that we have function $f(x)$ with period $\frac{13}{12}$ and another function $g(x)$ with period

$\frac{2}{21}$. What we need are two numbers of periods that we can multiply by the periods to get some

common, patched up period.

First, we can simplify the problem by multiplying each period by its denominator to find whole number periods. It is known that $f(x)$ has period of 13 (in 12 fundamental periods) and $g(x)$ a period of 2 (in 21 fundamental periods).

Now we can simply do what we did before and multiply both periods to find a period for the new combination function. Therefore, the combination function has a period of 26.

This suggests the following theorem.

Theorem 2:

If two periodic functions have rational periods, then any addition or multiplication combination of those functions (not composition) will also be periodic. If $f(x)$ is a periodic function and $g(x)$ is not a periodic function, then $g(f(x))$ is periodic and $f(g(x))$ is not.

Example 5: Find the fundamental period of the function $f(x) = \frac{\sin 3x}{\sin 7x}$.

For $\sin 3x$, $p_1 = \frac{2\pi}{3}$, for $\sin 7x$, $p_2 = \frac{\pi}{7}$

p_1 is 2π in 3 fundamental periods

p_2 is π in 7 fundamental periods

But 2π is a multiple of π . Thus, $p = 2\pi$.

Example 6: Find the fundamental period of the function $f(x) = \sin x + \sin 4x$

For $\sin x$, $p_1 = 2\pi$

For $\sin 4x$, $p_2 = \frac{\pi}{2}$

$$\text{LCM}\left(2\pi, \frac{\pi}{2}\right) = 2\pi, \quad p = 2\pi.$$

Note that the period of $f(ax+b) = \frac{P}{|a|}$.

If f is a periodic function with fundamental period P then $-f$ and $\frac{1}{f}$ are also a periodic function with period P . A constant function is periodic but does not have a fundamental period. Periods of trigonometric functions are given below:

S/N	Function	Period
1	$\sin(ax + b)$	$\frac{2\pi}{ a }$
2	$\cos(ax + b)$	$\frac{2\pi}{ a }$
3	$\tan(ax + b)$	$\frac{\pi}{ a }$
4	$\cot(ax + b)$	$\frac{\pi}{ a }$
5	$\sec(ax + b)$	$\frac{2\pi}{ a }$
6	$\csc(ax + b)$	$\frac{2\pi}{ a }$

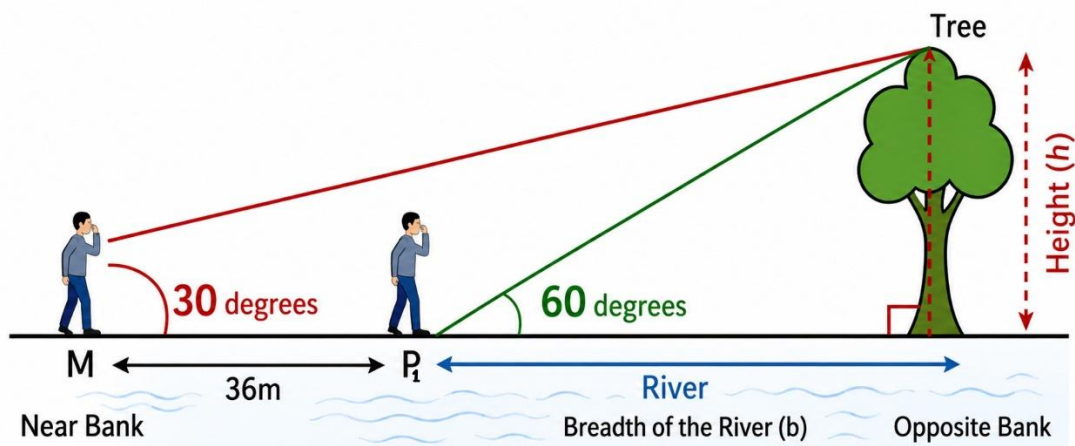
Standard results:

- Period of $|\sin(x)|, |\cos x|, |\tan x|, |\cot x|, |\sec x|, |\csc x|$ is π .
- Period of $|\sin(x)| + |\cos x|$ and $|\tan x| + |\cot x|$ is $\frac{\pi}{2}$.
- Period of $|\sin(x) + \cos x|$ and $|\sin(x) - \cos x|$ is π .
- Period of $|\tan x + \cot x|$ and $|\tan x - \cot x|$ is $\frac{\pi}{2}$.
- If $n \in \mathbb{N}$, the period of $\sin^{2n} x + \cos^{2n} x, \sec^{2n} x + \csc^{2n} x, \tan^{2n} x + \cot^{2n} x$ is $\frac{\pi}{2}$.
- For natural numbers n and m , the period of $a \sin^{2n} x + b \cos^{2m} x, a \sec^{2n} x + b \csc^{2m} x, a \tan^{2n} x + b \cot^{2m} x$ is π .



Application activity 1.1

- A man standing on the banks of a river observes that the angle subtended by a tree on the opposite bank is 60° . He walks 36 meters backward on the bank and observes the angle to be 30° .



What is the breadth of the river and the height of the tree?

2) Find the period of the following function:

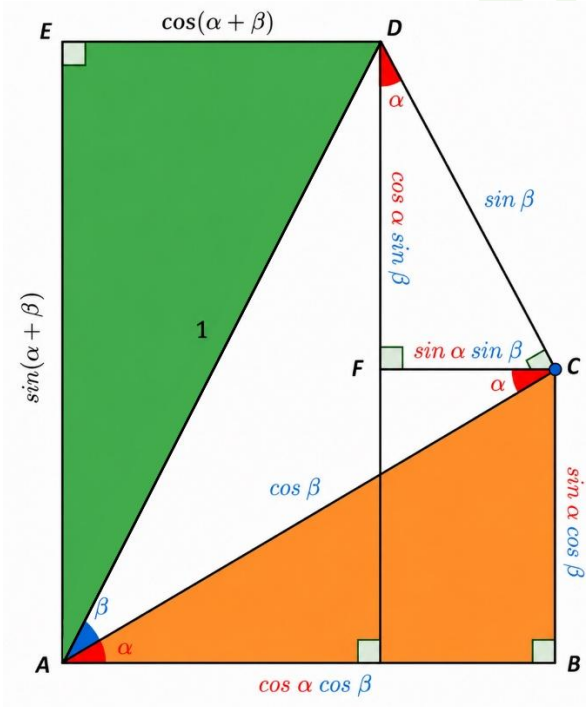
a) $f(x) = \cos(3x - 90)$ b) $g(x) = 2 \sin 5x$.

1.2 Addition and subtraction formulas



Activity 1.2

Step 1: Observe the figure below where ADE is a right-angled triangle with hypotenuse AD with length = 1 unit of length.



Step 2: Draw a right-angled triangle ACD rectangle in C. Then, find the length of AC and DC in function of the angle $\angle DAC = \beta$.

Step 3: Draw the right-angled triangle ACB rectangle in C and deduce the length of the side BC and AB in function of the angle $\angle CAB = \alpha$.

Step 4: It is clear that

$AE = \sin(\alpha + \beta) = BC + FD$. Express $\sin(\alpha + \beta)$ in function of α and β .

Step 5: Given that $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$, use the results from step 4 and deduce $\sin(\alpha - \beta)$.



Content summary

Based on the activity, to prove the sum and difference identities for sine, we will use the co-function identities:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \text{ and}$$

$$\sin(\alpha - \beta) = \sin(\alpha + (-\beta)) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

Sum and difference identities for tangent:

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)} = \frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta + \sin \alpha \sin \beta} = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Sum and difference identities for cotangent:

$$\cot(\alpha + \beta) = \frac{\cos(\alpha + \beta)}{\sin(\alpha + \beta)} = \frac{\cos \alpha \cos \beta - \sin \alpha \sin \beta}{\sin \alpha \cos \beta + \cos \alpha \sin \beta} = \frac{\cot \alpha \cot \beta - 1}{\cot \beta + \cot \alpha}$$

$$\cot(\alpha - \beta) = \frac{\cos(\alpha - \beta)}{\sin(\alpha - \beta)} = \frac{\cos \alpha \cos \beta + \sin \alpha \sin \beta}{\sin \alpha \cos \beta - \cos \alpha \sin \beta} = \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha}$$

Example 1: Use addition and subtraction formulae to find $\cos 75^\circ$

Solution

$$\cos 75^\circ = \cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Example 2:

Use addition and subtraction formulae to find $\sin \frac{\pi}{12}$

Solution

$$\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \sin \frac{\pi}{4} \cos \frac{\pi}{6} - \cos \frac{\pi}{4} \sin \frac{\pi}{6} = \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

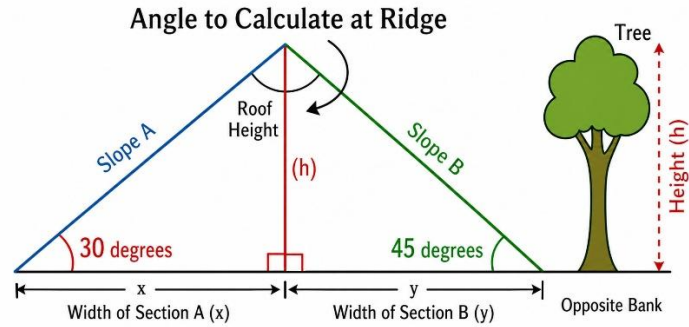


Application activity 1.2

1) Imagine you are an architect designing a roof with two different sloped sections that meet at a ridge.

One section of the roof has a slope of 30° , and the other has a slope of 45° .

You need to calculate the angle between the two sections of the roof at the ridge to ensure proper alignment and structural integrity.



2) Use addition formula to find $\sin 75^\circ$

1.3 Double angle formulae



Activity 1.3

For each of the following relations, replace y by x and give your result.

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\sin(x + y) = \sin x \cos y + \cos y \sin x$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}, \quad \cot(x + y) = \frac{\cot x \cot y - 1}{\cot y + \cot x}.$$



Content summary

This relation of trigonometric function of an angle and trigonometric function of its double angle is called the **fundamental relation of trigonometry**.

From this relation we can write $\cos^2 x = 1 - \sin^2 x$ and $\sin^2 x = 1 - \cos^2 x$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\cos 2x = 2 \cos^2 x - 1$$

$$\cos 2x = 1 - 2 \sin^2 x$$

$$\sin 2x = 2 \cos x \sin x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} \text{ and } \cot 2x = \frac{\cot^2 x - 1}{2 \cot x}$$

Example 1: Express $\cos 4x$ in function of $\sin x$ only

Solution

$$\begin{aligned}
 \cos 4x &= \cos 2(2x) = \cos^2 2x - \sin^2 2x = 1 - \sin^2 2x - \sin^2 2x = 1 - 2\sin^2 2x \\
 &= 1 - 2(2\sin x \cos x)^2 = 1 - 8\sin^2 x \cos^2 x = 1 - 8\sin^2 x(1 - \sin^2 x) \\
 &= 1 - 8\sin^2 x + 8\sin^4 x
 \end{aligned}$$

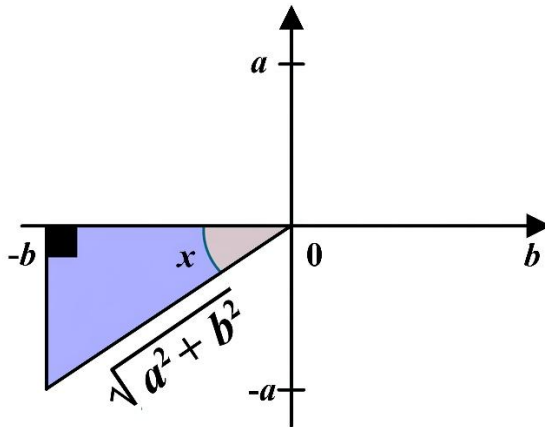
Example 2: Given that $\tan \frac{x}{b}$ and $\pi \leq x \leq \frac{3\pi}{2}$, evaluate

a) $\sin 2x$

b) $\tan 2x$

Solution

The given information produces the triangle shown below. Note the signs associated with a and b . The Pythagorean Theorem is used to find the hypotenuse.



a) $\sin 2x = 2\sin x \cos x = 2 \frac{-a}{\sqrt{a^2 + b^2}} \frac{-b}{\sqrt{a^2 + b^2}} = \frac{2ab}{\sqrt{a^2 + b^2}}$

b) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} = \frac{2 \frac{a}{b}}{1 - \left(\frac{a}{b}\right)^2} = \frac{2ab}{b^2 - a^2}.$



Application activity 1.3

- Express $\sin 4x$ in function of $\sin x$ and $\cos x$
- Express $\cos 8x$ in function of $\sin x$ and $\cos x$
- Evaluate exactly $2 \sin 15^\circ \cos 15^\circ$
- If $\cos A = \frac{\sqrt{2}+1}{2\sqrt{2}}$, find $\cos 2A$.
- If $\sin A = \frac{\sqrt{5}}{5}$, find $\sin 2A$, $\cos 2A$ and $\tan 2A$ if A
 - is acute
 - is obtuse
- Prove that $\sqrt{2\sqrt{2+\sqrt{2+2\cos 8x}}} = 2\cos x$.

1.3. Half-angle formulas



Activity 1.3

- Show that $\cos 2x = 1 - 2\sin^2 x$. By letting $\theta = 2x$, deduce the value of $\sin \frac{\theta}{2}$
- Show that $\cos 2x = 2\cos^2 x - 1$. By letting $\theta = 2x$, deduce the value of $\cos \frac{\theta}{2}$.
- Using results in 1 and 2, deduce the value of $\tan \frac{\theta}{2}$.



Content summary

Note that:

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}. \text{ We can write } \sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}.$$

$$\text{Also } \cos \frac{\theta}{2} = \pm \sqrt{\frac{\cos \theta + 1}{2}}. \text{ We can write } \cos \frac{x}{2} = \pm \sqrt{\frac{\cos x + 1}{2}}.$$

The half angle formulae are

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}} \text{ and } \cos \frac{x}{2} = \pm \sqrt{\frac{\cos x + 1}{2}}$$

$$\tan \frac{x}{2} = \frac{1 - \cos x}{\sin x} \text{ or } \tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} \text{ or } \tan \left(\frac{x}{2} \right) = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}}$$

$$\cot \left(\frac{x}{2} \right) = \pm \sqrt{\frac{1 + \cos x}{1 - \cos x}} \text{ or } \cot \left(\frac{x}{2} \right) = \frac{1 + \cos x}{\sin x} \text{ or } \cot \left(\frac{x}{2} \right) = \frac{\sin x}{1 - \cos x}$$

Example 1:

Using the half angle formula, find the exact value of $\cos 15^\circ$

Solution

15° is in first quadrant, then $\cos 15^\circ$ must be positive

$$\cos 15^\circ = \cos \left(\frac{1}{2} (30^\circ) \right) = \sqrt{\frac{1 + \cos 30^\circ}{2}} = \sqrt{\frac{1 + \frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2 + \sqrt{3}}{2}}$$



Application activity 1.4

1. If $\cos A = -\frac{7}{25}$, find the values of $\sin \frac{1}{2}A$, $\cos \frac{1}{2}A$ and $\tan \frac{1}{2}A$.
2. If $\tan 2A = \frac{7}{24}$, $0 < A < \frac{\pi}{4}$, find the value of $\tan A$.
3. Find the value of $\sin 22\frac{1}{2}^\circ$, $\cos 22\frac{1}{2}^\circ$ and $\tan 22\frac{1}{2}^\circ$.
4. Find the value of $\sin 7\frac{1}{2}^\circ$.

1.5 Transformation of product in sum and difference



Activity 1.5

From addition and subtraction formulae, evaluate two of the following:

1. $\sin(x + y) + \sin(x - y)$
2. $\sin(x + y) - \sin(x - y)$

3. $\cos(x + y) + \cos(x - y)$

4. $\cos(x + y) - \cos(x - y)$



Content summary

The formulae for transforming product into sum are

$$\cos x \cos y = \frac{1}{2} [\cos(x + y) + \cos(x - y)], \quad \sin x \sin y = -\frac{1}{2} [\cos(x + y) - \cos(x - y)]$$

$$\sin x \cos y = \frac{1}{2} [\sin(x + y) + \sin(x - y)], \quad \cos x \sin y = \frac{1}{2} [\sin(x + y) - \sin(x - y)]$$

Example:

Transform in sum the product $\sin 3x \cos 4x$.

Solution

$$\sin 3x \cos 4x = \frac{1}{2} [\sin(3x + 4x) + \sin(3x - 4x)] = \frac{1}{2} [\sin(7x) + \sin(-x)] = \frac{1}{2} [\sin 7x - \sin x]$$



Application activity 1.5

1. Transform in sum

a) $\sin x \cos 3x$

b) $\cos 12x \sin 9x$

c) $\sin 9x \sin 11x$

d) $2 \cos 5x \sin 3x$

e) $\cos \frac{5x}{2} \sin \frac{3x}{2}$

2. Prove that

a) $\sin(45^\circ + A) \sin(45^\circ - A) = \frac{1}{2} \cos 2A$.

b) $\sin 10^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{8}$.

1.6 Transformation of sum in product



Activity 1.6

Using the relations $x + y = p$ and $x - y = q$, express each of the formulae for transforming product in sum, in function of p and q .

$$\text{a) } \cos x \cos y = \frac{1}{2} [\cos(x + y) + \cos(x - y)] \quad \text{b) } \sin x \sin y = -\frac{1}{2} [\cos(x + y) - \cos(x - y)]$$



Content summary

The formulas for transforming sum into product are:

$$\cos p + \cos q = 2 \cos \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \cos p - \cos q = -2 \sin \frac{p+q}{2} \sin \frac{p-q}{2}$$

$$\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}, \quad \sin p - \sin q = 2 \cos \frac{p+q}{2} \sin \frac{p-q}{2}$$

Example: Transform in product the sum $\sin 3x + \sin 4x$

Solution: $\sin 3x + \sin 4x = 2 \sin \frac{3x+4x}{2} \cos \frac{3x-4x}{2} = 2 \sin \frac{7x}{2} \cos \frac{-x}{2} = 2 \sin \frac{7x}{2} \cos \frac{x}{2}.$



Application activity 1.6

1) Transform in product

- a) $\cos x + \cos 7x$
- b) $\sin 4x - \cos 9x$
- c) $\sin 3x + \sin x$
- d) $\cos 2x - \cos 4x$

2) Prove that

- a) $\frac{\sin A + \sin B}{\cos A + \cos B} = \tan \frac{A+B}{2}.$
- b) $\frac{\cos 2B - \cos 2A}{\sin 2A + \sin 2B} = \tan(A-B).$
- c) $\frac{\sin 5x - 2 \sin 3x + \sin x}{\cos 5x - \cos x} = -\tan x.$

1.7 Introduction to trigonometric equation



Activity 1.7

1. Find at least three angles whose sine is $\frac{1}{2}$.
2. Find at least three angles whose cosine is $\frac{\sqrt{2}}{2}$.
3. Find at least three angles whose tangent is $\frac{\sqrt{3}}{3}$.



Content summary

A trigonometric equation is an equation that involves a trigonometric function of a variable, usually an angle x , such as:

$\sin x = a$, $\cos x = b$, $\tan x = c$ where a , b and c are real numbers.

Trigonometric equations are used to:

- Model periodic phenomena (waves, oscillations, rotations).
- Solve problems in physics, engineering, and geometry.
- Determine angles satisfying specific trigonometric relationships.

Solution Methods:

- **Unit circle method:** visually find angles corresponding to a trigonometric value.
- **Algebraic method:** isolate the trig function and apply inverse functions.
- **Use of identities:** simplify complex equations (e.g., $\sin^2 x = 1 - \cos^2 x$).

The solutions of a trigonometric equation for which $0 \leq x \leq 2\pi$, are called **principle solutions** while the expression (involving integer k) of solution containing all values of the unknown angle is called the **general solution** of the trigonometric equation. When the interval of solution is not given, you are required to find general solution.

Also recall the following identities:

$$\sin \alpha = \sin(\alpha + 2k\pi), \quad k \in \mathbb{Z}$$

$$\sin \alpha = \sin(\pi - \alpha)$$

$$\cos \alpha = \cos(\alpha + 2k\pi), \quad k \in \mathbb{Z}$$

$$\cos \alpha = \cos(-\alpha)$$

$$\tan \alpha = \tan(\alpha + k\pi), \quad k \in \mathbb{Z}$$

$$\tan \alpha = \tan(\alpha + \pi)$$

Example 1:

Find the principal solutions of the following equations

$$\text{a) } \cos x = -\frac{\sqrt{3}}{2} \quad \text{b) } \sin x = \frac{1}{\sqrt{2}} \quad \text{c) } \sec x = 2$$

Solution

$$\text{a) } \cos x = -\frac{\sqrt{3}}{2} \text{ is negative} \Rightarrow x \text{ lies in the 2}^{\text{nd}} \text{ or 3}^{\text{rd}} \text{ quadrant.}$$

$$\text{Here } \cos x = -\cos \frac{\pi}{6} = \cos \left(\pi - \frac{\pi}{6} \right) \text{ or } \cos \left(\pi + \frac{\pi}{6} \right).$$

$$\Rightarrow \cos x = \cos \frac{5\pi}{6} \text{ or } \cos \frac{7\pi}{6}.$$

$$\text{Thus } x = \frac{5\pi}{6} \text{ or } x = \frac{7\pi}{6}.$$

$$\text{The principal solutions are } x = \frac{5\pi}{6} \text{ or } x = \frac{7\pi}{6}.$$

$$\text{b) } \sin x = \frac{1}{\sqrt{2}} \text{ is positive which implies } x \text{ lies in the 1}^{\text{st}} \text{ or 2}^{\text{nd}} \text{ quadrant.}$$

$$\text{Here } \sin x = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4} \text{ or } \sin \left(\pi - \frac{\pi}{4} \right).$$

$$\Rightarrow x = \frac{\pi}{4} \text{ or } x = \frac{3\pi}{4}.$$

$$\text{Thus } x = \frac{\pi}{4} \text{ or } x = \frac{3\pi}{4}.$$

$$\text{c) } \sec x = 2 \Leftrightarrow \frac{1}{\cos x} = 2, \sec x = 2 \Rightarrow \frac{1}{\cos x} = 2 \Rightarrow \cos x = \frac{1}{2}, \text{ since it is positive, thus } x \text{ lies in the 1}^{\text{st}} \text{ or 4}^{\text{th}} \text{ quadrant.}$$

$$\cos x = \frac{1}{2} = \cos \frac{\pi}{3} \quad \text{or} \quad \cos \left(2\pi - \frac{\pi}{3} \right) \Rightarrow x = \frac{\pi}{3} \quad \text{or} \quad x = \frac{5\pi}{3}$$

The principal solutions are $x = \frac{5\pi}{6}$ or $x = \frac{7\pi}{6}$.

Example 2: Solve for x in the set of real numbers: $\sin x = \frac{1}{2}$.

Solution

$$\sin x = \frac{1}{2}$$

$$\sin^{-1}(\sin x) = \sin^{-1}\left(\frac{1}{2}\right) \Rightarrow x = \sin^{-1}\left(\frac{1}{2}\right)$$

Here we need to know the angle whose sine is $\frac{1}{2}$. Using table of remarkable angles or

scientific calculator we find that $\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$ or $\frac{\pi}{6}$, then $x = \frac{\pi}{6}$.

This is not the only solution since we know that $\sin \alpha = \sin(\pi - \alpha)$.

So $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$ is another solution. Also $\sin \alpha = \sin(\alpha + 2k\pi)$.

Then in general we write $x \in \left\{ \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \right\}$.



Application activity 1.7

Find the principal solutions of the following equations:

a) $\sin x = \frac{\sqrt{3}}{2}$ b) $\tan x = -\sqrt{3}$ c) $\cot x = \frac{1}{\sqrt{3}}$

d) $\sqrt{3} \tan x = 1$ e) $2 \sin x - \sqrt{3} = 0$

1.8 Trigonometric equation reducible to the form: $\sin(x + \alpha) = k$, $\cos(x + \alpha) = k$, and $\tan(x + \alpha) = b$ for $|k| \leq 1$ and b a real number ($b \in \mathbb{R}$).



Activity 1.8

Solve:

(a) $\sin x \cos 45^\circ + \cos x \sin 45^\circ = \frac{\sqrt{2}}{2}$

(b) $\cos \alpha \cos 15^\circ + \sin \alpha \sin 15^\circ = 1$

(c) $\frac{\tan \theta + \tan 10^\circ}{1 - \tan \theta \tan 10^\circ} = \sqrt{3}$



Content summary

The first step in finding the solution is to match the structure of the given expression to a known compound angle formula. Look for patterns in the functions (sine, cosine, tangent, cotangent, secant, or cosecant) and the algebraic operation plus or minus.

Example 1: Solve the equation $\cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ = \frac{1}{2}$ for $-180^\circ < \theta < 180^\circ$.

Solution

Since $\cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ = \cos(\theta + 30^\circ)$

$$\cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ = \frac{1}{2} \Leftrightarrow \cos(\theta + 30^\circ) = \frac{1}{2}$$

That is $\theta + 30^\circ = \begin{cases} -60^\circ + 360^\circ k \\ 60^\circ + 360^\circ k \end{cases}, k \in \mathbb{Z}.$

So, $\theta = -90^\circ$ or 30° , remembering the given condition.

Example 2: Solve $\sin^2 x + \sin x \cos x = 0$ for $0^\circ \leq x \leq 360^\circ$.

Solution

$$\sin^2 x + \sin x \cos x = 0$$

$$\sin x(\sin x + \cos x) = 0$$

Either $\sin x = 0$ or $\sin x + \cos x = 0$.

$$\sin x = 0 \Rightarrow x = 180^\circ k, k \in \mathbb{Z}.$$

Considering the given condition:

$$k = 0, x = 0^\circ \quad k = 1, x = 180^\circ \quad k = 2, x = 360^\circ.$$

$$\sin x + \cos x = 0 \Rightarrow \tan x = -1 \Rightarrow x = -45^\circ + 180^\circ k, k \in \mathbb{Z}.$$

Considering the given condition:

$$k = 0: x = -45^\circ, \quad k = 1: x = -45^\circ + 180^\circ = 135^\circ, \quad k = 2: x = -45^\circ + 360^\circ = 315^\circ.$$

$$\text{Thus, } x \in S = \{0^\circ, 135^\circ, 180^\circ, 315^\circ, 360^\circ\}$$



Application activity 1.8

(1) Solve $2\cos^2 x - 5\sin x + 1 = 0$ for $x \in [0, \pi]$.

(2) Solve the equation $\cos^2 x = 1 - \sin^2 x$

1.9 Trigonometric equation reducible to the form $\sin nx = k$



Activity 1.9

Find general solution for each of the following trigonometric equations

a) $\cos 2x = \frac{1}{\sqrt{2}}$

b) $\sin \frac{x}{2} = -\frac{1}{2}$

c) $\sin mx + \sin nx = 0$ (Remember transformation of sum into product)

d) $\cos 4x - \cos 2x = 0$



Content summary

For the trigonometric equation reducible to the form $\sin nx = k$, remember to divide the period by n when determining the general solution.

Example 1: Solve in the set of real numbers $\cos 2x = \frac{\sqrt{3}}{2}$.

Solution: $\cos 2x = \frac{\sqrt{3}}{2}$

$$\cos^{-1}(\cos 2x) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \Leftrightarrow 2x = \begin{cases} \frac{\pi}{6} + 2k\pi \\ -\frac{\pi}{6} + 2k\pi \end{cases}, k \in \mathbb{Z}$$

[since $\cos \alpha = \cos(-\alpha)$ and $\cos \alpha = \cos(\alpha + 2k\pi)$]

$$\Rightarrow x = \begin{cases} \frac{\pi}{12} + k\pi \\ -\frac{\pi}{12} + k\pi \end{cases}, k \in \mathbb{Z}$$

Example 2: Solve in the set of real numbers $\sin 3x + \sin 5x = 0$

Solution

Transform $\sin 3x + \sin 5x$ in product:

$$\sin 3x + \sin 5x = 2 \sin \frac{8x}{2} \cos \left(-\frac{2x}{2}\right) = 2 \sin 4x \cos x$$

$$\sin 3x + \sin 5x = 0 \Leftrightarrow 2 \sin 4x \cos x = 0 \Leftrightarrow \sin 4x \cos x = 0 \Leftrightarrow \sin 4x = 0 \text{ or } \cos x = 0$$

$$\sin 4x = 0$$

Since sine is zero at 0 and π , we can write

$$\sin 4x = 0 \Leftrightarrow 4x = k\pi, k \in \mathbb{Z} \Rightarrow x = \frac{k\pi}{4}, k \in \mathbb{Z}$$

$$\cos x = 0$$

Since cosine is zero at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$, we can write $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$.

$$\text{Then, } x = \begin{cases} \frac{k\pi}{4} \\ \frac{\pi}{2} + k\pi \end{cases}, k \in \mathbb{Z}$$

Example 3: Solve in the set of real numbers $\tan 3x = 1$

Solution: $\tan 3x = 1$

$$\tan^{-1}(\tan 3x) = \tan^{-1}(1) \Leftrightarrow 3x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z} \quad [\text{since } \tan \alpha = \tan(\alpha + 2k\pi)]$$

$$\Rightarrow x = \frac{\pi}{12} + \frac{k\pi}{3}, k \in \mathbb{Z}$$

$$\text{Hence, } x = \frac{\pi}{12} + \frac{k\pi}{3}, k \in \mathbb{Z}.$$

Example 4:

$$\text{Solve } \sin \frac{x}{3} = -\frac{\sqrt{3}}{2} \text{ for } x \in [0, 2\pi]$$

Solution

$$\sin \frac{x}{3} = -\frac{\sqrt{3}}{2} \text{ for } x \in [0, 2\pi]$$

$$\frac{x}{3} = \begin{cases} -\frac{\pi}{3} + 2k\pi \\ \pi - \left(-\frac{\pi}{3}\right) + 2k\pi \end{cases}$$

$$\frac{x}{3} = \begin{cases} -\frac{\pi}{3} + 2k\pi \\ \frac{4\pi}{3} + 2k\pi \end{cases} \Rightarrow x = \begin{cases} -\pi + 6k\pi \\ 4\pi + 6k\pi \end{cases}$$

Since we are given the condition $x \in [0, 2\pi]$, we need to substitute k with some integers

(..., -2, -1, 0, 1, 2, ...). But doing this no value can be found in the given interval.

Thus, there is no solution, $S = \emptyset$

Example 5:

$$\text{Solve the equation } \sin 3x = \frac{1}{2} \text{ for } x \in [0, 2\pi].$$

Solution

$$\sin 3x = \frac{1}{2} \Rightarrow 3x = \begin{cases} \frac{\pi}{6} + 2k\pi \\ \frac{5\pi}{6} + 2k\pi \end{cases}, k \in \mathbb{Z}, \quad x = \begin{cases} \frac{\pi}{18} + \frac{2}{3}k\pi \\ \frac{5\pi}{18} + \frac{2}{3}k\pi \end{cases}, k \in \mathbb{Z}.$$

As the condition is $x \in [0, 2\pi]$, we need to find all possible values in the given interval. Since we need the positive angles, we will take k to be a positive integer or zero.

$$k = 0 \Rightarrow x = \begin{cases} \frac{\pi}{18} \\ \frac{5\pi}{18} \end{cases}, k = 1 \Rightarrow x = \begin{cases} \frac{\pi}{18} + \frac{2}{3}\pi = \frac{13\pi}{18} \\ \frac{5\pi}{18} + \frac{2}{3}\pi = \frac{17\pi}{18} \end{cases},$$

$$k = 2 \Rightarrow x = \begin{cases} \frac{\pi}{18} + \frac{4}{3}\pi = \frac{25\pi}{18} \\ \frac{5\pi}{18} + \frac{4}{3}\pi = \frac{29\pi}{18} \end{cases}, k = 3 \Rightarrow x = \begin{cases} \frac{\pi}{18} + \frac{6}{3}\pi = \frac{37\pi}{18} \\ \frac{5\pi}{18} + \frac{6}{3}\pi = \frac{41\pi}{18} \end{cases}.$$

For $k = 3$, the obtained angles fall out of the given interval. Looking at the obtained angles for $k = 3$ we see that they are equivalent to the angles obtained for $k = 0$. This means that if we continue, we will find the angles equivalent to previous angles.

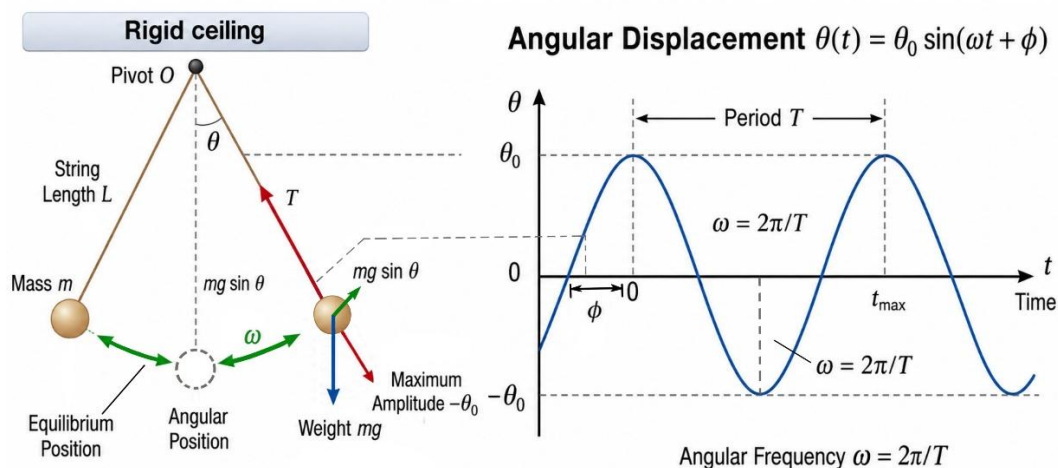
Thus, $x = \left\{ \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18} \right\}$.



Application activity 1.9

1) Solve: in the set of real numbers $\tan 6x = 1$

2) A simple pendulum of length L is oscillating with an angular frequency ω .



The displacement of the pendulum from its equilibrium position as a function of time t is described by the equation: $\theta(t) = \theta_0 \sin(\omega t + \phi)$, where:

- θ_0 is the maximum angular displacement (amplitude),
- ω is the angular frequency,
- ϕ is the phase angle,
- t is the time.

Assuming $\theta_0 = 30^\circ$, $\omega = 2 \text{ rad/s}$, and $\phi = 0$, determine the time at which the displacement $\theta(t)$ is 15° .

1.10 Trigonometric equation of the form $a \sin x + b \cos x = c$



Activity 1.10

Consider the equation $\sqrt{3} \cos x - \sin x = \sqrt{3}$.

1. Divide each term by $\sqrt{3}$.
2. Letting $\tan \alpha$ be the coefficient of $\cos x$, obtained in 1, find the value of α ($-\pi < \alpha \leq \pi$).
3. Replace the coefficient of $\cos x$, obtained in 1, by $\frac{\sin \alpha}{\cos \alpha}$ and find the expression of $\cos \alpha$.
4. Use addition formula to find new equation and deduce the value(s) of x .



Content summary

One of the methods of solving the equation $a \sin x + b \cos x = c$, we start by dividing each side by a . That is, $\sin x + \frac{b}{a} \cos x = \frac{c}{a}$. Now let $\tan \alpha = \frac{b}{a} \Rightarrow \alpha = \tan^{-1} \left(\frac{b}{a} \right)$.

Replace $\frac{b}{a}$ by $\frac{\sin \alpha}{\cos \alpha}$, multiply each side by $\cos \alpha$ and then use addition formula.

Example: Solve $3 \sin x + \sqrt{3} \cos x = 3$.

Solution: $3 \sin x + \sqrt{3} \cos x = 3 \Leftrightarrow \sin x + \frac{\sqrt{3}}{3} \cos x = 1$.

Let $\tan \alpha = \frac{\sqrt{3}}{3} \Rightarrow \alpha = \frac{\pi}{6}$.

$$\sin x + \frac{\sin \alpha}{\cos \alpha} \cos x = 1 \Leftrightarrow \sin x \cos \alpha + \sin \alpha \cos x = \cos \alpha \Leftrightarrow \sin(x + \alpha) = \cos \alpha$$

$$\Leftrightarrow \sin\left(x + \frac{\pi}{6}\right) = \cos \frac{\pi}{6} \Leftrightarrow \sin\left(x + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$x + \frac{\pi}{6} = \begin{cases} \frac{\pi}{3} + 2k\pi \\ \frac{2\pi}{3} + 2k\pi \end{cases} \Rightarrow x = \begin{cases} \frac{\pi}{6} + 2k\pi \\ \frac{\pi}{2} + 2k\pi \end{cases}, \quad k \in \mathbb{Z}$$



Application activity 1.10

Solve in $\mathbb{R}$

1. $\cos x + \sqrt{3} \sin x = \sqrt{3}$
2. $\cos x + \sin x = \sqrt{2}$
3. $\cos x - \sin x = -1$
4. $\sqrt{3} \cos x + \sin x = \sqrt{2}$
5. $2 \sin x + \sqrt{3} \cos x = 1 + \sin x$
6. $\sqrt{2} \sec x + \tan x = 1$

1.11 Trigonometric inequalities



Activity 1.11

On a trigonometric circle, shade the region containing the angle whose

1) sine is less than 0

2) cosine is greater than or equal to $\frac{1}{2}$

3) sine is greater than $\frac{\sqrt{3}}{2}$

4) cosine is less than or equal to $-\frac{\sqrt{2}}{2}$

..

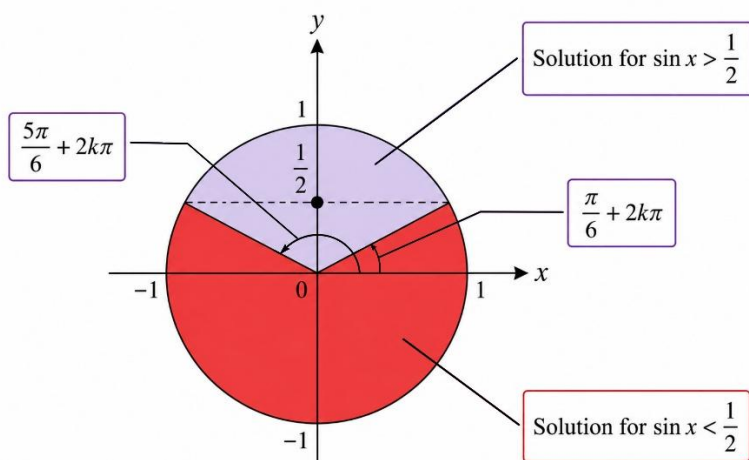


Content summary

When solving inequalities, first replace the inequality sign by equal sign and then solve. Find all no equivalent angles in $[0, 2\pi]$. Place these angles on a trigonometric circle. They will divide the circle into arcs. Choose the arcs containing the angles corresponding to the given inequality.

Example 1: Solve $\sin x < \frac{1}{2}$ and $\sin x > \frac{1}{2}$.

Solution: $\sin x = \frac{1}{2} \Rightarrow x = \begin{cases} \frac{\pi}{6} + 2k\pi \\ \frac{5\pi}{6} + 2k\pi \end{cases}$



Solution for $\sin x < \frac{1}{2}$

$$x \in \left] 0 + 2k\pi, \frac{\pi}{6} + 2k\pi \right[\cup \left] \frac{5\pi}{6} + 2k\pi, 2\pi + 2k\pi \right[, \quad k \in \mathbb{Z}$$

Or

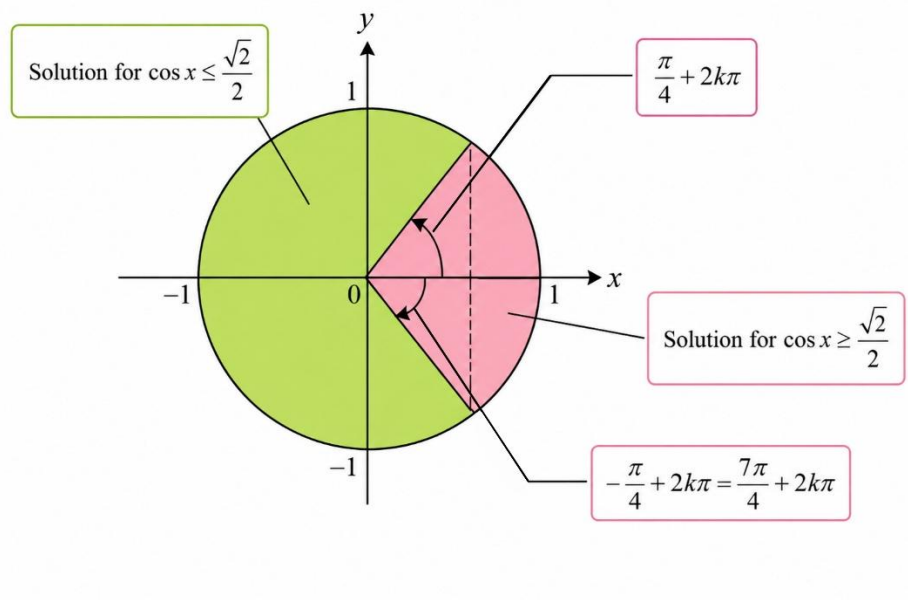
$$x \in \left] 0 + 2k\pi, \frac{\pi}{6} + 2k\pi \right[\cup \left] \frac{5\pi}{6} + 2k\pi, 2\pi + 2k\pi \right[, \quad k \in \mathbb{Z}$$

Solution for $\sin x > \frac{1}{2}$, $x = \left] \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \right[, \quad k \in \mathbb{Z}.$

Example 2: Solve $\cos x \geq \frac{\sqrt{2}}{2}$ for $x \in [0, 2\pi]$

Solution:

$$\cos x = \frac{\sqrt{2}}{2}, \quad x = \pm \frac{\pi}{4}$$



Since we are given the condition $x \in [0, 2\pi]$ the angle $-\frac{\pi}{4}$ will be replaced by its positive equivalent angle in the given interval, which is $\frac{7\pi}{4}$.

Thus, $S = \left[0, \frac{\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right]$

If the condition was not given the answer should be

$$S = \left[-\frac{\pi}{4} + 2k\pi, \frac{\pi}{4} + 2k\pi\right]$$

Example 3: Solve :

a) $\sin 2x \leq \frac{1}{2}$ b) $\sin 2x \geq \frac{1}{2}$.

Solution

a) $\sin 2x \leq \frac{1}{2}$

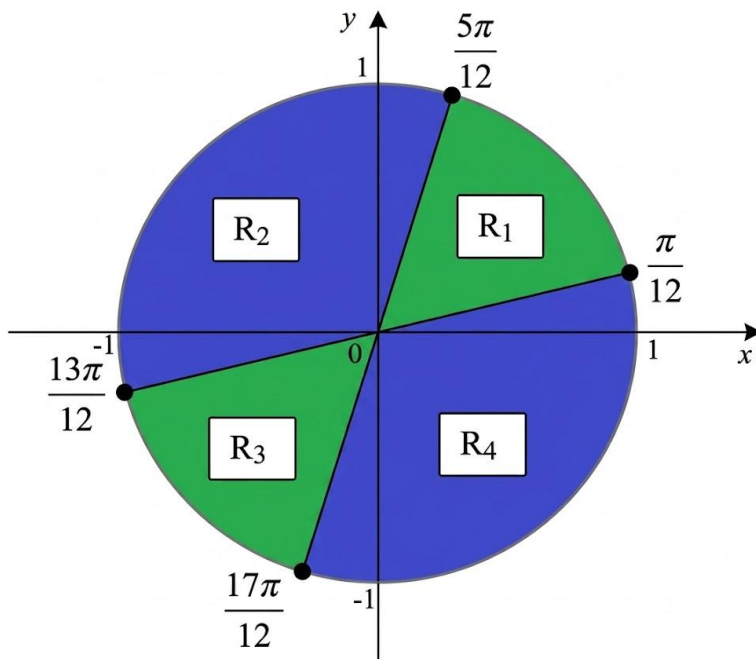
$$\sin 2x = \frac{1}{2}$$

$$2x = \begin{cases} \frac{\pi}{6} + 2k\pi \\ \frac{5\pi}{6} + 2k\pi \end{cases} \Rightarrow x = \begin{cases} \frac{\pi}{12} + k\pi \\ \frac{5\pi}{12} + k\pi \end{cases}$$

It is clear that there are more than two values in the interval $[0, 2\pi]$. To see them substitute k by difference integers starting by 0:

$$k=0, x = \begin{cases} \frac{\pi}{12} \\ \frac{5\pi}{12} \end{cases}, k=1, x = \begin{cases} \frac{\pi}{12} + \pi = \frac{13\pi}{12} \\ \frac{5\pi}{12} + \pi = \frac{17\pi}{12} \end{cases}, k=2, x = \begin{cases} \frac{\pi}{12} + 2\pi = \frac{25\pi}{12} \\ \frac{5\pi}{12} + 2\pi = \frac{29\pi}{12} \end{cases}.$$

The values that fall in the interval $[0, 2\pi]$ are $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}$ and $\frac{17\pi}{12}$.



We have four regions R_1, R_2, R_3 and R_4 . To find the regions that contain the solution we will take one value in each region and check if it satisfies the given inequality or not.

Start with the inequality $\sin 2x \leq \frac{1}{2} \therefore$

➤ For R_1 , take for example $\frac{\pi}{4}$:

$$\sin 2\frac{\pi}{4} = \sin \frac{\pi}{2} = 1 > \frac{1}{2}$$

It is inconstant to the given inequality.

➤ For R_2 , take for example $x = \pi$:

$$\sin 2\pi = 0 < \frac{1}{2}$$

It satisfies the given inequality.

➤ For R_3 , take for example $x = \frac{15\pi}{12}$.

$$\sin 2\frac{5\pi}{4} = \sin \frac{5\pi}{2} = 1 > \frac{1}{2}$$

It is inconstant to the given inequality.

➤ For R_4 , take for example $x = 0$: $\sin 0 = 0 < \frac{1}{2}$.

It satisfies the given inequality.

Then, the solution of the inequality $\sin 2x \leq \frac{1}{2}$ is the set of all angles found in region 2 and region 4. Remember that since no condition given, we will use equivalent angles property. That is

$$S = \left[2k\pi, \frac{\pi}{12} + 2k\pi \right] \cup \left[\frac{5\pi}{12} + 2k\pi, \frac{13\pi}{12} + 2k\pi \right] \cup \left[\frac{17\pi}{12} + 2k\pi, 2\pi(1+k) \right], k \in \mathbb{Z}$$

b. From the trigonometric circle in a), the solution of the inequality $\sin 2x \geq \frac{1}{2}$ is the set of all angles found in region 1 and region 2.

That is

$$S = \left[\frac{\pi}{12} + 2k\pi, \frac{5\pi}{12} + 2k\pi \right] \cup \left[\frac{13\pi}{12} + 2k\pi, \frac{17\pi}{12} + 2k\pi \right], k \in \mathbb{Z}$$

Notice

We can also use sign table

x	$\frac{\pi}{12}$ $\frac{5\pi}{12}$ $\frac{13\pi}{12}$ $\frac{17\pi}{12}$ 2π											
$\sin 2x - \frac{1}{2}$	-	-	0	+	0	-	0	+	0	-	-	

Now it is simple to write the solution set considering the given inequality.

Example 4:

Solve $\sin x + \sin 3x < -\sin 2x$

Solution

$$\sin x + \sin 3x < -\sin 2x$$

$$\sin x + \sin 3x + \sin 2x < 0$$

$$2 \sin 2x \cos(-x) + \sin 2x < 0$$

$$2 \sin 2x \cos(-x) + \sin 2x < 0 \quad [\text{Sum into product}]$$

$$2 \sin 2x \cos x + \sin 2x < 0$$

$$\sin 2x(2 \cos x + 1) < 0$$

First, we solve for $\sin 2x(2 \cos x + 1) = 0$

$$\sin 2x = 0 \Rightarrow x = \frac{k\pi}{2}$$

$$2 \cos x + 1 = 0 \Rightarrow \cos x = -\frac{1}{2} \Rightarrow x = \pm \frac{2\pi}{3} + 2k\pi$$

To construct sign table, we need to know all values of x that belong to the interval $[0, 2\pi]$.

For the first case, we have $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

For the second case, we have $\frac{2\pi}{3}, \frac{4\pi}{3}$.

Sign table x	0	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	2π
$\sin 2x$	0	+	0	-	-	-	0
$2 \cos x + 1$		+	+	0	-	-	+
$\sin 2x(2 \cos x + 1)$	0	+	0	-	0	+	0

Thus, $S = \left] \frac{\pi}{2} + 2k\pi, \frac{2\pi}{3} + 2k\pi \right[\cup \left] \pi + 2k\pi, \frac{4\pi}{3} + 2k\pi \right[\cup \left] \frac{3\pi}{2} + 2k\pi, 2\pi + 2k\pi \right[$, $k \in \mathbb{Z}$

Example 5:

Solve $\tan x + \cot x < -4$ for $x \in [0, 2\pi]$

Solution

$$\tan x + \cot x < -4$$

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} + 4 < 0$$

$$\frac{\sin x \sin x + \cos x \cos x + 4 \cos x \sin x}{\cos x \sin x} < 0$$

$$\frac{1 + 2(2 \cos x \sin x)}{\frac{2 \cos x \sin x}{2}} < 0$$

$$\frac{1 + 2 \sin 2x}{\frac{\sin 2x}{2}} < 0$$

$$\frac{2(1 + 2 \sin 2x)}{\sin 2x} < 0$$

$$\text{Case 1: } 1 + 2 \sin 2x = 0 \Rightarrow \sin 2x = -\frac{1}{2}$$

$$2x = \begin{cases} -\frac{\pi}{6} + 2k\pi \\ \frac{7\pi}{6} + 2k\pi \end{cases} \Rightarrow x = \begin{cases} -\frac{\pi}{12} + k\pi \\ \frac{7\pi}{12} + k\pi \end{cases}$$

We need values in interval $[0, \pi]$.

$$k = 0, x = \begin{cases} -\frac{\pi}{12} \\ \frac{7\pi}{12} \end{cases}, \quad k = 1, x = \begin{cases} \frac{11\pi}{12} \\ \frac{19\pi}{12} \end{cases}, \quad k = 2, x = \begin{cases} \frac{23\pi}{12} \\ \frac{31\pi}{12} \end{cases}$$

We take $\frac{7\pi}{12}$ and $\frac{11\pi}{12}$

$$\text{Case 2: } \sin 2x = 0 \Rightarrow x = \frac{k\pi}{2}$$

$$k = 0 : x = 0, \quad k = 1 : x = \frac{\pi}{2}, \quad k = 2 : x = \pi, \quad k = 3 : x = \frac{3\pi}{2}.$$

We take $0, \frac{\pi}{2}$ and π but for these values there is no solution since they make the denominator to be zero.

Sign table:

x	0	$\frac{\pi}{2}$	$\frac{7\pi}{12}$	$\frac{11\pi}{12}$	π				
$1+2\sin 2x$	+	+	0	-	0	+	+		
$\sin 2x$	0	+	0	-	-	-	-	0	
$\frac{1+2\sin 2x}{\sin 2x}$		+		-	0	+	0	-	

Thus, $S = \left[\frac{\pi}{2}, \frac{7\pi}{12} \right[\cup \left] \frac{11\pi}{12}, \pi \right[$.



Application activity 1.11

Solve

- $\sin x \leq \frac{\sqrt{3}}{2}$
- $\sin x \cos 2x > 0$
- $\sin 3x < -\frac{1}{2}$
- $\cos^2 x \geq 1$

1.12 Applications of trigonometry



Activity 1.12

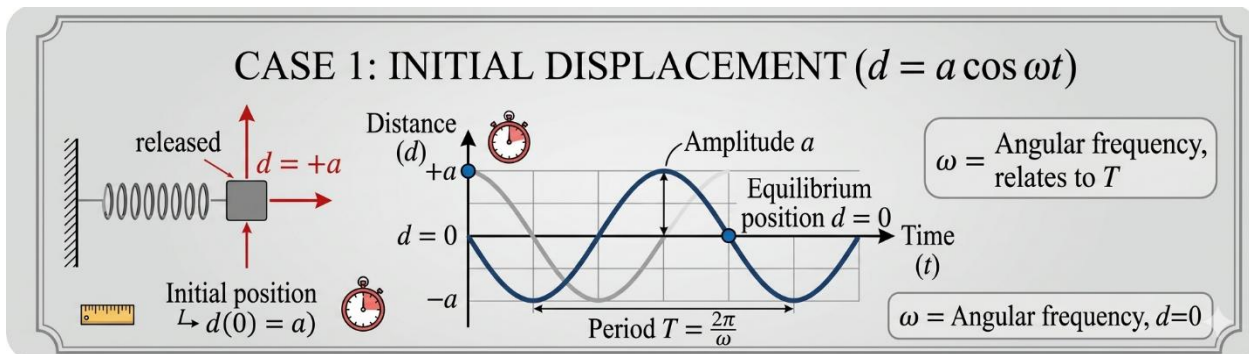
Discuss on how trigonometric theory is used in harmonic motion and refraction of a light.



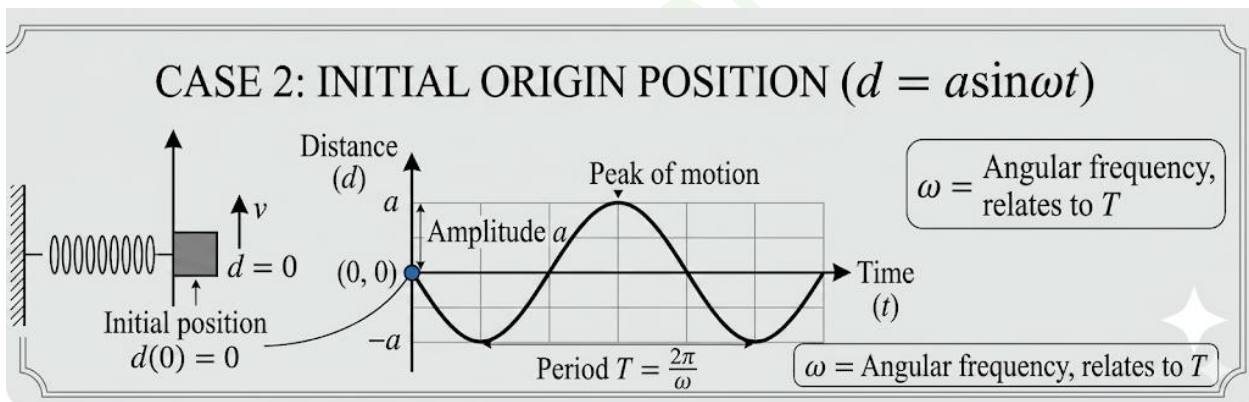
Content summary

Simple Harmonic motion:

An object that moves on a coordinate axis is in **simple harmonic motion** if its distance from the origin, d , at time t is given by either $d = a \cos \omega t$



or $d = a \sin \omega t$.



The motion has **amplitude** $|a|$, the maximum displacement of the object from its rest position. The period of the motion is $\frac{2\pi}{\omega}$, where $\omega > 0$.

The **period** gives the time it takes for the motion to go through one complete cycle.

In describing simple harmonic motion, the equation with the cosine function, $d = a \cos \omega t$, is used if the object is at its greatest distance from rest position, the origin, at $t = 0$.

By contrast, the equation with the sine function, $d = a \sin \omega t$, is used if the object is at its rest position, the origin at $t = 0$.

Example 1: If the instantaneous voltage in a current is given by the equation $E = 204 \sin 3680t$, where E is expressed in volts and t is expressed in seconds, find E if $t = 0.27$ seconds.

Solution

$$E = 204 \sin 3680t = 204 \sin [(3680)(0.27)] = 204 \sin 993.6E \approx 154 \text{ volts.}$$

Example 2:

The horizontal displacement d of the end of a pendulum is $d = K \sin 2\pi t$. Find K if $d = 12$ centimetres and $t = 3.25$ seconds.

Solution

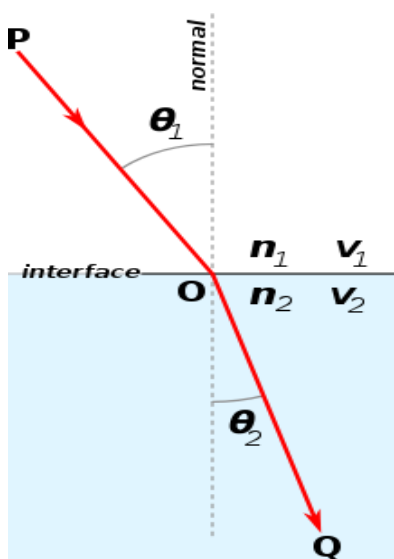
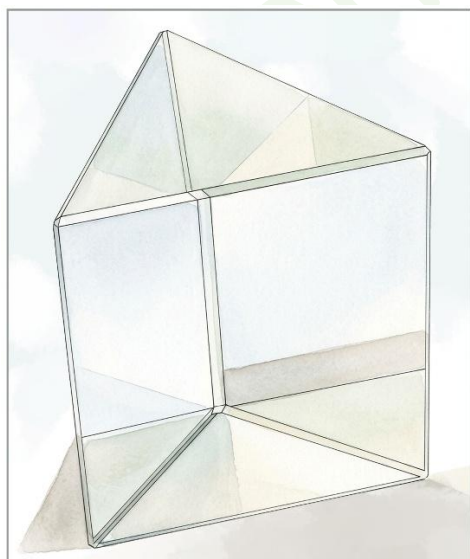
$$d = K \sin 2\pi t \Leftrightarrow 12 \approx K \sin [(2)(3.1415)(3.25)] \Leftrightarrow 12 \approx K \sin 20.42$$

$$K \approx \frac{12}{\sin 20.42} \approx 12$$

Refraction of a light

In optics, light changes speed as it moves from one medium to another (for example, from air into the glass of the prism). This speed change causes the light to be refracted and to enter the new medium at a different angle (Huygens principle).

A **prism** is a transparent optical element with flat, polished surfaces that refract light.



The degree of bending of the light's path depends on the angle that the incident beam of light makes with the surface, and on the ratio between the refractive indices of the two media (**Snell's law**).

If the light is traveling from a rarer region (lower n) to a denser region (higher n), it will bend **towards the normal** but if it is traveling from a denser region (higher n) to a rarer region (lower n), it will bend **away from the normal**.

Snell's law state that: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

Example 3:

Light travels from air into an optical fiber with an index of refraction of 1.44

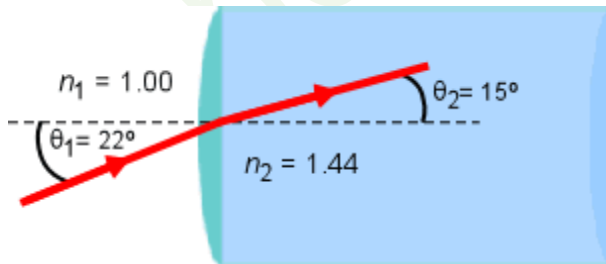
- (a) In which direction does the light bend?
- (b) If the angle of incidence on the end of the fiber is 22° , what is the angle of refraction inside the fiber?
- (c) Sketch the path of light as it changes media.

Solution

- a) Since the light is traveling from a rarer region (lower n) to a denser region (higher n), it will bend toward the normal.
- b) We will identify air as medium 1 and fiber as medium 2. Thus, $n_1 = 1.00$ (index of air), $n_2 = 1.44$ and $\theta_1 = 22^\circ$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \Leftrightarrow \sin 22 = 1.44 \sin \theta_2, \sin \theta_2 = \frac{\sin 22}{1.44}, \theta_2 = \sin^{-1}(0.26) \Rightarrow \theta_2 = 15^\circ$$

- c) The path of the light is shown in the figure below





Application activity 1.12

A ray of light is incident through glass, with refractive index 1.52, on an interface separating glass and water with refractive index 1.32. What is the angle of refraction if the angle of incidence of the ray in glass is 25° ?



END OF UNIT ASSESSMENT

Knowledge and understanding

1. State the sine and cosine addition formulas.
2. Derive the half-angle formula for sine and cosine.
3. Explain the relationship between the trigonometric functions of an angle in terms of the unit circle.

Skills

4. Express $\tan 4x$ in function of tangent
5. Given that θ is acute and $\sin \theta = \frac{5}{13}$, find the exact value of $\sin 2\theta$.
6. Prove the identity $\cot a - \tan a = 2 \cot 2a$
7. Find, without using calculator, the exact value of
 - a) $2 \sin 75^\circ \cos 75^\circ$
 - b) $\cos^2 \left(\frac{45^\circ}{2} \right) - \sin^2 \left(\frac{45^\circ}{2} \right)$
 - c) $\frac{2 \tan \left(\frac{135^\circ}{2} \right)}{1 - \tan^2 \left(\frac{135^\circ}{2} \right)}$
 - d) $1 - 2 \sin^2 15^\circ$
8. Transform in product
 - (a) $\cos 8x - \cos 9x$
 - (b) $\sin 3x + \sin 11x$
9. solve the given inequalities
 - (a) $\sin 5x < 0$

(b) $\cos 3x \leq \frac{\sqrt{3}}{2}$

(c) $\frac{2\cos x - 1}{2\cos x + 1} < 0$ for $0 \leq x \leq 2\pi$

10. Solve the following equations:

(a) $(2\sin x - 1)(\tan x - 1) = 0$ for $0 \leq x \leq \pi$

(b) $\cos 2x \cos x - \sin 2x \sin x = 0$ for $0 \leq x \leq 2\pi$.

Attitude and values

11) Why is it important to solve trigonometric equations step by step and check solutions within the specified range?

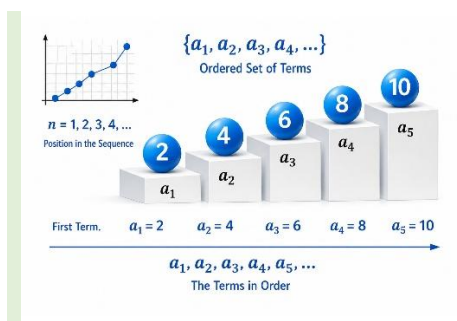
- A. To make the solution longer and more detailed
- B. To ensure accuracy and confirm that all solutions are valid within the given interval
- C. To avoid using calculators
- D. To simplify all equations into linear form

12. How can students develop persistence when solving complex trigonometric equations or identities?

- A. By skipping difficult steps and guessing answers
- B. By practicing regularly, breaking problems into smaller steps, and learning from mistakes
- C. By only solving easy problems first
- D. By memorizing formulas without understanding them

13. How does understanding real-world applications of trigonometry (e.g., in engineering, physics, or architecture) influence a student's perception of the subject?

- A. It makes trigonometry seem less important
- B. It shows that trigonometry is only useful in school
- C. It helps students appreciate the usefulness and relevance of trigonometry in real-life situations
- D. It makes the subject more confusing.



UNIT 2: SEQUENCES

Key unit competence: Use arithmetic, geometric, and harmonic sequences, including their convergence in analysing mathematical and real-life problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain sequence and differentiate convergent sequences from divergent sequences.
- Explain monotone sequences and their characteristics.
- State the definitions of arithmetic, geometric, and harmonic sequences.
- Explain the concepts of general term, arithmetic/geometric means, and sum of “n” terms.
- Apply sequences in real-life problems.

Skills:

Determine the general term of arithmetic and geometric sequences.

- Calculate arithmetic, geometric, and harmonic means.
- Compute the sum of the first “n” terms of arithmetic and geometric sequences.
- Analyze sequences to identify convergence, divergence, and monotonicity.
- Apply sequences to solve practical, real-life problems.

Attitudes and values:

- Develop accuracy, logical reasoning, and persistence in working with sequences.
- Appreciate the importance of sequences in modelling real-life situations.
- Show curiosity and willingness to explore patterns and relationships in numerical data.

Introduction

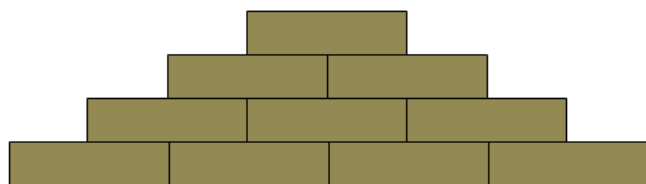
Consider a scientist doing an experiment; he/she is collecting data every day. Put u_1 to be the data collected the first day, u_2 be the data collected the second day, u_3 be the data collected the third day, and so on..., and u_n be the data collected after n days. Clearly we are generating a set of numbers with a very special characteristic. There is an order in the numbers; we actually have the first number, the second number and so on. A sequence is a set of real numbers with natural order.



Introductory activity

1. Suppose that you want to build a tower with blocks.
 - a) On a piece of paper, draw that tower starting by 15 blocks for the bottom row until you cannot add another row.
 - b) How many rows are there?
 - c) Write down the number of blocks that are in each row (from bottom row to top row).
 - d) In the numbers written down each number can be found by adding a constant number to the previous. Guess that constant number.

See the following picture:



2. Suppose that an insect population is growing in such a way that each new generation is 2 times as large as the previous generation. If there are 126 insects in the first generation, on a piece of paper, write down the number of insects that will be there in second, third, fourth,...,tenth generation.

2.1 Meaning of sequences



Activity 2.1

Look at the following sequences and identify the rule that is being followed.

1. 5, 10, 15, 20, 25, ...
2. 2, 4, 8, 16, 32, ...
3. 1, 1, 2, 3, 5, 8, 13, ...
4. 1, 3, 6, 10, 15, ...



Content summary

A sequence (or progression) is an ordered list of numbers (or objects) that follow a specific rule or pattern. Numbers in sequence are denoted $\{u_1, u_2, u_3, \dots, u_{n-1}, u_n, \dots\}$ and shortly $\{u_n\}$. We can

also write $\{u_1, u_2, u_3, \dots, u_{n-1}, u_n, \dots\}$. The dots are used to suggest that the sequence continues indefinitely, following the obvious pattern. The numbers $u_1, u_2, u_3, \dots, u_{n-1}, u_n, \dots$ in a sequence are called **terms** of the sequence. The natural number n is called **term number** and value u_n is called a **general term** of a sequence and the term u_1 is the **initial term**.

As a sequence continues indefinitely, it can be denoted as $\{u_n\}_{n=1}^{+\infty}$.

The number of terms of a sequence (possibly infinite) is called the **length of the sequence**.

Notice

- Sometimes, the term number, n , starts from 0. In this case terms of a sequence are $\{u_0, u_1, u_2, u_3, \dots, u_{n-1}, u_n, \dots\}$ and this sequence is denoted by $\{u_n\}_{n=0}^{+\infty}$. In this case the initial term is u_0 .
- A sequence can be finite, like the sequence $2, 4, 8, 16, \dots, 256$.
- Finite sequences are sometimes known as **strings** or **words** and infinite sequences as **streams**.

The empty sequence $\{\}$ is included in most notions of sequences but may be excluded depending on the context.

Usually, a numerical sequence is given by some formula $u_n = f(n)$, permitting you to find any term of the sequence by its number n ; this formula is called a **general term formula**.

This suggests the definition that a sequence (or **infinite sequence**) is a function whose domain is the set of positive integers.

Note that it is not always possible to give the numerical sequence by a general term formula; sometimes a sequence is given by description of its terms.

Example 1:

Numerical sequences:

- 1, 2, 3, 4, 5, ... a sequence of natural numbers;
- 2, 4, 6, 8, 10, ... a sequence of even numbers;

c) 1.4, 1.41, 1.414, 1.4142, ... a numerical sequence of approximate, defined more precisely values of $\sqrt{2}$.

For the last sequence it is impossible to give a general term formula, nevertheless this sequence is described completely.

Example 2:

List the first five term of the sequence $\{2^n\}_{n=1}^{+\infty}$.

Solution

Here, we substitute 1, 2, 3, 4, 5, ... into the formula 2^n . This gives $2^1, 2^2, 2^3, 2^4, 2^5, \dots$

Or, equivalently, 2, 4, 8, 16, 32, ...

Example 3: Express the following sequences in general notation

a) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$

b) $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$

Solution

a) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$

Beginning by comparing terms and term numbers:

Term number	1	2	3	4	...
Term	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$	...

In each term, the numerator is the same as the term number, and the denominator is one greater than the term number.

Thus, the n^{th} term is $\frac{n}{n+1}$ and the sequence may be written as $\left\{ \frac{n}{n+1} \right\}_{n=1}^{+\infty}$.

b) $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$

Beginning by comparing terms and term numbers:

Term number	1	2	3	4	...
Term	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	...

Or

Term number	1	2	3	4	...
Term	$\frac{1}{2^1}$	$\frac{1}{2^2}$	$\frac{1}{2^3}$	$\frac{1}{2^4}$	...

In each term, the denominator is equal to 2 powers the term number. We observe that the n^{th} term is $\frac{1}{2^n}$ and the sequence may be written as $\left\{ \frac{1}{2^n} \right\}_{n=1}^{+\infty}$.

Example 4:

A sequence is defined by $\{u_n\} = \begin{cases} u_0 = 1, & \text{if } n = 0 \\ u_{n+1} = 3u_n + 2, & \text{Otherwise} \end{cases}$

Determine u_1, u_2 and u_3

Solution:

Since $u_0 = 1$ and $u_{n+1} = 3u_n + 2$, replace n by 0, 1, 2 to obtain u_1, u_2, u_3 respectively.

$$n = 0, \quad u_{0+1} = u_1 = 3u_0 + 2 = 3 \times 1 + 2 = 5$$

$$n = 1, \quad u_{1+1} = u_2 = 3u_1 + 2 = 3 \times 5 + 2 = 17$$

$$n = 2, \quad u_{2+1} = u_3 = 3u_2 + 2 = 3 \times 17 + 2 = 53$$

Thus, $u_1 = 5, u_2 = 17, u_3 = 53$.



Application activity 2.1

A child saves money every month. In the first month, she saved 10 Frw.

Every new month, she saves 20 Frw more than the previous month.

- How much money does she save within the first 6 months?
- How much money does she save during the first 9 months?

2.2 Monotone sequences



Activity 2.2

For each of the following sequences, state whether the terms are arranged in ascending order, descending order, both, or neither.

1. $1, 2, 3, 4, 5, 6, \dots$

2. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

3. $1, -1, 1, -1, 1, \dots$

4. $2, 2, 2, 2, 2, 2, \dots$



Content summary

A sequence $\{u_n\}$ is said to be

- **Increasing** or in **ascending** order if $u_1 < u_2 < u_3 < \dots < u_n < \dots$
- **Nondecreasing** if $u_1 \leq u_2 \leq u_3 \leq \dots \leq u_n \leq \dots$
- **Decreasing** or in **descending** order if $u_1 > u_2 > u_3 > \dots > u_n > \dots$
- **Nonincreasing** if $u_1 \geq u_2 \geq u_3 \geq \dots \geq u_n \geq \dots$

A sequence that is either nondecreasing or nonincreasing is called **monotone**, and a sequence that is increasing or decreasing is called **strictly monotone**. Observe that a strictly monotone sequence is monotone, but not conversely.

In order for a sequence to be **increasing**, all pairs of successive terms, u_n and u_{n+1} , must satisfy $u_n < u_{n+1}$, or equivalently $u_n - u_{n+1} < 0$.

More generally, monotone sequences can be classified as follows:

Difference between successive terms	Classification
$u_n - u_{n+1} < 0$	Increasing

$u_n - u_{n+1} > 0$	Decreasing
$u_n - u_{n+1} \leq 0$	Nondecreasing
$u_n - u_{n+1} \geq 0$	Nonincreasing

If the terms in the sequence are all positive, then we can divide both sides of the inequality

$$u_n < u_{n+1} \text{ by } u_n \text{ to obtain } 1 < \frac{u_{n+1}}{u_n} \text{ or equivalently } \frac{u_{n+1}}{u_n} > 1.$$

Example 1:

Prove that the sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is an increasing sequence.

Solution

$$\text{Here, } u_n = \frac{n}{n+1} \text{ and } u_{n+1} = \frac{n+1}{n+2}.$$

$$\text{Thus, for } n \geq 1, u_n - u_{n+1} = \frac{n}{n+1} - \frac{n+1}{n+2} = \frac{n^2 + 2n - n^2 - 2n - 1}{(n+1)(n+2)} = -\frac{1}{(n+1)(n+2)} < 0.$$

This proves that the given sequence is **increasing**.

Alternative method,

We can show that the sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is increasing by examining the ratio of

$$\text{successive terms as follows } \frac{u_{n+1}}{u_n} = \frac{\frac{n+1}{n+2}}{\frac{n}{n+1}} = \frac{n+1}{n+2} \times \frac{n+1}{n} = \frac{n^2 + 2n + 1}{n^2 + 2n}.$$

Since the numerator exceeds the denominator, the ratio exceeds 1, that is $\frac{u_{n+1}}{u_n} > 1$ for $n \geq 1$.

This proves that the sequence is increasing.

Example 2:

The sequence 4, 4, 4, 4, ... is both neither decreasing nor increasing.

Example 3:

The sequence $-2, 2, -2, 2, -2, \dots$ is not monotonic.



Application activity 2.2

Imagine you're an investor tracking the daily closing price of a stock. You have data about how the stock price changes every day, and you need to determine if the stock price follows a monotone pattern (increasing or decreasing). Here are the monthly values of an investment for 6 months: Month 1: 2000Frw, Month 2: 2100 Frw, Month 3: 2150 Frw, Month 4: 2200 Frw, Month 5: 2250 Frw, and Month 6: 2300 Frw.

- (a) Determine if this investment value sequence is monotone increasing or monotone decreasing.
- (b) If the sequence is monotone increasing, predict the value of the investment for the next month, assuming the trend continues.

2.3 Arithmetic sequences: Meaning and common difference



Activity 2.3

In each of the following sequences, each term can be found by adding a constant number to the previous. Guess that constant number.

- 1. $\{3n + 2\}$
- 2. $\{16 - 5n\}$



Content summary

Sequences of numbers that follow a pattern of adding a fixed number from one term to the next are called **arithmetic sequences** or **arithmetic progressions**.

Example 1:

The following sequences are arithmetic sequences:

Sequence u_n : 5, 10, 15, 20, 25, 30, ... (adding 5)

Sequence v_n : 26, 31, 36, 41, 46, ... (adding 5)

Sequence w_n : 20,18,16,14,12,...,(taking away 2)

Common difference:

The fixed numbers that bind each sequence together are called the **common differences**.

Sometimes mathematicians use the letter ***d*** when referring to these types of sequences.

d can be calculated by subtracting any two consecutive terms in an arithmetic sequence. That is

$$d = u_{n+1} - u_n \text{ or } d = u_n - u_{n-1}.$$

Rule for 3 consecutive terms:

If three consecutive terms are in arithmetic sequence, the double of the middle term is equal to the sum of extreme terms.

That is for an arithmetic sequence u_{n-1}, u_n, u_{n+1} , we have $2u_n = u_{n-1} + u_{n+1}$.

Example 2:

The numbers 4, 6 and 8 are three consecutive terms of an arithmetic sequence because

$$2 \times 6 = 8 + 4 \Leftrightarrow 12 = 12.$$

Example 3:

If $\frac{1}{a+b}, \frac{1}{a+c}, \frac{1}{b+c}$ are 3 consecutive terms of an arithmetic progression, show that it will be

the same for a^2, b^2, c^2 .

Solution

If $\frac{1}{a+b}, \frac{1}{a+c}, \frac{1}{b+c}$ are 3 consecutive terms of an arithmetic progression, then

$$\frac{2}{a+c} = \frac{1}{a+b} + \frac{1}{b+c} \Leftrightarrow \frac{2}{a+c} = \frac{2b+c+a}{(a+b)(b+c)}$$

$$\Leftrightarrow 2(ab+ac+b^2+bc) = (a+c)(2b+a+c)$$

$$\Leftrightarrow 2ab+2ac+2b^2+2bc = 2ab+a^2+ac+2bc+ac+c^2$$

$$\Leftrightarrow 2b^2 = a^2 + c^2$$

Also a^2, b^2, c^2 are 3 consecutive terms of an arithmetic progression if $2b^2 = a^2 + c^2$.

Thus, if $\frac{1}{a+b}, \frac{1}{a+c}, \frac{1}{b+c}$ are 3 consecutive terms of an arithmetic progression, it will be the same for a^2, b^2, c^2 .

Example 4:

Form an arithmetic progression of three numbers such that the sum of its terms is 30 and the sum of the squares of its terms is 332.

Solution

Let the second term be x . The first term is $x-d$ and the third term is $x+d$ where d is the common difference.

$$\text{Now, } x-d + x + x+d = 30 \Rightarrow 3x = 30 \Rightarrow x = 10. \text{ Also, } (x-d)^2 + x^2 + (x+d)^2 = 332$$

$$\text{or } (10-d)^2 + 100 + (10+d)^2 = 332 \text{ or } 2d^2 = 32 \Rightarrow d = \pm 4.$$

Therefore, the progression is 6, 10, 14 or 14, 10, 6.

Example 5:

Calculate x so that the squares of $1+x, q+x$, and q^2+x will be three consecutive terms of an arithmetic progression where q is any given number.

Solution

We need to find x such that $(1+x)^2, (q+x)^2$, and $(q^2+x)^2$ form an arithmetic progression.

$$2(q+x)^2 = (1+x)^2 + (q^2+x)^2.$$

$$\Leftrightarrow 2(q^2 + 2qx + x^2) = 1 + 2x + x^2 + q^4 + 2xq^2 + x^2 \Leftrightarrow 2q^2 + 4qx + 2x^2 = 1 + 2x + x^2 + q^4 + 2xq^2 + x^2$$

$$\Leftrightarrow 2q^2 + 4qx = 1 + 2x + q^4 + 2xq^2 \Leftrightarrow 4qx - 2x - 2xq^2 = 1 - 2q^2 + q^4$$

$$\Leftrightarrow x(4q - 2 - 2q^2) = (1 - q^2)^2 \Leftrightarrow x = \frac{(1 - q^2)^2}{-2(1 - 2q + q^2)} \Leftrightarrow x = \frac{(1 - q)^2(1 + q)^2}{-2(1 - q)^2} \Leftrightarrow x = \frac{(1 + q)^2}{-2}n$$

$$\text{Thus, } x = -\frac{(1 + q)^2}{2}.$$



Application activity 2.3

- 1) Discuss real-world applications of arithmetic sequences in various fields.
- 2) Determine the common difference in the sequence $U_n = \{2n + 1\}$

2.4 General term of an arithmetic sequence



Activity 2.4

If $\{u_n\}$ is an arithmetic sequence with common difference d and initial term u_1 , then

$$u_2 = u_1 + d, \quad u_3 = u_2 + d = (u_1 + d) + d = u_1 + 2d.$$

Continue in this manner up to u_{10} and conclude the general formula could be used for u_n .



Content summary

The n^{th} term is called the general term u_n of an arithmetic sequence $\{u_n\}$ with common difference d and initial term u_1 is given by $u_n = u_1 + (n-1)d$

Generally, if u_p is any p^{th} term of a sequence then the n^{th} term is given by $u_n = u_p + (n-p)d$

Example 1:

If the first and tenth terms of an arithmetic sequence are 3 and 30, respectively, find the fiftieth term of the sequence.

Solution

$$u_1 = 3, \quad u_{10} = 30$$

$$\text{But } u_n = u_1 + (n-1)d, \quad u_n = u_1 + (n-1)d$$

$$u_{10} = u_1 + (10-1)d \quad u_{10} = u_1 + (10-1)d$$

$$\text{Then } 30 = 3 + (10-1)d \Leftrightarrow 30 = 3 + 9d \Rightarrow d = 3$$

$$\text{Now, } u_{50} = u_1 + (50-1)d = 3 + 49 \times 3 = 150$$

Thus,

The fiftieth term of the sequence is 150.

Example 2:

If the 3rd term and the 8th term of an arithmetic sequence are 5 and 15 respectively, find the common difference.

Solution

$$u_3 = 5, \quad u_8 = 15$$

Using the general formula: $u_n = u_p + (n - p)d$

$$u_3 = u_8 + (3 - 8)d.$$

$$5 = 15 - 5d \Leftrightarrow 5d = 15 - 5 \Rightarrow 5d = 10 \Rightarrow d = 2.$$

The common difference is 2.

Example 3:

Consider the sequence 5, 8, 11, 14, 17, ..., 47. Find the number of terms in this sequence.

Solution

We see that $u_1 = 5, u_n = 47$ and $d = 3$.

But we know that $u_n = u_1 + (n - 1)d$. This gives

$$47 = 5 + (n - 1)3 \Leftrightarrow 42 = 3n - 3 \Rightarrow n = 15.$$

This means that there are 15 terms in the sequence.

Example 4:

Consider the sequence which went from 20 to -26 with -2 as common difference. Find the number of terms.

Solution

We have

$$-26 = 20 + (n - 1)(-2) \Leftrightarrow -46 = -2n + 2 \Rightarrow n = 24.$$

This means that there are 24 terms in the sequence.



Application activity 2.4

You decide to start saving money for a big purchase, like a laptop, and you deposit money into your savings account every month. You begin by saving 50,000Frw in the first month. Every subsequent month, you save 10,000Frw more than the previous month.

- Identify the Arithmetic Sequence.
- Calculate the deposit for the 6th month.

2.5 Arithmetic means of a sequence



Activity 2.5

Suppose that you need to form an arithmetic sequence of 7 terms such that the first term is 2 and the seventh term is 20. Write down that sequence.



Content summary

If three or more than three numbers are in arithmetic sequence, then all terms lying between the first and the last numbers are called **arithmetic means**. If B is arithmetic mean between

A and C , then $B = \frac{A+C}{2}$.

To insert k terms called **arithmetic means** between two terms u_1 and u_n is to form an arithmetic sequence of $n = k + 2$ terms whose first term is u_1 and last term is u_n .

While there are several methods, we will use our n^{th} term formula $u_n = u_1 + (n-1)d$.

As u_1 and u_n are known, we need to find the common difference d taking $n = k + 2$ where k is the number of terms to be inserted.

Example 1:

Insert three arithmetic means between 7 and 23.

Solution

Here $k = 3$ and then $n = k + 2 = 5$, $u_1 = 7$ and $u_n = u_5 = 23$.

Then $u_5 = u_1 + (5-1)d \Leftrightarrow 23 = 7 + 4d \Rightarrow d = 4$

Now, insert the terms using $d = 4$, the sequence is 7, 11, 15, 19, 23.

Example 2:

Insert five arithmetic means between 2 and 20.

Solution

Here $k = 5$ and then $n = k + 2 = 5 + 2 = 7$, $u_1 = 2$ and $u_n = u_7 = 20$.

Then

$$u_7 = u_1 + (7 - 1)d \Leftrightarrow 20 = 2 + 6d \Rightarrow d = 3$$

Now, insert the terms using $d = 3$, the sequence is 2, 5, 8, 11, 14, 17, 20.



Application activity 2.5

1) Imagine you are an engineer designing a road that starts at sea level (elevation = 0 meters) and ends at a mountain top with an elevation of 600 meters. The road will have 5 points where the elevation increases gradually. Using the arithmetic sequence concept, calculate the elevations at the intermediate points.

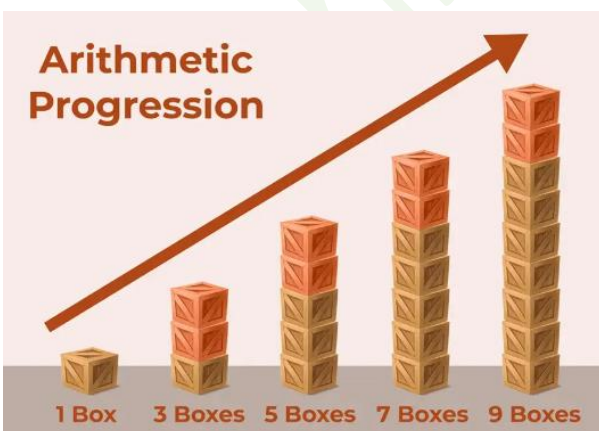
2) Insert 9 arithmetic means between 2 and 32. Note that First term is 2, last term is 32 and the total number of terms $n=11$.

2.6 Sum of n first terms of an arithmetic sequence



Activity 2.6

Consider a finite arithmetic sequence $\{u_n\} = u_1, u_2, u_3, \dots, u_n$ with common difference d .



u_n looks like the number of boxes on each column. The common difference $d=2$ boxes to be added to get the number of boxes for the next pile.

S_n is the sum of all boxes for a given n .

Let S_n denotes the sum of these terms (or the sum of all boxes). We have

$$\begin{aligned}
u_1 &= u_1 \\
u_2 &= u_1 + d \\
&\vdots \\
u_{n-1} &= u_1 + (n-2)d \\
u_n &= u_1 + (n-1)d
\end{aligned}$$

Sum up these terms in forward form and in reverse order. Deduce the expression of S_n .



Content summary

For any arithmetic sequence $\{u_n\} = u_1, u_2, u_3, \dots, u_n$, the sum of its n first terms is

$$S_n = \sum_{r=1}^n u_r = u_1 + u_2 + u_3 + \dots + u_n.$$

The sum of first n terms of a finite arithmetic sequence with initial term u_1 is given by

$$S_n = \frac{n}{2}(u_1 + u_n)$$

If the initial term is u_0 , the formula becomes $S_n = \frac{n+1}{2}(u_0 + u_n)$.

Example 1:

Calculate the sum of first 100 terms of the sequence 2, 4, 6, 8, ...

Solution

We see that the common difference is 2 and the initial term is $u_1 = 2$. We need to find $u_n = u_{100}$

$$u_{100} = 2 + (100-1)2 = 2 + 198 = 200$$

Now,

$$S_{100} = \frac{100}{2}(u_1 + u_{100}) = 50(2 + 200) = 10100.$$

Example 2:

Find the sum of first k even integers ($k \neq 0$).

Solution

$$u_1 = 0 \text{ and } d = 2$$

$$u_n = u_k$$

$$u_k = 0 + (k-1)2$$

$$u_k = 2(k-1)$$

$$S_n = S_k$$

$$S_k = \frac{k}{2}(0 + 2k - 2)$$

$$S_k = k(k-1)$$



Application activity 2.6

- 1) Suppose you are a financial planner tasked with managing the budget for a small company. The company expects its annual expenses to increase gradually by 200,000Frw each year over the next 5 years. In the first year, the expenses are expected to be 1,500,000 Frw. Calculate the total expenses over the next 5 years, considering the gradual increase.
- 2) The sum of the terms in the sequence 1, 8, 15, ... is 396. How many terms does the sequence contain?

2.7 Harmonic sequence



Activity 2.7

Consider the following arithmetic sequence: 2, 4, 6, 8, 10, 12, 14, 16.

Form another sequence whose terms are the reciprocals of the terms of the given sequence.

What can you say about the new sequence?



Content summary

Harmonic sequence is a sequence of numbers in which the reciprocals of the terms are in arithmetic sequence.

Example 1:

Example of harmonic sequence is $\frac{1}{3}, \frac{1}{6}, \frac{1}{9}, \dots$

If you take the reciprocal of each term from the above harmonic sequence, the sequence will become 3, 6, 9, ... which is an arithmetic sequence with a common difference of 3.

Example 2:

Another example of harmonic sequence is 6, 3, 2. The reciprocals of each term are $\frac{1}{6}, \frac{1}{3}$, and $\frac{1}{2}$ respectively which is an arithmetic sequence with a common difference of $\frac{1}{6}$.

Remark

To find the term of **harmonic sequence**, convert the sequence into arithmetic sequence then do the calculations using the arithmetic formulae. Then take the reciprocal of the answer in arithmetic sequence to get the correct term in harmonic sequence.

Example 3:

The 2nd term of an harmonic progression is $\frac{1}{6}$ and 6th term is $-\frac{1}{6}$. Find 20th term and n^{th} term.

Solution

In harmonic progression, $h_2 = \frac{1}{6}$ and $h_6 = -\frac{1}{6}$.

Thus, in the corresponding arithmetic progression $u_2 = 6$ and $u_6 = -6$

Or $u_6 = a_2 + 4d \Rightarrow 6 + 4d = -6$ or $d = -3$.

Hence $u_{20} = 6 + 18(-3) = -48 \Rightarrow h_{20} = -\frac{1}{48}$.

$u_n = 6 + (n-2)(-3) = 12 - 3n \Rightarrow h_n = \frac{1}{12-3n}$.

Notice: If three terms a, b, c are in harmonic progression, the middle one is said to be

Harmonic mean between the other two and $b = \frac{2ac}{a+c}$.

Example 4:

Insert 4 harmonic means between $\frac{2}{3}$ and $\frac{6}{19}$

Solution

Let the four harmonic means be h_1, h_2, h_3, h_4 .

Then $\frac{2}{3}, h_1, h_2, h_3, h_4, \frac{6}{19}$ are in harmonic progression

$\Rightarrow \frac{3}{2}, \frac{1}{h_1}, \frac{1}{h_2}, \frac{1}{h_3}, \frac{1}{h_4}, \frac{19}{6}$ are in arithmetic progression. Where $u_1 = \frac{3}{2}$ and $u_6 = \frac{19}{6}$

$u_6 = \frac{19}{6} \Rightarrow a_1 + 5d = \frac{19}{6}$ with d common difference.

$$\Rightarrow \frac{3}{2} + 5d = \frac{19}{6} \Rightarrow 5d = \frac{19}{6} - \frac{3}{2} \Rightarrow 5d = \frac{10}{6} \Rightarrow d = \frac{1}{3}$$

$$\Rightarrow \begin{cases} \frac{1}{h_1} = \frac{3}{2} + \frac{1}{3} = \frac{11}{6} \equiv 1^{st} \text{ term of arithmetic progression} \\ \frac{1}{h_2} = \frac{3}{2} + \frac{2}{3} = \frac{13}{6} \equiv 2^{nd} \text{ term of arithmetic progression} \\ \frac{1}{h_3} = \frac{3}{2} + \frac{3}{3} = \frac{15}{6} = \frac{5}{2} \equiv 3^{rd} \text{ term of arithmetic progression} \\ \frac{1}{h_4} = \frac{3}{2} + \frac{4}{3} = \frac{17}{6} \equiv 4^{th} \text{ term of arithmetic progression} \end{cases}$$

The four harmonic means are $\frac{6}{11}, \frac{6}{13}, \frac{6}{15}, \frac{6}{17}$

Example 5:

Find the n^{th} term of the series $2\frac{1}{2} + 1\frac{7}{13} + 1\frac{1}{9} + \frac{20}{23} + \dots$

Solution

The given series is $2\frac{1}{2} + 1\frac{7}{13} + 1\frac{1}{9} + \frac{20}{23} + \dots$

The reciprocal of the terms are $\frac{2}{5}, \frac{13}{20}, \frac{9}{10}, \frac{23}{20}, \dots$

There are in arithmetic progression, with the first term $\frac{2}{5}$ and the common difference

$$\frac{13}{20} - \frac{2}{5} = \frac{1}{4}.$$

The given series in arithmetic progression: n^{th} term of arithmetic progression:

$$u_n = \frac{2}{5} + (n-1)\frac{1}{4} = \frac{8+5n-5}{20} = \frac{5n+3}{20}.$$

Hence n^{th} term of the given harmonic progression is $h_n = \frac{1}{u_n}$ or $h_n = \frac{20}{5n+3}$.

The n^{th} term of the series $2\frac{1}{2} + 1\frac{7}{13} + 1\frac{1}{9} + \frac{20}{23} + \dots$ is $\frac{20}{5n+3}$.



Application activity 2.7

- 1) Find the 4^{th} and 8^{th} term of the harmonic sequence 6, 4, 3, ...
- 2) Insert two harmonic means between 3 and 10.
- 3) Find the term number of harmonic sequence $\frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{1}{\sqrt{5}}, \dots, \frac{\sqrt{5}}{13}$
- 4) The harmonic mean between two numbers is 3 and the arithmetic mean is 4. Find the numbers.
- 5) Find the n^{th} term of the series $4 + 4\frac{2}{7} + 4\frac{8}{13} + 5 + \dots$

2.8 Geometric sequences: Meaning and common ratio



Activity 2.8

Take a piece of paper with a square shape.

1. Cut it into two equal parts.
2. Write down a fraction corresponding to one part according to the original piece of paper
3. Take one part obtained in step 2) and cut it into two equal parts again and write down the fraction corresponding to that part.
4. Continue until you remain with a small piece of paper that you are not able to cut into two equal parts.

Observe the sequence of numbers you obtained and give the relationship between any two consecutive numbers.



Content summary

Sequences of numbers that follow a pattern of multiplying a fixed number from one term to the next are called **geometric sequences** or **geometric progression**.

Example 1:

The following sequences are geometric sequences:

Sequence u_n : 5, 10, 20, 40, 80, ...

Sequence v_n : 2, 1, $\frac{1}{2}$, $\frac{1}{4}$, ...

Sequence w_n : 1, -2, 4, -8, 16, ...

Common ratio

We can examine these sequences to greater depth, we must know that the fixed numbers that bind each sequence together are called the **common ratios**, denoted by the letter r .

r can be calculated by dividing any two consecutive terms in a geometric sequence. That is

$$r = \frac{u_{n+1}}{u_n} \text{ or } r = \frac{u_n}{u_{n-1}} \text{ or } u_n = ru_{n-1}.$$

Consecutive terms of a geometric sequence

If three terms are consecutive terms of a geometric sequence, the square of the middle term is equal to the product of extreme terms.

That is for a geometric sequence u_{n-1} , u_n , u_{n+1} , we have $u_n^2 = u_{n-1} \times u_{n+1}$.

Example 2:

6, 12, 24 are consecutive terms of a geometric sequence because $(12)^2 = 6 \times 24 \Leftrightarrow 144 = 144$.

Example 3:

Find b such that 8, b , 18 will be in geometric sequence.

Solution

$$b^2 = 8 \times 18 \Rightarrow b^2 = 144 \Rightarrow b = \pm\sqrt{144} = \pm 12.$$

Thus, 8, 12, 18 or 8, -12, 18 are in geometric sequence.

Example 4:

The product of three consecutive numbers in geometric progression is 27. The sum of the first two and nine times the third is -79. Find the numbers.

Solution

Let the three terms be $\frac{x}{a}, x, ax$.

The product of the numbers is 27. So $\frac{x}{a} \cdot x \cdot ax = 27 \Rightarrow x^3 = 27 \Rightarrow x = 3$.

$$\left(\frac{x}{a}\right) \cdot x \cdot (ax) = 27x^3 = 27 \Rightarrow x = \sqrt[3]{27} = 3.$$

The sum of the first two and nine times the third is -79:

$$\frac{x}{a} + x + 9ax = -79 \Rightarrow \frac{3}{a} + 3 + 27a = -79$$

$$27a^2 + 82a + 3 = 0 \Rightarrow a = -3 \quad \text{or} \quad a = -\frac{1}{27}.$$

The numbers are: $-1, 3, -9$ or $-81, 3, -\frac{1}{9}$.



Application activity 2.8

1. For what values of k , the numbers $1+k, \frac{5}{6}+k$, and $\frac{13}{18}+k$ are in geometric progression.
2. If $a^2 + b^2, ab + bc$ and $b^2 + c^2$ are in geometric progression, show that a, b, c are also in geometric progression.
3. If $\frac{1}{x+y}, \frac{1}{2y}, \frac{1}{y+z}$ are in arithmetic progression, show that x, y, z are in geometric progression.

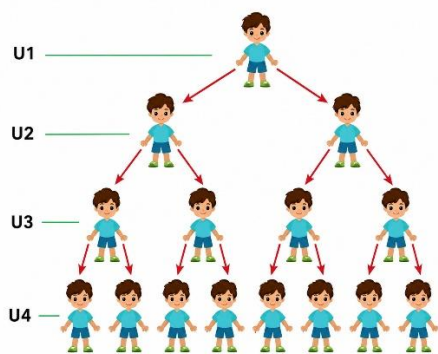
2.9 General terms of a geometric sequence



Activity 2.9

If $\{u_n\}$ is a geometric sequence with common ratio r and initial term u_1 , then

$$u_2 = u_1 r; \quad u_3 = u_2 r = u_1 r^2; \quad u_4 = u_3 r = u_1 r^3$$



Assume that a child goes to school with the flu virus and forgot to cover his mouth when two friends came. They leave, and the next day they also have the flu. Let's assume that each friend in turn spreads the virus to two of their friends by the same droplet spread the following day. Assuming this pattern continues, and each sick person infects 2 other friends. In the number of infected children by a disease, the first infected is $u_1 = 1$ and the ratio is $r = 2$

Continue in this manner up to u_{10} and conclude the general formula should be used for u_n



Content summary

The n^{th} term, u_n , of a geometric sequence $\{u_n\}$ with common ratio r and initial term u_1 is given by $u_n = u_1 r^{n-1}$.

Generally, if u_p is the p^{th} term of the sequence, then the n^{th} term is given by $u_n = u_p r^{n-p}$.

Example 1:

If the first and tenth terms of a geometric sequence are 1 and 4, respectively, find the nineteenth term.

Solution

$$u_1 = 1 \text{ and } u_{10} = 4$$

$$\text{But } u_n = u_1 r^{n-1}, \text{ then } 4 = 1r^9 \Leftrightarrow r = \sqrt[9]{4} \text{ or } r = 4^{\frac{1}{9}}.$$

Now,

$$u_{19} = u_1 r^{19-1} = 1 \left(\frac{1}{4^9} \right)^{18} = 16.$$

Thus, the nineteenth term of the sequence is 16.

Example 2:

If the 2nd term and the 9th term of a geometric sequence are 2 and $-\frac{1}{64}$ respectively, find the common ratio.

Solution

$$u_2 = 2, u_9 = -\frac{1}{64}$$

Using the general formula: $u_n = u_p r^{n-p}$.

$$u_2 = u_9 r^{2-9} \Leftrightarrow 2 = -\frac{1}{64} r^{-7} \Leftrightarrow 128 = -\frac{1}{r^7} \Leftrightarrow r^7 = -\frac{1}{128} \Leftrightarrow r = \sqrt[7]{-\frac{1}{128}} = -\frac{1}{2}$$

The common ratio is $r = -\frac{1}{2}$.

Example 3:

Find the number of terms in sequence 2, 4, 8, 16, ..., 256.

Solution

This sequence is geometric with common ratio of 2, $u_1 = 2$ and $u_n = 256$

But $u_n = u_1 r^{n-1}$, then $256 = 2 \times 2^{n-1} \Leftrightarrow 256 = 2^n$ or $2^8 = 2^n \Rightarrow n = 8$.

Thus, the number of terms in the given sequence is 8.



Application activity 2.9

1. Find the fourth term from the end of geometric sequence $8, 4, 2, \dots, \frac{1}{128}$.
2. The fifth term of a geometric sequence is $\frac{81}{32}$ and the ratio is 3, write the geometric sequence and its 8th term.

2.10 Geometric means of numbers



Activity 2.10

Suppose that you need to form a geometric sequence of 6 terms such that the first term is 1 and the sixth term is 243. Write down that sequence



Content summary

To insert k terms called **geometric means** between two terms u_1 and u_n is to form a geometric sequence of $n = k + 2$ terms whose the first term is u_1 and the last term is u_n . While there are several methods, we will use our n^{th} term formula $u_n = u_1 r^{n-1}$. As u_1 and u_n are known, we need to find the common ratio r taking $n = k + 2$ where k is the number of terms to be inserted.

Example 1:

Insert three geometric means between 3 and 48.

Solution

Here $k = 3$, then $n = 5$, $u_1 = 3$ and $u_n = u_5 = 48$

$$u_5 = u_1 r^{n-1} \Leftrightarrow 48 = 3r^4 \Rightarrow r = 2$$

Inserting three terms using common ratio $r = 2$ gives 3, 6, 12, 24, 48

Example 2:

Insert 6 geometric means between 1 and $-\frac{1}{128}$.

Solution

Here $k = 6$, then $n = 8$, $u_1 = 1$, and $u_n = u_8 = -\frac{1}{128}$

$$u_8 = u_1 r^{n-1}$$

$$u_8 = u_1 r^{n-1} \Leftrightarrow -\frac{1}{128} = 1r^7 \Leftrightarrow r^7 = -\frac{1}{128} \Leftrightarrow r^7 = -\frac{1}{(2)^7} \Leftrightarrow r = \left[-\frac{1}{(2)^7} \right]^{\frac{1}{7}} = -\frac{1}{2}$$

Inserting 6 terms using common ratio $r = -\frac{1}{2}$ gives $1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32}, \frac{1}{64}, -\frac{1}{128}$.



Application activity 2.10:

1. Insert 5 geometric means between $\frac{1}{4}$ and $\frac{1}{256}$.

2. Suppose that 4, 36, 324 are in geometric progression. Insert two more numbers in this sequence so that it again forms a geometric sequence.

2.11 Sum of n first terms of a geometric sequence



Activity 2.11

- 1) Consider a geometric sequence with initial term u_1 and common ratio r .

Let $s_n = u_1 + u_1r + u_1r^2 + \dots + u_1r^{n-1}$

- Multiply both sides of above equation by r and subtract new equation from old equation.
- Give the general formula for s_n .

Note that you will need the sum $S_{n-1} = 1 + 2 + \dots + n - 1$ which is $S_{n-1} = \frac{n-1}{2}(1 + n - 1) = \frac{n(n-1)}{2}$.

- 2) Suppose that we need the product of $u_1, u_2, u_3, \dots, u_n$. Then $P_n = u_1 \times u_2 \times u_3 \times \dots \times u_n$ or $P_n = u_1 \times u_1r \times u_1r^2 \times \dots \times u_1r^{n-1} \dots$

Expand this relation and demonstrate the general formula that should be applied for P_n .



Content summary

For finite geometric sequence $\{u_n\} = u_1, u_2, u_3, \dots, u_n$, the sum $\sum_{r=1}^n u_r = u_1 + u_2 + u_3 + \dots + u_n$ is the

sum of the first n terms of the sequence denoted by S_n . Thus, $S_n = u_1 + u_2 + u_3 + \dots + u_n = \sum_{r=1}^n u_r$

The sum of first n terms of a geometric sequence with initial term u_1 and common ratio r is

given by: $S_n = \frac{u_1(1-r^n)}{1-r}$ with $r \neq 1$. If the initial term is u_0 , then the formula is

$$S_n = \frac{u_0(1-r^{n+1})}{1-r} \text{ with } r \neq 1$$

If $r = 1$, $S_n = nu_1$

Also, the product of first n terms of a geometric sequence with initial term u_1 and common

ratio r is given by $P_n = (u_1)^n r^{\frac{n(n-1)}{2}}$

If the initial term is u_0 then $P_n = (u_0)^{n+1} r^{\frac{n(n+1)}{2}}$.

Example 1:

Find the sum of the first 20 terms of a geometric sequence if the first term is 1 and common ratio is 2.

Solution

Here $u_1 = 1, r = 2, n = 20$

Then,

$$S_{20} = \frac{1(1-2^{20})}{1-2} = \frac{1-2^{20}}{-1} = 1048575$$

Example 2:

Consider the sequence $\{u_n\} = 2, 6, 18, 54, \dots$

- Show that $\{u_n\}$ is a geometric sequence and find its first term and common ratio.
- Express $\{u_n\}$ in term of n .

Solution

- Check the ratio:

$$\frac{6}{2} = 3; \frac{18}{6} = 3; \frac{54}{18} = 3$$

Since the ratio is constant, the sequence is geometric

The first term: $u_1 = 2$; common ratio: $r = 3$

- using the general formula of geometric sequence $u_n = u_1 r^{n-1}$.

Since $u_1 = 2$, and $r = 3$ then $u_n = 2 \cdot 3^{n-1}$

Example 3:

Find the product of the first 20 terms of a geometric sequence if the first term is 1 and common ratio is 2.

Solution

$$P_n = (u_1)^n r^{\frac{n(n-1)}{2}}$$

Here, $u_1 = 1, r = 2, n = 20$.

$$\text{Then, } P_{20} = (1)^{20} 2^{\frac{20(19)}{2}} = 2^{190}.$$

The sum to infinity of a geometric series with first term u_1 and the common ratio r is

$$S_{\infty} = \frac{u_1}{1-r} \text{ provided } -1 < r < 1.$$

Example 4:

Given the geometric progression 16, 12, 9, ... Find the sum of terms up to infinity.

Solution

Here $u_1 = 16, r = \frac{12}{16} = \frac{3}{4}$

Thus $-1 < r < 1$ and hence the sum to infinity will exist

$$S_{\infty} = \frac{u_1}{1-r} = \frac{16}{1-\frac{3}{4}} = 64.$$

The sum to infinity is 64.

Example 5:

Express the recurring decimal $\overline{0.32}$ as a rational number.

Solution

$\overline{0.32} = \frac{32}{10^2} + \frac{32}{10^4} + \frac{32}{10^6} + \dots$ which is an infinite geometric series with first term $u_1 = 0.32$ and common ratio is $r = 0.01$. Since $-1 < r < 1$, the sum to infinity exists and equal to

$$\frac{u_1}{1-r} = \frac{0.32}{1-0.01} = \frac{0.32}{0.99} = \frac{32}{99}.$$

Therefore, $\overline{0.32} = \frac{32}{99}.$

**Application activity 2.11**

1. Find the sum of the first 8 terms of the geometric sequence 32, -16, 8, ...
2. Find the sum of the geometric sequence with the first term 0.99 and the common ratio is equal to the first term.

2.12 Convergence of sequences**Activity 2.12**

As n tends to plus infinity, evaluate the value each of the following sequences will tend to.

1. $\left\{ \frac{n^2 - 1}{n^3} \right\}$

2. $\left\{ \sqrt{n+1} - \sqrt{n} \right\}$

3. $\left\{ n^2 \right\}$



Content summary

A numerical sequence is said to be **convergent** if the limit exists whereas if the limit does not exist (or is infinity) the sequence is said to be **divergent**.

A number L is called a **limit** of a numerical sequence $\{u_n\}$ if $\lim_{n \rightarrow \infty} u_n = L$.

Example 1:

Determine whether the sequence $\left\{ \frac{n}{2n+1} \right\}_{n=1}^{+\infty}$ converges or diverges.

Solution

First, we find the limit of this sequence as n tends to infinity

$$\lim_{n \rightarrow +\infty} \frac{n}{2n+1} = \lim_{n \rightarrow +\infty} \frac{n}{n \left(2 + \frac{1}{n} \right)} = \lim_{n \rightarrow +\infty} \frac{1}{2 + \frac{1}{n}} = \frac{1}{2}$$

Thus, $\left\{ \frac{n}{2n+1} \right\}_{n=1}^{+\infty}$ converges to $\frac{1}{2}$.

Example 2:

Determine whether the sequence $\{8-2n\}_{n=1}^{+\infty}$ converges or diverges.

Solution

First, we find the limit of this sequence as n tends to infinity

$$\lim_{n \rightarrow +\infty} (8-2n) = 8-2(+\infty) = -\infty.$$

Thus, $\{8-2n\}_{n=1}^{+\infty}$ diverges.



Application activity 2.12

Which of the sequences converge, and which diverge? Find the limit of each convergent sequence.

1. $\{2 + (0.1)^n\}$
2. $\left\{ \frac{1-2n}{1+2n} \right\}$

$$3. \left\{ \frac{1-5n^4}{n^4+8n^3} \right\}$$

2.13 Application of sequences in solving real life problems



Activity 2.13

Discuss how sequences are used in real life problems.



Content summary

There are many applications of sequences. Sequences are useful in our daily lives as well as in higher mathematics. For example; the interest portion of monthly payments made to pay off an automobile or home loan, and the list of maximum daily temperatures in one area for a month are sequences. Sequences are used in **calculating interest, population growth, half-life and decay in radioactivity ...**

Example 1:

A child builds a **tower with blocks** uses 15 for the bottom row. Each row has 2 fewer blocks than the previous row. Suppose that there are 8 rows in the tower.

- How many blocks are used for the top row?
- What is the total number of blocks in the tower?

Solution

- The number of blocks in each row forms an arithmetic sequence with $u_1 = 15$ and $d = -2$

$n = 8, u_8 = u_1 + (8 - 1)(-2)$. There is just one block in the top row.

- Here we must find the sum of the terms of the arithmetic sequence formed with

$$u_1 = 15, n = 8, u_8 = 1, S_8 = \frac{8}{2}(15+1) = 64.$$

There are 64 blocks in the tower.

Example 2:

An **insect population is growing** in such a way that each new generation is 1.5 times as large as the previous generation. Suppose there are 100 insects in the first generation.

- a) How many will there be in the fifth generation?
- b) What will be the total number of insects in the five generations?

Solution

- a) The population can be written as a geometric sequence with $u_1 = 100$ as the first-generation population and common ratio $r = 1.5$. Then the fifth-generation population will be $u_5 = 100(1.5)^{5-1} = 506.25$. In the fifth generation, the population will number about 506 insects.
- b) The sum of the first five terms using the formula for the sum of the first n terms of a geometric sequence.

$$S_5 = \frac{100(1 - (1.5)^5)}{1 - 1.5} = 1318.75$$

The total population for five generations will be about 1319 insects.

The compound interest formula:

$A = P \left(1 + \frac{r}{k} \right)^{kt}$ with P = principle, t = time in years, r = annual rate, and k = number of periods per year. And Finally the interest $I = A - P$

The simple interest formula:

$I = Prt$ with I = total interest, P = principle, r = annual rate, and t = time in years.

Example 3:

Find the accumulated value of 15,000 Frw at 5% per year for 18 years using simple interest.

Solution

$$P = 15000, r = 0.05, t = 18$$

$$I = 15000(0.05)(18) = 13500.$$

A total of 13,500 Frw in interest will be earned.

Hence, the accumulated value in the account will be $13,500\text{Frw} + 15,000\text{ Frw} = 28,500\text{ Frw}$.

Example 4:

A mine worker discovered an ore sample containing 500 mg of radioactive material. It is discovered that radioactive material has a half-life of 1 day. Find the amount of radioactive material in the sample at the beginning of the 7th day.

Solution

Half-life of one-day means that half of the amount remains after 1 day.

Begin of day 1: 500 mg	Begin of day 2: 250 mg	Begin of day 3: 125 mg	...
End of day 1: 250 mg	End of day 2: 125 mg	End of day 3: 62.5 mg	...

Decide to either work with the "beginning" of each day, or the "end" of each day, as each can yield the answer. Only the starting value and number of terms will differ. We will use "beginning":

$$u_n = u_1 r^{n-1}$$

$$u_8 = 500 \left(\frac{1}{2} \right)^{7-1} = 7.8125 \text{ mg}$$



Application activity 2.12

- 1) A ball is dropped from a height of 10 m and after each bounce, returns to a height which is 84% of the previous height. Calculate the total distance travelled by the ball before coming to rest.
- 2) If Linda deposits 1300 Frw in a bank at 7% interest compounded annually, how much will be in the bank 17 years later?



END OF UNIT ASSESSMENT

Knowledge and understanding

1. Define a sequence and a series. How do they differ in mathematical notation?
2. Given the following patterns, identify which is Arithmetic, which is Geometric, and which is neither:
 - (a) 2, 6, 18, 54....
 - (b) 10, 7, 4, 1,
 - (c) 1, 4, 9, 16, ...

Skills

3. Which of the following sequences converge, and which diverge? Find the limit of each convergent sequence.

a) $\left\{ \sqrt{\frac{2n}{n+1}} \right\}$

b) $\frac{n}{2^n}$

4. A mathematical child negotiates a new pocket money deal with her unsuspecting father in which she receives 1 Frw on the first day of the month, 2 Frw on the second day, 4 Frw on the third day, 8 Frw on the fourth day, 16 Frw on the fifth day, ... until the end of the month. How much would the child receive during a month of 30 days?
5. The culture of bacteria doubles every 2 hours. If there are 500 bacteria at the beginning, how many bacteria will there be after 24 hours?
6. You complain that the hot tub in your hotel suite is not hot enough. The hotel tells you that they will increase the temperature by 10% each hour. If the current temperature of the hot tub is 75° F, what will be the temperature of the hot tub after 3 hours, to the *nearest tenth* of a degree?
7. Consider the infinite geometric series $\sum_{n=1}^{\infty} 10 \left(1 - \frac{3x}{2} \right)^n$
 - a. For what values of x does a sum to infinity exist?
 - b. Find the sum of the series if $x = 1.3$

8. Let the sequence $\{u_n\}$, $\forall n \in \mathbb{N}_0 : u_0 = 9$ and $u_{n+1} = \frac{1}{2}u_n - 3$
- Calculate u_1, u_2, u_3 , and u_4 .
 - $\forall n \in \mathbb{N}_0$, set $v_n = u_n + 6$ show that the sequence $\{v_n\}$ is geometric and determines the common ratio and 1st term.
 - Express v_n and u_n in function of n .
 - Calculate the sum $s_n = v_0 + v_1 + \dots + v_n$ in function of n and deduce
- $$s_n = u_0 + u_1 + \dots + u_n$$

Attitude and values

9) A financial institution aggressively markets a high-interest loan featuring compound interest to vulnerable individuals who clearly do not understand exponential growth. From an ethical and professional standpoint, how should this business practice be evaluated?

- **A)** It is ethical because it is the sole responsibility of the borrower to understand the mathematical terms of a contract before signing it.
- **B)** It is unethical because it exploits a lack of mathematical/financial literacy, violating the values of fairness, transparency, and social responsibility.
- **C)** It is ethical because maximizing profit for the institution is the primary goal, regardless of the borrower's understanding of compound sequences.
- **D)** It is neutral because ethics do not apply to standard financial mathematical applications or loan structures.

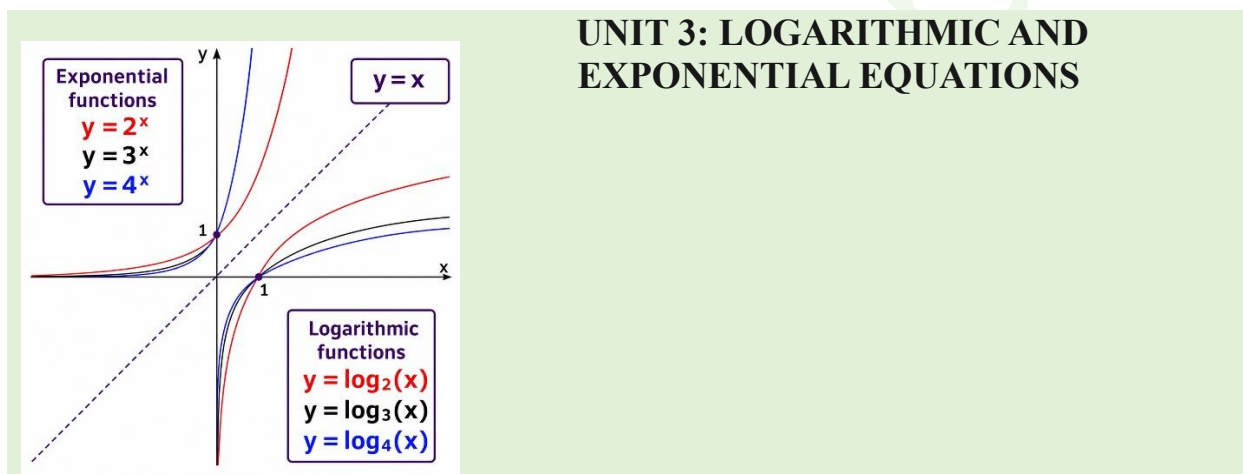
10) When sharing research or statistical trends with the public, a researcher decides to present only the final "sum" or the ultimate result, intentionally hiding the step-by-step sequence of events or data points over time. How does this action impact the core values of honesty and academic transparency?

- **A)** It enhances clarity because the public only needs to see final numbers rather than being confused by a long sequence of intermediate terms.
- **B)** It is acceptable as long as the final sum is mathematically accurate, even if the underlying sequence contains conflicting trends.
- **C)** It compromises honesty and transparency because hiding the full sequence can mask critical fluctuations, misrepresent data trends, and manipulate public perception.
- **D)** It has no impact on transparency because sequences and final sums always communicate identical messages to an audience.

11) You observe a close friend repeatedly taking out short-term, high-interest loans where the debt compounds exponentially each month, arguing that they will "worry about the total sum

later." Which response best demonstrates a responsible attitude toward mathematical sequences and financial planning?

- **A)** Encourage them to keep borrowing, since the initial terms of the sequence (u_1) look small and manageable.
- **B)** Advise them to pause and model the monthly growth using the geometric or compound sequence formula to visually understand how rapidly the debt will grow over time, prioritizing long-term financial safety over short-term relief.
- **C)** Remain silent because analyzing the future terms of a financial sequence is unnecessary unless the bank demands immediate payment.
- **D)** Suggest they look for a different bank that calculates the sequence using a higher interest rate so they can reach the final sum quicker.



Key unit competence: Apply exponential or logarithmic functions and equations in solving mathematics and real-life problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain exponential and logarithmic functions and identify their properties.
- Explain the relationship between exponential and logarithmic functions.
- State and apply the laws of exponents and logarithms.
- Describe the nature of exponential growth and decay processes.
- Recognize real-life situations that can be modelled by exponential and logarithmic equations.

Skills:

- Construct and interpret graphs of exponential and logarithmic functions.
- Simplify and manipulate exponential and logarithmic expressions.
- Solve exponential and logarithmic equations analytically and graphically.

- Apply exponential and logarithmic models to real-life problems such as interest, population growth, radioactive decay, mortgages, earthquakes, carbon dating, and alcohol-risk analysis.
- Interpret solutions in context and draw meaningful conclusions

Attitudes and values:

- Appreciate the usefulness of exponential and logarithmic models in real-life problem-solving.
- Show persistence and logical reasoning when solving equations and modeling data.
- Develop accuracy and critical thinking in mathematical computations.
- Demonstrate teamwork, communication, and a positive attitude toward applying mathematics in daily life.

Introduction

People such as scientists, sociologists and town planners are often more concerned with the rate at which a particular quantity is growing than with its current size. The director of education is more concerned with the rate at which the school population is increasing or decreasing than with what the population is now, because he/she has to plan for the future and ensure that there are enough (and not too many) school places available to meet demand each year. The scientists may need to know the rate at which a colony of bacteria is growing rather than how many of the bacteria exists at this moment, or the rate at which a liquid is cooling rather than the temperature of the liquid now, or the rate at which a radioactive material is decaying rather than how many atoms current exist.

One thing that each of these populations has in common, is that their rate of increase is proportional to the size of the population at any time. Exponential and logarithmic equations are really relevant in our day to day activities. The above events show us the areas where this unit finds use in our daily activities.



Introductory activity

A bacteria culture starts with **1 bacterium**. If it **doubles every hour**, complete the table:

Hour (t)	Number of Bacteria
0	1
1	
2	
3	
4	
5	
6	

Which function describes the number of bacteria over time?

3.1 Exponential functions



Activity 3.1

Draw the graph of $y = 2^x$ for $-2 \leq x \leq 3$

In the same plane, sketch the graph of $y = 2$ and $y = -3$. How many times does any horizontal line cross the curve of $y = 2^x$? How can you conclude?



Content summary

For any positive number $a > 0$, there is a function $f: \mathbb{R} \rightarrow (0, \infty)$ called an exponential function that is defined as $f(x) = a^x$. For example, $f(x) = 3^x$ and $f(x) = \left(\frac{3}{4}\right)^x$ are exponential functions.

The constant a is called the base and the variable x is called exponent.

When the $a = e$, the exponential function $f(x) = e^x$ is called **natural exponential function**.

Example 1:

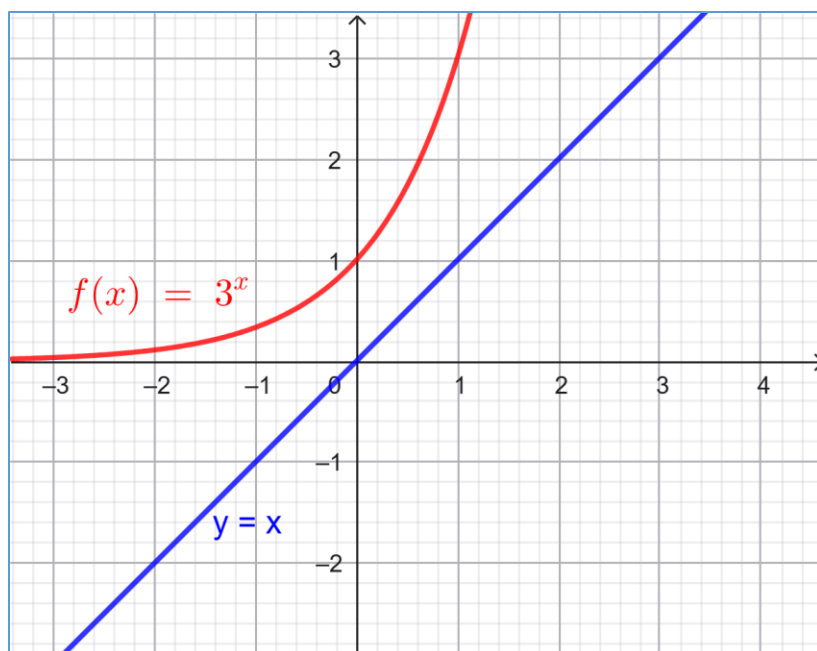
In the same Cartesian plane, sketch the curve of the function $f(x) = 3^x$ for $-2 \leq x \leq 2$ and its inverse $f^{-1}(x)$ with the first bisector.

Solution

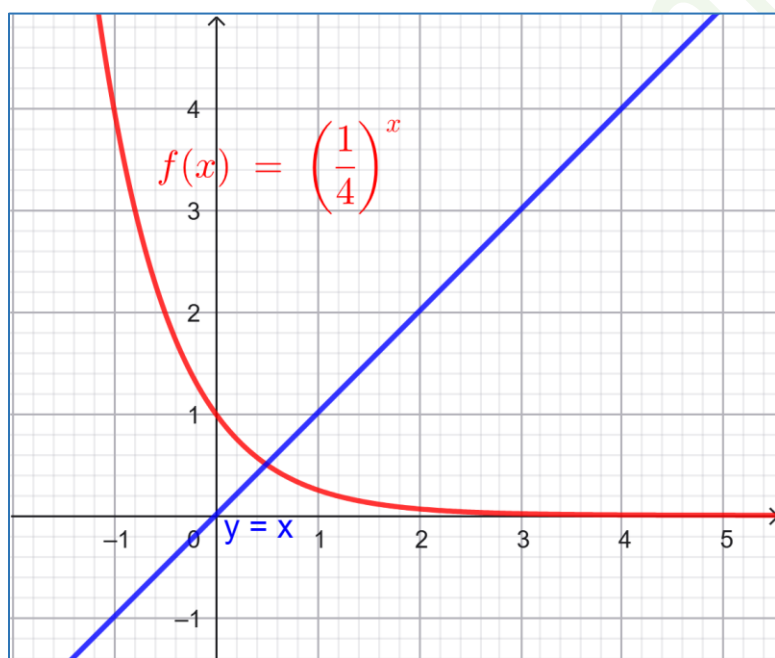
Table of coordinates of $f(x) = 3^x$

x	-2	-1.6	-1.2	-0.8	-0.4	0	0.4	0.8	1.2	1.6	2
y	0.1	0.2	0.3	0.4	0.6	1.0	1.6	2.4	3.7	5.8	9.0

a) For $a > 1$, the function is **strictly increasing**. As x increases, $y = a^x$ increases rapidly.



b) When $0 < a < 1$ there is an exponential decay.



This function is decreasing rapidly.

Basic rules for exponents

For $a > 0$ and $a \neq 1, m, n \in \mathbb{R}$

$$1) \quad a^m \times a^n = a^{m+n}$$

$$2) \quad a^m \div a^n = a^{m-n}$$

$$4) \quad a^{-n} = \frac{1}{a^n}$$

$$3) \quad (a^m)^n = a^{m \times n}$$

$$5) \quad a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$6) \quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$



Application activity 3.1

Sketch the following functions in Cartesian plane.

$$1. \quad y = \left(\frac{1}{2}\right)^x, -3 \leq x \leq 3$$

$$2. \quad y = \left(\frac{1}{3}\right)^x, -2 \leq x \leq 2$$

3.2 Logarithmic functions



Activity 3.2

Reflect $y = 2^x$ on the line $y = x$ and name the new curve $g(x)$. How is this new function called?

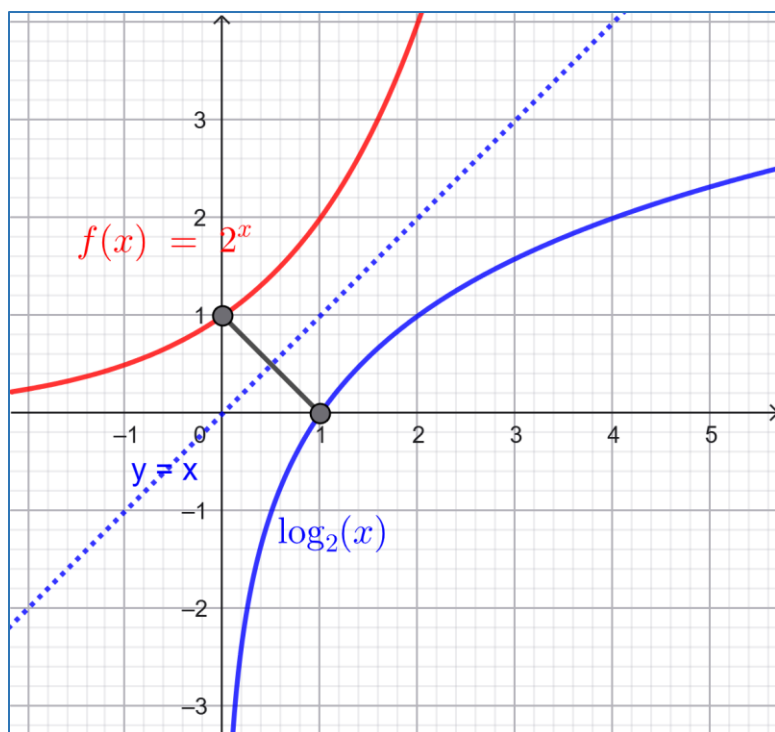


Content summary

Table of coordinates of $f^{-1}(x)$

x	0.1	0.2	0.3	0.4	0.6	1.0	1.6	2.4	3.7	5.8	9.0
y	-2	-1.6	-1.2	-0.8	-0.4	0	0.4	0.8	1.2	1.6	2

The graph of the two functions is given below.



Behavior of logarithmic function $y = f(x) = \log_a x$

- In standard mathematics, a logarithmic function with a negative base ($a < 0$) is undefined for real numbers.
- When $0 < a < 1$ The function is **strictly decreasing**. As x increases, y decreases.
- For $a > 1$, the function is **strictly increasing**. As x increases, y also increases.

Basic rules for logarithms

$\forall x, y \in]0, +\infty[, a \in]0, +\infty[\setminus \{1\}$:

a. $\log_a xy = \log_a x + \log_a y$

b. $\log_a \frac{1}{y} = -\log_a y$

c. $\log_a \frac{x}{y} = \log_a x - \log_a y$

d. $\log_a x^r = r \log_a x$

e. $\log_a x = \frac{\log_b x}{\log_b a}$

f. $\log_a a = 1$

g. $\log_a 1 = 0$

h. If $y = \log_a x$, then $x = a^y$

i. $a^{\log_a b} = b$

Example 1:

Write $2^6 = 64$ in logarithmic form.

Solution

$$2^6 = 64 \Leftrightarrow 2^6 = 2^{\log_2 64} \Rightarrow \log_2 64 = 6$$

Example 2:

Write $\log_m b = c$ in exponential form.

Solution

$$\log_m b = c \Rightarrow b = m^c$$

Example 3:

Find x if $\log_2 32 = x$

Solution

$$\log_2 32 = x \Rightarrow 32 = 2^x$$

But $32 = 2^5$.

$$\text{So } 32 = 2^x \Leftrightarrow 2^5 = 2^x \Rightarrow x = 5$$

Example 4:

Find the numerical value of $\log_3 \sqrt[3]{9}$

Solution

$$\text{Let } y = \log_3 \sqrt[3]{9}, \text{ then } 3^y = \sqrt[3]{9}$$

$$\Leftrightarrow 3^y = 9^{\frac{1}{3}} \Leftrightarrow 3^y = 3^{2\left(\frac{1}{3}\right)} \Leftrightarrow 3^y = 3^{\frac{2}{3}} \Rightarrow y = \frac{2}{3}.$$

$$\text{Hence, } \log_3 \sqrt[3]{9} = \frac{2}{3}$$

There are two common bases for logarithms, 10 and e . The number e is called Neperian number and it is irrational number whose approximate value $e \approx 2.718281828$. Note that this number can be expressed using limit as $e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$. The logarithmic function with base 10 is called decimal logarithm denoted as $f(x) = \log_{10} x$ or simply $f(x) = \log x$. When the base is e , the logarithmic function is called natural logarithm denoted as $f(x) = \ln x$.

You should find a "ln" button on your scientific calculator which will evaluate logarithms to base e and "log" button to evaluate logarithms to base 10.

Basic rules for logarithms

$$\forall x, y \in]0, +\infty[, a \in]0, +\infty[\setminus \{1\} :$$

$$\text{a. } \ln xy = \ln x + \ln y.$$

$$\text{e. } \ln e = 1.$$

$$\text{f. } \ln 1 = 0.$$

$$\text{b. } \ln \frac{x}{y} = \ln x - \ln y.$$

$$\text{c. } \ln x^y = y \ln x.$$

$$\text{d. } \log_a x = \frac{\ln x}{\ln a}.$$

..

$$\text{g. If } y = \ln x, \text{ then } x = e^y.$$

$$\text{h. } e^{\ln x} = x.$$



Application activity 3.2

Sketch the following functions in Cartesian plane with their inverses

$$\text{a) } y = \left(\frac{1}{2}\right)^x, -3 \leq x \leq 3$$

$$\text{b) } y = \left(\frac{1}{3}\right)^x, -2 \leq x \leq 2$$

3.3 Exponential equations



Activity 3.3

Suppose $P(t) = 100e^{0.012t}$ gives the population of a city (in thousands) after t years. When will the population be 200, 000?



Content summary

An exponential equation is an equation where the variable appears in the exponent.

Equation of the form $a^{f(x)} = a^{g(x)}$:

If the bases are identical ($a > 0, a \neq 1$), then their exponents must be equal:

$$f(x) = g(x)$$

Example 1: $2^{3x-1} = 32$

Rewrite 32 as a power of 2: $2^{3x-1} = 2^5$ $3x-1=5$ and $x=2$

Equation of the form $a^{f(x)} = b^{g(x)}$ (where $a \neq b$)

The power rule of logarithms, $\log(m^n) = n \log(m)$.

Example 2:

$$2^{3x} = 3^{2x-1}$$

Solution

Given that $2^{3x} = 3^{2x-1}$, taking logarithms of both sides and applying logarithmic laws give

$$3x \log 2 = (2x-1) \log 3 \Leftrightarrow 3x \log 2 = 2x \log 3 - \log 3$$

$$\Leftrightarrow 3x \log 2 - 2x \log 3 = -\log 3 \Leftrightarrow x(3 \log 2 - 2 \log 3) = -\log 3$$

$$\Leftrightarrow x = \frac{-\log 3}{3 \log 2 - 2 \log 3} \Rightarrow x \approx 9.327$$

3) Equation of the form $A(a^x)^2 + B(a^x) + C = 0$ or $Aa^{2x} + Ba^x + C = 0$

Substituting $y = a^x$ (where $y > 0$) turns the expression into $Ay^2 + By + C = 0$

Example 3:

Solve the equation $2(5^{2x}) - 5^x = 6$.

Solution

Let $y = 5^x$, with $y > 0$. Then $2y^2 - y = 6$ or $2y^2 - y - 6 = 0$.

$$(2y+3)(y-2) = 0 \Rightarrow y = -1\frac{1}{2} \text{ or } y = 2.$$

The value $y = -1\frac{1}{2}$ is to be excluded since $y = 5^x$ must be positive.

$$\text{So } y = 2 \text{ gives } 5^x = 2 \Rightarrow x = \log_5 2 = \frac{\log 2}{\log 5} = 0.431.$$

Equation of the form $a^{f(x)} = b^{f(x)}$ (where $a \neq b$ and $a, b > 0$)

Since two different numbers raised to the same power can only be equal if that power evaluates to zero (because $a^0 = 1$ and $b^0 = 1$), you solve these by setting the structural exponent to zero.

If $a^{f(x)} = b^{f(x)}$, then: $f(x) = 0$

Example 4:

Solve for x: $5^{2x-4} = 7^{2x-4}$

Solution:

Because the bases 5 and 7 are completely different but the powers match, the exponent must equal 0: $2x - 4 = 0$

Solve for x and find: $2x = 4 \Rightarrow x = 2$.

Equating the Bases

If you can write both sides of the equation using the same base, you can use the property:

If $b^x = b^y$ (where $b > 0$ and $b \neq 1$), then $x = y$.

Example 5: Solve for x: $2^{x+3} = 16$

Solution: $16 = 2^4$

$$2^{x+3} = 2^4$$

Since the bases are identical, set the exponents equal to each other: $x + 3 = 4$

$$x = 1$$

Example 6: Fractional or Reciprocal Bases

$$\text{Solve for } x: \left(\frac{1}{3}\right)^{x-2} = 9$$

Solution:

Express both sides using a common base of 3, remembering that $\frac{1}{3} = 3^{-1}$ and $9 = 3^2$:

$$(3^{-1})^{x-2} = 3^2$$

Apply the power-to-a-power rule $(a^m)^n = a^{m \cdot n}$: $3^{-1(x-2)} = 3^2$

$$3^{-x+2} = 3^2$$

Equate the exponents: $-x + 2 = 2$

$$-x = 0 \Rightarrow x = 0$$

Using Logarithms (When Bases Differ)

When the bases cannot be easily rewritten to match, take the logarithm (common log $\log$ or natural log $\ln$) of both sides. This allows you to bring the exponent down using the power property: $\log(a^b) = b \cdot \log(a)$.

Example 7: Unmatched Bases

$$\text{Solve for } x: 5^x = 12$$

Solution:

Take the natural logarithm ($\ln$) of both sides: $\ln(5^x) = \ln(12)$

Move the exponent x to the front using the power property: $x \cdot \ln(5) = \ln(12)$

$$\text{Isolate } x \text{ by dividing by } \ln(5): x = \frac{\ln(12)}{\ln(5)}$$

$$\text{Evaluate using a calculator: } x \approx \frac{2.4849}{1.6094} \approx 1.544.$$

Reducible to Quadratic Form (Substitution)

Some exponential equations resemble quadratic equations and can be solved by substituting a temporary variable (often u).

Example 8: Quadratic Style

$$\text{Solve for } x: 9^x - 4(3^x) + 3 = 0$$

Solution:

$$\text{Notice that } 9^x = (3^2)^x = (3^x)^2.$$

$$\text{Rewrite the equation: } (3^x)^2 - 4(3^x) + 3 = 0$$

Let $u = 3^x$.

Substitute u into the equation: $u^2 - 4u + 3 = 0$

Factor the quadratic equation: $(u - 3)(u - 1) = 0$

So, $u = 3$ or $u = 1$

Substitute 3^x back in for u to solve for x :

○ Case 1: $3^x = 3 \Rightarrow 3^x = 3^1 \Rightarrow x = 1$

○ Case 2: $3^x = 1 \Rightarrow 3^x = 3^0 \Rightarrow x = 0$

The solutions are $x = 1$ and $x = 0$.



Application activity 3.3

1. Find the general solution of the following equation $9^{\cos x} - 2 \times 3^{\cos x} + 1 = 0$.
2. Solve for x : $9^x - 2 \times 3^{x+1} = 27$.

3.4 Logarithmic equations



Activity 3.4

Prove each of the following logarithmic laws

1. $\log_a(m^p) = p \log_a m$ (Power Law)

2. $\log_a b = \frac{\log_c b}{\log_c a}$ (Change of Base Law)



Content summary

A logarithmic equation is an equation in which the desired variable is inside one or more logarithms.

Type 1: Single Logarithmic Term

These equations contain only one logarithm on one side of the equation and a constant on the other.

Structure: $\log_b(f(x)) = c$

When solving, convert the equation directly into its equivalent exponential form: $b^c = f(x)$, and then solve for x .

Example 1: Solve for x : $\log_2(3x - 1) = 3$

Solution: Convert to exponential form: $2^3 = 3x - 1$

Simplify the constant: $8 = 3x - 1$

Isolate x : $9 = 3x \Rightarrow x = 3$

Check: $\log_2(3(3) - 1) = \log_2(8) = 3$. The solution is valid.

Type 2: Equating Two Logarithms (Same Base)

Structure: $\log_b(f(x)) = \log_b(g(x))$

Solving Strategy: Use the **Property of Equality for Logarithmic Functions**. If the bases match, their arguments must be equal: $f(x) = g(x)$.

Example 2: Solve for x : $\ln(x^2 - 6) = \ln(x)$

Solution: Set the arguments equal to each other: $x^2 - 6 = x$

Rearrange into a standard quadratic equation: $x^2 - x - 6 = 0$

Factor the quadratic: $(x - 3)(x + 2) = 0$

This gives potential solutions of $x = 3$ or $x = -2$.

Check for extraneous solutions:

If $x = 3$, the argument $\ln(3)$ is defined.

If $x = -2$, the argument $\ln(-2)$ is **undefined** (you cannot take the log of a negative number).

Therefore, $x = -2$ is extraneous. The only valid solution is $x = 3$.

Note: it is better to set the condition before solving: $\ln(x^2 - 6) = \ln(x)$, this means $x^2 - 6 > 0$ and $x > 0$. These conditions show directly that $x = -2$ must be rejected.

Type 3: Multiple Logarithmic Terms

These equations contain multiple log terms on one or both sides, often combined with constants.

Structure: $\log_b(f(x)) + \log_b(g(x)) = c$ or $\log_b(f(x)) - \log_b(g(x)) = \log_b(h(x))$

Solving Strategy: Use the laws of logarithms (Product, Quotient, or Power properties) to condense the multiple logs into a single log term. Once condensed, treat it as a Type 1 or Type 2 equation.

Key rules:

Product Property: $\log_b(M) + \log_b(N) = \log_b(M \cdot N)$

Quotient Property: $\log_b(M) - \log_b(N) = \log_b\left(\frac{M}{N}\right)$

Power Property: $k \cdot \log_b(M) = \log_b(M^k)$

Example 3: Solve for x: $\log_3(x) + \log_3(x-8) = 2$

Use the product property to condense the left side: $\log_3(x(x-8)) = 2 \Rightarrow \log_3(x^2 - 8x) = 2$

Convert the condensed equation to exponential form (Type 1): $3^2 = x^2 - 8x$

$$9 = x^2 - 8x$$

Set to zero and solve the quadratic: $x^2 - 8x - 9 = 0$

$$(x-9)(x+1) = 0$$

Potential solutions: $x = 9$ or $x = -1$.

Check: Substituting $x = -1$ into the original equation creates $\log_3(-1)$, which is undefined.

Thus, $x = 9$ is the unique solution.

Type 4: Quadratic Form (Substitution)

These equations feature a logarithmic term raised to a power, creating a structure that mirrors a standard quadratic equation.

Structure: $A(\log_b x)^2 + B(\log_b x) + C = 0$

Solving strategy: Use u-substitution. Let $u = \log_b x$, solve the resulting quadratic equation for u, and then substitute $\log_b x$ back in to find x.

Example 4: Solve for x: $(\log x)^2 - 3\log x + 2 = 0$

Solution:

Let $u = \log x$. Rewrite the equation: $u^2 - 3u + 2 = 0$

Factor and solve for u: $(u-2)(u-1) = 0 \Rightarrow u = 2$ or $u = 1$

Substitute $\log x$ back in for u (assuming base 10):

○ Case 1: $\log x = 2 \Rightarrow 10^2 = x \Rightarrow x = 100$

○ Case 2: $\log x = 1 \Rightarrow 10^1 = x \Rightarrow x = 10$

Both solutions yield positive arguments, so $x = 100$ and $x = 10$ are both valid.

Note: Always check your solutions in the *original* equation. Arguments of logarithms must strictly be greater than zero ($f(x) > 0$). Any solution that results in taking the logarithm of zero or a negative number is **extraneous** and must be discarded.

Example 5: Solve $\log_3(7x+3) = \log_3(5x+9)$.

Solution:

Drop the logarithms to get $7x+3 = 5x+9$.

Finish solving by subtracting $5x$ from each side, subtracting 3 from each side, and finally dividing each side by 2 to get $x = 3$.

Example 6: Solve $\log_2(5x+7) = 5$.

Solution:

Rewrite the problem in exponential form by moving the base of the logarithm to the other side.

$$5x+7 = 2^5.$$

Simplify the problem by raising 2 to the fifth power.

$$5x+7 = 32.$$

Solve for x by subtracting 7 from each side and then dividing each side by 5 to get $x = 5$.

Example 7: Solve $\log_4 x + \log_4(x-12) = 3$.

Solution:

$$\log_4 x(x-12) = 3.$$

$$x(x-12) = 4^3.$$

$$x^2 - 12x - 64 = 0$$

$$(x+4)(x-16) = 0 \Rightarrow x = -4 \text{ and } x = 16.$$

Only one answer $x = 16$ is acceptable when it is plugged back in.



Application activity 3.4

1. Given that $\log_m x = p$, express each of the following in terms of p

a) $\log_m(x^4)$

b) $\log_m\left(\frac{1}{x^2}\right)$

c) $\log_m(mx)$

2. Solve for x

$$(a) \ln(x^2 - 1) = \ln(4x - 1) - 2 \ln 2$$

$$(b) 2 \log_2 x + \log_x 2 = 3$$

$$(c) \begin{cases} 2 \ln x + 3 \ln y = -2 \\ 3 \ln x + 5 \ln y = -4 \end{cases}$$

$$(d) \begin{cases} \ln(xy) = 7 \\ \ln \frac{x}{y} = 1 \end{cases}$$

3.5 Applications of exponential and logarithmic equations

3.5.1 Interest and Population growth problems.



Activity 3.5.1

In a laboratory, for experiment, we start with 2 cells in a dish. The number of cells in the dish doubles every 30 minutes.

- How many cells are in the dish after four hours?
- After what time are there 2^{13} cells in the dish?

After $10\frac{1}{2}$ hours there are 2^{22} cells in the dish and an experiment fluid is added which eliminates half of the cells. How many cells are left?



Content summary

To calculate exactly how an investment grows with compounding, we use the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

Where: **A** = the future value of the investment/loan, including interest; **P** = the principal investment amount (the initial deposit); **r** = the annual interest rate (decimal format, so 5% becomes 0.05); **n** = the number of times interest is compounded per year (e.g., 12 for monthly, 4 for quarterly, 1 for annually); **t** = the time the money is invested or borrowed for, in years.

It is clear that $t = \frac{\ln\left(\frac{A}{P}\right)}{n \cdot \ln\left(1 + \frac{r}{n}\right)}$.

Example 1:

Jack operates an account with a certain bank which pays a compound interest rate of 13.5% per annum. He opened the account at the beginning of the year with 500,000 FRW and deposits the same amount of money at the beginning of every year. Calculate how much he will receive at the end of 9 years. After how long will the money have accumulated to FRW 3.32 million?

Solution

The compound interest formula:

The 1st deposit will be

$$500000 + \frac{500000 \times 13.5}{100} = 500000 \left(1 + \frac{13.5}{100}\right) \text{ or } 500000 + \frac{500000 \times 13.5}{100} = 500000 \times 1.135.$$

The 2nd deposit will grow to $500000 \times (1.135)^2$

The 3rd deposit will grow to $500000 \times (1.135)^3$

The nth deposit will grow to $500000 \times (1.135)^n$

So, the 9th deposit will grow to $500000 \times (1.135)^9$

The total sum

$$\begin{aligned} S_9 &= 500000 \times (1.135) + 500000 \times (1.135)^2 + 500000 \times (1.135)^3 + \dots + 500000 \times (1.135)^9 \\ &= 500000 \left[1.135 + (1.135)^2 + (1.135)^3 + \dots + (1.135)^9 \right] \end{aligned}$$

From $S_n = u_1 \left(\frac{1-r^n}{1-r} \right)$, we get

$$S_9 = 500000 \left[1.135 \left(\frac{1 - (1.135)^9}{1 - 1.135} \right) \right] = 8,936,281 \text{ or}$$

$$S_9 = \frac{500000 \times 1.135 \times 2.125811278}{0.135} = \frac{500000 \times 1.135 \times 2.125811278}{0.135} = 8,936,281.$$

Finding how long it will take the money to accumulate to 3,320,000 FRW

$$S_n = 3320000$$

$$\Rightarrow 500000 \left[1.135 \left(\frac{1 - (1.135)^n}{1 - 1.135} \right) \right] = 3320000$$

$$\Rightarrow \frac{1-(1.135)^n}{1-1.135} = \frac{3320000}{500000 \times 1.135}$$

$$\Leftrightarrow \frac{1-(1.135)^n}{-0.135} = \frac{3320000}{500000 \times 1.135}$$

$$\Leftrightarrow 1-(1.135)^n = -\frac{332 \times 0.135}{50 \times 1.135}$$

$$\Leftrightarrow (1.135)^n - 1 = \frac{332 \times 0.135}{50 \times 1.135}$$

$$\Leftrightarrow (1.135)^n - 1 = 0.7897$$

$$(1.135)^n = 0.7897 + 1, \quad (1.135)^n = 1.7897.$$

Introducing logarithm to base 10 on both sides gives

$$n \log(1.135) = \log(1.7897); \quad n = \frac{\log(1.7897)}{\log(1.135)}; \quad n \approx 4.6.$$

Hence it will take 4.6 years for the amount to accumulate to 3.32 million FRW

Example 2:

A man deposits 800,000 FRW into his savings account on which interest is 15% per annum. If he makes no withdrawals, after how many years will his balance exceed 8 million FRW?

Solution

Here, the interest rate will be compound such that amount is $P \left(1 + \frac{r}{100}\right)^n$, where n is period of

$$\text{time. } 8000000 = 800000 \left(1 + \frac{15}{100}\right)^n \Leftrightarrow 10 = (1 + 0.15)^n \Leftrightarrow 10 = (1.15)^n.$$

$$\log 10 = \log(1.15)^n \Leftrightarrow 1 = n \log(1.15) \Rightarrow n = \frac{1}{\log(1.15)} \approx 16.5 \text{ years}.$$

Population growth: Just like the compound interest formula, the basic continuous model for population growth relies on an initial value, a rate, and time: $N(t) = N_0 \cdot e^{rt}$

Where:

- $N(t)$ = the final population after a certain time.
- N_0 = the initial population size (at time $t = 0$).
- e = Euler's number (the mathematical constant approximately equal to 2.71828).
- r = the annual population growth rate (expressed as a decimal).
- t = the time period over which the population grows (usually in years).

Note: If the growth rate is negative ($r < 0$), the exact same formula models population decline or decay.

After isolating the time, we find that $t = \frac{\ln\left(\frac{N(t)}{N_0}\right)}{r}$.

Example 3:

According to United Nation data, the world population at the beginning of 1975 was approximately 4 billion and growing at rate of about 2% per year. Assuming an exponential growth model, estimate the world population at the beginning of the year 2020.

Solution:

Let t be time (in years) elapsed from the beginning of 1975.

$P(t)$ be world population in billions. Since the beginning of 1975 corresponds to $t = 0$, it follows from the given data that $P_0 = P(0) = 4$ (billions). Since the growth rate is 2% ($k = 0.02$), it flows that the world population at time will be $P(t) = P_0 e^{kt} = 4e^{0.02t}$.

Since the beginning of the year 2020 corresponds to $t = 45$ ($2020 - 1975 = 45$), it follows that the world population by the year 2020 will be $P(45) = 4e^{0.02 \times 45}$ (billion) or

$P(45) = 4e^{0.9}$ (billion) $= 4(2.459603)$ (billion) $= 9.838412$ (billion). Which is a population of approximately 9.8 billion. A population whose rate of decrease is proportional to the size of the population at any time obeys a law of the forms $P = Ae^{-kt}$. The negative sign on exponent indicates that the population is decreasing. This is known as exponential decay.



Application activity 3.5.1

The population of certain city is known to grow at a rate proportional to the number of people living in the city. If after ten years the population has trebled and if after 20 years the population is 180000. Then find the number of people initially living in the city.

3.5.2 Radioactive decay problems



Activity 3..5.2

The amount, $A(t)$ gram of radioactive material in a sample after t years is given by

$$A(t) = 80 \left(2^{-\frac{t}{100}} \right).$$

- Find the amount of material in the original sample.
- Calculate the half-life of the material (the **half-life** is the time taken for half of the original material to decay).

Calculate the time taken for the material to decay 1 gram.



Content summary

If a quantity has an exponential growth model, then the time required for it to double in size is called the **doubling time**. Similarly, if a quantity has an exponential decay model, then the time required for it to reduce in value by half is called the **halving time**. For radioactive elements, halving time is called **half-life**.

Doubling and halving times depend only on the growth rate and not on the amount present initially.

Doubling time for a quantity with an exponential growth model ($k > 0$) is $T = \frac{1}{k} \ln 2$ and

halving time for a quantity with an exponential decay model ($k < 0$) is $T = -\frac{1}{k} \ln 2$.

Example 1:

The radioactive element carbon-14 has a half-life of 5750 years. If 100 grams of this element are present initially, how much will be left after 1000 years?

Solution:

As $T = -\frac{1}{k} \ln 2$, the decay constant is

$$k = -\frac{1}{5750} \ln 2 = -\frac{1}{5750} \cdot 0.693147181 = -0.000120547 \approx -0.00012.$$

Radioactive decay obeys a law of the forms $P(t) = e_0^{-kt}$.

Thus, if we take $t = 0$ to be the present time, then $P_0 = P(0) = 100$, thus the amount of carbon-14 after 1000 years will be $P(1000) = 100e^{-0.00012(1000)} = 100e^{-0.12} \simeq 100(0.88692) = 88.692$. Thus about 88.692 grams of carbon-14 will remain.



Application activity 3.5.2

A radioactive substance has a half-life of 1620 years.

- If its mass is now 4 g (grams), how much will be left 810 years from now?
- Find the time t_1 when 1.5 g of the substance remains.

3.5.3 Mortgage problems



Activity 3.5.3

Conduct research in the library, on the internet, or by speaking with a bank officer. Write down the meaning of the following terms when you get a loan from the bank:

- i) The periodic payment (P)
- ii) The annual interest rate (r)
- iii) The mortgage amount (M)
- iv) The number of years t to cover the mortgage
- v) The number (n) of payments per year
- vi) Among all these elements/components, which is the most useful for the client to be informed about by the bank once he/she is given the mortgage loan?



Content summary

When a person gets a loan (mortgage) from the bank, the mortgage amount M , the number of payments or the number t of years to cover the mortgage, the amount of the payment P , how often the payment is made or the number n of payments per year, and the interest rate r , it is proved that all the 5 components are related by the following formula

$$P = \frac{\frac{rM}{n}}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}.$$

If P is monthly payment

$$P = \frac{M \left(\frac{r}{12} \right)}{1 - \left[\frac{1}{1 + \frac{r}{12}} \right]^{12t}}.$$

Example 1:

A businesswoman wants to apply for a mortgage of 75,000 Frw dollars with an interest of 8% per month that runs for 20 years. How much interest will she pay over the 20 years?

Solution:

Substituting for $M = 75,000$, $r = 0.08$, $t = 20$, $n = 12$ in the equation

$$P = \frac{\frac{rM}{n}}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}$$

$$P = \frac{\frac{(0.08)(75,000)}{12}}{1 - \left(1 + \frac{0.08}{12}\right)^{-(12)(20)}} = 627.33$$

Each month she will be paying 627.33 Frw.

The total amount she will pay is $627.33 \text{ Frw} \times 12 \times 20 \text{ Frw} = 150,559.2 \text{ Frw}$.

The interest will be $150,559.2 \text{ Frw} - 75,000 \text{ Frw} = 75,559.2 \text{ Frw}$.



Application activity 3.5.3

A bank can offer a mortgage at 10% interest rate to be paid back with monthly payments for 20 years. After analysis, a potential borrower finds that she can afford a monthly payment of 200,000 FRW. How much mortgage can she ask for?

3.5.4 Earthquake problems and Carbon dating problems



Activity 3.5.4

Do the research in the library or explore internet to find out how Charles Richter tried to compare the magnitude of two earthquakes by the use of logarithmic function.



Content summary

Magnitudes of **earthquakes** are measured using the **Richter scale**. On this scale, the

magnitude R of an earthquake is given by $R = \log\left(\frac{I}{I_0}\right)$ where I_0 is a fixed standard intensity

used for comparison, and I is the intensity of earthquakes being measured.

- Show that if an earthquake measures $R = 3$ on Richter scale, then its intensity is 1000 times the standard, that is, $I = 1000I_0$.
- The San Francisco earthquake of 1906 registered $R = 8.2$ on Richter scale. Express its intensity in terms of the standard intensity.

- c) How many times more intense is an earthquake measuring $R = 8$ than on measuring $R = 4$?

Solution:

- (a) If an earthquake measure $R = 3$ on Richter scale, then

$\log\left(\frac{I}{I_0}\right) = 3 \Rightarrow \frac{I}{I_0} = 10^3 \Leftrightarrow I = 10^3 I_0 \Leftrightarrow I = 1000 I_0$. Therefore, intensity is 1000 times the standard, that is $I = 1000 I_0$.

- (b) The San Francisco earthquake of 1906 registered $R = 8.2$ on Richter scale. It means that

$\log\left(\frac{I}{I_0}\right) = 8.2$ or $\frac{I}{I_0} = 10^{8.2} \Leftrightarrow I = 10^{8.2} I_0$ or which expresses its intensity in terms of the standard intensity.

- (c) Let E_1, E_2 be earthquakes measuring $R = 8$ and $R = 4$ respectively.

For E_1 : $R = 8 \Rightarrow \frac{I}{I_0} = 10^8 \Leftrightarrow I = 10^8 I_0$,

For E_2 : $R = 4 \Rightarrow \frac{I}{I_0} = 10^4 \Leftrightarrow I = 10^4 I_0$

Intensity of E_1 is $I_1 = 10^8 I_0$ (1)

Intensity of E_2 is $I_2 = 10^4 I_0$ (2)

The ratio of two above equations yields $\frac{I_1}{I_2} = \frac{10^8 I_0}{10^4 I_0} = 10^4 \Rightarrow I_1 = 10^4 I_2 \Leftrightarrow I_1 = 10000 I_2$.

An earthquake measuring $R = 8$ is 10000 times more intense than one measuring $R = 4$.



Application activity 3.5.4

The earthquake that took place in Ecuador in April 2016 was of magnitude 7.8 on Richter's scale. How intense was that earthquake compared to the one that took place in:

- a) The Mexico City in 1985, which was of magnitude 8.1 on Richter's.

b) San Francisco in 1906, which was of magnitude 6.9 on Richter's scale.

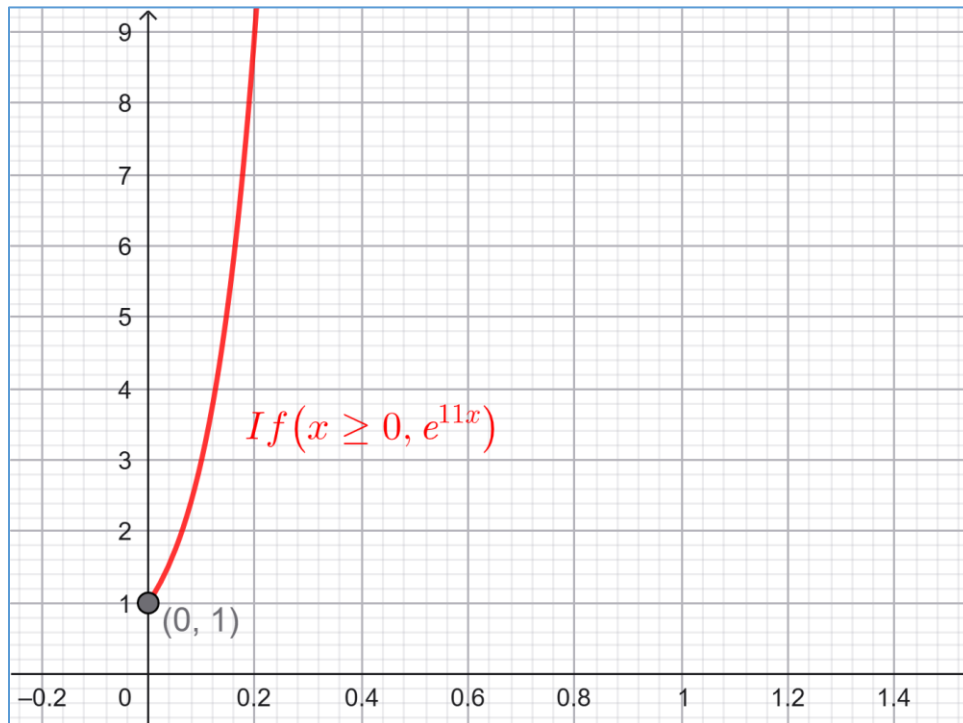
3.5.5 Problems about alcohol and risk of car accidents.



Activity 3.5.5

a) Discuss the dangers caused by drivers who drink alcohol in excess.

b) The following graph shows the risk of a car accident with respect to the driver's blood concentration of alcohol:



i) What is the risk when there is no alcohol in the blood? Why is that risk not 0?

ii) Comment on the variation of the risk with respect to the concentration of alcohol in the driver's blood.

iii) Write down approximately the type of the equation that can be used to model the risk.



Content summary

Science shows that the concentration of alcohol in a person's blood is measurable. Recent medical research suggests that the risk R (given as a percent) of having an accident while driving can be

modelled by an equation of the type $R(x) = R_0 e^{kx}$ where x is the variable concentration of alcohol in the blood and k is a constant.

Example 1:

The risk R of having an accident while driving is modelled by the equation $R(x) = 2e^{kx}$ where x is the concentration of alcohol in the driver's blood. Suppose that a concentration of alcohol in the blood of 0.06 results in 4% risk ($R = 4$) of an accident.

- Find the value of the constant k in the equation $R(x) = 2e^{kx}$
- Using this value of k , what is the risk if the concentration of alcohol is 0.08?
- Using the same value of k , what concentration of alcohol corresponds to a risk of 100%?
- If the law stipulates that anyone with a risk of having an accident of 10% or more should not drive, at what concentration of alcohol should the driver be arrested and charged?

Solution:

From the equation $R(x) = 2e^{kx}$, substituting

$$(a) \quad 4 = 2e^{k(0.06)} \Leftrightarrow e^{0.06k} = 2 \Rightarrow 0.06k = \ln 2 \Rightarrow k = \frac{\ln 2}{0.06} = 11.552453.$$

$$(b) \quad \text{The equation becomes } R(x) = 2e^{11.552453x}.$$

Now, $R(0.08) = 2e^{11.552453 \times 0.08} = 0.5$. Therefore, the risk is 5%.

$$(c) \quad 100 = 2e^{11.552453x} \Leftrightarrow 50 = e^{11.552453x} \Rightarrow x = \frac{\ln 50}{11.552453} = 0.33.$$

$$(d) \quad 10 = 2e^{11.552453x} \Leftrightarrow 5 = e^{11.552453x} = \frac{\ln 5}{11.552453} = 0.1393036274 \simeq 0.14.$$

Thus, the concentration of alcohol that the driver is arrested and charged is 0.14.



Application activity 3.5.5

Suppose that the risk R of having an accident while driving a car is modelled by the equation $R(x) = 6e^{kx}$. Suppose the concentration of alcohol of 0.04 results in 10% of risk of accident, what is the risk if the concentration is 0.06 and what concentration yields to 100% of risk of accident.



End of unit assessment

Knowledge and understanding

1. The formula for continuous compounding interest is $A = Pe^{rt}$. If you want to find the time t it takes for an investment to double, show how the natural logarithm is used to isolate t .
2. In the Richter scale equation, what happens to the intensity if the magnitude increases by exactly 2 units?

Skills

3. Solve for x
 - a) $\ln(x-2)(x-1) = \ln(2x+8)$
 - b) $\begin{cases} x^2 + y^2 = 130 \\ \ln x + \ln y = \ln 63 \end{cases}$
 - c) $2^{x-1} - 2^{x-3} = 2^{3-x} - 2^{1-x}$
 - d) $e^{4x} - 13e^{2x} + 36 = 0$
4. Find the numerical value
 - a) $\log_2 32$
 - b) $\log_4 8$
 - c) $\log_6 7$
 - d) $\log_5 \sqrt{125}$
5. A bank pays compound interest on money invested in an account. After n year a sum of 2000 will rise to 2000×1.08^n Frw
 - a) How much money is in the account after three years?
 - b) After how many years will the original 2000 Frw have nearly doubled

6. The population of a country grows according to the law $P = Ae^{0.06t}$ where P million is the population at time t years and A is a constant. Given that at time $t = 0$, the population is 27.3 million, calculate the population when
- $t = 10$
 - $t = 20$
 - $t = 25$
7. The law of cooling for a bath of water is $\theta = Ae^{-0.05t}$ where θ is excess of temperature of the water over the temperature of the bathroom at time t minutes and A is a constant. Given that at time t the temperature of the water is 60°C and that the bathroom has a constant temperature of 15°C , calculate the value of t when the temperature of the water is
- 50°C
 - 35°C
 - 27°C
8. At a time $t = 0$, one bacteria is placed in a culture in a laboratory. The number of bacteria doubles every 10 minutes

Time t in minutes	0	10	20	30	40	...
Number of bacteria	1	2	4	8	16	...

- Draw a graph to show the growth of the bacteria from $t = 0$, to $t = 120$, minutes
Use a scale of 1cm to 10 minutes across the page and 5 cm to 1000 units up the page.
- Use your graph to estimate the time taken to reach 800 bacteria.

Attitude and values

9) Many natural phenomena, like population growth or the spread of a virus, follow an exponential model. **Why is it crucial for citizens and leaders to understand the "speed" of exponential growth compared to linear growth?**

A) It helps people predict exactly when a crisis will end.

B) It prevents overreacting, as exponential growth slows down naturally over time.

C) It allows for early, proactive decisions before a situation escalates rapidly.

D) It proves that linear models are always incorrect and should not be used.

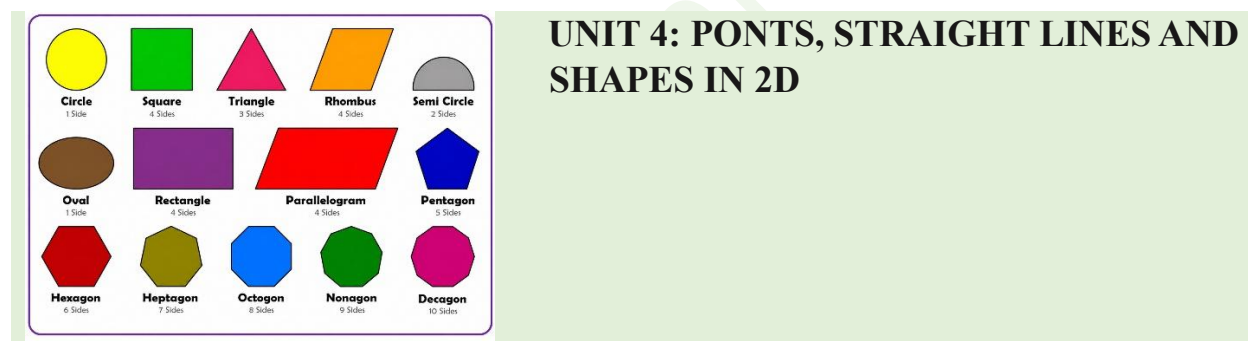
10) Logarithmic scales (like the Richter scale for earthquakes or pH for acidity) compress huge ranges of numbers into small, manageable scales. **How does this mathematical simplification help us communicate complex scientific data to the public?**

A) It makes the data easier to understand, allowing the public to respond appropriately to risks.

B) It hides the true danger of natural events so the public does not panic.

C) It forces scientists to use smaller numbers so they don't have to do difficult math.

D) It ensures that only experts can interpret the data correctly.



Key unit competence: Apply the meaning of coordinates in solving problems involving points, lines, and shapes in 2D space.

Learning objectives:

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain key concepts: Cartesian coordinates, vector/parametric equations, and geometric shapes (circle and quadrilaterals).
- Write the formulas for distance, midpoint, angles, and equations of lines, and circles.
- Identify the different forms of linear equations and their interrelationships.
- Describe the positional relationships between points, lines and circles.

Skills

- Represent points, lines and circles in the Cartesian plane
- Convert between different forms of linear equations (Cartesian, vector, parametric).
- Calculate distances, midpoints, angles in 2D.

- Determine the position of points relative to lines and circles (e.g., on, inside/outside).
- Analyse the relationship between lines (parallel or intersecting)
- Solve geometric problems involving collinearity and shortest distances.

Attitudes and value

- Collaborate effectively to tackle multi-step geometric challenges
- Appreciate the power of coordinates in algebra, geometry and engineering.
- Value precision and accuracy in formulating equations and performing calculations.
- Develop spatial reasoning and visualization skills for 2D problems.

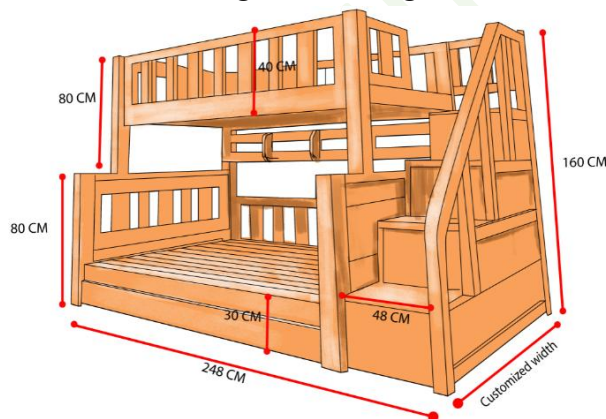
Introduction

In this unit, you will explore how points, straight lines, and shapes can be represented and analyzed in a two-dimensional Cartesian plane. By learning how coordinates describe the exact position of points, you will be able to plot locations, form equations of lines, and study geometric figures such as circles. These skills allow you to calculate distances, determine angles, examine relationships between geometric objects, and solve real-life problems such as designing maps, planning roads, creating digital graphics, and modelling structures in engineering. Understanding how algebra and geometry work together through coordinates strengthens your ability to visualize space, reason logically, and apply mathematics to everyday contexts and technical fields.



Introductory activity:

Observe a bed design on the diagrams below:



- Write down parts that can represent points, lines, angles, planes and shapes on these objects and explain your answer in each case.
- What are essential data required to draw a line on a sheet of paper or a chalk board (2D)?

4.1 Points in 2D

4.1.1 Cartesian coordinates of a point



Activity 4.1.1

In Lower Secondary, you studied points on a Cartesian plane.

- 1) What do coordinates of a point in 2D mean?
- 2) Draw a Cartesian plane and locate the points $A(1,2)$, $B(5,4)$
- 3) Represents these points in xy plane
- 4) Draw a line segment from A to B.

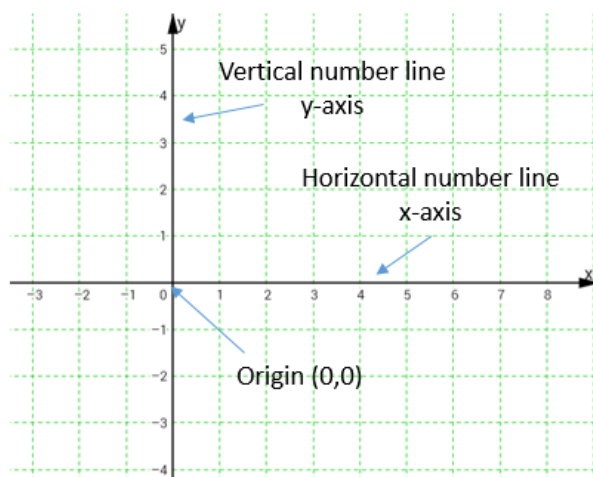


Content Summary

A **point** in 2D is the fundamental object that represents a specific, exact location on a flat plane. It has no size, no width, and no height. It is defined solely by its position. A point is not a thing, but a place. We indicate the position of a point by placing a dot with a pencil. In coordinate geometry, points are located on the plane using their coordinates - two numbers that show where the point is positioned with respect to two number line “axes” at right angles to each other.

The position of a point on the Cartesian plane is represented by an ordered pair or coordinates (x, y) . The first number, x , called the x -coordinate and the second number, y , is called the y -coordinate.

Coordinate Plane or Cartesian Plane



The origin is indicated by the ordered pair or coordinates $(0, 0)$.

To get to the point (x, y) in cartesian plane, we start from the origin. If x is positive, then we move x units right from the origin otherwise if x is negative, then we move x units left from the origin. Then, if y is positive, we move y units up otherwise if y is negative, we move y units down

Example:

Represent the points $M(2, 1.5)$, $L(-3, 1.5)$ and $N(-2, -3)$ in Cartesian plane.

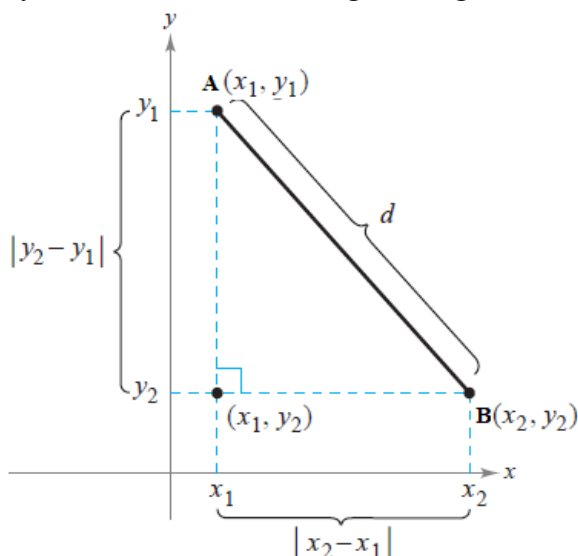
Solution

- Point M has coordinates $M(2, 1.5)$. To get to point M, we move 2 units to the right of x-axis from the origin (positive side) and 1.5 units up on y-axis from the origin (positive side).
- Point L is represented by the coordinates $L(-3, 1.5)$. To get to point L, we move 3 units to the left of x-axis from the origin (negative side) and 1.5 units up on y-axis from the origin (positive side)
- Point N has coordinates $N(-2, -3)$. To get to point N, we move 2 units to the left of x-axis from the origin (negative) and 3 units down on y-axis from the origin (negative side).

Distance between two points

The distance between two points is the length of the straight-line segment that connects them.

If $A = (x_1, y_1)$ and $B = (x_2, y_2)$ are two points, then the length of the line joining points A and B , they can be used to form a right triangle, as shown in the figure below.



The length of the vertical side of the triangle is $|y_2 - y_1|$, and the length of the horizontal side is $|x_2 - x_1|$. By Pythagorean Theorem, it follows that:

$$\text{and } d(A, B) = \| \overline{AB} \| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

i.e the distance, is: $d(AB) = |AB| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Example 1: The length of a line segment joining $A = (3, y)$ and $B = (7, 5)$ is 5 units. Find the possible values of y .

Solution:

$$d(AB) = |AB| = \sqrt{(7-3)^2 + (5-y)^2} = 5$$

$$d(AB) = |AB| = \sqrt{16 + 25 - 10y + y^2} = 5$$

$$16 + 25 - 10y + y^2 = 25$$

$$y^2 - 10y + 16 = 0 \Rightarrow y = 2 \text{ or } y = 8.$$

The possible values of y are 8 and 2.

Example 2: Consider the points $C(k, -2)$ and $D(0, 1)$ in cartesian plane.

a) Find the value of k if distance between the point C and D is 5 units.

Solution

$$d(C, D) = \sqrt{k^2 + 9}$$

$$\sqrt{k^2 + 9} = 5 \Leftrightarrow k^2 + 9 = 25 \Leftrightarrow k^2 = 16 \Rightarrow k = \pm 4$$

Thus, the values of k are -4 and 4.

**Application activity 4.1.1**

- Without plotting, state which quadrant (*I, II, III, or IV*) the following points lie in:
 - $P = (5, 2)$
 - $Q = (-3, 7)$
- Without plotting, state which axis the following point lies on:
 - $T = (0, 9)$
 - $U = (-\sqrt{2}, 0)$
- A point M has an x -coordinate of -2 and lies on the same horizontal line as point $N = (4, 3)$. What are the coordinates of point M ?
- A point P has a y -coordinate of -4 and lies on the same vertical line as point $Q = (-1, 5)$. What are the coordinates of point P ?
- Represent the points $A(-3, 2)$, $B(4, 2)$, $C(-3, -2)$, $D(3, -1)$ in xy -plane.
- Find the distance between the following pairs of points
 - $(2, 3)$ and $(5, 7)$.
 - $(a, 0)$ and $(0, b)$.
- Find the value of k if the distance between the following points is 5 units.
 $A = (3, k)$ and $B = (0, k)$

4.1.2 Midpoint of a line segment**Activity 4.1.2**

Three friends: Pascal, Steve, and Benjamin live on the same side of a street from Nyabugogo to Ruyenzi. Steve's house is halfway between Pascal's and Benjamin's houses. If the locations of Pascal can be given by the coordinates $(2, 5)$ and Benjamin's house at $(4, 7)$.

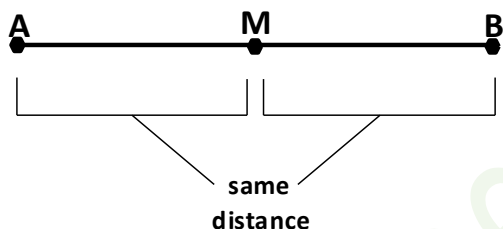
- Draw a Cartesian plan and locate Pascal's and Benjamin's houses
- Show the line segment joining the locations of Pascal's and Benjamin's houses;
- Given that Steve's house is in the halfway, locate his house, estimate the coordinate of that location and explain how to find it.

Content summary

A line segment from point A to point B , denoted $[AB]$ is the set of all points on the part of the line that joins A and B , including A and B . The midpoint of this segment is point M such that the distance $[AM] = [MB]$ and it is given by given by $M = \frac{1}{2}(A + B)$

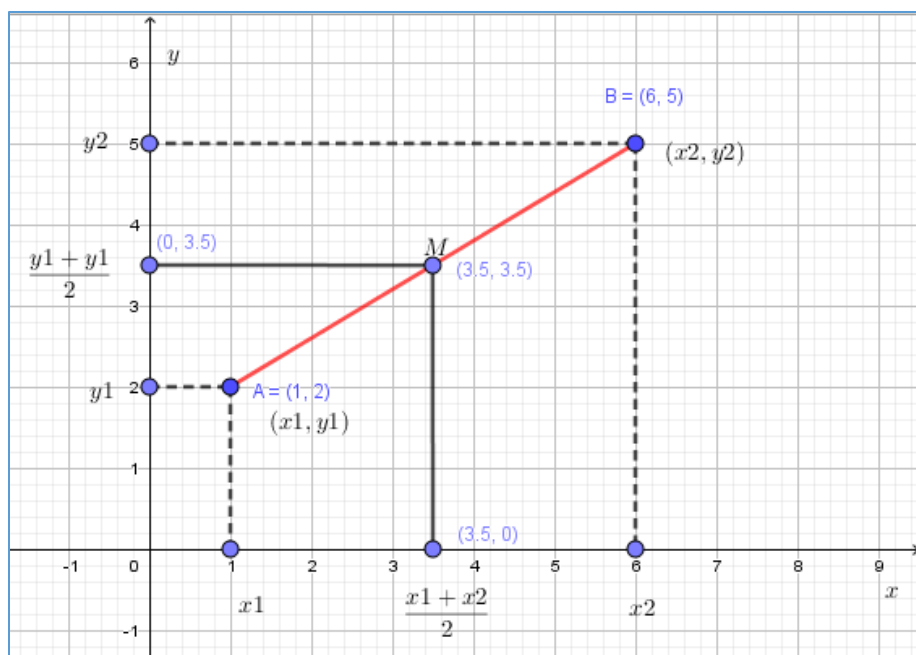
The midpoint of a segment is a point on that line segment which maintains the same distance from both of the endpoints of that line segment.

For example, consider segment to the left.



It has endpoints named A and B. The midpoint of the segment is labelled M. It is **at** the same distance from each of the endpoints.

Suppose that the coordinates of the line segment $\overline{AB}$, are $A(x_1, y_1)$ and $B(x_2, y_2)$.



The coordinates of the midpoint, are obtained using the midpoint formula, such that

The midpoint formula, is given by: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Example:

1) Find the midpoint of the segment joining points $A(3,0)$ and $B(1,8)$.

Solution

The midpoint is $M = \frac{1}{2}(A + B) = \frac{1}{2}(4, 8) = (2, 4)$

2) Find the coordinates of the midpoints of the straight line joining the following pairs of point:
 $(3, 7), (5, 9)$

Solution:

$$M = \frac{1}{2}(x_1 + x_2, y_1 + y_2) = \frac{1}{2}(3 + 5, 7 + 9) = \frac{1}{2}(8, 16) = (4, 8)$$



Application activity 4.1.2

- Find the midpoint of the following pairs of points
 - $(2, 3)$ and $(5, 7)$.
 - $(-3, -2)$ and $(-6, 7)$.
- Find the value of x_1 if the coordinate of midpoint between the points $(x_1, 2)$ and $(3, 4)$ be $(1, 3)$.
- Find the value of k , if the coordinate of midpoint between points is $(2, 3)$.
 $A = (4, k)$ and $B = (0, k)$

4. M is a mid-point of a straight-line joining point $A(10, 5)$ to point B. If M has coordinates $(6, 4)$, find the coordinates of B.
5. Given the triangle with vertices at $A(2, 4)$, $B(4, -2)$ and $C(8, 12)$ and that L is the mid-point of $\overline{AB}$ and M is the mid-point of $\overline{BC}$, find:
 - a) the coordinate of point L
 - b) the coordinate of point M
 - c) the distance LM.

4.2 Lines in 2D

4.2.1 Determination of equation of a straight line passing through a point and parallel to a direction vector

Activity 4.2.1

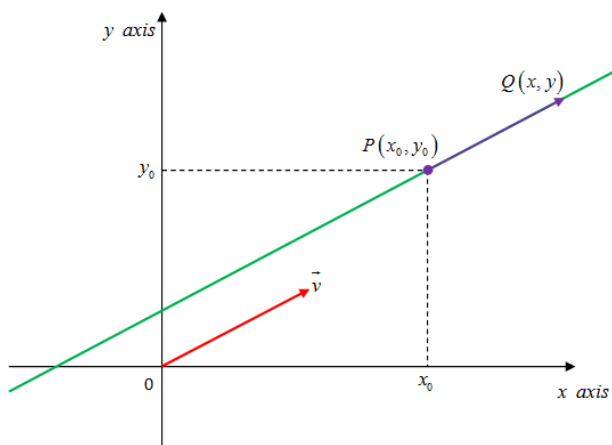
Given the vector $\vec{v} = (2, -3)$.

- a) Determine the form of a vector $\vec{w} = (x, y)$ parallel to $\vec{v}$ and passing to the point $P(1, 6)$.
- b) Give 2 examples of such vectors $\vec{w}$.
- c) If D is a line with the direction vector $\vec{w}$, what should be the equation of D?

Content summary

To determine the equation of the line passing through the point $P(x_0, y_0)$ and parallel to the direction vector, $\vec{v} = (a, b)$ we will use our knowledge that parallel vectors are scalar multiples.

The vector $\overrightarrow{OP}$ is called the **position vector** P .



Thus, the vector through $P(x_0, y_0)$

and any other point $Q(x, y)$ on the line is the product of a scalar and the direction vector $\vec{v} = (a, b)$.

$$\overrightarrow{PQ} = r\vec{v} \text{ with } r \text{ a parameter and } \overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

Hence, the **vector equation** of the line that is parallel to the vector $\vec{v} = (a, b)$.

and which passes through the point P with position vector $\vec{OP} = (x_0, y_0)$ is given as

$\vec{OQ} = \vec{OP} + r\vec{v}$ where $Q(x, y)$ is any point of the line.

$$(x, y) = (x_0, y_0) + r(a, b) \quad \text{or} \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + r \begin{pmatrix} a \\ b \end{pmatrix}$$

The **parametric equations** of the line that is parallel to the vector $\vec{v} = (a, b)$.

$\vec{v} = (a, b)$ and which passes through the point P with position vector $\vec{OP} = (x_0, y_0)$ are given by:

$$\begin{cases} x = x_0 + ra \\ y = y_0 + rb \end{cases}$$

The **Cartesian equation** is found after eliminating parameter r .

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}$$

This can be expanded to the standard form $bx - ay = bx_0 - y_0$.

Example

Find the vector equation of the straight line that is parallel to the vector $\vec{u} = (2, -1)$ and which passes through the point with position vector $\vec{v} = (3, 2)$.

Solution

The vector equation is, $(x, y) = (3, 2) + r(2, -1)$ r is a parameter

The parametric equations:

$$\begin{cases} x = 3 + 2r \\ y = 2 - r \end{cases}$$

The Cartesian equation:

$$\frac{x-3}{2} = \frac{y-2}{-1} \text{ or } 2y - 4 = -x + 3 \text{ or } x + 2y = 7$$

In general

The **vector equation** of the line can be rewritten as

$$(x, y) = (x_0, y_0) + r(a, b) \quad \text{or} \quad x\vec{i} + y\vec{j} = x_0\vec{i} + y_0\vec{j} + r(a\vec{i} + b\vec{j}) \text{ where}$$

$$\{\vec{i} = (1, 0), \vec{j} = (0, 1)\} \text{ form the standard basis of } \mathbb{R}^2.$$

Application activity 4.2.1.

1. Find the vector, parametric and Cartesian equation of the line passing through the point with position vector $(2, -3)$ and parallel to the line $(x, y) = (3, 5) + r(1, 6)$
2. Find the Cartesian equations of the lines whose vector equations are given below. Give your answers in the form $y = mx + c$

a. $(x, y) = (-1, 2) + r(-2, 3)$

b. $x\vec{i} + y\vec{j} = 3\vec{i} + 2\vec{j} + r(3\vec{i} - \vec{j})$

4.2.2 Determination of an equation of a straight line given 2 points and direction vector.

Activity 4.2.2

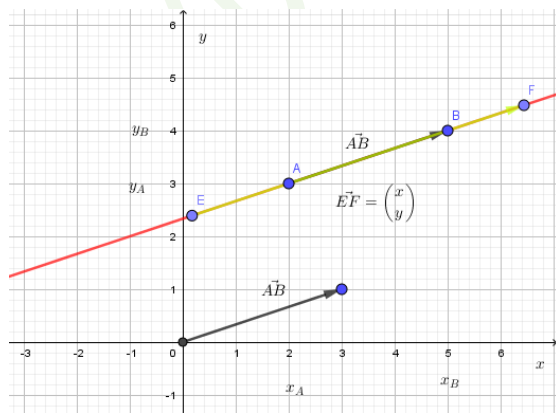
Consider the line passing through points $A(1, 4)$ and $B(3, -2)$.

- a) Determine the vector $\overrightarrow{AB}$
- b) Determine the form of a vector $\vec{w} = (x, y)$ parallel to $\overrightarrow{AB}$ and passing to the point $B(3, -2)$
- c) Give 2 examples of vectors like $\vec{w}$.
- d) If D is a line with the direction vector $\vec{w}$, what should be the equation of D?

Content summary

Two points $A(x_A, y_A)$ and $B(x_B, y_B)$ form a vector $\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$. A vector parallel to

$\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$ and passing to the point $B(x_B, y_B)$ is of the form $\vec{w} = (x, y) = r\overrightarrow{AB} + A$



Therefore, **the vector equation of the line is:**

$$(x, y) = \overrightarrow{OA} + r\overrightarrow{AB} \text{ or } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_A \\ y_A \end{pmatrix} + r \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}, \text{ where } \overrightarrow{AB} \text{ is the direction vector.}$$

The parametric equations of the line are

$$\begin{cases} x = x_A + r(x_B - x_A) \\ y = y_A + r(y_B - y_A) \end{cases}$$

The Cartesian equation is obtained in this way:

$$x = x_A + r(x_B - x_A), \quad r = \frac{x - x_A}{x_B - x_A}$$

$$y = y_A + r(y_B - y_A), \quad r = \frac{y - y_A}{y_B - y_A}$$

Equalizing the value of r we find the Cartesian equation:

$$\frac{x - x_A}{x_B - x_A} = \frac{y - y_A}{y_B - y_A}$$

This can be expanded to the standard form:

$$x(y_B - y_A) - y(x_B - x_A) = x_A(y_B - y_A) - y_A(x_B - x_A).$$

Example

Find the vector equation of the straight line passing through $A(3, 2)$ and $B(4, 1)$.

Solution

The direction vector is, $\overrightarrow{AB} = (1, -1)$ r is a parameter

The vector equation is

$$(x, y) = \overrightarrow{OA} + r\overrightarrow{AB} \quad (x, y) = \overrightarrow{OA} + r\overrightarrow{AB} \text{ or } (x, y) = (3, 2) + r(1, -1)$$

The parametric equations: is

$$\begin{cases} x = 3 + r \\ y = 2 - r \end{cases}$$

The Cartesian equation, is

$$\frac{x-3}{1} = \frac{y-2}{-1} \text{ or } x + y = 5$$

Application activity 4.2.2

- 1) Determine the equation of a straight line passing through $P(2, 4)$ and $B(3, -2)$.
- 2) Find the vector equation of the straight line that passes through the point with position vector $2\vec{i} + 3\vec{j}$ and which is perpendicular to the line. $x\vec{i} + y\vec{j} = 3\vec{i} + 2\vec{j} + r(\vec{i} - 2\vec{j})$

4.2.3 Equation of a straight line given its gradient

Activity 4.2.3

Determine the slope and the y-intercept in the equations below. Hence, draw the lines in the same graph.

a) $y = \frac{3}{2}x - 2$

b) $y = -3x + \frac{5}{2}$

Content summary

Consider a line having gradient m and passing through the point $P_1(x_1, y_1)$. Suppose that the point $P(x, y)$ is another point on the line. Then the gradient of the line is the rate at which the line rises (or falls) vertically for every unit across to the right. It is defined by the change in y to the change in x .

$$m = \frac{y - y_1}{x - x_1}$$

Thus, the equation of the line in point-slope form is defined by

$$y - y_1 = m(x - x_1)$$

From the equation above, if we take any other point $P_2(x_2, y_2)$ that lies on this line, then:

Vector equation

$$\overrightarrow{OP} = \overrightarrow{OP_1} + r\overrightarrow{P_1P_2} \text{ or } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + r \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix},$$

$\overrightarrow{P_1P_2}$ is the direction vector.

Parametric equations

$$\begin{cases} x = x_1 + r(x_2 - x_1) \\ y = y_1 + r(y_2 - y_1) \end{cases}$$

Cartesian equation

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$

Example:

Write the equation of the line that has slope -3 that passes through the $(-1, 7)$

Solution

$y - y_1 = m(x - x_1)$ use point-slope form

$y - 7 = -3(x - (-1))$ substitute $(-1, 7)$ for (x_1, y_1) and -3 for m .

$$y - 7 = -3(x + 1)$$

Note:

The general equation of the line is $Ax + By + C = 0$, where A, B, C are constants,

And $A \neq 0, B \neq 0$.

- If $A = 0$ the line is horizontal.
- If $B = 0$ the line is vertical.
- If $C = 0$ the line passes through the origin.

On the other hand, the line has slope $m = \frac{-A}{B}$ and the intercept of $y, b = \frac{-C}{B}$

- If two lines are parallel, then their slopes/gradients are equal.
Therefore $m_1 = m_2$
Thus, the equations $Ax + By + C = 0$ and $Ax + By + D = 0$ are parallel.
- If two lines are perpendicular, then the product of their slopes/gradients is equal to -1 .
Therefore $m_1 \times m_2 = -1$.
Thus, the equations $Ax + By + C = 0$ and $Bx - Ay + D = 0$ are perpendicular.

Example:

1) Find the equation for the line that contains the point $(5, 1)$ and is parallel to $y = \frac{3}{5}x + 3$.

Step1: Identify the slope of the given line

$$y = \frac{3}{5}x + 3 \Rightarrow m = \frac{3}{5}$$

Step2: Write the equation of the line through point $(5, 1)$ and $m = \frac{3}{5}$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{3}{5}(x - 5)$$

$$y = \frac{3}{5}x - 3 + 1$$

$$y = \frac{3}{5}x - 2$$

2. Find the equation for the line that contains the point $(0, -2)$ and is perpendicular to $y = 5x + 3$?

Step1: Identify the slope of the given line and write its negative reciprocal.

$$y = 5x + 3 \Rightarrow m = 5 \text{ and it has negative reciprocal } \frac{-1}{5}$$

Step2: Write the equation of the line through point $(0, -2)$ and $m = \frac{-1}{5}$

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = \frac{-1}{5}(x - 0)$$

$$y = \frac{-1}{5}x - 2$$

Application activity 4.2.3

- Write the equation in point-slope form for the line through the given point that has the given slope
 - $(3, -4); m = 6$
 - $(-2, -7); m = \frac{-3}{2}$
 - $(-5, 2); m = 0$
- Is $y - 5 = 2(x - 1)$ an equation of a line passing through $(4, 11)$? Explain.
- Write an equation of the line that contains the point $(-3, -5)$ and the same slope as $y + 2 = 7(x + 3)$

4.2.4 Intersecting lines, perpendicular lines or parallel lines

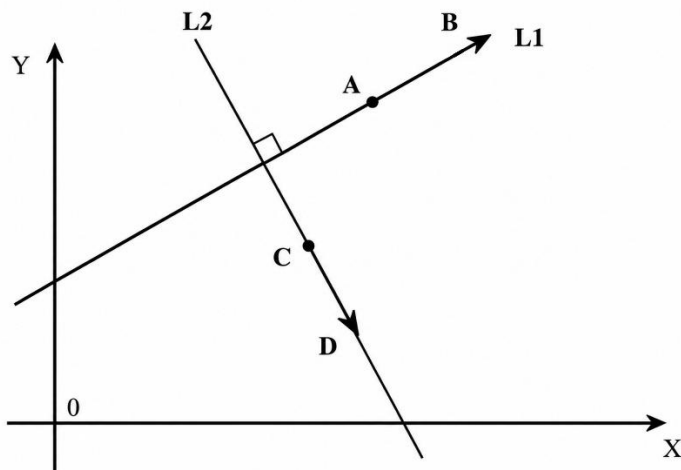
Activities 4.2.4

- In the same Cartesian plane plot the straight lines containing the following points $(0, 4)$; $(\frac{1}{2}, 2)$ and $(\frac{1}{2}, 2)$; $(4, 1)$

What is your opinion about the two lines in the Cartesian plane?
- In the same Cartesian plane plot the straight lines containing the following points $(-3, -1)$; $(0, 3)$ and $(-2, -4)$; $(1, 0)$

Content summary

a) Orthogonality and intersection of two lines



Here L_1 and L_2 are perpendicular

To write a straight line perpendicular to a given straight line we proceed as follows:

Step I: Interchange the coefficients of x and y in equation $ax + by + c = 0$.

Step II: Interchange the sign between the terms in x and y of equation i.e., If the coefficient of x and y in the given equation are of the same signs make them of opposite signs and if the coefficient of x and y in the given equation are of the opposite signs make them of the same sign.

Step III: Replace the given constant of equation $ax + by + c = 0$ by an arbitrary constant.

For example, the equation of a line perpendicular to the line $7x + 2y + 5 = 0$ is $2x - 7y + c = 0$ again, the equation of a line, perpendicular to the line $9x - 3y = 1$ is $3x + 9y + k = 0$

Note:

Assigning different values to k in $bx - ay + k = 0$, we shall get different parallel straight lines each of which is perpendicular to the line $ax + by + c = 0$. Thus, we can have a family of straight lines perpendicular to a given straight line.

b) Intersection point of two lines

Let the equations of two intersecting straight lines be

$$a_1x + b_1y + c_1 = 0 \dots\dots(i) \quad \text{and} \\ a_2x + b_2y + c_2 = 0 \dots\dots(ii)$$

Suppose the above equations of two intersecting lines intersect at $P(x_1, y_1)$. Then (x_1, y_1) will satisfy both the equations (i) and (ii).

Therefore, $a_1x_1 + b_1y_1 + c_1 = 0$ and $a_2x_1 + b_2y_1 + c_2 = 0$

Solving the above two equations by using the method of cross-multiplication, we get,

$$\frac{x_1}{b_1c_2 - b_2c_1} = \frac{y_1}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\text{Therefore, } x_1 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \text{ and } y_1 = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}, a_1b_2 - a_2b_1 \neq 0$$

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right), a_1b_2 - a_2b_1 \neq 0$$

Notes:

To find the coordinates of the point of intersection of two non-parallel lines, we solve the given equations simultaneously and the values of x and y so obtained determine the coordinates of the point of intersection.

$$\text{If } a_1b_2 - a_2b_1 = 0 \text{ then } a_1b_2 = a_2b_1 \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} \Rightarrow -\frac{a_1}{b_1} = -\frac{a_2}{b_2}$$

i.e., the **slope of line (i) equals to the slope of line (ii)**

Therefore, in this case the straight lines (i) and (ii) are parallel and hence they do not intersect at any real point.

Example

Find the coordinates of the point of intersection of the lines $2x - y + 3 = 0$ and $x + 2y - 4 = 0$

Solution:

We know that the co-ordinates of the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and

$$a_2x + b_2y + c_2 = 0 \text{ are } \left(\frac{b_1c_1 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right), a_1b_2 - a_2b_1 \neq 0$$

Given equations are

$$2x - y + 3 = 0 \dots (i)$$

$$x + 2y - 4 = 0 \dots (ii)$$

Here $a_1 = 2$, $b_1 = -1$, $c_1 = 3$, $a_2 = 1$, $b_2 = 2$ and $c_2 = -4$

$$\left(\frac{(-1)(3) - (2)(3)}{(2)(2) - (1)(-1)}, \frac{(3)(-1) - (-4)(2)}{(2)(2) - (1)(-1)} \right)$$

$$\Rightarrow \left(\frac{-3 - 6}{4 + 1}, \frac{-3 + 8}{4 + 1} \right) \Rightarrow \left(\frac{-9}{5}, \frac{5}{4} \right)$$

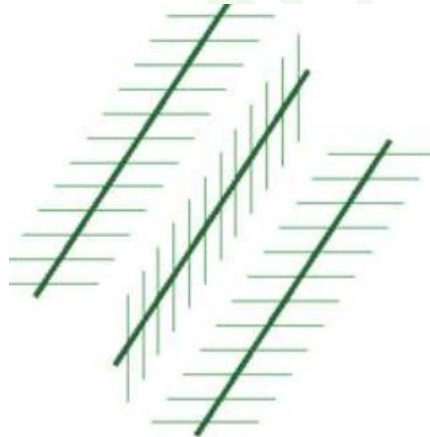
Therefore, the co-ordinates of the point of intersection of the lines $2x - y + 3 = 0$ and

$$x + 2y - 4 = 0 \text{ are } \left(\frac{-9}{5}, \frac{5}{4} \right)$$

Alternatively, by solving simultaneous equations $2x - y + 3 = 0 \dots (i)$ and $x + 2y - 4 = 0 \dots (ii)$

using different methods you get the same answer. $\left(\frac{-9}{5}, \frac{5}{4} \right)$.

c) Equation of a Line Parallel to a given line



These are three parallel antennas

Let, $ax + by + c = 0$ ($b \neq 0$) be the equation of the given straight line.

Now, convert the equation $ax + by + c = 0$ to its slope-intercept form.

$$ax + by + c = 0$$

$$\Rightarrow by = -ax - c$$

Dividing both sides by b , [$b \neq 0$] we get, $y = -\frac{a}{b}x - \frac{c}{b}$ which is the slope-intercept form.

Now comparing the above equation to slope-intercept form ($y = mx + b$) we get, the slope of the line $ax + by + c = 0$ is $(-\frac{a}{b})$.

Since the required line is parallel to the given line, the slope of the required line is also $(-\frac{a}{b})$.

Let k (an arbitrary constant) be the intercept of the required straight line. Then the equation of the straight line is $y = -\frac{a}{b}x + k$

$$\Rightarrow by = -ax + bk$$

$$\Rightarrow ax + by = \lambda, \text{ Where } \lambda = bk \text{ } \lambda = \text{another arbitrary constant.}$$

Note:

(i) Assigning different values to λ in $ax + by = \lambda$, we shall get different straight lines each of which is parallel to the line $ax + by + c = 0$. Thus, we can have a family of straight lines parallel to a given line.

(ii) To write a line parallel to a given line we keep the expression containing x and y same and simply replace the given constant by a new constant λ . The value of λ can be determined by some given conditions.

To get it clear let us compare the equation $ax + by = \lambda$ with equation $ax + by + c = 0$. It follows that to write the equation of a line parallel to a given straight line we simply need to replace the given constant by an arbitrary constant, the terms with x and y remain unaltered. For example, the equation of a straight line parallel to the straight line;

$$7x - 5y + 9 = 0 \text{ is } 7x - 5y + \lambda = 0 \text{ where } \lambda \text{ is an arbitrary constant.}$$

Example

Find the equation of the straight line which is parallel to $5x - 7y = 0$ and passing through the point $(2, -3)$.

Solution:

The equation of any straight line parallel to the line $5x - 7y = 0$ is $5x - 7y + \lambda = 0$

(i) [Where λ is an arbitrary constant].

If the line (i) passes through the point $(2, -3)$ then we shall have,

$$5.2 - 7(-3) + \lambda = 0 \Rightarrow 10 + 21 + \lambda = 0$$

$$\Rightarrow 31 + \lambda = 0 \Rightarrow \lambda = -31$$

Therefore, the equation of the required straight line is $5x - 7y + 31 = 0$

d) Angles between two lines

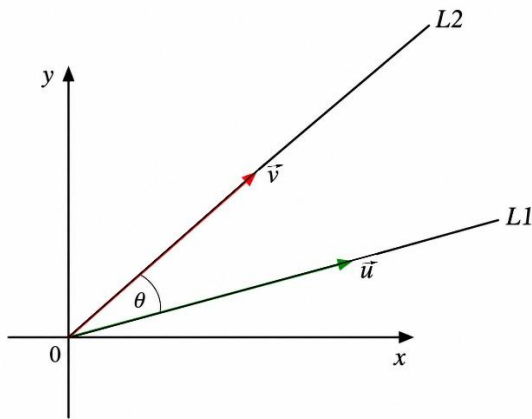
Geometrically, the scalar product of two vectors $\vec{u}(a,b)$ and $\vec{v}(c,d)$ of plane is given by $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$. Where θ is the angle between vectors $\vec{u}$ and $\vec{v}$. From this relation we

have $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$ or $\cos \theta = \frac{a \cdot c + b \cdot d}{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}}$

We deduce that the angle θ between two lines l_1 and l_2 with direction vectors $\vec{u} = (a,b)$

$\vec{u} \cdot \vec{v} = ac + bd$ and $\vec{v} = (c,d)$ respectively is given by $\theta = \cos^{-1} \left(\frac{a \cdot c + b \cdot d}{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}} \right)$ Where

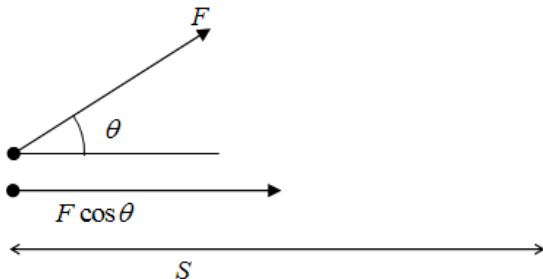
$\cos^{-1}$ denote the inverse function of cosine.



We can illustrate the scalar product in terms of work done by the force on the body:

We saw that if the constant force F and the displacement S are in the same direction, the work W done by the force on the body is $W = \vec{F} \cdot \vec{S}$

If the force does not act in the direction in which motion occurs but an angle θ to it, then the work done is defined as the product of the component of the force in the direction of motion and the displacement in that direction. $W = \|\vec{F}\| \cdot \|\vec{S}\| \cos \theta$



Note:

- Two lines are perpendicular if the angle between them is a right angle.
- Two lines are parallel and with the same direction if the angle between them is a multiple of a zero angle.
- Two lines are parallel and with the opposite direction if the angle between them is a multiple of a straight angle.

Example:

Find the angle between vectors $\vec{u} = (3, 0)$ and $\vec{v} = (5, 5)$

Solution

Let α be the angle between these two vectors

$$\alpha = \cos^{-1} \left(\frac{3 \cdot 5 + 0 \cdot 5}{\sqrt{3^2 + 0^2} \sqrt{5^2 + 5^2}} \right) = \cos^{-1} \left(\frac{\sqrt{2}}{2} \right)$$

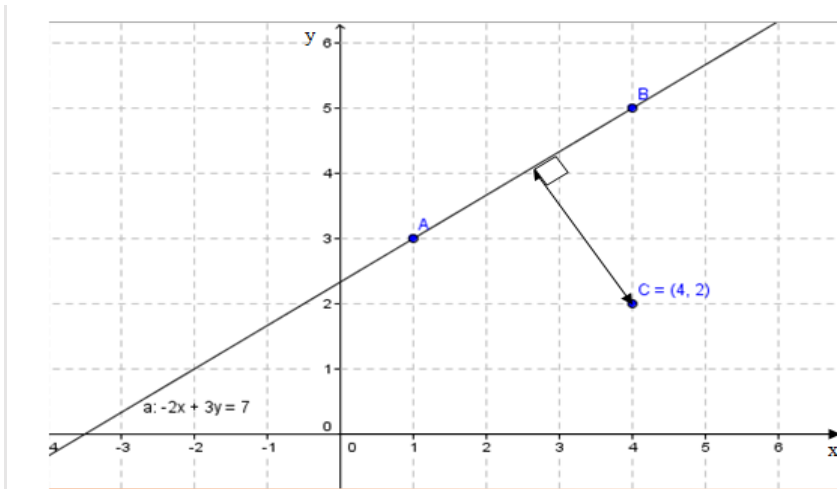
$$\alpha = 45^\circ$$

Application activities 4.2.4

1. Find the equation of a straight line that passes through the point $(-2, 3)$ and perpendicular to the straight line $2x + 4y + 7 = 0$
2. Find the equation of the straight line which passes through the point of intersection of the straight lines $x + y + 9 = 0$ and $3x - 2y + 2 = 0$ and perpendicular to the line $4x + 5y + 1 = 0$
3. Find the equation of the straight line passing through the point $(5, -6)$ and parallel to the straight line $3x - 2y + 10 = 0$.
4. Calculate the dot product and the angle formed by the following vectors: $\vec{u} = (3, 4)$, $\alpha = \frac{\pi}{4}$ and $\vec{v} = (-8, 6)$

4.2.5 Distance between a point and line and distance between two lines**Activities 4.2.5**

- 1) Find the shortest distance between the point $C(4, 2)$ and the line $a: -2x + 3y = 7$



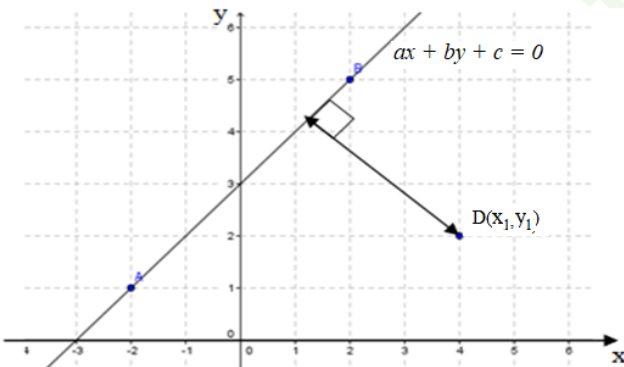
2) In one Cartesian plane plot the straight lines containing the following points

$$L1: (0, 4); \left(\frac{1}{2}, 2\right) \text{ and } L2: \left(\frac{1}{2}, 2\right); (4, 1)$$

What is the distance between these two lines?

Content summary

Distance between a point and a line



The perpendicular distance from a point $D(x_1, y_1)$ to the line $ax + by + c = 0$ is given by

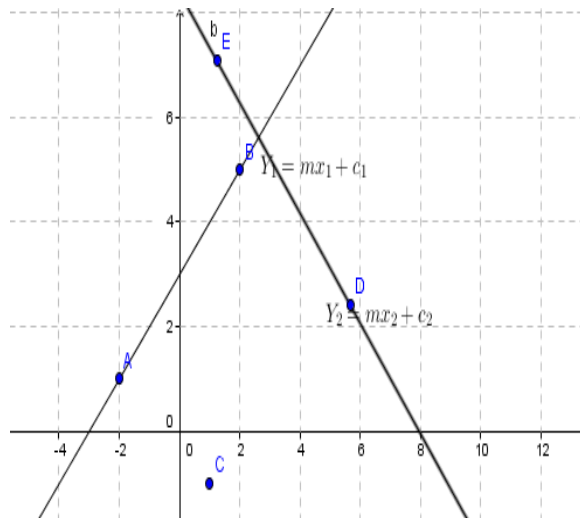
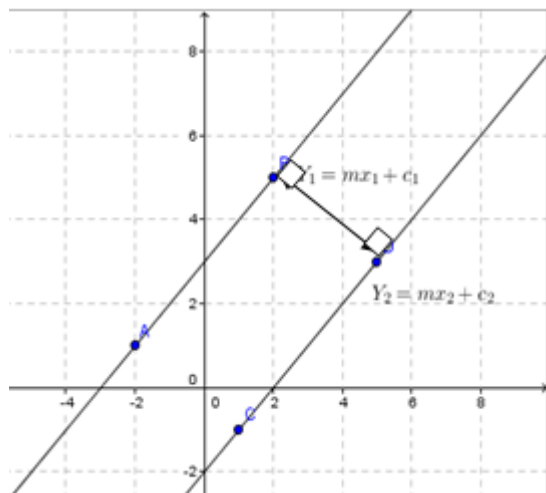
$$\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$$

Example

Find the shortest distance from the line $2x + y - 1 = 0$ to the point $(3, 2)$

Solution: The distance is $\frac{2(3) + 1(2) - 1}{\sqrt{(2)^2 + (1)^2}} = \frac{7}{\sqrt{5}} = \frac{7\sqrt{5}}{5}$.

Distance between two parallel lines and two intersecting lines



$y_1 = mx_1 + c_1$ and $y_2 = mx_2 + c_2$ are Parallel $y_1 = mx_1 + c_1$ and $y_2 = mx_2 + c_2$ are intersecting

If they intersect, the distance is **0** at the point of intersection.

If they're parallel: the distance between them will be given by the following formula

$$d = \frac{|c_1 - c_2|}{\sqrt{1 + m^2}}$$

Example

What is the distance between the parallel lines $3x + 4y = 9$ and $6x + 8y = 15$?

Solution:

First of all, we find the slopes of the two lines. We convert the equations into slope intercept form

$$3x + 4y = 9 \Rightarrow 4y = -3x + 9 \Rightarrow y = -\frac{3}{4}x + \frac{9}{4}$$

$$6x + 8y = 15 \Rightarrow 8y = -6x + 15 \Rightarrow y = -\frac{6}{8}x + \frac{15}{8} \Rightarrow y = -\frac{3}{4}x + \frac{15}{8}$$

The slope m of the lines is $-\frac{3}{4}$

Hence both lines are parallel

y Intercept of the first line is $\frac{9}{4}$

y Intercept of the second line is $\frac{15}{8}$

Difference between the y- intercepts is $\left| \frac{9}{4} - \frac{15}{8} \right| = \left| \frac{18}{8} - \frac{15}{8} \right| = \frac{3}{8}$

$$\sqrt{1+m^2} = \sqrt{1+\frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

Distance between the lines $= \frac{3}{8} \div \frac{5}{4} = \frac{3}{8} \times \frac{4}{5} = \frac{3}{10}$.

Alternatively, the distance between two parallel lines $L_1 \equiv ax + by + c_1 = 0$ and $L_2 \equiv ax + by + c_2$

is given by $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$. Using the previous example, $3x + 4y = 9$ and $6x + 8y = 15$ the distance

between the two lines is calculated taking $3x + 4y = 9$ and $3x + 4y = \frac{15}{2}$.

Therefore, the distance is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}} = \frac{\left| 9 - \frac{15}{2} \right|}{\sqrt{3^2 + 4^2}} = \frac{\frac{3}{2}}{5} = \frac{3}{10}$.

Application activities 4.2.5

1. What is the distance between the parallel lines $3x + 4y = 9$ and $6x + 8y = -15$
2. How far is the line $3x - 4y + 15 = 0$ from the origin?
3. The distance between the parallel lines $3x + 4y + 5 = 0$ and $6x - 8y + 5 = 0$ is?
4. What is the distance of the point $(-2, 3)$ from the line $x - y = 5$?
5. Calculate the angles of the triangle with vertices: $A = (6, 0)$, $B = (3, 5)$ and $C = (-1, -1)$
6. Find the value of k if the angle between $\vec{u} = (k, 3)$ and $\vec{v} = (2, -4)$ is 45°
7. Given the vectors $\vec{u} = (2, k)$ and $\vec{v} = (3, -2)$. Calculate the value of k so that the vectors $\vec{u}$ and $\vec{v}$ are: a) Perpendicular. b) Parallel. c) Make an angle of 60° .

4.3 Circle in 2D

4.3.1 Cartesian equation

Activity 4.3.1

In xy plane,

1. Use a compass to draw a circle centered at point $C(2, 1)$ and passing through the point $P(3, 0)$
2. Draw a line r that is a segment from C to P and find its length;

3. Use the distance between two points formula to calculate the distance $d(c, p) = r$
4. Deduce the distance between the points $M(x, y)$ and the center $C(2, 1)$ to calculate the radius (r) of the circle.

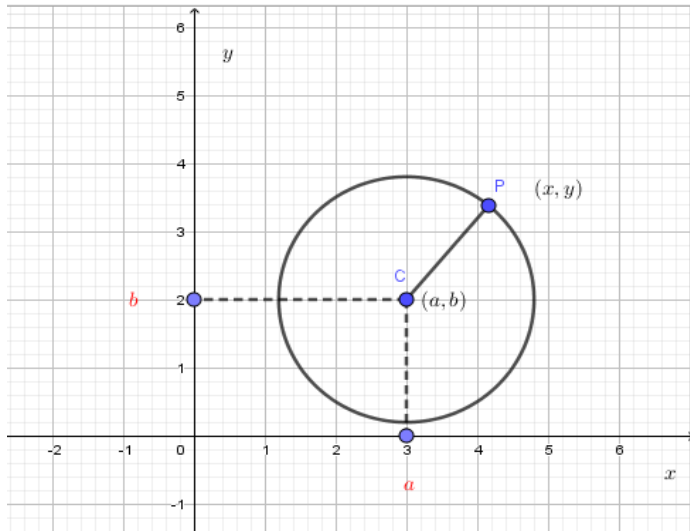
Content summary

A circle is the locus of all points (x, y) that are equidistant from a fixed given point. The fixed point is called the centre of the circle, and the distance from the centre to any point on the circle is called the radius of the circle.

Let $P(x, y)$ be one point of the circle of Centre $C(a, b)$ and radius r . The distance between the center and the point P is: $d = r = \sqrt{(x-a)^2 + (y-b)^2}$. Squaring the two sides of the equation we find the equation $(x-a)^2 + (y-b)^2 = r^2$

This is the equation of a circle of centre $C(a, b)$ and radius r .

$$C \equiv (x-a)^2 + (y-b)^2 = r^2$$



Expanding this equation, we have $C \equiv x^2 + y^2 - 2ax - 2by + a^2 + b^2 = r^2$.

Supposing that $k = -2a, l = -2b$ and $m = a^2 + b^2 - r^2$; then

General equation of the circle $C \equiv x^2 + y^2 + kx + ly + m = 0$

In this case the centre is given by; $C = \left(-\frac{k}{2}, -\frac{l}{2}\right)$ The radius is given by $r = \frac{1}{2}\sqrt{k^2 + l^2 - 4m}$

If the coordinates of the point C are $(0,0)$, then the center is at origin. In this case the equation of the circle $C \equiv x^2 + y^2 = r^2$.

Example 1: Find the equation of the circle centered at origin and with radius 4

Solution:

Centre is $C(0,0)$, radius is $r = 4$. Then the equation is

$$(x-0)^2 + (y-0)^2 = 4^2 \text{ or } x^2 + y^2 - 16 = 0$$

Example 2: Find the equation of the circle centred at point $A(3,5)$ and passing through point $B(3, 0)$

Solution:

Centre is $C(3, 5)$, radius is the distance from point A to point B , $r = \sqrt{0+25} = 5$. Then the equation is or $x^2 + y^2 - 6x - 10y = -9$

Example 3: Find the centre and the radius of the circle with equation $(x+4)^2 + (y-3)^2 - 16 = 0$

Solution:

Compare the given equation with $(x-a)^2 + (y-b)^2 = r^2$ we can rewrite it as, $(x-(-4))^2 + (y-3)^2 = 4^2$. Then the centre is $(-4, 3)$ and the radius is $r = 4$

Example 4: Find the centre and the radius of the circle with equation $x^2 + y^2 - 4x - 2y = 4$

Solution:

$$x^2 + y^2 - 4x - 2y = 4 \Leftrightarrow x^2 + y^2 - 4x - 2y - 4 = 0$$

The centre is given by $C\left(-\frac{-4}{2}, -\frac{-2}{2}\right) = (2,1)$ and the radius is

$$r = \frac{1}{2} \sqrt{(-4)^2 + (-2)^2 - 4 \times (-4)} = \frac{1}{2} \sqrt{36} = 3$$

Another method:

Rearranging the given equation

$$\Leftrightarrow x^2 - 4x + 4 - 4 + y^2 - 2y + 1 - 1 = 4 \quad r = \sqrt{2^2 + 1^2 - (-4)} = \sqrt{9} = 3$$

$$\Leftrightarrow (x^2 - 4x + 4) + (y^2 - 2y + 1) = 4 + 5$$

$$\Leftrightarrow (x-2)^2 + (y-1)^2 = 3^2$$

The centre is $(2,1)$ and the radius is 3.

Application activity 4.3.1:

1. Find the equation of the circle with centre $(3, 2)$ and radius 4.
2. Find the equation of the circle whose diameter is a line segment joining the points $(3, -2)$ and $(3, 6)$
3. Find the centre and the radius of the circle with equation $x^2 + y^2 - 4x + 8y = 12$
4. Find the centre and the radius of the circle with equation $3x^2 + 3y^2 - 18x - 12y - 27 = 0$

4.3.2 Parametric equation of a circle

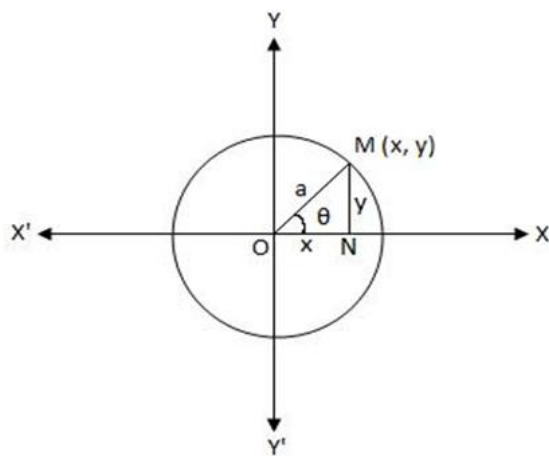
Activity 4.3.2

- a) Find the centre and the radius of circle $x^2 + y^2 = 19$
- b) Given that (a, b) is the center of the circle and r its radius, use the result from (a) and complete the following formula:
$$\begin{cases} x = a + r \cos \theta \\ y = b + r \sin \theta \end{cases}$$
- c) What do you notice from solution (b)?

Content summary

From the figure, a circle with centre at the origin and radius a , the equation of a circle is given by PYTHAGORAS'THEOREM

$$x^2 + y^2 = a^2 \quad \text{or} \quad x^2 + y^2 = r^2; (r = a)$$

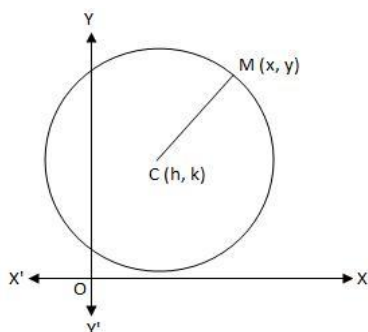


Expressing the equation of a circle using a parameter θ , we write parametric equations

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \text{ for } 0 \leq \theta \leq 2\pi$$

For a circle with centre at the point (h, k) and radius r , the equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$



Hence, we write parametric equations

$$\begin{cases} x = h + r \cos \theta \\ y = k + r \sin \theta \end{cases} \quad \text{for } 0 \leq \theta \leq 2\pi$$

Noted: the angle θ is given by

$$\diamond \begin{cases} \cos \theta = \frac{x-h}{r} \\ \sin \theta = \frac{y-k}{r} \end{cases} \quad \text{with } C(h, k) \text{ and } r.$$

$$\diamond \begin{cases} \cos \theta = \frac{x}{r} \\ \sin \theta = \frac{y}{r} \end{cases} \quad \text{with } C(0, 0) \text{ and } r.$$

Therefore, the formula obtained from solution (b) is called the parametric equation of a circle of radius r and centred at the point P with coordinate (a, b) .

Example:

Find the centre and the radius of the circles with equations. Hence, write parametric equations for each.

a) $(x-2)^2 + y^2 = 25$

b) $x^2 + y^2 - 4x - 2y = 4$

c) $x^2 + y^2 = 4$

Solution

a. i) $(x-2)^2 + y^2 = 25$

Comparing with $(x-a)^2 + (y-b)^2 = r^2$

$(x-2)^2 + (y-0)^2 = 5^2$. Therefore; the centre $C(2, 0)$ and $r = 5$

ii) The parametric equations are:

$$\begin{cases} x = 2 + 5 \cos \theta \\ y = 5 \sin \theta \end{cases} \quad \text{for } 0 \leq \theta \leq 2\pi$$

b. i) $x^2 + y^2 - 4x - 2y = 4$

By rearranging like terms and completing terms

$$(x^2 - 4x + 4) + (y^2 - 2y + 1) = 4 + 4 + 1$$

$$(x-2)^2 + (y-1)^2 = 3^2$$

Therefore; the centre $C(2,1)$ and $r = 3$

ii) The parametric equations are:

$$\begin{cases} x = 2 + 3\cos\theta \\ y = 1 + 3\sin\theta \end{cases} \text{ for } 0 \leq \theta \leq 2\pi$$

c. i) $x^2 + y^2 = 4$

Comparing with $(x-a)^2 + (y-b)^2 = r^2$

$$(x-0)^2 + (y-0)^2 = 2^2$$

Therefore; the centre $C(0,0)$ and $r = 2$

ii) The parametric equations are:

$$\begin{cases} x = 2\cos\theta \\ y = 2\sin\theta \end{cases} \text{ for } 0 \leq \theta \leq 2\pi$$

Application activity 4.3.2

1. State the equation of the circle that can satisfy the given conditions. Hence, write parametric equations for each.

a) $c(0,0)$, $r = 5$

b) $C(4,-2)$, $r = 8$

c) $c(-4,-2)$ and passes through $P(1,3)$

2. Find the equation of the circle passing through the points $(0, 1)$, $(4, 3)$, $(1,-1)$.

3. Given the equations of a circle

$$x^2 + y^2 - 10x - 8y + 5 = 0 \quad (1)$$

$$x^2 + y^2 - 6x + 8y + 25 = 0 \quad (2)$$

a) Express each equation below in the form $(x-a)^2 + (y-b)^2 = r^2$

b) Write the coordinates centre and find the radius of the circle.

c) Find parametric equations

4.3.3 Intersection of a line and a circle

Activity 4.3.3

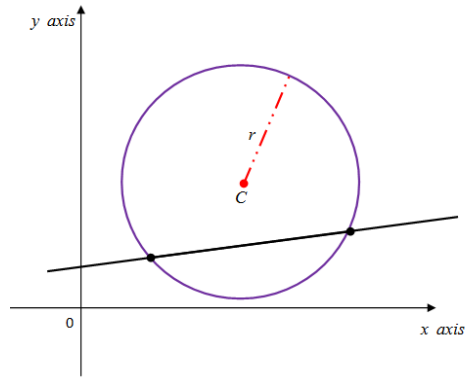
Consider the circle with equation $x^2 + y^2 - 10x - 6y + 25 = 0$ and the lines $y = x$, $x + y = 2$ and $y = 6$. In Cartesian plane

1. Sketch that circle and these lines
2. Find the intersection of the circle and the lines.
3. What can you say about the intersection of the circle and each line?

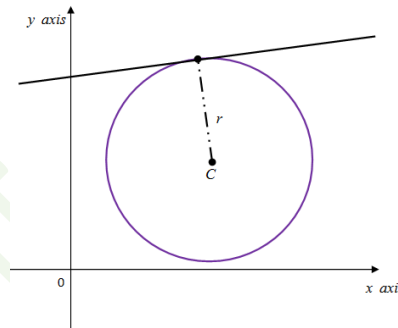
Content summary

Intersection of a line and a circle occurs in three possible situations:

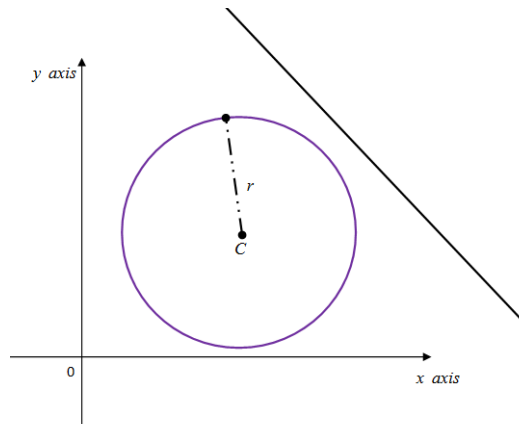
The line cuts the circle in two distinct points



The line touches the circle at one point (tangent to the circle)



The line neither cuts nor touches the circle



Given the equations of a line and a circle we can solve the two equations simultaneously.

If

- There are two distinct roots then the line cuts the circle in two distinct points and the intersection is the set of these two points. In this case part of the line is a chord of the circle and the distance from the centre of the circle to the line is less than the radius of the circle.

- There is a repeated root then the line touches the circle and the intersection is this root. In this case the line is tangent to the circle and the distance from the centre of the circle to the line is equal to the radius of the circle.
- There is no real root then the line neither cuts nor touches the circle. In this case there is no intersection and the distance from the centre of the circle to the line is greater than the radius of the circle.

Examples 1:

Show that part of the line $3y = x + 5$ is a chord of the circle $x^2 + y^2 - 6x - 2y - 15 = 0$ and find the length of this chord.

Solution

Method 1

$$y^2 - 5y + 4 = 0$$

Either $y = 4$ or $y = 1$

If $y = 4, x = 7$

$y = 1, x = -2$

Thus, the line cuts the circle in 2 distinct points $(7, 4)$ and $(-2, 1)$

Solve simultaneously the system

$$\begin{cases} 3y = x + 5 \\ x^2 + y^2 - 6x - 2y - 15 = 0 \end{cases}$$

$$x = 3y - 5$$

$$(3y - 5)^2 + y^2 - 6(3y - 5) - 2y - 15 = 0$$

$$9y^2 - 30y + 25 + y^2 - 18y + 30 - 2y - 15 = 0$$

$$y^2 - 5y + 4 = 0$$

Either $y = 4$ or $y = 1$

If $y = 4, x = 7$ $y = 1, x = -2$

Thus, the line cuts the circle in 2 distinct points $(7, 4)$ and $(-2, 1)$

The length of the cord is the distance between points $(7, 4)$ and $(-2, 1)$

i.e. the length of the chord is

$$l = \sqrt{81 + 9} = \sqrt{90}$$

$$= 3\sqrt{10} \text{ units}$$

Method 2

$$\text{Circle: } x^2 + y^2 - 6x - 2y - 15 = 0$$

$$\text{Line: } 3y = x + 5 = x - 3y + 5 = 0$$

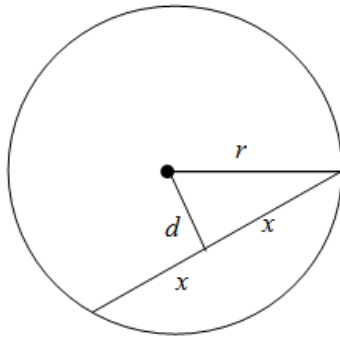
The centre of the circle is $\left(-\frac{-6}{2}, -\frac{-2}{2}\right) = (3, 1)$

The radius is $\frac{1}{2}\sqrt{36 + 4 + 60} = 5$

The distance from the centre of the circle to the line is

$$\begin{aligned}\frac{3 - 3 + 5}{\sqrt{1 + 9}} &= \frac{5}{\sqrt{10}} = \frac{5\sqrt{10}}{10} \\ &= \frac{1}{2}\sqrt{10} \text{ units}\end{aligned}$$

Since this distance is less than the radius of the circle, then the line cuts the circle in two points and hence part of the line is a chord of the circle.



From the diagram:

r is the radius of the circle equals to 5

d is the distance from the centre of the circle to the line equals $\frac{1}{2}\sqrt{10}$

$2x$ is the length of the chord

$$\text{Now, } 2x = 2\sqrt{r^2 - d^2} = 2\sqrt{25 - \frac{10}{4}} = \sqrt{90} = 3\sqrt{10}$$

Example 2:

Find the intersection of the circle $x^2 + y^2 - 14x + 12y + 69 = 0$ and the line $x + y = 1$

Solution

Solve simultaneously the system

$$\begin{cases} x^2 + y^2 - 14x + 12y + 69 = 0 & (1) \\ x + y = 1 & (2) \end{cases}$$

From (2): $y = 1 - x$ (3)

plug (3) into (1):

$$x^2 + 1 - 2x + x^2 - 14x + 12 - 12x + 69 = 0$$

$$2x^2 - 28x + 82 = 0$$

$$x^2 - 14x + 41 = 0$$

$$\Delta = 196 - 164 = 32$$

Since $\Delta > 0$, there are two points of intersection.

$$x_1 = \frac{14 + 4\sqrt{2}}{2} = 7 + 2\sqrt{2} \Rightarrow y_1 = 1 - x_1 = -6 - 2\sqrt{2}$$

$$x_2 = \frac{14 - 4\sqrt{2}}{2} = 7 - 2\sqrt{2} \Rightarrow y_2 = 1 - x_2 = -6 + 2\sqrt{2}$$

The points of intersection are $(7 + 2\sqrt{2}, -6 - 2\sqrt{2})$ and $(7 - 2\sqrt{2}, -6 + 2\sqrt{2})$.

Application activity 4.3.3

Find the intersection point of the circle and the lines.

Consider the following equation of circle and the line:

(1) $x^2 + y^2 - 14x + 12y + 69 = 0$ and line $y = -2$

(2) $x^2 + y^2 - 14x + 12y + 69 = 0$ and line $y = x$

(3) $x^2 + y^2 - 14x + 12y + 69 = 0$ and line $x + y = 1$



END OF UNIT ASSESSMENT

Assessing knowledge and understanding:

1) Answer by True or false:

If a line has equation $2x - 3y + 6 = 0$, then its slope is $\frac{2}{3}$

Explanation: In the form $Ax + By + C = 0$, the slope $m = -\frac{A}{B}$. Here $m = -\frac{2}{-3} = \frac{2}{3}$.

2) Which of the following is the **distance** between points $A(3, -1)$ and $B(-1, 4)$?

A. $\sqrt{(3+1)^2 + (-2-4)^2}$

B. $\sqrt{(3-1)^2 + (-2+4)^2}$

C. $\sqrt{(3-(-1))^2 + (-2-4)^2}$

D. $\sqrt{(3-(-1))^2 + (-2-(-4))^2}$

$$AB = \sqrt{(3-(-1))^2 + (-2-4)^2} = \sqrt{4^2 + (-6)^2} = \sqrt{16+36} = \sqrt{52} = 2\sqrt{13}.$$

3) Match each concept in **Column A** with its correct formula/description in **Column B**.

A	B
1 Distance between two points (x_1, y_1) and (x_2, y_2)	A. $y - y_1 = m(x - x_1)$
2 Coordinates of the Midpoint of the segment joining (x_1, y_1) and (x_2, y_2)	B. $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
3 Point-slope form of a line with slope m through the point (x_1, y_1)	C. $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
4 Standard equation of a circle with centre (h, k) and radius r	D. $(x - h)^2 + (y - k)^2 = r^2$

Assessing skills

4) Determine the distance between the pairs of the points below:

a) $(2, 1); (8, 7)$ b) $(-3, -8); (-3, -2)$

5) Find the centre and radius of each of the following circles

a) $(x - 3)^2 + y^2 = 25$ b) $x^2 + y^2 + 2x - 2y = 2$

6) On the floor, tiles make a Cartesian plane. Three points ball were placed on points A , B and C with coordinates $(1, 0)$, $(4, 4)$ and $(13, 5)$ respectively. By how many units does the distance from ball A to ball C exceed the distance from ball A to ball B ?

7) $L(5, 1)$ is the midpoint of the straight-line joining point $C(p, -3)$ to point $D(7, q)$

Find p and q .

8) Find the Cartesian equations of the lines whose vector equations are given below. Give your answers in the form $y = mx + c$

a) $(x, y) = (3, 2) + r(1, 4)$ b) $(x, y) = (-1, 2) + r(-2, 3)$.

Assessing attitudes and values:

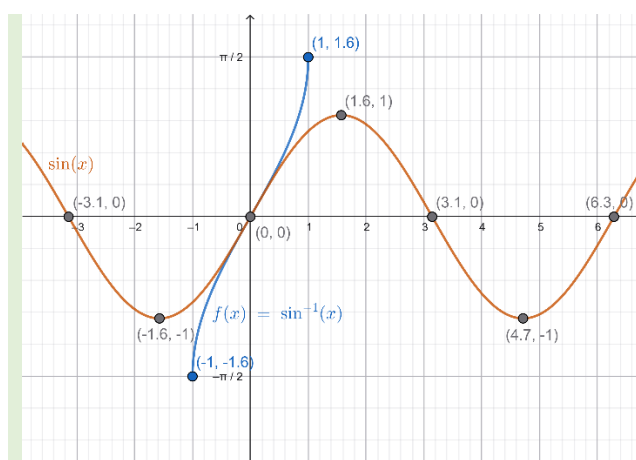
9) Comment by True or false: A good mathematician always checks their calculations when working with coordinates to ensure accuracy, even when the problem seems simple.

10) Which of the following best shows *collaboration* when solving a geometric problem involving distance and midpoints?

- A. Working alone because others might slow you down.
- B. Sharing your method with others and comparing approaches to improve understanding.
- C. Letting others do all the work while you just write the answer.
- D. Ignoring your group and submitting your own solution.

11) Match each attitude/value in **Column A** with the correct description in **Column B**.

A (Attitudes/Values)	B (description)
1. Appreciation of coordinate geometry	A. Using diagrams, sketches, and visual thinking to understand positions of shapes and lines.
2. Spatial reasoning	B. Recognizing how coordinate geometry helps in real-life fields such as engineering, architecture, and digital design.
3. Responsible problem-solving	C. Taking care to organize work neatly, label diagrams, and justify steps clearly.



UNIT 5: TRIGONOMETRIC AND INVERSE TRIGONOMETRIC FUNCTIONS

Key unit competence: Use trigonometric and inverse trigonometric functions in solving mathematical and real-life problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain trigonometric functions and their inverses.
- State and describe the domain and range of six trigonometric functions and their inverses.
- Explain the concepts of parity and periodicity in trigonometric functions.

- Identify and determine limits of trigonometric and inverse trigonometric functions.
- Recall and apply rules for derivatives of sine, cosine, tangent, cotangent, secant, and cosecant functions.

Skills:

- Compute and sketch graphs showing the domain, range, and period of trigonometric and inverse trigonometric functions.
- Determine parity and periodicity from trigonometric equations and graphs.
- Evaluate limits involving trigonometric and inverse trigonometric functions.
- Differentiate trigonometric and inverse trigonometric functions accurately.
- Find successive derivatives of sine, cosine, and other trigonometric functions.
- Apply derivatives of trigonometric functions to solve practical problems in refraction, simple harmonic motion, and optimization.
- Interpret and communicate results of trigonometric models in real contexts.

Attitudes and values:

- Develop precision, patience, and logical reasoning when solving trigonometric problems.
- Show curiosity and critical thinking in exploring patterns of trigonometric derivatives.
- Appreciate the importance of trigonometric models in understanding physical phenomena and real-world applications.
- Demonstrate collaboration, consistency, and a positive attitude during group and individual tasks.

5.0. Introduction

The techniques in trigonometry are used for finding relevance in navigation particularly satellite systems and astronomy, naval and aviation industries, oceanography, land surveying, and in cartography (creation of maps). Now those are the scientific applications of the concepts in trigonometry, but most of the mathematics we study would seem (on the surface) to have little real-life application. Trigonometry is really relevant in our day to day activities. In this unit we will see how we can use trigonometry to resolve problems we might encounter.



Introductory activity

1. Complete the table below

Function	Domain	Range (Principal value branches)	Function	Domain	Range
$\sin x$			$\sin^{-1} x$		
$\cos x$			$\cos^{-1} x$		
$\tan x$			$\tan^{-1} x$		
$\cot x$			$\cot^{-1} x$		
$\sec x$			$\sec^{-1} x$		
$\csc x$			$\csc^{-1} x$		

2. Find the derivatives of the function in the table above.

3. Evaluate the following limit $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

5.1. Introduction on trigonometric functions and their inverses



Activity 5.1

Identify trigonometric functions and inverse trigonometric function.

$\sin x$, $\tan x$, $\cot x$, $\cos^{-1} x$, $\sec x$, $\csc x$, $\sin^{-1} x$, $\cos x$, $\csc^{-1} x$.



Content summary

The study of trigonometry revolves around the relationship between the angles and sides of a right-angled triangle. This relationship is expressed through two primary sets of functions as follows.

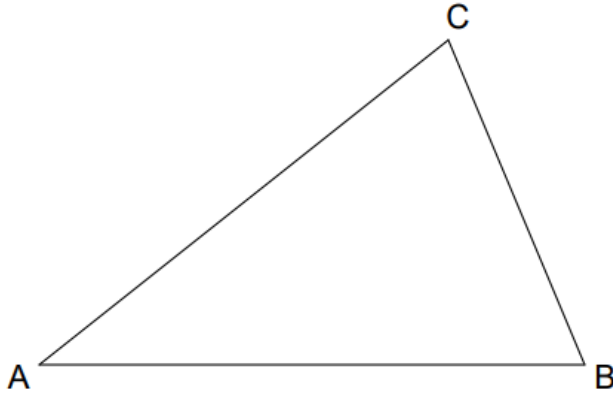
- Trigonometric functions: These functions map an angle (often denoted as θ) to a ratio of two sides of a triangle.
- Inverse trigonometric functions: These functions perform the opposite operation. They map a ratio back to its corresponding angle. They are essential for solving triangles when the side lengths are known but the angles are not.

Function	Formula	Ratio in Right Triangle	Inverse function
Sine (sin)	$\sin \theta$	opposite / hypotenuse	$\arcsin (x)$
Cosine (cos)	$\cos \theta$	adjacent / hypotenuse	$\arccos (x)$
Tangent (tan)	$\tan \theta$	opposite / adjacent	$\arctan (x)$
Cosecant (csc)	$\csc \theta = 1/\sin \theta$	hypotenuse / opposite	$\operatorname{arccsc} (x)$
Secant (sec)	$\sec \theta = 1/\cos \theta$	hypotenuse / adjacent	$\operatorname{arcsec} (x)$
Cotangent (cot)	$\cot \theta = 1/\tan \theta$	adjacent / opposite	$\operatorname{arccot} (x)$

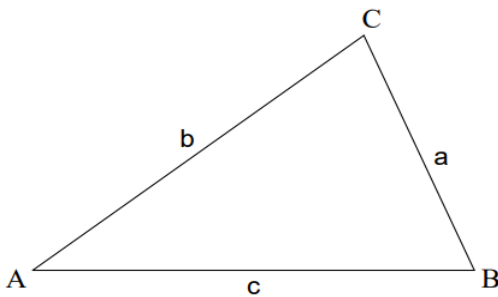
Note that the einverse trigonometric functions can also denoted as

$$f(x) = \sin^{-1} x, f(x) = \cos^{-1} x, f(x) = \tan^{-1} x, f(x) = \cot^{-1} x, f(x) = \sec^{-1} x, f(x) = \csc^{-1} x.$$

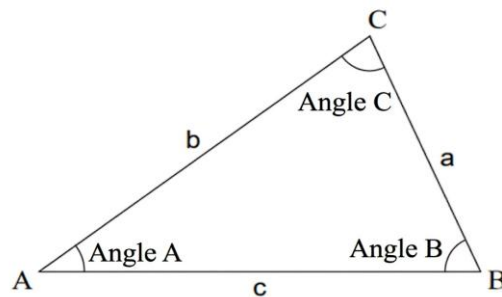
Trigonometric functions and their inverses can be used to solve a triangle. It means to determine all its unknown angles and sides. There is a standard way of labelling triangles which makes it relatively easy to describe what has to be calculated. The vertices are labelled A, B and C.



We use lower case letters to denote the lengths of the sides. It is agreed to denote the length of the side opposite angle A by a, the length of the side opposite angle B by b and that of the side opposite angle C by c.



and



Sine Rule: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ or $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

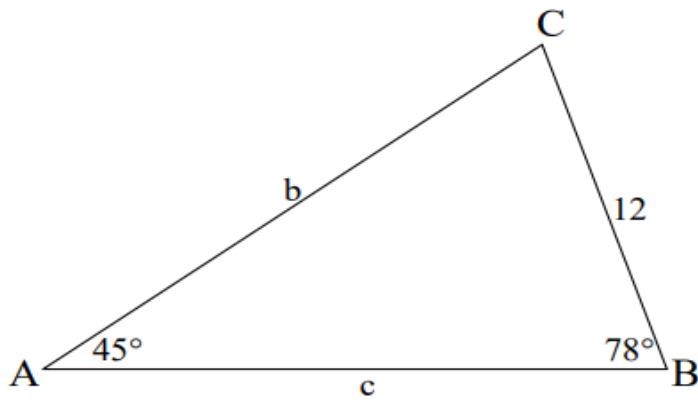
$$a^2 = b^2 + c^2 - 2bc \cos A$$

Cosine Rule: $b^2 = a^2 + c^2 - 2ac \cos B$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Example 1:

If $A=45^\circ$, $B=78^\circ$ and $a=12\text{ cm}$. Determine C, b and c.

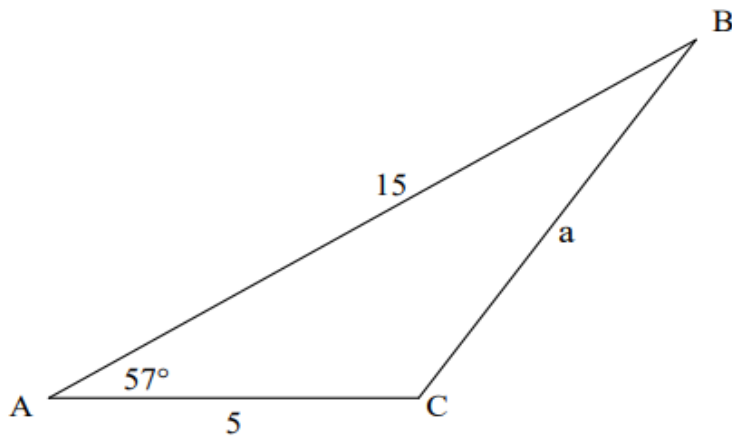


Solution:

Using sine rule, we have $\frac{\sin 45^\circ}{12} = \frac{\sin 78^\circ}{b} = \frac{\sin C}{c}$.

From the first equality, $b = \frac{12 \sin 78^\circ}{\sin 45^\circ} = 16.599 \approx 16.6$ and $C = 180^\circ - 45^\circ - 78^\circ$.

Example 2: Solve the triangle below.



Solution:

By using cosine rule we have,

$$a^2 = b^2 + c^2 - 2bc \cos A = 5^2 + 15^2 - 2 \times 5 \times 15 \cos 57^\circ$$

$$\Rightarrow a = \sqrt{5^2 + 15^2 - 2 \times 5 \times 15 \cos 57^\circ} = 12.973 \approx 13$$

$$c^2 = a^2 + b^2 - 2ab \cos C \Rightarrow C = \cos^{-1} \left(\frac{13^2 + 5^2 - 15^2}{2 \times 13 \times 5} \right) = 103.8^\circ.$$

The angle B is given by $B = 180^\circ - 13^\circ - 103.8^\circ = 63.2^\circ$.



Application activity 5.1

Imagine a surveyor looking at the top of the triangle's peak (Vertex C) from point A. They know the vertical height (altitude) of the triangle and the horizontal distance to that peak. The vertical height (CD) from the base to vertex C is 11.6 units. The horizontal distance from point A to the point directly below the peak (D) is also 11.6 units. Find the measure of angle A (the angle of elevation).

5.2. Domain and range of six trigonometric functions



Activity 5.2

State the values of x where the following functions are not defined:

1. $y = \sin x$

4. $y = \cot x$

2. $y = \cos x$

5. $y = \sec x$

3. $y = \tan x$

6. $y = \csc x$

..



Content summary

Trigonometric functions relate the angles of a right triangle to the ratios of its sides, or points on the unit circle.

Inverse trigonometric functions return the angle corresponding to a given trigonometric ratio.

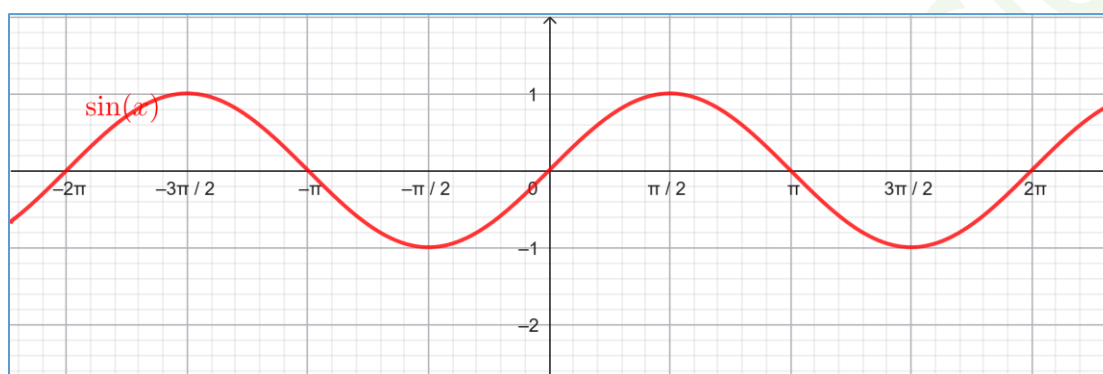
Function	Inverse	Domain for inverse	Range for inverse
$\sin x$	$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos x$	$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan x$	$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot x$	$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

$\sec x$	$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi], x \neq \frac{\pi}{2}$
$\csc x$	$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right], x \neq 0$

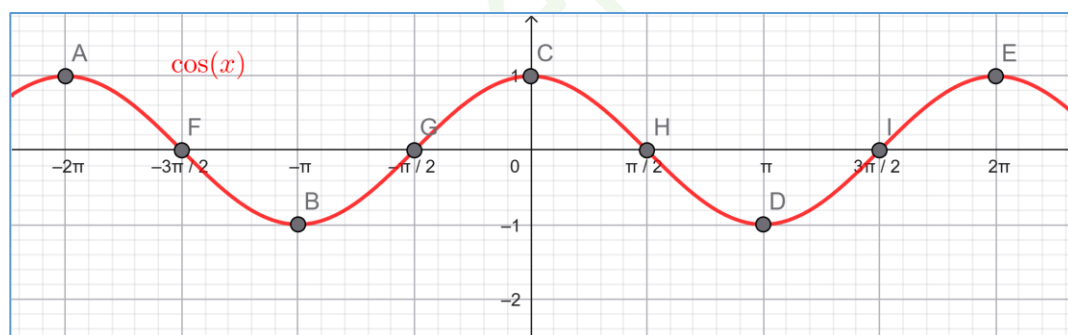
Cosine and sine

$\sin x$ and $\cos x$ are functions which are defined for all positive and negative values of x even for $x=0$. Thus, the domain of $\sin x$ and $\cos x$ is the set of real numbers. The range of $\sin x$ and $\cos x$ is $[-1, 1]$.

The graph of $y = f(x) = \sin x$



The graph of $y = f(x) = \cos x$



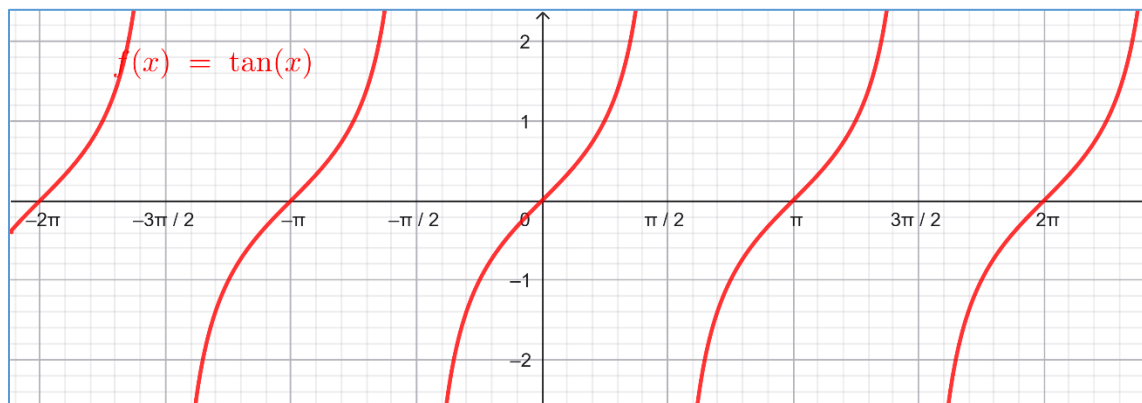
Tangent and Cotangent

Function $\tan x$ is not defined for $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$. Generally, $\tan x$ is not defined for

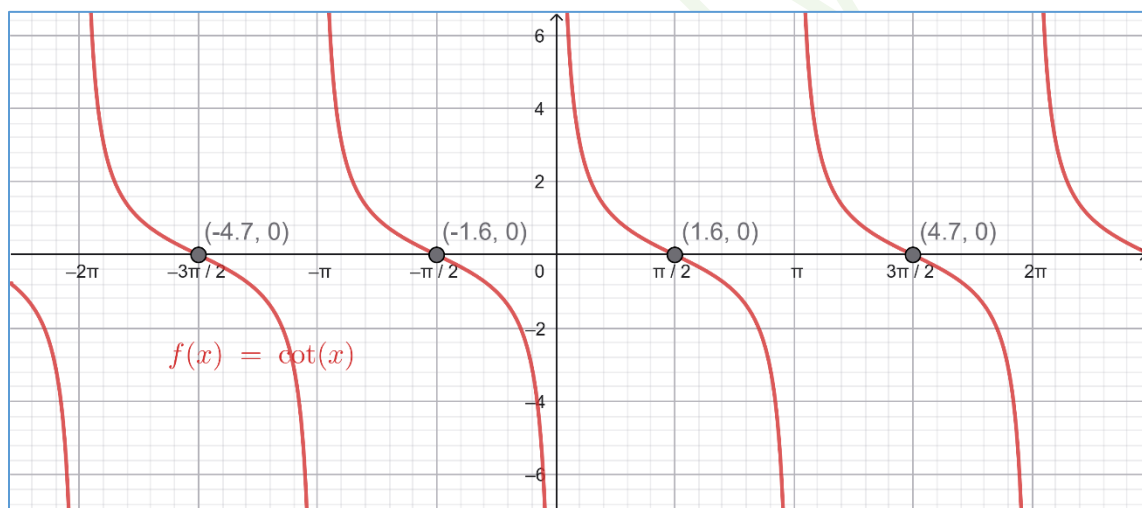
$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$. Thus, domain of $\tan x$ is $\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \right\}, k \in \mathbb{Z}$. The range of $\tan x$ is the set of real numbers. Function $\cot x$ is not defined for $x = 0, \pm\pi, \pm2\pi, \dots$. Generally, $\cot x$ is not defined for $x = k\pi, k \in \mathbb{Z}$.

Thus, domain of $\cot x$ is $\mathbb{R} \setminus \{k\pi\}, k \in \mathbb{Z}$. The range of $\cot x$ is the set of real numbers.

The graph of $y = f(x) = \tan x$



The graph of $y = f(x) = \cot x$



Secant and Cosecant:

Function $\sec x$ is not defined for $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$. Generally, $\sec x$ is not defined for

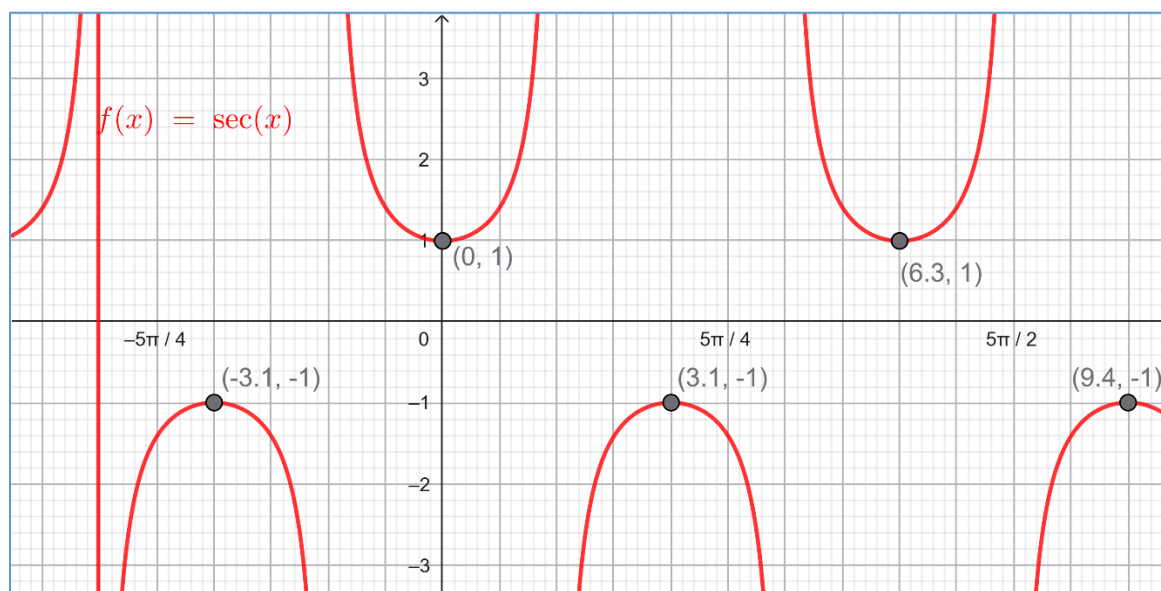
$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$. Thus, like tangent, domain of $\sec x$ is $\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \right\}, k \in \mathbb{Z}$.

Since $\sec x = \frac{1}{\cos x}$ and range of cosine is $[-1, 1]$, $\frac{1}{\cos x}$ will vary from negative infinity to -1 or from 1 to positive infinity. Thus, the range of $\sec x$ is $]-\infty, -1[\cup]1, +\infty[$

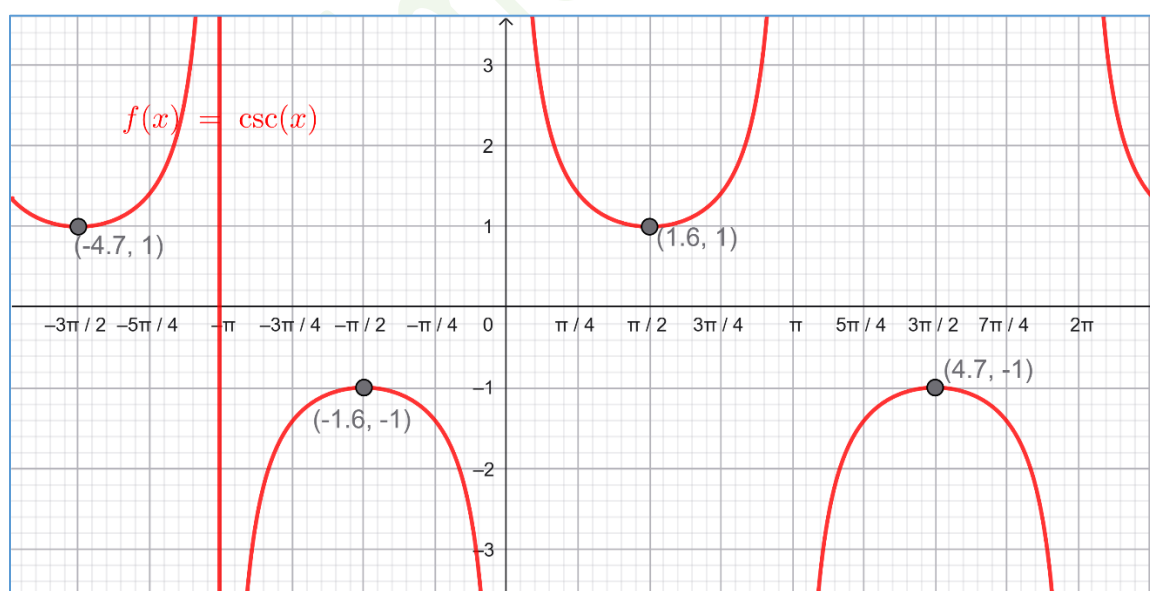
Function $\csc x$ is not defined for $x = 0, \pm\pi, \pm2\pi, \dots$. Generally, $\csc x$ is not defined for $x = k\pi, k \in \mathbb{Z}$. Thus, like cotangent, domain of $\csc x$ is $\mathbb{R} \setminus \{k\pi\}, k \in \mathbb{Z}$.

Since $\csc x = \frac{1}{\sin x}$ and range of sine is $[-1, 1]$, $\frac{1}{\sin x}$ will vary from negative infinity to -1 or from 1 to positive infinity. Thus, the range of $\csc x$ is $]-\infty, -1] \cup [1, +\infty[$.

The graph of $y = f(x) = \sec x$



The graph of $y = f(x) = \csc x$





Application activity 5.2

Find the domain of definition for each of the following functions

1. $f(x) = \sin x + \cos x$

2. $f(x) = \sin \frac{1}{x}$

3. $f(x) = \cos \left(\frac{x+1}{x} \right)$

4. $f(x) = \frac{1}{x} + \sin 2x$

5. $f(x) = \cos x + \tan x$

6. $f(x) = \cos \frac{\sqrt{x}}{x}$

5.3. Domain of inverses for trigonometric functions



Activity 5.3

Use properties of inverse functions and state the values of x where the following functions are not defined:

1) $y = \sin^{-1} x$ 2) $y = \cos^{-1} x$ 3) $y = \tan^{-1} x$

4) $y = \cot^{-1} x$ 5) $y = \sec^{-1} x$ 6) $y = \csc^{-1} x$



Content summary

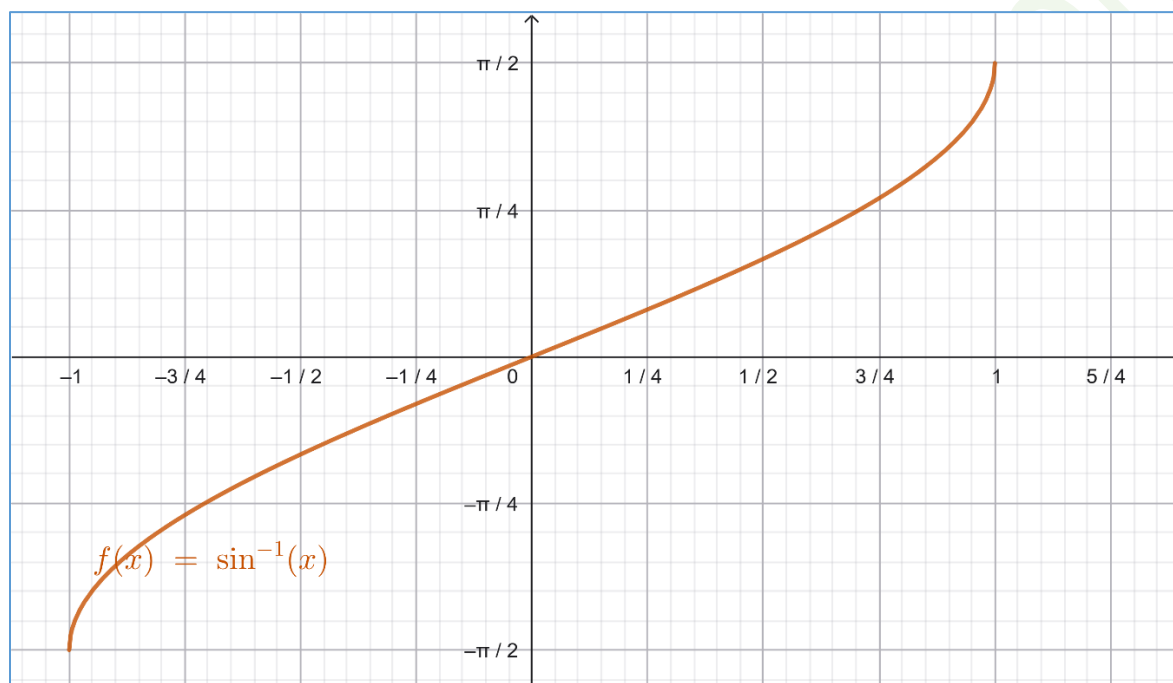
Inverse sine and inverse cosine

The functions $\sin x$ and $\cos x$ are defined on the entire interval $(-\infty, \infty)$. They have the inverses called **inverse sine** and **inverse cosine** denoted by $\sin^{-1} x$ and $\cos^{-1} x$ respectively.

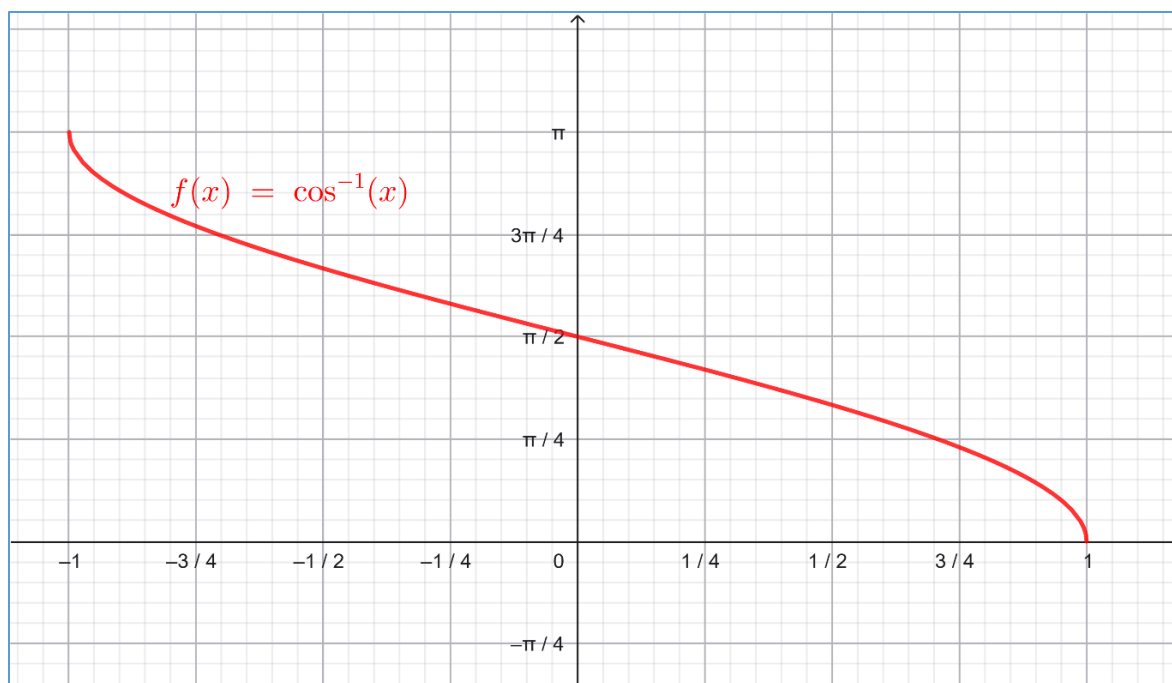
Note that the symbols $\sin^{-1} x$ and $\cos^{-1} x$ are never used to denote $\frac{1}{\sin x}$ and $\frac{1}{\cos x}$ respectively.

If desired $\frac{1}{\sin x}$ and $\frac{1}{\cos x}$ can be written as $\csc x$ and $\sec x$ respectively. In other literature, $\sin^{-1} x$ and $\cos^{-1} x$ are called **arcsine of x** and **arccosine of x** and they are denoted by $\arcsin x$ and $\arccos x$ respectively.

The graph of $y = f(x) = \arcsin x$



The graph of $y = f(x) = \arccos x$



Remark

The inverses of the trigonometric functions are not functions; they are relations. The reason why they are not functions is that for a given value of x , there are an infinite number of angles at which the trigonometric functions take on the value of x . Thus, the range of the inverses of the trigonometric functions must be restricted to make them functions. Without these restricted ranges, they are known as the inverse trigonometric relations.

To define $\sin^{-1} x$ and $\cos^{-1} x$ we restrict the domain of $\sin x$ and $\cos x$ to the interval

$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$ respectively to obtain a one-to-one function.

There are other ways to restrict the domain of $\sin x$ and $\cos x$ to obtain one-to-one functions, we might have required that $\frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$ and $\pi \leq x \leq 2\pi$ (or $\frac{-5\pi}{2} \leq x \leq \frac{-3\pi}{2}$ and $-2\pi \leq x \leq -\pi$) respectively.

Because $\sin x$ (restricted) and $\sin^{-1} x$; $\cos x$ (restricted) $\cos^{-1} x$ are inverses each other, it follows that:

- $\sin^{-1}(\sin y) = y$ if $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$; $\sin(\sin^{-1} x) = x$ if $-1 \leq x \leq 1$

- $\cos^{-1}(\cos y) = y$ if $0 \leq y \leq \pi$; $\cos(\cos^{-1} x) = x$ if $-1 \leq x \leq 1$

From these relations we obtain the following important results:

Theorem 5.1

- If $-1 \leq x \leq 1$ and $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$, then $y = \sin^{-1} x$ and $\sin y = x$ are equivalent.
- If $-1 \leq x \leq 1$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, then $y = \cos^{-1} x$ and $\cos y = x$ are equivalent.

Example 1:

Find

a) $\sin^{-1}\left(\frac{1}{2}\right)$

b) $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

Solution

a) Let $y = \sin^{-1}\left(\frac{1}{2}\right)$.

From Theorem 5.1 this equation is equivalent to $\sin y = \frac{1}{2}$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

The only value satisfying these conditions is $y = \frac{\pi}{6}$, so $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$.

b) Let $y = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$.

From Theorem 5.1 this equation is equivalent to $\sin y = -\frac{1}{\sqrt{2}}$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

The only value satisfying these conditions is $y = -\frac{\pi}{4}$, so $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4}$.

Example 2:

Simplify the function $\cos(\sin^{-1} x)$.

Solution

Express cosine in terms of sine to take advantage of the simplification $\sin^{-1}(\sin x) = x$.

Thus, we start by the identity $\cos^2 \theta = 1 - \sin^2 \theta$ and substitute $\theta = \sin^{-1} x$ to obtain

$$\cos^2(\sin^{-1} x) = 1 - \sin^2(\sin^{-1} x).$$

Or by taking square root $|\cos(\sin^{-1} x)| = \sqrt{1 - \sin^2(\sin^{-1} x)}$

$$\text{or } |\cos(\sin^{-1} x)| = \sqrt{1 - x^2}.$$

Since $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$, it follows that $\cos(\sin^{-1} x)$ is nonnegative.

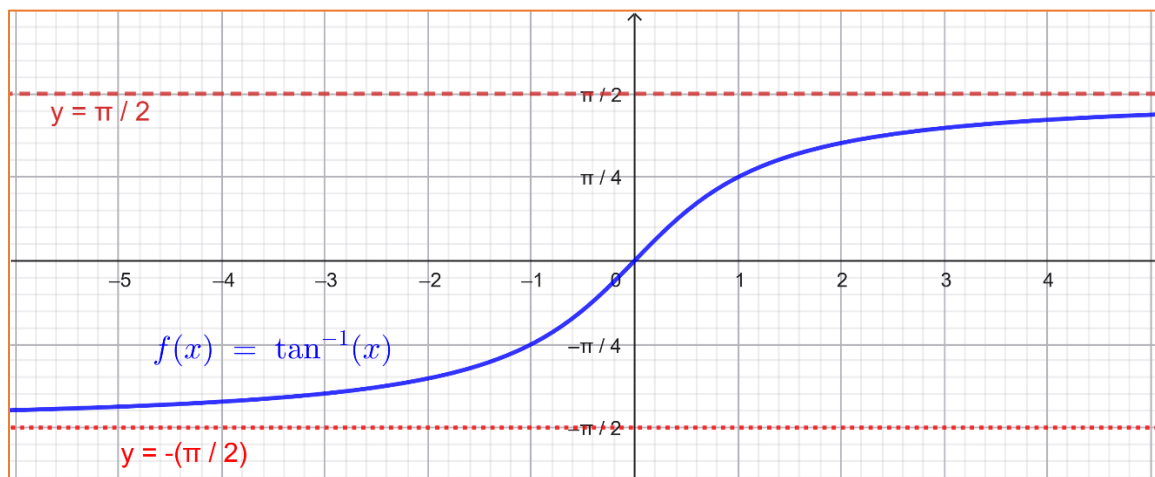
Thus, we can drop the absolute value and write $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Inverse tangent

Tangent of x , denoted $\tan x$, is a function which is defined for all positive and negative values

of x except $\pm 90^\circ, \pm 270^\circ, \dots$. To define $\tan x$, we restrict the domain of $\tan x$ to $\left(-\frac{\pi}{2}, +\frac{\pi}{2}\right)$.

The graph of $y = f(x) = \arctan x$



Because $\tan x$ (restricted) and $\tan^{-1} x$ are inverse each other, it follows that

- $\tan^{-1}(\tan y) = y$ if $-\frac{\pi}{2} < y < \frac{\pi}{2}$; $\tan(\tan^{-1} x) = x$ if $-\infty < x < +\infty$.

From these relations we obtain the following important results:

Example 3:

If $-\infty < x < +\infty$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$, then $y = \tan^{-1} x$ and $\tan y = x$ are equivalents.

Example 4:

Simplify the function $\sec^2(\tan^{-1} x)$.

Solution

The idea is to express secant in terms of $\tan x$ to take the advantage of simplification

$$\tan(\tan^{-1} x) = x.$$

Let $\theta = \tan^{-1} x$, in the identity $\sec^2 \theta = 1 + \tan^2 \theta$, we get

$$\sec^2(\tan^{-1} x) = 1 + \tan^2(\tan^{-1} x) = 1 + x^2$$

Thus, $\sec^2(\tan^{-1} x) = 1 + x^2$.

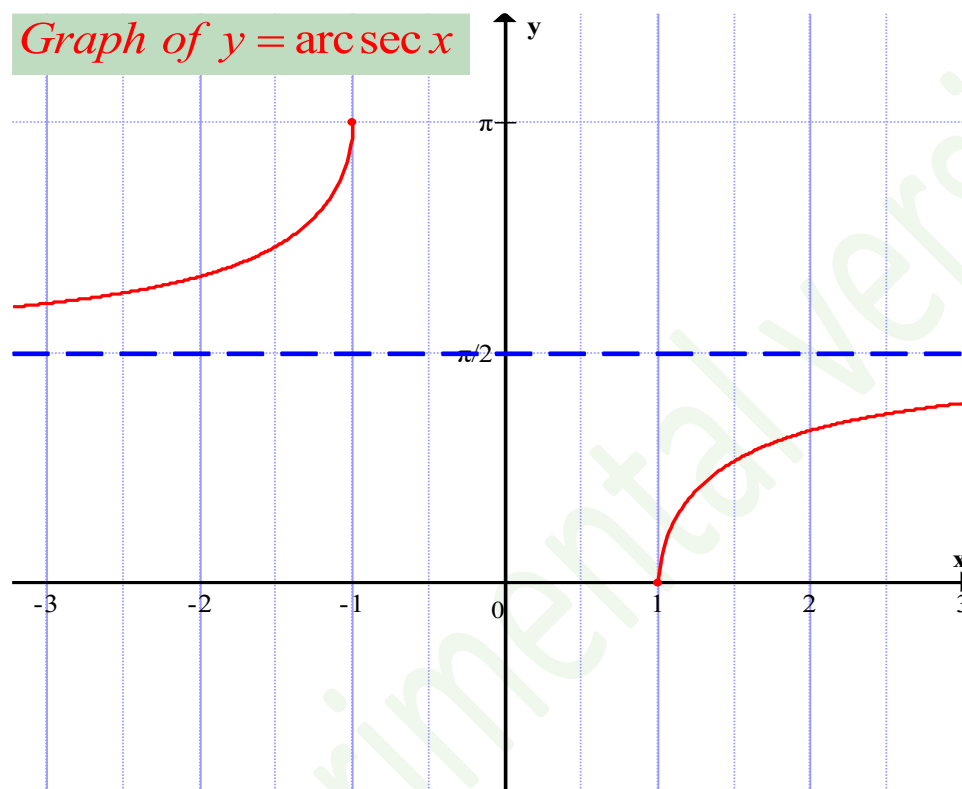
Inverse secant

The inverse secant, denoted $\sec^{-1} x$, is defined to be the inverse of restricted secant function.

For $f(x) = \sec x$, $0 \leq x < \frac{\pi}{2}$ or $\frac{\pi}{2} < x \leq \pi$.

If we let $y = \sec^{-1} x$, then we find that $x \leq -1$ or $x \geq 1$ and $0 \leq y < \frac{\pi}{2}$ or $\frac{\pi}{2} < y \leq \pi$.

Thus, the domain of $\sec^{-1} x$, is $(-\infty, -1] \cup [1, +\infty)$ and the range is $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{\pi}{2}, \pi\right]$



Theorem 2:

If $x \leq -1$ or $x \geq 1$ and if $0 \leq y < \frac{\pi}{2}$ or $\frac{\pi}{2} < y \leq \pi$, then $y = \sec^{-1} x$ and $\sec y = x$ are equivalent statements.

Example 5:

Simplify $\tan^2(\sec^{-1} x)$

Solution

We know that $\sec^2 \theta = \tan^2 \theta + 1$, then $\tan^2 \theta = \sec^2 \theta - 1$.

Putting $\theta = \sec^{-1} x$ we have $\tan^2(\sec^{-1} x) = \sec^2(\sec^{-1} x) - 1 = x^2 - 1$

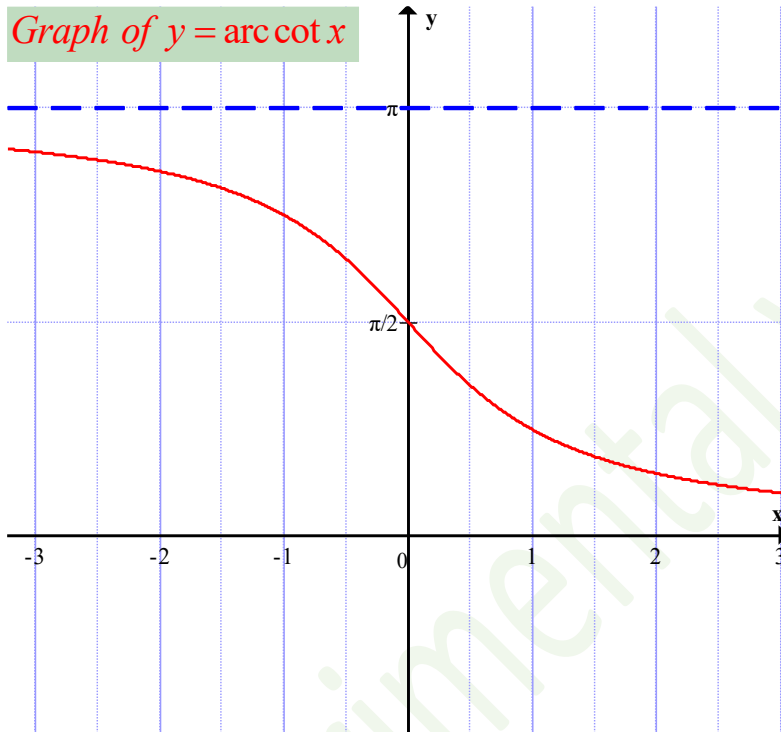
Thus, $\tan^2(\sec^{-1} x) = x^2 - 1$

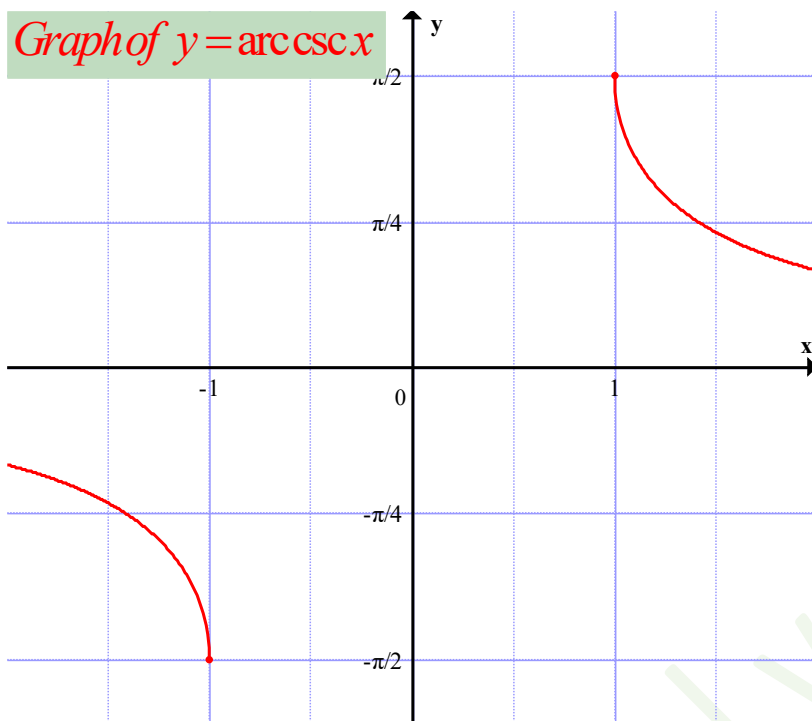
Inverse cotangent and inverse cosecant

We will summarize their properties briefly

$y = \cot^{-1} x$ is equivalent to $x = \cot y$ if $0 < y < \pi$ and $-\infty < x < +\infty$

$y = \csc^{-1} x$ is equivalent to $x = \csc y$ if $-\frac{\pi}{2} \leq y < 0$ or $0 < y < \frac{\pi}{2}$ and $|x| \geq 1$





Notice

If α and β are acute complementary angles, then from basic trigonometry, $\sin \alpha$ and $\sin \beta$ are equal. Let us write $x = \sin \alpha = \cos \beta$ so that $\alpha = \sin^{-1} x$ and $\beta = \cos^{-1} x$.

Since $\alpha + \beta = \frac{\pi}{2}$, we obtain the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

Similarly, we can obtain identities

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}.$$

$$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}.$$

Remark

$$\sin^{-1}(-x) = -\sin^{-1} x$$

$$\tan^{-1}(-x) = -\tan^{-1} x$$

$$\sec^{-1}(-x) = \pi + \sec^{-1} x, \text{ if } x \geq 1$$

Example 6:

For which values of x is true that

a) $\tan^{-1}(\tan x) = x$

b) $\tan(\tan^{-1} x) = x$

c) $\csc^{-1}(\csc x) = x$

d) $\csc(\csc^{-1} x) = x$

Solution

The values of x are:

a) $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

b) $-\infty < x < +\infty$.

c) $-\frac{\pi}{2} \leq x < 0$ or $0 < x \leq \frac{\pi}{2}$.

d) $|x| \geq 1$.



Application activity 5.3

Find the domain of definition of the following functions

1. $f(x) = \frac{1}{x} + \sin^{-1} 2x$

2. $f(x) = \cos^{-1} x + \tan^{-1} x$

3. $f(x) = \cos^{-1} \frac{\sqrt{x}}{x}$

4. $f(x) = \sin^{-1} \frac{1}{x}$

5.4. Range of inverses of trigonometric functions



Activity 5.4

Complete the following table:

Function	Range
$\sin^{-1} x$	
$\cos^{-1} x$	
$\tan^{-1} x$	
$\cot^{-1} x$	

$\sec^{-1} x$	
$\csc^{-1} x$	



Content summary

It has the inverse called **inverse tangent** and is denoted by $\tan^{-1} x$.

Function	Range
$\sin^{-1} x$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[0, \pi]$
$\tan^{-1} x$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$(0, \pi)$
$\sec^{-1} x$	$[0, \pi], x \neq \frac{\pi}{2}$
$\csc^{-1} x$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right], x \neq 0$



Application activity 5.4

Answer the following questions by True or false:

1. $\sin^{-1}(x)$ always lies in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. $\cos^{-1}(x)$ can take negative values.

3. $\tan^{-1}(x)$ lies strictly between $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

4. $\sec^{-1}(x)$ can take the value $\frac{\pi}{2}$.

5. $\csc^{-1}(x)$ never takes the value 0.

5.5. Parity of trigonometric functions



Activity 5.5

For the function

1. $f(x) = \frac{\sin x}{x}$, find $f(-x)$, $-f(x)$ and compare the two results to $f(x)$.
2. $g(x) = \frac{\cos x}{x}$, find $g(-x)$, $-g(x)$ and compare the two results to $g(x)$.
3. $h(x) = \sin x + \cos x$, find $h(-x)$, $-h(x)$ and compare the two results to $h(x)$.



Content summary

Even function

A function $f(x)$ is said to be **even** if the following conditions are satisfied

- $\forall x \in \text{Dom}f, -x \in \text{Dom}f$
- $f(x) = f(-x)$

The graph of such function is **symmetric about the vertical axis**, i.e. $x=0$

Trigonometric Function	Type of Function	Algebraic Property
$\cos x$	Even	$\cos(-x) = \cos x$
$\sec x$	Even	$\sec(-x) = \sec x$

Example 1:

The function $\cos x$ is an even function since $\forall x \in \mathbb{R}, -x \in \mathbb{R}$ and

$$f(-x) = \cos(-x) = \cos x = f(x).$$

Odd function

A function $f(x)$ is said to be **odd** if the following conditions are satisfied

- $\forall x \in \text{Dom}f, -x \in \text{Dom}f$

- $f(-x) = -f(x)$

The graph of such function looks the same when rotated through half a revolution about 0. This is called **rotational symmetry**.

Trigonometric Function	Type of Function	Algebraic Property
$\sin x$	Odd	$\sin(-x) = -\sin x$
$\csc x$	Odd	$\csc(-x) = -\csc x$
$\tan x$	Odd	$\tan(-x) = -\tan x$
$\cot x$	Odd	$\cot(-x) = -\cot x$

Example 2:

The function $\sin x$ is an odd function since $\forall x \in \mathbb{R}, -x \in \mathbb{R}$ and

$$f(-x) = \sin(-x) = -\sin x = -f(x)$$



Application activity 5.5

Study the parity of the following functions

1. $f(x) = \frac{x^2}{\cos x}$.

2. $f(x) = x + \sin 4x$.

3. $f(x) = \sqrt[3]{x} + \sin x$

4. $f(x) = \frac{\tan x}{x+1}$.

5.6. Limits of trigonometric functions



Activity 5.6

1. Evaluate:

a) $\lim_{x \rightarrow 0} \sin x$ b) $\lim_{x \rightarrow 0} x \sin x$ c) $\lim_{x \rightarrow 0} \cos x$ d) $\lim_{x \rightarrow 0} \frac{1}{x}$ e) $\lim_{x \rightarrow 0} \frac{\cos x}{x}$.

2. Consider the function $f(x) = \frac{\sin x}{x}$ where x is in radians. Use calculator to complete the following tables

x	$\frac{\sin x}{x}$
1	
0.9	
0.8	
0.7	
0.6	
0.5	
0.4	
0.3	
0.2	
0.1	
0.01	
0.001	
0.0001	

x	$\frac{\sin x}{x}$
-1	
-0.9	
-0.8	
-0.7	
-0.6	
-0.5	
-0.4	
-0.3	
-0.2	
-0.1	
-0.01	
-0.001	
-0.0001	

a) From results in 2), what is the limit of $\frac{\sin x}{x}$ as x approaches 0 from the right side.

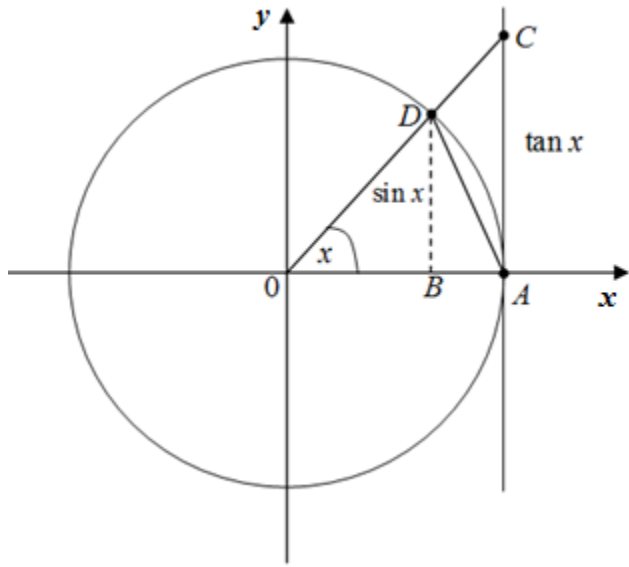
b) From results in 2), what is the limit of $\frac{\sin x}{x}$ as x approaches 0 from the left side.

c) What can you say about $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?



Content summary

There is another way to prove that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Let us consider the unit circle below:



Assume $0 < x < \frac{\pi}{2}$. Since the circle is unit circle the radius $\overline{OA} = 1$.

The area of the triangle OAD is $\frac{1}{2} \overline{OA} \times \sin x = \frac{1}{2} x$.

The area of the sector OAD :

Recall that the area of a sector with subtended angle measuring x radian is $A = \frac{\theta}{2} r^2$, where r

is the radius. The subtended angle of sector OAD is x and radius is r the area of the sector

OAD is $\frac{1}{2} \overline{OA}^2 \times x = \frac{1}{2} x$.

The area of the triangle OAC is $\frac{1}{2} \overline{OA} \tan x = \frac{1}{2} \tan x$

From the figure we see that $\text{area of } \triangle OAD \leq \text{area of sector } OAD \leq \text{area of } \triangle OAC$

Or $\frac{1}{2} \sin x \leq \frac{1}{2} x \leq \frac{1}{2} \tan x \Leftrightarrow \sin x \leq x \leq \frac{\sin x}{\cos x}$.

Dividing by $\sin x$, we get $1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x}$

Taking the inverse [remember that when taking the inverse, the order of the inequality must be

changed], we get $\cos x \leq \frac{\sin x}{x} \leq 1$.

Taking limit as x approaches 0 , we get $1 \leq \lim_{x \rightarrow 0} \frac{\sin x}{x} \leq 1$.

Using **Squeeze theorem**, since $\cos x \leq \frac{\sin x}{x} \leq 1$ and $\cos x \leq \frac{\sin x}{x} \leq 1$.

Thus, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

This result will help us to find limits of some other trigonometric functions

Example 1:

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \frac{0}{0}$.

Solution

$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \frac{0}{0}$ Indeterminate case (I.C.)

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2} \sin \frac{x}{2}}{x} \quad \left[1 - \cos x = 2 \sin^2 \frac{x}{2} \right]$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{x} \lim_{x \rightarrow 0} \sin \frac{x}{2} = \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{2 \frac{x}{2}} \lim_{x \rightarrow 0} \sin \frac{x}{2} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \lim_{x \rightarrow 0} \sin \frac{x}{2} = 1 \times 0 = 0$$

Thus, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$.

Example 2:

Evaluate $\lim_{x \rightarrow 0} \frac{x}{\sin x}$.

Solution

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = \frac{0}{0} \text{ I.C.}$$

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{x}{x \frac{\sin x}{x}} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1.$$

Thus, $\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1.$

Example 3:

Evaluate $\lim_{x \rightarrow 0} \frac{\tan x}{x}.$

Solution

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \frac{0}{0} \text{ I.C.}$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x}}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \times 1 = 1$$

Thus, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1.$

Example 4:

Evaluate $\lim_{x \rightarrow 0} \frac{\cot x}{x}.$

Solution

$$\lim_{x \rightarrow 0} \frac{\cot x}{x} = \frac{\infty}{0} = \infty$$

Or

$$\lim_{x \rightarrow 0} \frac{\cot x}{x} = \lim_{x \rightarrow 0} \cot x \lim_{x \rightarrow 0} \frac{1}{x} = \infty \times \infty = \infty$$

Left- and right-hand limits: $\frac{\cot x}{x} = \frac{\frac{\cos x}{\sin x}}{x} = \frac{\cos x}{x \sin x}$

x	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$
$\cos x$	+	1	+
x	-	0	+

$\sin x$	-	0	+
$x \sin x$	+	0	+
$\frac{\cos x}{x \sin x}$	+	$\frac{0}{0}$ ∞	+

Thus, $\lim_{x \rightarrow 0^+} \frac{\cot x}{x} = +\infty$

and $\lim_{x \rightarrow 0^-} \frac{\cot x}{x} = +\infty$ and hence $\lim_{x \rightarrow 0} \frac{\cot x}{x} = +\infty$ exists.

When finding limits of trigonometric functions sometimes we need to change the variable.

Example 5:

Evaluate $\lim_{x \rightarrow \frac{\pi}{3}} \frac{1 - 2 \cos x}{x - \frac{\pi}{3}}$

Solution

$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{1 - 2 \cos x}{x - \frac{\pi}{3}} = \frac{1 - 2 \times \frac{1}{2}}{\frac{\pi}{3} - \frac{\pi}{3}} = \frac{0}{0} \text{ I.C}$$

Let $x - \frac{\pi}{3} = t \Rightarrow x = t + \frac{\pi}{3}$. If $x \rightarrow \frac{\pi}{3}$, $t \rightarrow 0$.

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{1 - 2 \cos \left(t + \frac{\pi}{3} \right)}{t} &= \lim_{t \rightarrow 0} \frac{1 - 2 \left(\cos t \cos \frac{\pi}{3} - \sin t \sin \frac{\pi}{3} \right)}{t} = \lim_{t \rightarrow 0} \frac{1 - 2 \left(\frac{1}{2} \cos t - \frac{\sqrt{3}}{2} \sin t \right)}{t} \\ &= \lim_{t \rightarrow 0} \frac{1 - \cos t + \sqrt{3} \sin t}{t} = \lim_{t \rightarrow 0} \left(\frac{1 - \cos t}{t} + \frac{\sqrt{3} \sin t}{t} \right) = \lim_{t \rightarrow 0} \frac{1 - \cos t}{t} + \lim_{t \rightarrow 0} \frac{\sqrt{3} \sin t}{t} \\ &= \lim_{t \rightarrow 0} \frac{1 - \cos t}{t} + \sqrt{3} \lim_{t \rightarrow 0} \frac{\sin t}{t} = 0 + \sqrt{3} \times 1 = \sqrt{3}. \end{aligned}$$

Example 6:

Evaluate $\lim_{x \rightarrow 3} \frac{x - 3}{\sin \pi x}$.

Solution

$$\lim_{x \rightarrow 3} \frac{x-3}{\sin \pi x} = \frac{0}{0} \quad \text{I.C}$$

If $t = x - 3 \Rightarrow x = t + 3$. If $x \rightarrow 3, t \rightarrow 0$.

$$\lim_{x \rightarrow 3} \frac{x-3}{\sin \pi x} = \lim_{t \rightarrow 0} \frac{t}{\sin(\pi t + 3\pi)}$$

$$= \lim_{t \rightarrow 0} \frac{t}{\sin \pi t \cos 3\pi + \cos \pi t \sin 3\pi} = \lim_{t \rightarrow 0} \frac{t}{-\sin \pi t} \quad [\text{Since } \cos 3\pi = -1, \sin 3\pi = 0]$$

$$= -\lim_{t \rightarrow 0} \frac{\frac{t}{\pi t}}{\frac{\sin \pi t}{\pi t}} = -\lim_{t \rightarrow 0} \frac{\frac{1}{\pi}}{\frac{\sin \pi t}{\pi t}} = -\frac{1}{\pi}.$$



Application activity 5.6

Find the limit of the following functions

1. $\lim_{\theta \rightarrow \frac{\pi}{4}} (\theta \tan \theta)$

2. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{1 - \sin x}$

3. $\lim_{t \rightarrow 0} \frac{1 - \cos t}{\sin t}$

4. $\lim_{t \rightarrow 0} \frac{\sin 3t}{t}$

5.7. Limits of inverse of trigonometric functions



Activity 5.7

1) Find the exact value of

a) $\sin^{-1}(-1)$

b) $\tan^{-1}(1)$

c) $\cot^{-1}(-1)$

d) $\sec^{-1}(-2)$

e) $\csc^{-1}(-2)$

2) Evaluate the following limits

a) $\lim_{x \rightarrow 1} \cot^{-1}\left(\frac{2x-3}{x}\right)$

b) $\lim_{x \rightarrow 1} \sin^{-1}\left(\frac{1+x}{2x}\right)$

c) $\lim_{x \rightarrow 0} \cos^{-1}\left(\frac{\sqrt{x+1}-1}{x}\right)$

d) $\lim_{x \rightarrow -1} \tan^{-1}\left(\frac{1-x^2}{2x+2}\right)$



Content summary

In this section we will use l'Hôpital's rule.

Recall that if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$ we remove this indeterminate case by

differentiating function $f(x)$ and $g(x)$ and then evaluate the limit. That is if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$ then we evaluate $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$. We do this until the indeterminate case is removed.

Example

Evaluate $\lim_{x \rightarrow +\infty} \sin^{-1}\left(\frac{1}{x}\right)$

Solution

$$\lim_{x \rightarrow +\infty} \sin^{-1}\left(\frac{1}{x}\right) = \sin^{-1}\left(\frac{1}{\infty}\right) = \sin^{-1}(0) = 0$$

Example

Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{\sin^{-1}(1-x)}$

Solution

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{\sin^{-1}(1-x)} = \frac{0}{0} \text{ I.C}$$

Remove this indeterminate case by l'Hôpital's rule

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{\sin^{-1}(1-x)} = \lim_{x \rightarrow 1} \frac{2x}{\sqrt{1-(1-x)^2}} = \lim_{x \rightarrow 1} \frac{2x}{\sqrt{2x-x^2}} = \frac{2}{\sqrt{2-1}} = 2$$



Application activity 5.7

Evaluate the following limits

1) $\lim_{x \rightarrow 1} \left(\frac{\pi}{2} - \tan^{-1} \frac{1}{x} \right)$ 2) $\lim_{x \rightarrow 1^-} (x-2) \tan^{-1} \frac{1}{1-x}$ 3) $\lim_{x \rightarrow 0} \frac{\sin x}{\sin^{-1} x}$

4) $\lim_{x \rightarrow 1} \frac{\cos^{-1} x - \tan^{-1}(1-x)}{x^2 - 1}$ 5) $\lim_{x \rightarrow 0} \frac{\sin^{-1} 5x}{x}$ 6) $\lim_{x \rightarrow +\infty} x \tan^{-1} \frac{2}{x}$

5.8. Derivative of sine and cosine



Activity 5.8

- 1) Using definition of derivative, find the derivative of $\cos x$ and $\sin x$.
- 2) Use result in 1) and relation $\cos x = \sin\left(\frac{\pi}{2} - x\right)$ to find the derivative of $\cos x$.



Content summary

The functions $f(x) = \sin x$ and $f(x) = \cos x$ are differentiable on the set of real numbers. In addition, $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$.

After differentiation of composite functions, if u is another function, then $(\sin u)' = u' \cos u$ and $(\cos u)' = -u' \sin u$

Example 1:

Find the derivative of $f(x) = \sin(3x^2 + 4)$

Solution

$$f'(x) = (3x^2 + 4)' \cos(3x^2 + 4) = 6x \cos(3x^2 + 4)$$

Example 2:

Find the derivative of $f(x) = \cos(3x)$.

Solution

$$f'(x) = -(3x)' \sin(3x) = -3 \sin(3x)$$



Application activity 5.8

Find the derivative of the following functions

1. $f(x) = \sin(x^2 + 3)$
2. $f(x) = \tan^4 3x$
3. $f(x) = \cos 3x^2$
4. $f(x) = \cos^3 2x$.

5.9. Successive derivative for sine and cosine



Activity 5.9

Consider the function $f(x) = \cos 2x$. Find

1. $f'(x)$
2. The derivative of the function obtained in 1.
3. The derivative of the function obtained in 2.
4. The derivative of the function obtained in 3.
5. The derivative of the function obtained in 4.



Content summary

We have seen that the derivative of a function of x is in general also a function of x . This new function may also be differentiable, in which case the derivative of the first derivative is called the **second derivative** of the original function.

Similarly, the derivative of the second derivative is called the **third derivative** and so on.

The successive derivatives of a function f are **higher order derivatives** of the same function.

We denote higher order derivatives of the same function as follows:

The second derivative is:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = f''(x) = y''$$

The third derivative is:

$$\frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^3} = f'''(x) = y'''$$

And the n^{th} derivative is:

$$\frac{d}{dx}\left(\frac{d^{n-1}y}{dx^{n-1}}\right) = \frac{d^ny}{dx^n} = f^{(n)}(x) = y^{(n)}$$

Example 1:

Find the n^{th} derivative of $y = \sin x$.

Solution

$$y' = \cos x = \sin\left(x + \frac{\pi}{2}\right)$$

$$y'' = -\sin x = \sin\left(x + \frac{2\pi}{2}\right)$$

$$y''' = -\cos x = \sin\left(x + \frac{3\pi}{2}\right)$$

⋮

$$y^{(n)} = \sin\left(x + \frac{n\pi}{2}\right)$$

Thus, if $y = \sin x$, $y^{(n)} = \sin\left(x + \frac{n\pi}{2}\right)$.

Example 2:

Find the n^{th} derivative of $y = \cos x$

Solution

$$y' = -\sin x = \cos\left(x + \frac{\pi}{2}\right)$$

$$y'' = -\cos x = \cos\left(x + \frac{2\pi}{2}\right)$$

$$y''' = \sin x = \sin\left(x + \frac{3\pi}{2}\right)$$

⋮

$$y^{(n)} = \cos\left(x + \frac{n\pi}{2}\right)$$

Thus, if $y = \cos x$ then $y^{(n)} = \cos\left(x + \frac{n\pi}{2}\right)$



Application activity 5.9

1. Find the second derivative of

a) $y = \sin 2x \sin 3x$.

b) $y = \arctan \sqrt{x}$.

c) $y = \sin^2 x$.

d) $y = \tan^2 x$.

2. Find the third derivative of:

a) $y = x \arctan x$.

b) $y = \sin 2x \cos 3x$.

c) $y = \frac{\sin 2x}{\sin 3x}$.

d) $y = \tan x \tan 2x$.

3. Find the n th derivative of:

a) $y = \sin x + \cos x$.

b) $y = \cos 2x$.

5.10. Derivative of tangent and cotangent



Activity 5.10

1. Use rule for derivative of a quotient and the relation $\tan x = \frac{\sin x}{\cos x}$ to find the derivative of $\tan x$.

2. Use result in 1) and relation $\cot(x) = \left[\frac{\pi}{2} - x \right]$ to find the derivative of $\cot x$.



Content summary

The function $f(x) = \tan x$ is differentiable on $\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \right\}, k \in \mathbb{Z}$ and the function

$f(x) = \cot x$ is differentiable on $\mathbb{R} \setminus \{k\pi\}, k \in \mathbb{Z}$.

In addition, $(\tan x)' = \frac{1}{\cos^2 x} = \sec^2 x = 1 + \tan^2 x, \forall x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$.

Thus, $(\tan x)' = 1 + \tan^2 x$.

If u is another function then,

$$(\tan u)' = \frac{u'}{\cos^2 u} = u' \sec^2 u = u'(1 + \tan^2 u).$$

Thus, $(\tan u)' = u'(1 + \tan^2 u), \forall x \neq k\pi, k \in \mathbb{Z}.$

$$(\cot x)' = \frac{-1}{\sin^2 x} = -\csc^2 x = -(1 + \cot^2 x).$$

Thus, $(\cot x)' = -(1 + \cot^2 x).$

If u is another function, then, $(\cot u)' = \frac{-u'}{\sin^2 u} = -u' \csc^2 u = -u'(1 + \cot^2 u).$

Thus, $(\cot u)' = -u'(1 + \cot^2 u).$

Example 1: Find the derivative of $f(x) = x^2 \tan x$.

Solution

$$f'(x) = (x^2)' \tan x + x^2 (\tan x)' = 2x \tan x + x^2 \sec^2 x.$$

Example 2: Find the derivative $f(x) = \cot x^2$.

Solution

$$f'(x) = -(x^2)' \csc^2 x^2 = -2x \csc^2 x^2.$$



Application activity 5.10

Find the derivative of the following functions

1. $f(x) = x \tan x$.
2. $f(x) = \tan(3x + 2)$.
3. $f(x) = \cot(x^2 - 5)$.
4. $f(x) = \sin x \cot 4x$.

5.11. Derivative of secant and cosecant



Activity 5.11

1. Use rule for derivative of reciprocal of a function and relation $\sec x = \frac{1}{\cos x}$ to find the derivative of $\sec x$.
2. Use rule for derivative of reciprocal of a function and relation $\csc x = \frac{1}{\sin x}$ to find the derivative of $\csc x$.



Content summary

$(\sec x)' = \sec x \tan x$ and $(\csc x)' = -\csc x \cot x$. If u is another function, then
 $(\sec u)' = u' \sec u \tan u$ and $(\csc u)' = -u' \csc u \cot u$.

Example 1: Find the derivative of $f(x) = \sec(2x + 1)$.

Solution

$$f'(x) = 2 \sec(2x + 1) \tan(2x + 1)$$

Example 2: Find the derivative of $f(x) = \csc(x^2 + 1)$

Solution

$$f'(x) = -2x \csc(x^2 + 1) \cot(x^2 + 1)$$



Application activity 5.11

Find the derivative of the following functions

1. $f(x) = \sec(3x + 2)$
2. $f(\theta) = \theta^3 \csc 2\theta$
3. $f(x) = \sec^4 3x$

5.12. Derivative of inverse sine



Activity 5.12

It is known that $f(x) = \sin^{-1} x$ for $x \in [-1, 1]$ and $x = \sin y$ for $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ where $y = f(x)$.

Use rule for derivative of composite functions to find the derivative of $f(x) = \sin^{-1} x$, the inverse of sine function.



Content summary

$$\forall x \in]-1, 1[, (\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$\text{If } u \text{ is another function } (\sin^{-1} u)' = \frac{u'}{\sqrt{1-u^2}}$$

Example 1: Find the derivative of $f(x) = \sin^{-1} x^3$.

Solution

$$f'(x) = \frac{(x^3)'}{\sqrt{1-(x^3)^2}} = \frac{3x^2}{\sqrt{1-x^6}}$$

Example 2:

Find the derivative of $f(x) = \cos^{-1}(2x+1)$.

Solution

$$f'(x) = \frac{-(2x+1)'}{\sqrt{1-(2x+1)^2}} = \frac{-2}{\sqrt{1-4x^2-4x-1}} = \frac{-1}{\sqrt{-x^2-x}}$$

Example 3:

Find the derivative of $y = \sin^{-1}(1-x^2)$

Solution

$$y' = \frac{-2x}{\sqrt{1-(1-x^2)^2}} = \frac{-2x}{\sqrt{-x^4+2x^2}}$$

Example 4:

Find the derivative of $y = 3 \cos^{-1}(x^2 + 0.5)$

Solution

$$y' = 3 \frac{-2x}{\sqrt{1-(x^2+0.5)^2}} = \frac{-6x}{\sqrt{0.75-x^2-x^4}}$$

Example 5:

Find the derivative of $y = (x^2 + 1) \sin^{-1} 4x$.

Solution

$$y' = (2x) \sin^{-1} 4x + (x^2 + 1) \frac{4}{\sqrt{1-(4x)^2}} = \frac{4(x^2 + 1)}{\sqrt{1-16x^2}} + 2x \sin^{-1} 4x$$

**Application activity 5.12**

Find the derivative of the following functions

1. $f(x) = \arcsin(1-x)$
2. $f(x) = \sin^{-1} \sqrt{2x}$

5.13. Derivative of inverse cosine**Activity 5.13**

We know that $f(x) = \cos^{-1} x$ for $x \in [-1, 1]$ and $x = \cos y$ for $y \in [0, \pi]$ where $y = f(x)$. Use rule for derivative of composite functions to find the derivative of $\cos^{-1} x$, the inverse of cosine function.

**Content summary**

$$\forall x \in]-1, 1[, (\cos^{-1} x)' = \frac{-1}{\sqrt{1-x^2}}.$$

$$\text{If } u \text{ is another function } (\cos^{-1} u)' = \frac{-u'}{\sqrt{1-u^2}}.$$

**Application activity 5.13**

1. $f(x) = \cos^{-1} \frac{1}{x}$
2. $f(x) = \cos^{-1} x^2$

5.14. Differentiation of inverse tangent and inverse cotangent



Activity 5.14

1. We know that $f(x) = \tan^{-1} x$ for $x \in \mathbb{R}$ and $x = \tan y$ for $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ where $y = f(x)$.

Use rule of derivative for composite functions to find the derivative of

2. $\tan^{-1} x$, the inverse of tangent function
3. We also know that $f(x) = \cot^{-1} x$ for $x \in \mathbb{R}$ and $x = \cot y$ for $y \in]0, \pi[$ where $y = f(x)$. Use rule of derivative for composite functions to find the derivative of $\cot x$, the inverse of cotangent function.



Content summary

$$(\tan^{-1} x)' = \frac{1}{1+x^2} \text{ and } (\cot^{-1} x)' = \frac{-1}{1+x^2}$$

$$\text{If } u \text{ is another function, } (\tan^{-1} u)' = \frac{u'}{1+u^2} \text{ and } (\cot^{-1} u)' = \frac{-u'}{1+u^2}$$

Example 1: Find the derivative of $f(x) = \cot^{-1} 3x$

Solution

$$f'(x) = \frac{-(3x)'}{1+(3x)^2} = \frac{-3}{1+9x^2}$$

Example 2: Find the derivative of $f(x) = \sin^{-1}(3x^2 - 1)$

Solution

$$f'(x) = \frac{(3x^2 - 1)'}{\sqrt{1 - (3x^2 - 1)^2}} = \frac{6x}{\sqrt{1 - (3x^2 - 1)^2}}.$$



Application activity 5.14

Find the derivative of the following functions

1. $f(x) = \cot^{-1} \sqrt{x}$
2. $f(x) = \cos^{-1} \frac{1}{x} - \cot^{-1} x$
3. $f(x) = \cot^{-1} \sqrt{x-1}$

5.15. Derivative of inverse secant and inverse cosecant



Activity 5.15

1. We know that $f(x) = \sec^{-1} x$ for $x \leq -1$ or $x \geq 1$ and $x = \sec y$ for $y \in [0, \pi], y \neq \frac{\pi}{2}$ where $y = f(x)$. Use rule for derivative of composite functions to find the derivative of $\sec^{-1} x$, the inverse of secant function.
2. We know that $f(x) = \csc^{-1} x$ for $x \leq -1$ or $x \geq 1$ and $x = \csc y$ for $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], y \neq 0$ where $y = f(x)$. Use rule for derivative of composite functions to find the derivative of $\csc^{-1} x$, the inverse of cosecant function.



Content summary

$$(\sec^{-1} x)' = \frac{1}{x\sqrt{x^2-1}} \text{ and } (\csc^{-1} x)' = \frac{-1}{x\sqrt{x^2-1}}.$$

$$\text{If } u \text{ is another function, } (\sec^{-1} u)' = \frac{u'}{u\sqrt{u^2-1}} \text{ and } (\csc^{-1} u)' = \frac{-u'}{u\sqrt{u^2-1}}.$$

Example 1:

Find the derivative of $f(x) = \sec^{-1} 2x$

Solution

$$f'(x) = \frac{2}{2x\sqrt{4x^2-1}} = \frac{1}{x\sqrt{4x^2-1}}$$

Example 2:

Find the derivative of $f(x) = \csc^{-1} \sqrt{x}$

Solution

$$f'(x) = \frac{-\frac{1}{2\sqrt{x}}}{\sqrt{x}\sqrt{(\sqrt{x})^2 - 1}} = \frac{-1}{2x\sqrt{x^2 - 1}}$$



Application activity 5.15

Find the derivative of the following functions

1. $f(x) = \sec^{-1}(2x+1)$
2. $f(x) = \tan^{-1} \sqrt{x^2 - 1} + \csc^{-1} x$
3. $f(x) = \sec^{-1} 5x$
4. $f(x) = \csc^{-1}(x^2 + 1), \quad x > 0$

5.16. Applications of trigonometric functions: refraction of light in a prism



Activity 5.16

A ray of light enters a diamond from the air ($n_{\text{air}} = 1.00$). The refractive index of diamond is very high, $n_{\text{diamond}} = 2.42$.

- (a) Calculate the Critical Angle θ_c for light attempting to leave the diamond and enter the air.
- (b) If a light ray inside the diamond hits the surface at an angle of 30° , will it escape into the air or stay trapped inside the diamond?



Content summary

In physics, specifically in geometrical optics, inverse trigonometric functions are essential for calculating how light behaves as it moves between different materials (refraction) or reflects within a substance (total internal reflection).

When light passes from one medium (like air) to another (like water or glass), it changes speed and bends. This relationship is defined by **Snell's Law**:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

- n_1 and n_2 : The refractive indices of the two materials.
- θ_1 : The angle of incidence (the incoming ray).
- θ_2 : The angle of refraction (the bent ray).

To find the exact angle at which the light bends (θ_2), we rearrange the formula to solve for the

$$\text{angle: } \sin(\theta_2) = \frac{n_1 \sin(\theta_1)}{n_2} \Rightarrow \theta = \arcsin\left(\frac{n_1 \sin(\theta_1)}{n_2}\right).$$

When light travels from a denser medium (higher n) to a less dense medium (lower n), it bends away from the normal. If the angle of incidence is increased enough, the refracted ray will eventually graze the surface at exactly 90° . This specific angle of incidence is called the **critical angle**.

$$\text{Since } \theta_2 = 90^\circ, \text{ the formula becomes } n_1 \sin(\theta_c) = n_2 \Rightarrow \theta_c = \arcsin\left(\frac{n_2}{n_1}\right).$$

Example:

A diver shines a powerful flashlight from the bottom of a pool ($n_{\text{water}} = 1.33$) toward the surface. The light hits the surface at an angle of incidence of 40° relative to normal.

- (a) Find the Angle of refraction in the air ($n_{\text{air}} = 1.00$).
- (b) Calculate the critical angle (θ_c) for the water-air boundary to determine if the light can escape the water.

Solution:

$$(a) \quad n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \Leftrightarrow 1.33 \cdot \sin(40^\circ) = 1.00 \cdot \sin(\theta_2) \Rightarrow \sin(\theta_2) = 0.8549 \Rightarrow \theta_2 = \sin^{-1} 0.8549 = 58.75^\circ.$$

$$(b) \quad \sin(\theta_c) = \frac{n_2}{n_1} \Leftrightarrow \sin(\theta_c) = \frac{1.00}{1.33} \approx 0.7519 \Rightarrow \theta_c = \sin^{-1}(0.7519) = \theta_c \approx 48.75^\circ.$$

Since our incident angle 40° is less than the critical angle 48.75° , the light successfully refracts into the air at 58.75° .



Application activity 5.16

A fiber optic cable consists of a glass core ($n_{\text{core}} = 1.50$) surrounded by a plastic cladding ($n_{\text{cladding}} = 1.40$). For the cable to work, light must stay inside the core.

- Find the critical angle (θ_c) at the boundary between the core and the cladding.
- A light signal hits the boundary at an incident angle of 75° . Use the inverse sine to determine the maximum angle at which light can hit the boundary and still stay trapped (the critical angle).

5.17. Applications of trigonometric functions: simple harmonic motion problems



Activity 5.17

Discuss how differentiation of trigonometric functions is used to find the velocity, acceleration and jerk of a moving object knowing the function representing its position.



Content summary

The motion of an object or weight bobbing freely up and down with no resistance on the end of a spring is an example of simple harmonic motion. The motion is periodic and repeats indefinitely, so we represent it using trigonometric functions. If we have the function representing the position, say $s(t)$, then

- The velocity of the object is $v = \frac{ds}{dt}$
- The acceleration of the object is $a = \frac{d^2s}{dt^2}$
- The jerk of the object is $a = \frac{d^3s}{dt^3}$

Example:

A weight hanging from a spring is stretched down 5 units beyond its rest position and released at time $t = 0$ to bob up down. Its position at any later time t is $s = 5 \cos t$. What is its velocity, acceleration and jerk at time t ?

Solution

Position: $s = 5 \cos t$

Velocity (derivative of function representing the position):

$$v = \frac{ds}{dt} = \frac{d}{dt}(5 \cos t) = -5 \sin t$$

Acceleration (derivative of function representing the velocity):

$$a = \frac{dv}{dt} = \frac{d}{dt}(-5 \sin t) = -5 \cos t$$

Jerk (derivative of function representing the acceleration):

$$j = \frac{da}{dt} = \frac{d}{dt}(-5 \cos t) = 5 \sin t$$



Application activity 5.17

A mass is attached to a spring and undergoes simple harmonic motion with an amplitude of 10 cm and a period of 4 seconds. Assuming the mass starts at its maximum positive displacement ($x = 10 \text{ cm}$ at $t = 0$), find the time t it takes for the mass to reach a displacement of 5 cm for the first time.

5.18. Applications of trigonometric functions: Optimization



Activity 5.18

Suppose you have two equal wooden beams, each of length $L = 10$ meters, joined at a hinge. You want to spread them apart at an angle θ and connect their ends with a third beam to form a triangle. What angle θ will result in the maximum area?



Content summary

Optimization involves finding the maximum or minimum value of a function. Trigonometric functions are essential for optimization when the variables involve angles, rotations, or periodic motion. Here are the primary ways trigonometry is used to optimize real-world systems:

1. Optimizing projectile range

To achieve the maximum horizontal distance (range) for a projectile (like a football or a satellite launcher), we must find the optimal launch angle θ .

The range R is given by the formula: $R = \frac{v^2 \sin(2\theta)}{g}$.

The goal is to maximize R . We take the derivative with respect to θ and set it to zero:

$\frac{dR}{d\theta} = \frac{v^2}{g} \cdot 2 \cos(2\theta) = 0$. The Optimization: $\cos(2\theta) = 0$ occurs when $2\theta = 90^\circ$, meaning $\theta = 45^\circ$ is the optimal angle for maximum range.

2. Illumination and Light Intensity

Engineers use trigonometry to determine the best height to hang a streetlamp to maximize the illumination on a specific point on the ground.

The intensity of light I at a point is proportional to the cosine of the angle of incidence ϕ and inversely proportional to the square of the distance r :

$$I = \frac{k \cos \phi}{r^2}.$$

By expressing r and ϕ in terms of the height h of the lamp, we can use the inverse tangent to find the specific height that provides the brightest light for a given horizontal distance.



Application activity 5.18

A projectile is fired with speed $u = 20 \text{ m/s}$. The angle of projection is modeled as a function of a variable x by: $\theta = \tan^{-1}(x)$. The horizontal range is: $R = \frac{u^2 \sin(2\theta)}{g}$. Find the value of x that maximizes the range, and hence determine the optimal angle.



End of unit assessment

Knowledge and understanding

1. Explain why we must restrict the domain of $\sin x$ to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ to define its inverse, $\arcsin x$.
2. State the domain and range of $g(x) = \arcsin x$.

Skills

3. Suppose that the amount of money in a bank account is given by $P(t) = 500 + 100 \cos t - 150 \sin t$ where t is in years. During the first 10 years in which the account is open when is the amount of money in the account increasing?
4. The equation $s = 2 - 2 \sin t$ gives the position of a body moving on a coordinate line (s in metres, t in seconds). Find the body's velocity, speed, acceleration, and jerk at time $t = \frac{\pi}{4}$ sec.
5. A weight is attached to a spring and reaches its equilibrium position ($x = 0$). It is then set in motion resulting in a displacement of $x = 10 \cos t$. Find the spring's velocity when $t = 0$, $t = \frac{\pi}{3}$ and $t = \frac{3\pi}{4}$.
6. Assume that a particle's position on the x -axis is given by $x = 3 \cos t + 4 \sin t$, where x is measured in metres and t is measured in seconds. Find the particle's velocity when $t = 0$, $t = \frac{\pi}{2}$ and $t = \pi$.
7. Suppose that a piston is moving straight up and down and that its position at time t sec is $s = A \cos(2\pi bt)$ with A and b positive. The value of A is the amplitude of the motion, and b is the frequency (number of times the piston moves up and down each second). What effect does doubling the frequency have on the piston's velocity, acceleration and jerk? (Once you find out, you will know why some machinery breaks when you run it too fast).
8. Evaluate:

(a) $\lim_{x \rightarrow 2} \sin^{-1} \frac{\sqrt{x^2 - 1}}{x}$

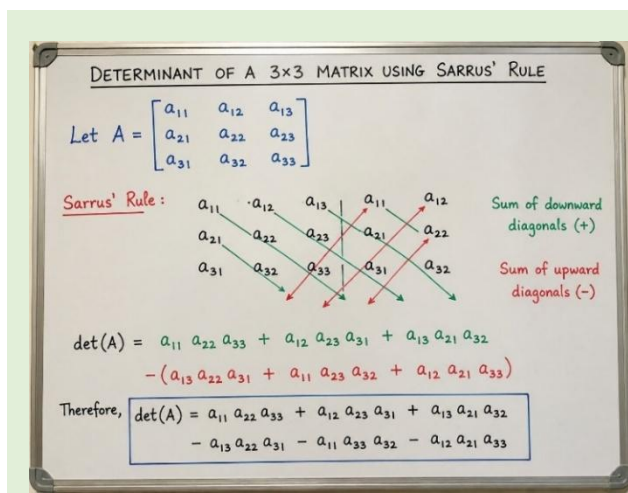
(b) $\lim_{x \rightarrow 0} \tan^{-1}(x - 1)$

(c) $\lim_{x \rightarrow \frac{\pi}{3}} \tan^{-1}(\tan 2x)$

$$(d) \lim_{x \rightarrow -1} \sec^{-1} \frac{x-1}{\sqrt{x^2+1}}$$

Attitude and values

9) In the study of inverse trigonometric functions, why is the habit of maintaining "exact form" (e.g., using $\frac{\pi}{6}$ instead of 0.523) considered a superior professional value compared to early decimal approximation?



UNIT 6: MATRIX AND DETERMINANTS OF ORDER 3

Key unit competence: Apply properties of matrices, determinants of order 3 and operations on linear transformations in solving problems in $\mathbb{R}^3$.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain square matrices of order 3 and their types (diagonal, scalar, identity, zero, symmetric, etc.).
- Recall the rules and properties of matrix addition, subtraction, multiplication, and transpose.
- Explain the concept of matrix inverse and its properties.
- Demonstrate understanding of determinants of order 3 and their properties.
- Recognize the application of matrices in solving systems of linear equations from real-life problems.

Skills:

Perform addition, subtraction, and multiplication of 3×3 matrices correctly.

- Compute the transpose of a matrix and verify its properties.
- Calculate the determinant of a 3×3 matrix.
- Determine the inverse of a matrix and apply its properties.
- Solve systems of three linear equations using matrices and interpret solutions in real-life contexts.

- Apply matrices to model and solve practical problems.

Attitudes and values:

- Demonstrate accuracy, precision, and logical reasoning in computations.
- Appreciate the usefulness of matrices in modeling and solving real-life problems.
- Develop persistence and analytical thinking in solving matrix-related problems.

Introduction

A matrix is a rectangular arrangement of numbers, expressions, symbols which are arranged in rows and columns.

Matrices play a virtual role in the projection of a three dimensional image into a two dimensional image. Matrices and their inverse are used by programmers for coding or encrypting a message. Matrices are applied in the study of electrical circuits, quantum mechanics and optics. A message is made as a sequence of numbers in a binary format for communication and it follows code theory for solving. Hence with the help of matrices, those equations are solved. Matrices are used for taking seismic surveys.



Introductory activity

Farmer Kalisa made three shopping trips to the Ruhango Market to stock his farm, purchasing different quantities of cocks (x), rabbits (y), and goats (z) each time: on the first day, he paid 85,000 Frw for 5 cocks, 4 rabbits, and 1 goat; on the second day, he paid 105,000 Frw for 3 cocks, 6 rabbits, and 2 goats; and on the final day, he paid 77,000 Frw for 2 cocks, 2 rabbits, and 3 goats. Using this data, construct a system of linear equations and apply algebraic methods to determine the individual unit price for one cock, one rabbit, and one goat.

a) Arrange what Kalisa bought according to their types in a simple table as follows

Cocks (x)	Rabbits (y)	Goats (z)	Total Price (Frw)

b) Discuss and explain in your own words how can you determine the cost of 1 Cocks, 1 Rabbit and the cost of 1 goat.

6.1. Introduction on square matrices of order 3



Activity 6.1

Imagine you are managing an electronics business with three branches in your town: Branch A, Branch B, and Branch C. Each branch sells Smartphones, Tablets, and Laptops. At the end of the day, you recorded the following sales data:

- Branch A: Sold 5 Smartphones, 2 Tablets, and 4 Laptops.
- Branch B: Sold 3 Smartphones, 6 Tablets, and 1 Laptop.
- Branch C: Sold 8 Smartphones, 0 Tablets, and 2 Laptops.

Represent in array S the quantities sold, where each row represents a branch and each column represents a product type.



Content summary

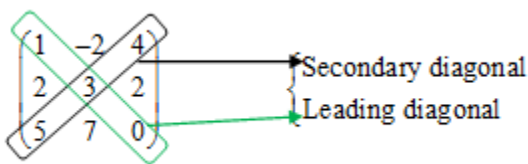
A square matrix is formed by the same number of rows and columns.

The elements of the form (a_{ij}) of $n \times n$, where the two subscripts i and j are equal, constitute the **principal diagonal** (or **leading diagonal** or **main diagonal** or **major diagonal** or **primary diagonal**). The **secondary diagonal** (or **minor diagonal** or **antidiagonal** or **counterdiagonal**) of $n \times n$ matrix is formed by the elements with $i + j = n + 1$.

Square matrix of order three has the form

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Example 1: Matrix of order three





Application activity 6.1

Consider the transformation $f(x, y, z) = (2x + 3y, x - y + 2z, 4x + y - z)$

Rewrite this transformation in the form

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Where a, b, c, d, e, f, g, h , and i are constants.

6.2.Types of matrices



Activity 6.2

Name the following matrices

$$\begin{array}{ll} \text{a) } \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} & \text{b) } \begin{pmatrix} a & b & c \\ 0 & b & d \\ 0 & 0 & e \end{pmatrix} \end{array}$$

$$\begin{array}{ll} \text{c) } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & \text{d) } \begin{pmatrix} a & 0 & 0 \\ c & b & 0 \\ d & 0 & c \end{pmatrix} \end{array}$$

$$\text{e) } \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$



Content summary

Upper Triangular Matrix

In an upper triangular matrix, the elements located below the leading diagonal are zeros.

Example 1:

$$\begin{pmatrix} 1 & -2 & 4 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{pmatrix}$$

Lower Triangular Matrix

In a lower triangular matrix, the elements above the leading diagonal are zeros.

Example 2:

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 5 & 7 & 9 \end{pmatrix}$$

Diagonal Matrix

In a diagonal matrix, all the elements above and below the leading diagonal are zeros.

Example 3:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

Scalar Matrix

A scalar matrix is a diagonal matrix in which the leading diagonal elements are equal.

Example 4:

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

Identity Matrix or Unity matrix

An identity matrix (noted by **I**) is a diagonal matrix in which the leading diagonal elements are equal to 1.

Example 5:

Identity matrix D of order three

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Toeplitz matrix

An $n \times n$ matrix A is Toeplitz if the element $A_{i,j}$ depends only on the difference $(i - j)$. Formally:

$$A_{i,j} = a_{i-j}.$$

This means that all elements along any diagonal are identical. A general 4×4 Toeplitz matrix looks like this:

$$A = \begin{bmatrix} a_0 & a_{-1} & a_{-2} \\ a_1 & a_0 & a_{-1} \\ a_2 & a_1 & a_0 \end{bmatrix} \text{ or } X = \begin{pmatrix} 3 & 2 & 3 \\ 5 & 3 & 2 \\ 1 & 5 & 3 \end{pmatrix}$$

Vandermonde matrix

A **Vandermonde matrix** is a type of matrix characterized by having terms of a geometric progression in each row.

$$V = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{bmatrix} \text{ Or } N = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 6 \\ 4 & 9 & 36 \end{pmatrix} \text{ Or } M = \begin{pmatrix} 1 & 4 & 16 \\ 1 & 3 & 9 \\ 1 & 9 & 81 \end{pmatrix}$$

Equality of matrices

Two matrices are equal if the elements of the two matrices that occupy the same position are equal.

$$\text{If } \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}, \text{ then } \begin{matrix} a_{11} = b_{11} & a_{12} = b_{12} & a_{13} = b_{13} \\ a_{21} = b_{21} & a_{22} = b_{22} & a_{23} = b_{23} \\ a_{31} = b_{31} & a_{32} = b_{32} & a_{33} = b_{33} \end{matrix}.$$



Application activity 6.2

1. Give the name to the matrices below:

$$(a) V = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{pmatrix}, (b) T = \begin{pmatrix} 8 & 5 & 2 \\ 7 & 8 & 5 \\ 4 & 7 & 8 \end{pmatrix}.$$

2. Given matrices A and B such that $A = \begin{pmatrix} 2x+1 & 2 \\ y+1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & z \\ 0 & t+1 \end{pmatrix}$

Find the value of x , y , z and t if matrix A equals to matrix B .

6.3. Addition on matrices and properties



Activity 6.3

Consider the matrices $A = \begin{pmatrix} 2 & -4 & 12 \\ 1 & 0 & -4 \\ 5 & 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 6 & 4 \\ 1 & 7 & 8 \\ 3 & 21 & 3 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 2 & 4 \\ 3 & -2 & 0 \end{pmatrix}$.

Find:

1. $2A + B$.
2. $A + (-A)$. Give any comment.
3. $A + B$ and $B + A$. From the results give your comment.
4. $(A + B) + C$ and $A + (B + C)$. Give your comment.
5. Interchange/switch the rows and column of matrix A , B and C .



Content summary

Given two matrices of the same dimension, $A = (a_{ij})$ and $B = (b_{ij})$, the matrix sum is defined as: $A + B = (a_{ij}) + (b_{ij})$. That is, the resultant matrix's elements are obtained by adding the elements of the two matrices that occupy the same position.

If $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$, then

$$A + B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{pmatrix}$$

Example 1:

Consider the matrices $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, find $A + B$

Solution

$$A+B = \begin{pmatrix} 2+1 & 0+0 & 1+1 \\ 3+1 & 0+2 & 0+1 \\ 5+1 & 1+1 & 1+0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 2 \\ 4 & 2 & 1 \\ 6 & 2 & 1 \end{pmatrix}.$$

Properties

1) Closure

The sum of two matrices of order three is another matrix of order three

2) Associative

$$3) \quad (A+B)+C = A+(B+C).$$

4) Additive identity

$A+0=A$, where 0 is the zero-matrix of the same dimension.

5) Additive inverse

$$A+(-A)=0$$

The opposite matrix of A is $-A$.

6) Commutative

$$A+B=B+A.$$

Scalar multiplication

Given a matrix, $A = (a_{ij})$, and a real number, $k \in \mathbb{R}$, the product of a real number by a matrix is a matrix of the same dimension as A , and each element is multiplied by k .

$$kA = (ka_{ij})$$

$$\text{If } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \text{ then } kA = \begin{pmatrix} ka_{11} & ka_{12} & ka_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{pmatrix}$$

Example 7.1

Consider the matrix $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, find $2A$

Solution

$$2A = 2 \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 2 \\ 6 & 0 & 0 \\ 10 & 2 & 2 \end{pmatrix}.$$

Properties:

- 1) $\alpha(\beta A) = (\alpha\beta)A$, $A \in M_{m \times n}$, $\alpha, \beta \in \mathbb{R}$.
- 2) $\alpha(A+B) = \alpha A + \alpha B$, $A, B \in M_{m \times n}$, $\alpha \in \mathbb{R}$.
- 3) $(\alpha + \beta)A = \alpha A + \beta A$, $A \in M_{m \times n}$, $\alpha, \beta \in \mathbb{R}$.
- 4) $1A = A$, $A \in M_{m \times n}$.



Application activity 6.3

The crops of maize, rice, and beans in three Rwandan districts of Eastern province within two agricultural seasons are given in the following tables:

Season 1:

Districts Crops (in ton)	Ngoma	Kayonza	Gatsibo
Maize	150	350	230
Rice	120	90	270
Beans	200	230	240

Season 2:

Districts Crops (in ton)	Ngoma	Kayonza	Gatsibo
Maize	253	450	245
Rice	160	99	263
Beans	240	330	246

Find the total crops for three districts within two agricultural seasons.

Solution:

Convert the crops in matrix notation

Let A be the matrix of crops of season 1, $A = \begin{pmatrix} 150 & 350 & 230 \\ 120 & 90 & 270 \\ 200 & 230 & 260 \end{pmatrix}$, and

Let B be the matrix of crops of season 2, $B = \begin{pmatrix} 253 & 450 & 245 \\ 160 & 99 & 263 \\ 240 & 330 & 246 \end{pmatrix}$.

The total crops for three districts within two agricultural seasons are given by the sum of two matrices, that is $A + B$

$$A + B = \begin{pmatrix} 150 & 350 & 230 \\ 120 & 90 & 270 \\ 200 & 230 & 260 \end{pmatrix} + \begin{pmatrix} 253 & 450 & 245 \\ 160 & 99 & 263 \\ 240 & 330 & 246 \end{pmatrix} = \begin{pmatrix} 403 & 800 & 475 \\ 280 & 189 & 533 \\ 440 & 560 & 506 \end{pmatrix}$$

6.4.Subtraction on matrices and properties**Activity 6.4**

Consider the matrices $A = \begin{pmatrix} 2 & -4 & 12 \\ 1 & 0 & -4 \\ 5 & 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 6 & 4 \\ 1 & 7 & 8 \\ 3 & 21 & 3 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 2 & 4 \\ 3 & -2 & 0 \end{pmatrix}$.

Find:

- $2A - B$.
- $A - B$ and $B - A$. From the results give your comment.
- $(A - B) - C$ and $A - (B - C)$. Give your comment.



Content summary

If $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$, then

$$A - B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{21} - b_{21} & a_{22} - b_{22} & a_{23} - b_{23} \\ a_{31} - b_{31} & a_{32} - b_{32} & a_{33} - b_{33} \end{pmatrix}$$

Example 1:

Consider the matrices $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, find $A - B$.

Solution

$$A - B = \begin{pmatrix} 2-1 & 0-0 & 1-1 \\ 3-1 & 0-2 & 0-1 \\ 5-1 & 1-1 & 1-0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -2 & -1 \\ 4 & 0 & 1 \end{pmatrix}.$$



Application activity 6.4

If $A = \begin{pmatrix} 1 & -2 & 6 \\ 2 & 2 & -5 \\ 0 & 4 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 8 & 9 \\ 14 & 4 & 6 \\ 0 & 6 & 7 \end{pmatrix}$ and $C = \begin{pmatrix} 13 & -2 & 0 \\ 10 & 0 & 3 \\ 9 & 1 & -5 \end{pmatrix}$.

Evaluate:

(a) $A - B$.

(b) $C - B$.

6.5. Transpose of matrices and properties



Activity 6.5

Consider the matrices $A = \begin{pmatrix} 1 & 3 & 1 \\ 1 & -2 & 2 \\ 3 & -2 & 0 \end{pmatrix}$, and $B = \begin{pmatrix} 12 & 3 & -1 \\ 3 & -2 & 0 \\ -4 & -1 & 0 \end{pmatrix}$.

1. Interchange/switch the rows and columns of matrix A and B
2. Add two matrices obtained in 1
3. Add A and B
4. Interchange/switch the rows and columns of matrix obtained in 3
5. What can you say about result in 2 and 4?
6. Interchange/switch the rows and columns of matrix A twice. What can you conclude?



Content summary

Given matrix A , the transpose of matrix A , noted A^t , is another matrix where the elements in the columns and rows have switched. In other words, the rows become the columns, and the columns become the rows.

$$\text{If } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \text{ then } A^t = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}.$$

Example 1:

$$A = \begin{pmatrix} 1 & 3 & 6 \\ 0 & 2 & 0 \\ 3 & 5 & 8 \end{pmatrix}, \quad A^t = \begin{pmatrix} 1 & 0 & 3 \\ 3 & 2 & 5 \\ 6 & 0 & 8 \end{pmatrix}$$

Properties of transpose of matrices

Let A, B be matrices of order three

1. $(A^t)^t = A$
2. $(A+B)^t = A^t + B^t$

$$3. (\alpha \times A)^t = \alpha \times A^t, \alpha \in \mathbb{R}$$

A symmetric matrix

An $n \times n$ matrix A is to be symmetric if $A^t = A$

$$\text{Example: } A = \begin{pmatrix} 3 & 5 \\ 5 & 7 \end{pmatrix} \text{ or } B = \begin{pmatrix} 2 & 3 & 5 \\ 3 & 4 & 1 \\ 5 & 1 & 7 \end{pmatrix}$$

A skew-symmetric matrix

An $n \times n$ matrix A is said to be skew - symmetric if $A^t = -A$

$$\text{Example: } B = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}, A = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 4 \\ 1 & -4 & 0 \end{pmatrix}$$

Theorem: If A is $n \times n$ matrix, then we can write any square matrix as a sum of symmetric and skew symmetric matrix i.e $A = S + T$ where S is a symmetric matrix and T is a skew-symmetric matrix

$$\text{Proof: } A = S + T \dots\dots\dots(1)$$

$$(A = S + T)^t \Rightarrow A^t = S^t + T^t$$

$$\Rightarrow A^t = S - T \dots\dots\dots(2)$$

Putting equation (1) and (2) together and solve yields

$$\begin{cases} A = S + T \\ A^t = S - T \end{cases} \Rightarrow S = \frac{A + A^t}{2} \text{ and } T = \frac{A - A^t}{2}$$

$$\text{Decompose } A = \begin{pmatrix} 1 & 3 & 8 \\ 4 & 7 & 8 \\ 0 & 0 & 1 \end{pmatrix} \text{ as a sum of symmetric matrix and skew symmetric matrix.}$$

Solution: $A' = \begin{pmatrix} 1 & 4 & 0 \\ 3 & 7 & 0 \\ 8 & 8 & 1 \end{pmatrix}$

$$S = \frac{A + A'}{2} = \frac{1}{2} \left\{ \begin{pmatrix} 1 & 3 & 8 \\ 4 & 7 & 8 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 0 \\ 3 & 7 & 0 \\ 8 & 8 & 1 \end{pmatrix} \right\} = \begin{pmatrix} 1 & \frac{7}{2} & 4 \\ \frac{7}{2} & 7 & 4 \\ 4 & 4 & 1 \end{pmatrix} :$$

$$S = \frac{A - A'}{2} = \frac{1}{2} \left\{ \begin{pmatrix} 1 & 3 & 8 \\ 4 & 7 & 8 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 4 & 0 \\ 3 & 7 & 0 \\ 8 & 8 & 1 \end{pmatrix} \right\} = \begin{pmatrix} 0 & -\frac{1}{2} & 4 \\ \frac{1}{2} & 0 & 4 \\ -4 & -4 & 0 \end{pmatrix}$$

$$A = S + T = \begin{pmatrix} 1 & \frac{7}{2} & 4 \\ \frac{7}{2} & 7 & 4 \\ 4 & 4 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -\frac{1}{2} & 4 \\ \frac{1}{2} & 0 & 4 \\ -4 & -4 & 0 \end{pmatrix}$$



Application activity 6.5

1. Consider matrices $A = \begin{pmatrix} 0 & 4 & 2 \\ 1 & 3 & 6 \\ 3 & -2 & 8 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 1 \\ -4 & 0 & 3 \\ 6 & 2 & 5 \end{pmatrix}$. Evaluate

a) $(A + B)^t$

b) $3A' + B$

c) $(-3B + 4A)^t$

2. Find the value of x in $M = \begin{pmatrix} 1 & 2 & x^2 \\ 4 & 1 & 0 \\ 1 & x+3 & 8 \end{pmatrix}$ if $M^t = \begin{pmatrix} 1 & 4 & 1 \\ 2 & 1 & 1 \\ 4 & 0 & 8 \end{pmatrix}$.

3. Decompose $A = \begin{pmatrix} 2 & 3 & 4 \\ 6 & 5 & 0 \\ 1 & 7 & 8 \end{pmatrix}$ as a sum of symmetric matrix and skew symmetric matrix.

6.6. Multiplication of matrices



Activity 6.6

Consider the matrices $B = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 3 \\ -1 & 6 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 3 \\ -1 & 6 & 4 \end{pmatrix}$ find $A \times B$.



Content summary

Two matrices A and B can be multiplied together if and only if the number of columns of A is equal to the number of rows of B .

$$M_{m \times n} \times M_{n \times p} = M_{m \times p}.$$

The element, c_{ij} , of the product matrix is obtained by multiplying every element in row i of matrix A by each element of column j of matrix B and then adding them together. This multiplication is called **ROCO** (row, column).

If $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$, then

$$\begin{aligned}
 A \times B &= \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \\
 &= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{pmatrix}
 \end{aligned}$$

Example 1:

Consider matrices $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, find $A \times B$.

Solution

$$\begin{aligned}
 A \times B &= \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 \times 1 + 0 \times 1 + 1 \times 1 & 2 \times 0 + 0 \times 2 + 1 \times 1 & 2 \times 1 + 0 \times 1 + 1 \times 0 \\ 3 \times 1 + 0 \times 1 + 0 \times 1 & 3 \times 0 + 0 \times 2 + 0 \times 1 & 3 \times 1 + 0 \times 1 + 0 \times 0 \\ 5 \times 1 + 1 \times 1 + 1 \times 1 & 5 \times 0 + 1 \times 2 + 1 \times 1 & 5 \times 1 + 1 \times 1 + 1 \times 0 \end{pmatrix} \\
 &= \begin{pmatrix} 3 & 1 & 2 \\ 3 & 0 & 3 \\ 7 & 3 & 6 \end{pmatrix}
 \end{aligned}$$



Application activity 6.6

If $A = \begin{pmatrix} 1 & -2 & 6 \\ 2 & 2 & -5 \\ 0 & 4 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 8 & 9 \\ 14 & 4 & 6 \\ 0 & 6 & 7 \end{pmatrix}$, and $C = \begin{pmatrix} 13 & -2 & 0 \\ 10 & 0 & 3 \\ 9 & 1 & -5 \end{pmatrix}$.

Evaluate

- 1) $A \times B$
- 2) $A \times C$
- 3) $B \times C$

6.7. Properties of multiplication of matrices



Activity 6.7

Consider the matrices $A = \begin{pmatrix} 3 & 0 & 1 \\ 0 & -1 & 1 \\ -2 & 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & 0 & 1 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ -1 & 1 & 0 \end{pmatrix}$.

Find:

1. $A \times B$ and $B \times A$
2. $(A \times B)^t$ and $B^t \times A^t$
3. $(A \times B) \times C$ and $A \times (B \times C)$
4. $A \times (B + C)$ and $A \times B + A \times C$

Comment on your results



Content summary

Properties of Matrices Multiplication

Let A, B, C be matrices of order three

- 1) **Associative:** $A \times (B \times C) = (A \times B) \times C$
- 2) **Multiplicative Identity:** $A \times I = A$, where I is the identity matrix with the same order as matrix A .
- 3) **Not Commutative:** $A \times B \neq B \times A$
- 4) **Distributive:** $A \times (B + C) = A \times B + A \times C$
- 5) $(A \times B)^t = B^t \times A^t$

Example 1:

Given the matrices: $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

Find

a) The product $A \times B$

b) The product $B \times A$

Solution

a)

$$A \times B = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 \times 1 + 0 \times 1 + 1 \times 1 & 2 \times 0 + 0 \times 2 + 1 \times 1 & 2 \times 1 + 0 \times 1 + 1 \times 0 \\ 3 \times 1 + 0 \times 1 + 0 \times 1 & 3 \times 0 + 0 \times 2 + 0 \times 1 & 3 \times 1 + 0 \times 1 + 0 \times 0 \\ 5 \times 1 + 1 \times 1 + 1 \times 1 & 5 \times 0 + 1 \times 2 + 1 \times 1 & 5 \times 1 + 1 \times 1 + 1 \times 0 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 \\ 3 & 0 & 3 \\ 7 & 3 & 6 \end{pmatrix}$$

b)

$$B \times A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 0 \times 3 + 1 \times 5 & 1 \times 0 + 0 \times 0 + 1 \times 1 & 1 \times 1 + 0 \times 0 + 1 \times 1 \\ 1 \times 2 + 2 \times 3 + 1 \times 5 & 1 \times 0 + 2 \times 0 + 1 \times 1 & 1 \times 1 + 2 \times 0 + 1 \times 1 \\ 1 \times 2 + 1 \times 3 + 0 \times 5 & 1 \times 0 + 1 \times 0 + 0 \times 1 & 1 \times 1 + 1 \times 0 + 0 \times 1 \end{pmatrix} = \begin{pmatrix} 7 & 1 & 2 \\ 13 & 1 & 2 \\ 5 & 0 & 1 \end{pmatrix}$$

Example 2:

Given matrices $A = \begin{pmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}$. Find the product AB . What is your

observation?

Solution

$$AB = \begin{pmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1-2+1 & 2-4+2 & 3-6+3 \\ -3+4-1 & -6+8-2 & -9+12-3 \\ -2+2+0 & -4+4+0 & -6+6+0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Observation: If $AB = 0$, it does not necessarily follow that $A = 0$ or $B = 0$

Example 3:

Given matrices $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 0 \\ -1 & -4 & 1 \end{pmatrix}$. Find the product AB and BA . What

is your observation?

Solution

$$AB = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 0 \\ -1 & -4 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 0 \\ 0 & -4 & 2 \end{pmatrix}$$
$$BA = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 0 \\ -1 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 0 \\ 0 & -4 & 2 \end{pmatrix} \Rightarrow AB = BA$$

Observation: The given matrices commute in multiplication.

Notice

- If $AB = 0$, it does not necessarily follow that $A = 0$ or $B = 0$.
- **Commuting matrices in multiplication**

In general, the multiplication of matrices is not commutative, i.e., $AB \neq BA$, but we can have the case where two matrices A and B satisfy $AB = BA$. In this case A and B are said to be **commuting**.

Trace of matrix

The sum of the entries on the leading diagonal of a square matrix, A , is known as the **trace** of that matrix, noted $tr(A)$.

Example 4:

1. Trace of $\begin{pmatrix} 1 & -2 & 4 \\ 2 & 3 & 2 \\ 5 & 7 & 2 \end{pmatrix} = 1 + 3 + 2 = 6$

2. Trace of $\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = 1 + 1 = 2$

Properties of trace of matrix

1. $tr(A+B) = tr(A) + tr(B)$
2. $tr(\alpha A) = \alpha tr(A)$
3. $tr(A) = tr(A)^t$
4. $tr(AB) = tr(BA)$
5. $tr(ABC) = tr(BCA) = tr(CAB)$, cyclic property.
6. $tr(ABC) \neq tr(ACB)$, arbitrary permutations are not allowed.



Application activity 6.7

1. Consider the matrices $A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 2 & 0 \\ -1 & 1 & 0 \end{pmatrix}$.

Find:

- a) $A \times B$ and $B \times A$
- b) $A \times (B \times C)$ and $(A \times B) \times C$
- c) $A \times (B + C)$ and $A \times B + A \times C$
- d) $tr(AB)$

2. Consider the matrix $A = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$, find $A \times A^t$

6.8. Matrix of linear transformation in 3 dimensions



Activity 6.8

Let $f(x, y, z) = (x + z, y - z, 2x)$ and the standard basis of $\mathbb{R}^3$ is

$\{\vec{e}_1 = (1, 0, 0), \vec{e}_2 = (0, 1, 0), \vec{e}_3 = (0, 0, 1)\}$. Find:

1. $f(\vec{e}_1)$

2. $f(\vec{e}_2)$

3. $f(\vec{e}_3)$

Form matrix whose j^{th} column is $f(\vec{e}_j)$, $j = 1, 2, 3$.



Content summary

Every linear transformation $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ can be identified with a matrix of order three, $[f]_e = (a_{ij})$, whose j^{th} column is $f(\vec{e}_j)$ where $f(\vec{e}_j), j = 1, 2, 3$ is the standard basis of $\mathbb{R}^3$. The matrix $[f]_e$ is called matrix representation of f relative to the standard basis $\vec{e}_j$.

Example 1:

Find the matrix of f relative to the standard basis if

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (4x - 2z, 2x + y, z + y)$$

Solution

The standard basis of $\mathbb{R}^3$ is $\{\vec{e}_1 = (1, 0, 0), \vec{e}_2 = (0, 1, 0), \vec{e}_3 = (0, 0, 1)\}$.

$$f(\vec{e}_1) = (4, 2, 0)$$

$$f(\vec{e}_2) = (0, 1, 1)$$

$$f(\vec{e}_3) = (-2, 0, 1)$$

Then the matrix of f relative to the standard basis is $[f]_e = \begin{bmatrix} 4 & 0 & -2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

What is the procedure if the given basis is not standard? The following is the general method.

To find the matrix of a linear mapping f relative to any basis $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$, we follow the following steps:

1. Find $f(\vec{e}_j), j = 1, 2, 3$.
2. Equate $f(\vec{e}_j)$ to $\vec{e}_i a_{ij}$ to find the values of a_{ij} .
3. The matrix of f is $[f]_e = (a_{ij})$ where $\begin{cases} i = \text{number of row} \\ j = \text{number of column} \end{cases}$

Example 2:

Consider the following linear mapping defined on $\mathbb{R}^3$ by $f(x, y, z) = (4x - 2z, 2x + y, z + y)$.

Then Calculate its matrix relative to the basis $\{\vec{e}_1 = (1, 1, 1), \vec{e}_2 = (-1, 0, 1), \vec{e}_3 = (0, 1, 1)\}$.

Solution

$$f(\vec{e}_1) = (4 - 2, 2 + 1, 1 + 1) = (2, 3, 2)$$

$$f(\vec{e}_2) = (-4 - 2, -2 + 0, 1 + 0) = (-6, -2, 1)$$

$$f(\vec{e}_3) = (0 - 2, 0 + 1, 1 + 1) = (-2, 1, 2)$$

$$f(\vec{e}_j) = \vec{e}_i a_{ij}$$

$$f(\vec{e}_1) = \vec{e}_i a_{i1}$$

$$\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} a_{11} + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} a_{21} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} a_{31}$$

$$\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{11} \\ a_{11} \end{pmatrix} + \begin{pmatrix} -a_{21} \\ 0 \\ a_{21} \end{pmatrix} + \begin{pmatrix} 0 \\ a_{31} \\ a_{31} \end{pmatrix}$$

$$\begin{cases} a_{11} - a_{21} = 2 \\ a_{11} + a_{31} = 3 \\ a_{11} + a_{21} + a_{31} = 2 \end{cases} \Rightarrow \begin{cases} a_{11} = 1 \\ a_{21} = -1 \\ a_{31} = 2 \end{cases}$$

$$f(\vec{e}_2) = \vec{e}_2 a_{i2}$$

$$\begin{pmatrix} -6 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} a_{12} + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} a_{22} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} a_{32}$$

$$\begin{pmatrix} -6 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{12} \\ a_{12} \end{pmatrix} + \begin{pmatrix} -a_{22} \\ 0 \\ a_{22} \end{pmatrix} + \begin{pmatrix} 0 \\ a_{32} \\ a_{32} \end{pmatrix}$$

$$\begin{cases} a_{12} - a_{22} = -6 \\ a_{12} + a_{32} = -2 \\ a_{12} + a_{22} + a_{32} = 1 \end{cases} \Rightarrow \begin{cases} a_{12} = -3 \\ a_{22} = 3 \\ a_{32} = 1 \end{cases}$$

$$f(\vec{e}_3) = \vec{e}_i a_{i3}$$

$$\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} a_{13} + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} a_{23} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} a_{33}$$

$$\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} a_{13} \\ a_{13} \\ a_{13} \end{pmatrix} + \begin{pmatrix} -a_{23} \\ 0 \\ a_{23} \end{pmatrix} + \begin{pmatrix} 0 \\ a_{33} \\ a_{33} \end{pmatrix}$$

$$\begin{cases} a_{13} - a_{23} = -2 \\ a_{13} + a_{33} = 1 \\ a_{13} + a_{23} + a_{33} = 2 \end{cases} \Rightarrow \begin{cases} a_{13} = -1 \\ a_{23} = 1 \\ a_{33} = 2 \end{cases}$$

The matrix of f is given by $[f]_e = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, therefore, $[f]_e = \begin{pmatrix} 1 & -3 & -1 \\ -1 & 3 & 1 \\ 2 & 1 & 2 \end{pmatrix}$

Theorems

- Let $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ be the basis of E and let f be any operator on E . Then, for any vector $\vec{v} \in E$, $[f]_e \cdot [\vec{v}]_e = [f(\vec{v})]_e$. That is, if we multiply the coordinate vector of $\vec{v}$ by matrix representation of f , we obtain the coordinate vector of $f(\vec{v})$.
- Let $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ be bases of E , U and V respectively. Let $f: E \rightarrow U$ and $g: U \rightarrow V$ be linear mappings. Then $[g \circ f]_{e_i}^{g_i} = [g]_{f_i}^{g_i} [f]_{e_i}^{f_i}$. That is, relative to the appropriate bases, the matrix representation of the composition of two linear mappings is equal to the product of the matrix representations of the individual mappings.
- For any $f, g \in L(E)$ and any scalar $\alpha \in K$,
 - i. $[g + f]_e = [g]_e + [f]_e$ and
 - ii. $[\alpha g]_e = \alpha [g]_e$.

Example 3: Matrices representation of linear transformation f and g are $A = \begin{pmatrix} 0 & -4 & 3 \\ -1 & 1 & 0 \\ -1 & 4 & -2 \end{pmatrix}$ and

$B = \begin{pmatrix} 3 & 0 & 4 \\ 1 & 5 & -1 \\ 2 & 1 & 2 \end{pmatrix}$ respectively. Find matrix representation of

- a) $4f$
- b) $2f + 3g$
- c) $f \circ g$

Solution

$$\text{a) } [4f] = 4 \begin{pmatrix} 0 & -4 & 3 \\ -1 & 1 & 0 \\ -1 & 4 & -2 \end{pmatrix} = \begin{pmatrix} 0 & -16 & 12 \\ -4 & 4 & 0 \\ -4 & 16 & -8 \end{pmatrix}$$

$$\text{b) } [2f + 3g] = 2 \begin{pmatrix} 0 & -4 & 3 \\ -1 & 1 & 0 \\ -1 & 4 & -2 \end{pmatrix} + 3 \begin{pmatrix} 3 & 0 & 4 \\ 1 & 5 & -1 \\ 2 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 9 & -8 & 18 \\ 1 & 17 & -3 \\ 4 & 11 & 2 \end{pmatrix}$$

$$\text{c) } [f \circ g] = \begin{pmatrix} 0 & -4 & 3 \\ -1 & 1 & 0 \\ -1 & 4 & -2 \end{pmatrix} \begin{pmatrix} 3 & 0 & 4 \\ 1 & 5 & -1 \\ 2 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -17 & 10 \\ -2 & 5 & -5 \\ -3 & 18 & -12 \end{pmatrix}$$



Application activity 6.8

1. Find matrix representation of the transformation

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (3x + 2y, 2z - y, z - x)$$

a) Relative to the standard basis of $\mathbb{R}^3$

b) Relative to the basis $\{\vec{e}_1 = (1, 1, 1), \vec{e}_2 = (-1, 1, 1), \vec{e}_3 = (0, 1, 1)\}$.

2. Matrices representation of linear transformation f and g are $A = \begin{bmatrix} 3 & 4 & 1 \\ -1 & 2 & 0 \\ 4 & -5 & -3 \end{bmatrix}$ and

$$B = \begin{pmatrix} 3 & 0 & 4 \\ 1 & 5 & -1 \\ 2 & 1 & 2 \end{pmatrix} \text{ respectively. Find matrix representation of}$$

a) $4f - 6g$

b) $f \circ g$

c) $g \circ f$

6.9.Determinants of order 3 and properties



Activity 6.9

Evaluate the following operations by considering the direction of arrows

a)

$$\begin{vmatrix} 1 & -4 & 2 \\ 3 & 6 & 1 \\ 5 & 0 & 1 \end{vmatrix}$$

b)

$$\begin{vmatrix} 10 & 2 & 4 \\ -6 & 5 & 2 \\ 0 & 3 & 1 \end{vmatrix}$$



Content summary

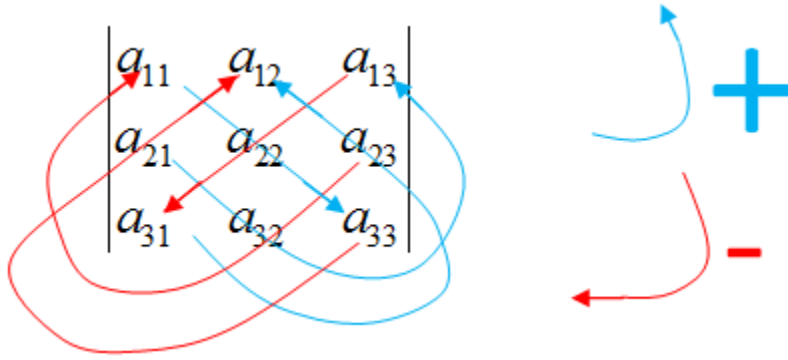
Consider an arbitrary 3x3 matrix, $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$. The determinant of A is calculated by

rule of SARRUS

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

The terms with a **positive sign** are formed by the elements of the **principal diagonal** and those of the **parallel diagonals** with its corresponding **opposite vertex**.

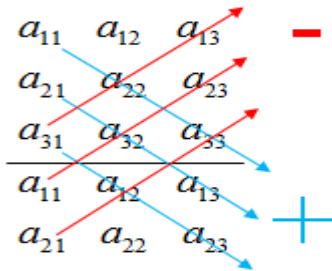
The terms with a **negative sign** are formed by the elements of the **secondary diagonal** and those of the **parallel diagonals** with its corresponding **opposite vertex**.



$$\det(A) = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{23}a_{12} - a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{33}a_{21}a_{12}.$$

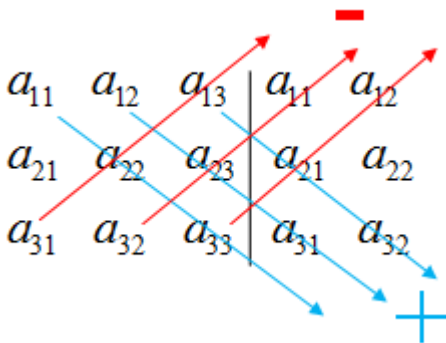
Or we can work as follow:

To calculate the 3x3 determinant we rewrite the first two rows below the determinant (or first two columns to the right of the determinant).



$$\det(A) = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33}$$

Or



$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

As multiplication of real numbers is commutative, the three are the same.

Example 1:

$$\begin{vmatrix} 3 & 2 & 1 \\ 0 & 2 & -5 \\ -2 & 1 & 4 \end{vmatrix} = 3 \times 2 \times 4 + 0 \times 1 \times 1 + (-2) \times (-5) \times 2 - 1 \times 2 \times (-2) - (-5) \times 1 \times 3 - 4 \times 0 \times 2$$

$$= 24 + 0 + 20 + 4 + 15 - 0 = 63$$

General rule for $n \times n$ matrices (minor and cofactor)

General method of finding the determinant of matrix with $n \times n$ dimension

($2 \times 2, 3 \times 3, 4 \times 4, 5 \times 5, \dots$) is the use of cofactors.

Minor

An element, a_{ij} , to the value of the determinant of order $n-1$, obtained by deleting the row i and the column j in the matrix is called a **minor**.

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & [5] & 4 \\ 3 & 6 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix}$$

Cofactor

The **cofactor** of the element a_{ij} is its minor prefixing:

The + sign if $i + j$ is **even**.

The - sign if $i + j$ is **odd**.

$$\begin{vmatrix} 1 & 2 & 1 \\ [2] & 5 & 4 \\ 3 & 6 & 2 \end{vmatrix} \rightarrow - \begin{vmatrix} 2 & 1 \\ 6 & 2 \end{vmatrix}$$

The value of a determinant is equal to the sum of the products of the elements of a line (row or column) by its corresponding cofactors:

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Example 2:

$$|A| = \begin{vmatrix} 3 & 2 & 1 \\ 0 & 2 & -5 \\ -2 & 1 & 4 \end{vmatrix} = 3 \begin{vmatrix} 2 & -5 \\ 1 & 4 \end{vmatrix} - 2 \begin{vmatrix} 0 & -5 \\ -2 & 4 \end{vmatrix} + 1 \begin{vmatrix} 0 & 2 \\ -2 & 1 \end{vmatrix}$$

$$= 3(8 + 5) - 2(0 - 10) + 1(0 + 4) = 39 + 20 + 4 = 63$$

Note that we choose only one line (row or column).

Properties of determinant

- 1) Matrix A and its transpose A' have the same determinant.

$$|A'| = |A|$$

- 2) If the matrix A has similar lines or columns, then $|A| = 0$.

Example 3:

- 3) All elements of a line are zero.

Example 4:

$$|A| = \begin{vmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

- 4) When the elements of a line are a linear combination of the others the determinant is zero.

Example 5:

$$|A| = \begin{vmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \\ 3 & 5 & 6 \end{vmatrix} = 0$$

We see that $r_3 = r_1 + r_2$

5) A triangular matrix determinant is the product of the leading diagonal elements.

Example 6:

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 3 & 5 & 6 \end{vmatrix} = 2 \times 2 \times 6 = 24$$

6) If a determinant switches two parallel lines, its determinant changes sign.

Example 7:

$$|A| = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 0 \\ 3 & 5 & 6 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 3 & 5 & 6 \end{vmatrix}$$

7) If the elements of a line are added to the elements of another parallel line previously multiplied by a real number, the value of the determinant is unchanged.

Example 8:

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 0 \\ 3 & 5 & 6 \end{vmatrix} = 16, \quad c_3 = 2c_1 + c_2 + c_3 : \begin{vmatrix} 2 & 1 & 7 \\ 1 & 2 & 4 \\ 3 & 5 & 17 \end{vmatrix} = 16$$

8) If a determinant is multiplied by a real number, any line can be multiplied by the above-mentioned number, but only one.

Example 9:

$$2 \times \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 0 \\ 3 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 2 \times 2 & 1 & 2 \\ 2 \times 1 & 2 & 0 \\ 2 \times 3 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 4 & 1 & 2 \\ 2 & 2 & 0 \\ 6 & 5 & 6 \end{vmatrix} = 32$$

$$2 \times \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 0 \\ 3 & 5 & 6 \end{vmatrix} = 2 \times 16 = 32$$

9) If all the elements of a line are formed by two addends, the above mentioned determinant decomposes in the sum of two determinants.

Example 10:

$$\begin{vmatrix} 2 & 1 & 2 \\ a+b & a+c & a+d \\ 3 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 2 \\ a & a & a \\ 3 & 5 & 6 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 \\ b & c & d \\ 3 & 5 & 6 \end{vmatrix}$$

10) The determinant of a product equals the product of the determinants.

$$|A \times B| = |A| \times |B| \text{ or } \det(A \times B) = \det(A) \times \det(B)$$

Example 11:

$$\text{Let } A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 8 \\ 3 & 2 & 3 \end{pmatrix}, B = \begin{pmatrix} 3 & -1 & 1 \\ 0 & 4 & 1 \\ 3 & 2 & 2 \end{pmatrix}$$

$$A \times B = \begin{pmatrix} 6 & 9 & 5 \\ 30 & 30 & 22 \\ 18 & 11 & 11 \end{pmatrix}$$

$$|A \times B| = \begin{vmatrix} 6 & 9 & 5 \\ 30 & 30 & 22 \\ 18 & 11 & 11 \end{vmatrix} = 72$$

$$|A| = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & 8 \\ 3 & 2 & 3 \end{vmatrix} = 24, \quad |B| = \begin{vmatrix} 3 & -1 & 1 \\ 0 & 4 & 1 \\ 3 & 2 & 2 \end{vmatrix} = 3$$

$$|A| \times |B| = 24 \times 3 = 72$$

10) **Power Rule:** For any integer exponent n: $\det(A^n) = (\det(A))^n$

11) **The Scalar Property:** $\det(kA) = k^3 \det(A)$ for a 3×3 matrix.

Example 12:

Let A be a 3×3 diagonal matrix defined as:

$$A = \begin{pmatrix} x & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

If $\det((2A)^3) = 1,728,000$, find the value of the unknown x (where x is a real number).

Solution:

$$\det(A) = x \cdot 2 \cdot 3 = 6x$$

$$\det(2A) = 2^3 \cdot \det(A)$$

$$\det(2A) = 8 \cdot (6x) = 48x$$

$$\det((2A)^3) = (\det(2A))^3$$

$$\det((2A)^3) = (48x)^3 = 1,728,000.$$

$$(48x)^3 = 1,728,000$$

$$48x = \sqrt[3]{1,728,000}$$

$$48x = 120$$

$$x = \frac{5}{2} = 2.5$$

Verification: $A = \begin{pmatrix} 2.5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \Rightarrow \det(A) = 2.5 \cdot 2 \cdot 3 = 15$

$$\det(2A) = 2^3 \cdot \det(A) = 8 \cdot 15 = 120$$

$$\det((2A)^3) = (120)^3 = 1,728,000.$$

Therefore, the value $x = 2.5$ is correct.

Example 13: Imagine a venture capital firm in Rwanda that manages an investment portfolio distributed across three core economic sectors: **FinTech** (x), **Agribusiness**, and **E-commerce**.

Every year, the firm applies a specific target growth multiplier to each sector. To find the projected value of the entire fund after several years, analysts compute the power of the growth matrix.

Let F be the annual growth factor matrix:

$$F = \begin{pmatrix} x & 0 & 0 \\ 0 & 1.2 & 0 \\ 0 & 0 & 1.5 \end{pmatrix}$$

- **Agribusiness** has a fixed annual growth factor of 1.2 (a 20% annual return).

- **E-commerce** has a fixed annual growth factor of 1.5 (a 50% annual return).
- **FinTech** (x) is a new, volatile sector with an unknown annual growth factor x that needs to be determined based on performance goals.

The Problem: The firm is looking at a **3-year compound projection** (represented by F^3).

The risk management team calculates a portfolio scaling metric known as the **Total Expansion Volume**, which is mathematically equivalent to the determinant of the 3-year compounded matrix, $\det(F^3)$. If the target portfolio expansion volume over 3 years (noted as $\det(F^3) = (\det(F))^3$) must equal exactly **2.7**, what must the annual growth factor x of the FinTech sector be to meet this strategic goal?

Solution:

$$\det(F) = x \cdot 1.2 \cdot 1.5 = 1.8x. \text{ Therefore, } \det(F^3) = (1.8x)^3$$

The firm requires the total expansion volume to be 2.7: $(1.8x)^3 = 2.7$ which implies that

$$x = \frac{5}{3\sqrt[3]{10}} \approx 0.774$$

Financial Interpretation

An annual growth factor of $x \approx 0.774$ means the FinTech sector experiences a **negative return** of approximately -22.6% per year (since $0.774 - 1 = -0.226$).

The Insight for the Board: Even if the risky FinTech sector loses about 22.6% of its value annually ($x \approx 0.774$), the high performance of Agribusiness ($+20\%$) and E-commerce ($+50\%$) provides strong enough compounding asset volume to ensure the overall fund still hits its total 3-year expansion target of 2.7.



Application activity 6.9

1) Consider the following matrices:

$$A = \begin{pmatrix} 12 & 0 & 1 \\ 34 & 0 & 2 \\ -3 & 0 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 2 & 5 \end{pmatrix}, C = \begin{pmatrix} 6 & 7 & 6 \\ 2 & 4 & 8 \\ 1 & 3 & 9 \end{pmatrix}, D = \begin{pmatrix} -3 & 5 & 1 \\ 2 & 10 & 1 \\ 1 & 8 & 1 \end{pmatrix},$$

Find:

a. $|A|, |B|, |C|$ and $|D|$

b. $|BC|$

c. $|CD|$

6.10. Matrix inverse



Activity 6.10

Consider the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix}$,

- 1) Calculate the determinant of A .
- 2) Replace every element in matrix A by its cofactor to find a new matrix called cofactor matrix.
- 3) Find the transpose of the cofactor matrix.
- 4) Multiply the inverse value of determinant obtained in 1 by the matrix obtained in 3.
- 5) Multiply matrix A by matrix obtained in 4. Discuss about your result.



Content summary

Calculating matrix inverse of matrix, A , is to find matrix A^{-1} such that, $A.A^{-1} = A^{-1}.A = I$,

Where I is identity matrix.

The matrix inverse of matrix A is equal to the inverse value of its determinant multiplied by the adjoint matrix.

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

Where $\text{adj}(A)$ is the **adjoint** (also called **adjugate**) matrix which is the transpose of the cofactor matrix. The cofactor matrix is found by replacing every element in matrix A with its cofactor.

Example 1:

Find the inverse of the following matrix $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$.

Solution

We find its inverse as follows:

The determinant of the given matrix is $|A| = 3$.

Cofactor of each element: $c(2) = \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0$, $c(3) = -\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1$, $c(5) = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$,

$c(0) = -\begin{vmatrix} 3 & 0 \\ 5 & 1 \end{vmatrix} = -3$, $c(0) = \begin{vmatrix} 2 & 1 \\ 5 & 1 \end{vmatrix} = -3$, $c(1) = -\begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} = 3$,

$c(1) = \begin{vmatrix} 3 & 0 \\ 5 & 1 \end{vmatrix} = 3$, $c(0) = -\begin{vmatrix} 2 & 0 \\ 5 & 1 \end{vmatrix} = -2$, $c(1) = \begin{vmatrix} 2 & 0 \\ 3 & 0 \end{vmatrix} = 0$.

The cofactor matrix is $A = \begin{pmatrix} 0 & -3 & 3 \\ 1 & -3 & -2 \\ 0 & 3 & 0 \end{pmatrix}$, and then $adj(A) = \begin{pmatrix} 0 & 1 & 0 \\ -3 & -3 & 3 \\ 3 & -2 & 0 \end{pmatrix}$.

The matrix inverse of A is $A^{-1} = \frac{1}{\det(A)} adj(A) = \frac{1}{3} \begin{pmatrix} 0 & 1 & 0 \\ -3 & -3 & 3 \\ 3 & -2 & 0 \end{pmatrix}$.

Therefore, $A^{-1} = \begin{pmatrix} 0 & \frac{1}{3} & 0 \\ -1 & -1 & 1 \\ 1 & \frac{-2}{3} & 0 \end{pmatrix}$.

Note: Inverse Matrix: If A is invertible, its determinant is the reciprocal of $\det(A)$.

$$\det(A^{-1}) = \frac{1}{\det(A)}$$



Application activity 6.10

Find the inverse of the following matrices

1) $A = \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 2 & 5 \end{pmatrix}$

2) $B = \begin{pmatrix} 11 & -8 & 1 \\ 0 & -6 & 2 \\ 3 & 2 & 7 \end{pmatrix}$

3) $C = \begin{pmatrix} 6 & 7 & 6 \\ 2 & 4 & 8 \\ 1 & 3 & 9 \end{pmatrix}$

4) $D = \begin{pmatrix} -3 & 5 & 1 \\ 2 & 10 & 11 \\ 1 & 8 & 1 \end{pmatrix}$

6.11. Properties of Matrix inverse



Activity 6.11

Consider the matrices $A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix}$ find

1. $(AB)^{-1}$ and $B^{-1}A^{-1}$
2. $(A^{-1})^{-1}$
3. $(4A)^{-1}$ and $\frac{1}{4}A^{-1}$
4. $(A^t)^{-1}$ and $(A^{-1})^t$
5. $A^{-1}A$ and AA^{-1} , comment on the results.



Content summary

Properties of the Inverse Matrix

For two invertible matrices A and B

- 1) $(A.B)^{-1} = B^{-1}.A^{-1}$
- 2) $(A^{-1})^{-1} = A.$
- 3) $(\alpha A)^{-1} = \alpha^{-1}A^{-1}.$
- 4) $(A^t)^{-1} = (A^{-1})^t.$

Example 1:

Consider matrix $A = \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 3 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 & 0 \\ 2 & -2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, find

- i. A^{-1} and B^{-1}
- ii. $(AB)^{-1}$
- iii. $(3B)^{-1}$
- iv. $(B^t)^{-1}$

Solution

$$\text{a) } |A|=3, \text{ } Adj(A)=\begin{pmatrix} 0 & 0 & 1 \\ -3 & 6 & 1 \\ -3 & 3 & 1 \end{pmatrix}, \text{ } A^{-1}=\begin{pmatrix} 0 & 0 & \frac{1}{3} \\ -1 & 2 & \frac{1}{3} \\ -1 & 1 & \frac{1}{3} \end{pmatrix}$$

$$|B|=-2, \text{ } Adj(B)=\begin{pmatrix} -2 & -1 & 0 \\ -2 & 0 & 0 \\ 4 & 1 & -2 \end{pmatrix}, \text{ } B^{-1}=\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ -2 & -\frac{1}{2} & 1 \end{pmatrix}$$

$$\text{b) } (AB)^{-1}=B^{-1}A^{-1}=\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ -2 & -\frac{1}{2} & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & \frac{1}{3} \\ -1 & 2 & \frac{1}{3} \\ -1 & 1 & \frac{1}{3} \end{pmatrix}=\begin{pmatrix} -\frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{3} \\ -\frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix}$$

$$\text{c) } (3B)^{-1}=\frac{1}{3}B^{-1}=\frac{1}{3}\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ -2 & -\frac{1}{2} & 1 \end{pmatrix}=\begin{pmatrix} \frac{1}{3} & \frac{1}{6} & 0 \\ \frac{1}{3} & 0 & 0 \\ -\frac{2}{3} & -\frac{1}{6} & \frac{1}{3} \end{pmatrix}$$

$$\text{d) } (B^t)^{-1}=(B^{-1})^t=\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ -2 & -\frac{1}{2} & 1 \end{pmatrix}^t=\begin{pmatrix} 1 & 1 & 2 \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}$$



Application activity 6.11

Consider of the following matrices $A = \begin{pmatrix} 3 & 7 & 2 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -2 & 1 \\ 3 & 1 & 6 \\ -1 & 1 & 1 \end{pmatrix}$

find

1) A^{-1} and B^{-1} 2) $(A^{-1})^{-1}$

3) $(10A)^{-1}$ 4) $(A^t)^{-1}$

6.12. Application of matrices in everyday life: Solving a system of 3 linear equations delivered from real life problems.



Activity 6.12

Consider the following system of 3 linear equations in 3 unknowns.

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = c_1 \\ a_{21}x + a_{22}y + a_{23}z = c_2 \\ a_{31}x + a_{32}y + a_{33}z = c_3 \end{cases}$$

1. Rewrite this system in matrix form

2. If we pre-multiply (multiply to the left) both sides of the equality obtained in 1) by

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}^{-1}, \text{ what will be the new equality?}$$



Content summary

The solution of the following system of 3 linear equations in 3

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = c_1 \\ a_{21}x + a_{22}y + a_{23}z = c_2 \\ a_{31}x + a_{32}y + a_{33}z = c_3 \end{cases}$$

unknowns by matrix inversion method is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}B$, provided that A^{-1} exists,

where $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ and $B = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$.

Notice

- If at least, one of c_i is different of zero the system is said to be **non-homogeneous** and if all c_i are zero the system is said to be **homogeneous**.
- The set of values of x, y, z that satisfy all the equations of system (1) is called **solution of the system**.
- For the homogeneous system, the solution $x = y = z = 0$ is called **trivial solution**. Other solutions are **non-trivial solutions**.
- Non- homogeneous system cannot have a trivial solution as at least one of x, y, z is not zero.

Alternative method: Cramer's rule

Consider the system

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = c_1 \\ a_{21}x + a_{22}y + a_{23}z = c_2 \\ a_{31}x + a_{32}y + a_{33}z = c_3 \end{cases}$$

We use Cramer's rule as follows:

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \Delta_x = \begin{vmatrix} c_1 & a_{12} & a_{13} \\ c_2 & a_{22} & a_{23} \\ c_3 & a_{32} & a_{33} \end{vmatrix}, \Delta_y = \begin{vmatrix} a_{11} & c_1 & a_{13} \\ a_{21} & c_2 & a_{23} \\ a_{31} & c_3 & a_{33} \end{vmatrix}, \Delta_z = \begin{vmatrix} a_{11} & a_{12} & c_1 \\ a_{21} & a_{22} & c_2 \\ a_{31} & a_{32} & c_3 \end{vmatrix}.$$

Therefore, $x = \frac{\Delta_x}{\Delta}$, $y = \frac{\Delta_y}{\Delta}$, and $z = \frac{\Delta_z}{\Delta}$.

Notice:

- The solution $\frac{b}{0}, b \neq 0$ means impossible
- The solution $\frac{0}{0}$ means indeterminate

Example 1:

In the shop, x: is the cost (in USD) of one a Mouse, y is the cost (in USD) of one Calculator and z is the cost (in USD) of one Pack of Pens.

1) If I buy exactly 1 mouse, 1 calculator, and 1 pack of pens, I pay \$6. Write the related equation.

Invoice 2: The Bulk Trade-In Shipment

2) If I purchase 2 mice and 1 desktop calculator and I sell 1 pack of pens I pay only 1\$. Write the related equation.

3) If I buy 3 mice, 2 desktop calculator, and 1 pack of pens, I pay exactly \$10. Write the equation.

4) Form the system of the simultaneous linear equations obtained.

5) write it in the form of a matrix product $AX = B$ where $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and B the column matrix of

the amount paid.

6) Use the determinants to find the cost of each item x, y, and z.

Solution:

From (1) to (4) we get:

$$(4) \begin{cases} x + y + z = 6 \\ 2x + y - 1z = 1 \\ 3x + 2y + z = 10 \end{cases}$$

$$(5) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ 10 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 6 \\ 1 \\ 10 \end{pmatrix}$$

We find the inverse of A.

Matrix A is invertible if its determinant is not zero.

$$6) \det(A) = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix} = 1 + 4 - 3 - 3 + 2 - 2 = -1 \neq 0, \text{ then } A \text{ has inverse.}$$

We have seen that the adjoint (or adjugate) matrix and determinant of a matrix are used to find its inverse.

$$\text{Then, } A^{-1} = \begin{pmatrix} -3 & -1 & 2 \\ 5 & 2 & -3 \\ -1 & -1 & 1 \end{pmatrix} \text{ and } X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 & -1 & 2 \\ 5 & 2 & -3 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Therefore, the set of the solution $S = \{(1, 2, 3)\}$.

By Cramer method:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix} = -1, \quad \Delta_x = \begin{vmatrix} 6 & 1 & 1 \\ 1 & 1 & -1 \\ 10 & 2 & 1 \end{vmatrix} = -1, \quad \Delta_y = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 1 & -1 \\ 3 & 10 & 1 \end{vmatrix} = -2, \quad \Delta_z = \begin{vmatrix} 1 & 1 & 6 \\ 2 & 1 & 1 \\ 3 & 2 & 10 \end{vmatrix} = -3,$$

$$x = \frac{\Delta_x}{\Delta} = \frac{-1}{-1} = 1, \quad y = \frac{\Delta_y}{\Delta} = \frac{-2}{-1} = 2, \quad \text{and } z = \frac{\Delta_z}{\Delta} = \frac{-3}{-1} = 3.$$

Therefore, $S = \{(1, 2, 3)\}$.

We find the exact price of each item:

- **Mouse (x):** \$1 USD.
- **Desktop Calculator (y):** \$2 USD.
- **Pack of Gel Pens (z):** \$3 USD.

Note: The inverse of a square matrix can be obtained using system of equations.

Example 2: Find the inverse of the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix}$.

Solution: Suppose that its inverse is given by $A^{-1} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$.

We know that $AA^{-1} = I$, , then

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{cases} a+b+c=1 \\ 2a+b-c=0 \\ 3a+2b+c=0 \end{cases} \quad (1)$$

$$\Rightarrow \begin{cases} d+e+f=1 \\ 2d+e-f=0 \\ 3d+2e+f=1 \end{cases} \quad (2)$$

$$\Rightarrow \begin{cases} g+h+i=1 \\ 2g+h-i=0 \\ 3g+2h+i=1 \end{cases} \quad (3)$$

We solve these three systems to find value of $a, b, c, d, e, f, g, h,$ and i .

$$\begin{cases} a+b+c=1 \\ 2a+b-c=0 \\ 3a+2b+c=0 \end{cases} \quad (1) \Rightarrow \begin{cases} a=-3 \\ b=5 \\ c=-1 \end{cases}$$

$$\begin{cases} d+e+f=1 \\ 2d+e-f=0 \\ 3d+2e+f=1 \end{cases} \quad (2) \Rightarrow \begin{cases} d=-1 \\ e=2 \\ f=-1 \end{cases}$$

$$\begin{cases} g+h+i=1 \\ 2g+h-i=0 \\ 3g+2h+i=1 \end{cases} \quad (3) \Rightarrow \begin{cases} g=2 \\ h=-3 \\ i=1 \end{cases}$$

$$\text{Hence, } A^{-1} = \begin{pmatrix} -3 & -1 & 2 \\ 5 & 2 & -3 \\ -1 & -1 & 1 \end{pmatrix}.$$

The system of linear equations is said to be inconsistent if there is no solution that satisfies all the equations simultaneously. In this case the determinant of the coefficient $\Delta=0$. The system has unique solution if $\Delta \neq 0$ and it has more than one solution if $\Delta = \Delta_x = \Delta_y = \Delta_z = 0$.



Application activity 6.12

An automobile company uses three types of steel, S_1, S_2 and S_3 for producing three different types of cars, *Car 1, Car 2 and Car 3*. Steel requirements (in tons) for each of car and total available steel of all three types is summarized in the following table.

Type of steel	Type of car			Total steel available
	Car 1	Car 2	Car 3	
S_1	2	3	4	29
S_2	1	1	2	13
S_3	3	2	1	16

Determine the number of cars of each type which can be manufactured.



End of unit assessment

Knowledge and understanding

1. Define a matrix. What is meant by the order (or dimension) of a matrix?
2. State the necessary condition for two matrices, A and B, to be added or subtracted.

Skills

3. Find the matrix of the following map relative to the canonical basis

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (2x + y, y - z, 2x + 4y)$$

4. Let f be a linear operator on $\mathbb{R}^3$ defined by $f(x, y, z) = (2y + z, x - 4y, 3x)$

a) Find the matrix of f in the basis $\{e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)\}$.

b) Verify that $[f]_e[v]_e = [f(v)]_e$ for any vector $v \in \mathbb{R}^3$.

$$5. \text{ If } A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -1 & 0 & -2 \end{pmatrix},$$

- a) Show that $A^3 - 4A + 7I = O$ where I, O are the unit and the null matrix of order 3 respectively. Use this result to find A^{-1} .

- b) Find the matrix X such that $AX = \begin{pmatrix} 2 \\ 1 \\ -7 \end{pmatrix}$

6. Given $A = \begin{pmatrix} 2 & -2 & -1 \\ 1 & 1 & -2 \\ 3 & 1 & -3 \end{pmatrix}$, find A^3 and hence solve the equations

a) $\begin{cases} 2x - 2y - z = -18 \\ x + y - 2z = -2 \\ 3x + y - 3z = -10 \end{cases}$

b) $\begin{cases} x + 7y - 5z = 9 \\ x + y - z = 3 \\ x + 4y - 2z = 6 \end{cases}$

7. Find $\det(A)$, in terms of t , the determinant of the matrix $A = \begin{pmatrix} 2-t & 1 & 3 \\ 1 & 1-t & 1 \\ -1 & -1 & -2-t \end{pmatrix}$

8. If A is a square matrix of order 3 such that $\det A = x$, find the value of

b) $\det(A^2)$ b) $\det(A^n)$, $n \in \mathbb{Z}$ c) $\det(2A)$ d) $\det(mA)$, $m \in \mathbb{R}$

9. For what value of λ and μ the following system of equations

$$\begin{cases} 2x + 3y + 5z = 9 \\ 7x + 3y - z = 1 \\ 4x + 3y + \lambda z = \mu \end{cases}$$

will have

- a) No solution.
- b) Unique solution.
- c) More than one solution.

Assessing attitudes and values

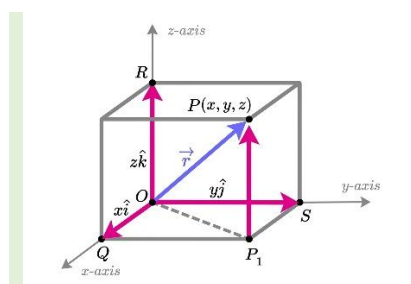
10) Read and choose the correct answer:

(a) A matrix is essentially a way to organize large amounts of information into a single structure of rows and columns. How does developing the habit of categorizing data in matrix dimensions best help you improve your personal or academic life?

- **A.** It trains your mind to structure unstructured tasks, allowing you to prioritize schoolwork, manage your study timetable, and categorize personal goals efficiently.
- **B.** It guarantees that you will never make a mistake in data entry or lose any physical documents.
- **C.** It teaches you that data should only be analyzed using technology rather than manual, logical thinking.
- **D.** It proves that academic organization is completely different from and unrelated to everyday personal habits.

(b) When working in a group to model a local community project (such as tracking market prices or managing school resources) using matrices, which attitude is most critical for achieving an accurate and fair mathematical outcome?

- **A.** Allowing one student to do all the work to ensure the matrix calculations are perfectly uniform.
- **B.** Valuing each team member's input to ensure data from different sources is represented accurately, promoting collective responsibility and shared accountability.
- **C.** Changing the numbers in the matrix arbitrarily if the initial calculation doesn't match what the group wanted to see.
- **D.** Focusing only on finishing the calculations quickly, ignoring whether the underlying data reflects real community needs.



UNIT 7: VECTOR SPACE $\mathbb{R}^3$ OF REAL NUMBERS

Key unit competence: Apply operations of vectors in three dimensions in solving mathematical and real-life problems involving magnitude, direction, work, area, and volume.

Learning objectives:

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain vectors, scalar (dot) product, vector (cross) product, and mixed product.
- State the properties of vector operations in 3D.
- Identify sub vector spaces, linear combination and bases.
- Recall formulas for magnitude, the angle between vectors, work, area, and volume.

Skills

- Represent vectors geometrically and algebraically in 3D.
- Perform operations on vectors (addition, scalar multiplication, dot and cross products).
- Calculate the magnitude of a vector, the angle between two vectors, and direction cosines.
- Determine if a set of vectors is linearly independent, spans a space, or forms a basis.
- Compute the coordinate vector relative to a given basis.
- Apply dot and cross products to calculate work done, areas of parallelograms/triangles, and volumes of parallelepipeds/tetrahedrons.
- Solve real-world problems involving force, displacement, and spatial geometry.

Experimental version

- Value the power of vector notation in simplifying complex spatial problems.
- Appreciate the interplay between geometric intuition and algebraic computation.
- Show perseverance and accuracy in vector calculations.
- Collaborate effectively in hands-on and problem-solving activities.

Introduction

This unit introduces the fundamental ideas of a set of real numbers in a space of three dimensions (3D), which form the foundation of many topics in geometry. The purpose of this unit is to help you understand how to represent a real vector in 3D, to make operations in 3D, what vector space and sub-vector space are in 3D, and basis and dimension, and to help you get skills in applying scalar and vector products in physics and in geometry.

The key words to be explained in this unit are the following: vector space, vector space of real numbers, sub vector space, dot product, cross product, basis and dimension of a vector space, and linear transformation in real vector space.

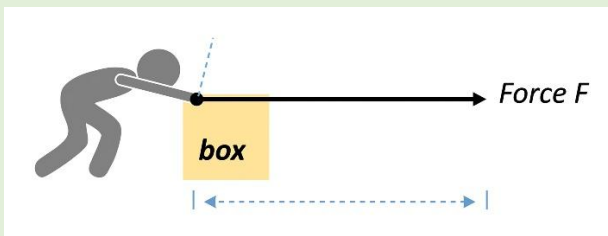


Introductory activity:

A vector space is a collection of objects with directions and magnitudes called vectors, which may be added together and multiplied by numbers, called scalars in this context.

In physics, forces are vectors, and forces add as vectors do.

- a) Refer to the addition of forces and discuss the properties of addition of vectors.



- b) What happens when a vector is multiplied by a scalar (real number)?

- c) State whether each of the following is a scalar or a vector:

- i. Velocity ii. Force iii. Mass.

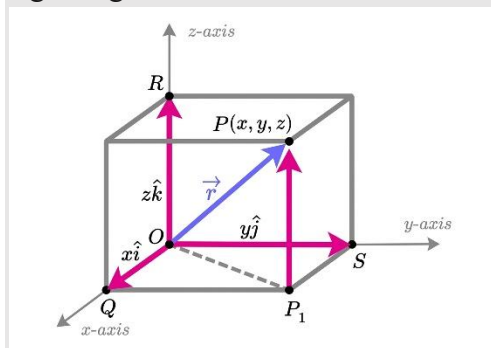
7.1 Vector space $\mathbb{R}^3$

7.1.1 Vector position in $\mathbb{R}^3$



Activity 7.1.1

We know that the position of a point in space is determined by three numbers x, y, z obtained with reference to three straight lines x -axis, y -axis and z -axis respectively) intersecting at right angles.



1. Draw the cartesian plane with 3 axis and locate the point P in the following way: Along x -axis measure, $OS = 3$ units measure $SN = 2$ units parallel to y -axis then measure $NP(x, y, z) = 2$ units parallel to z -axis. P is the point of coordinate $(3, 2, 2)$ in space.
2. Show the vector A (from the origine to the point A)
3. In the same space, present the point $B(1, 3, 2)$ and show the vector B (from the origine to the point B)
4. Join points **A** and **B** with arrow from A to B to get the vector $\overrightarrow{AB}$
5. Can you find the position of the vector $\overrightarrow{AB}$ when it is projected to the origine?



Content Summary

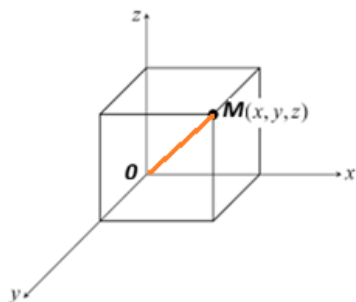
Physical forces and velocities are not confined to the plane, so it is natural to extend the concept of vectors from two-dimensional space to three-dimensional space.

In space, points and vectors are denoted by ordered triples. Taking Cartesian axes x, y and z , any point M in three-dimensional space can be represented by giving x, y and z coordinates.

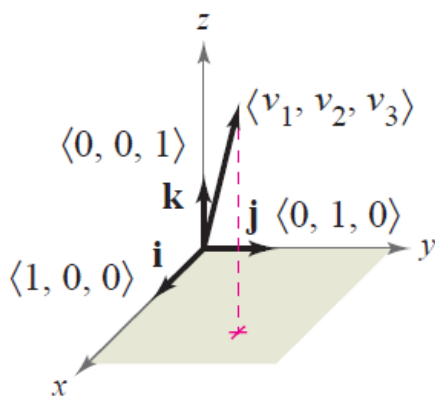
Denoting unit vectors along these axes by $\vec{i}$, $\vec{j}$ and $\vec{k}$, respectively, we can write the vector

from 0 to $M(x, y, z)$ as $\overrightarrow{OM} = x\vec{i} + y\vec{j} + z\vec{k} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. In this case x, y, z are the components of

the vector $\overrightarrow{OM}$ in space.



In space, vectors are denoted by ordered triples $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.



The set of vectors of space is denoted by V .

Let the point 0 be fixed, as common origin of all representatives. This point 0 will be called the origin of the space E and define a bijection of the set of points of the space E on the set V of vectors of space.

The set of vectors of space with origin 0 is denoted by E_0 and $E_0 = \{\overrightarrow{0a} : a \in E\}$

The vector $\overrightarrow{OP}$ joining the origin, 0, to the point P is called the **position vector** of P with respect to 0, or simply the position vector of P .

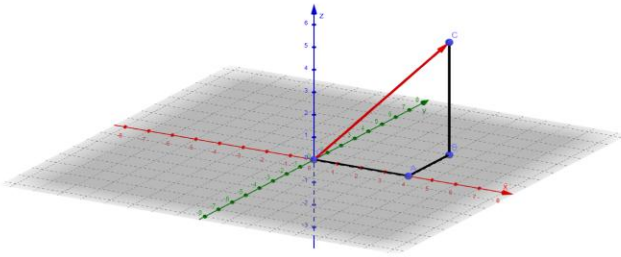
We sometimes denote the position vector of P by $\vec{p}$. That is $\overrightarrow{0P} = \vec{p}$

The **zero vector** is $(0,0,0)$ denoted by $\vec{0}$.

Example 1.

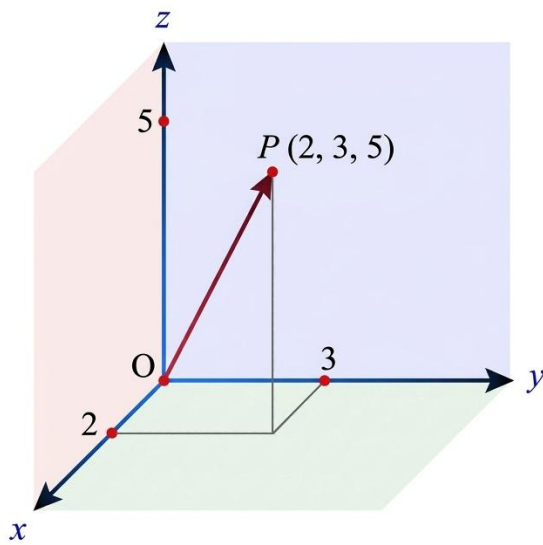
In $\mathbb{R}^3$, allocate the position of vector $\vec{v} = (4,3,5)$

Solution: Move 4 units in +X direction, move 3 units in +Y direction; Move 5 units in +Z direction (toward viewer). Join the origine and the point. Mark this final point as $C(4,3,5)$



Example 2:

Vector $\vec{P} = \overrightarrow{OP}$ where $p(2,3,5)$



Vector joining two points:

Taking Cartesian axes x, y and z , any point M in three-dimensional space can be represented by giving x, y and z coordinates. Denoting unit vectors along these axes by $\vec{i}, \vec{j}$, and $\vec{k}$ respectively,

we can write the vector from 0 to $M(x, y, z)$ as $\overrightarrow{OM} = x\vec{i} + y\vec{j} + z\vec{k} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

In this case x, y, z are the components of the vector $\overrightarrow{OM}$ in space.

Given two points $A(x_A, y_A, z_A)$ and $B(x_B, y_B, z_B)$ the vector $\overrightarrow{AB}$ is given by the vector

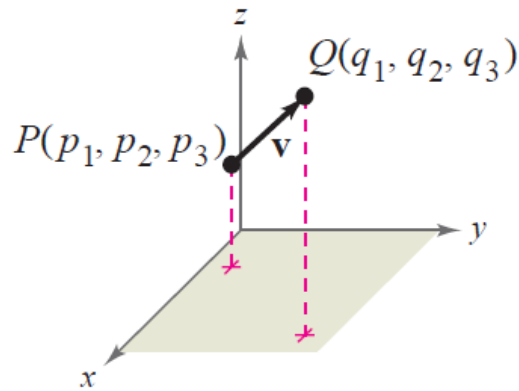
$$\overrightarrow{OB} - \overrightarrow{OA} = \left(x_B \vec{i} + y_B \vec{j} + z_B \vec{k} \right) - \left(x_A \vec{i} + y_A \vec{j} + z_A \vec{k} \right) = (x_B - x_A) \vec{i} + (y_B - y_A) \vec{j} + (z_B - z_A) \vec{k}$$

Therefore, the position vector of $\overrightarrow{AB}$ is of the following form:

$$\overrightarrow{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \\ z_B - z_A \end{pmatrix}.$$

Therefore, the vector joining point A and point B is denoted by $\overrightarrow{AB}$ and its components are found by subtracting the coordinates of point A from the coordinates of point B . For example, the components of vector $\overrightarrow{AB}$ defined by two points (a_1, a_2, a_3) and (b_1, b_2, b_3) are given by $(b_1, b_2, b_3) - (a_1, a_2, a_3)$. Then a vector in space may be described by ordered triple of coordinates (a, b, c) . The point is called the **initial point** or **tail** of $\overrightarrow{AB}$ and B is called the **terminal point** or **tip**.

If V is represented by the directed line segment from $P(p_1, p_2, p_3)$ to $Q(q_1, q_2, q_3)$, as shown in Figure, the **component form** of V is produced by subtracting the coordinates of the initial point from the corresponding coordinates of the terminal point $\mathbf{v} = (v_1, v_2, v_3) = (q_1 - p_1, q_2 - p_2, q_3 - p_3)$.



Example 3:

Find vectors $\overrightarrow{AB}$ and $\overrightarrow{BA}$ given that $A(1, 2, 3)$ and $B(2, 1, 0)$

Solution:

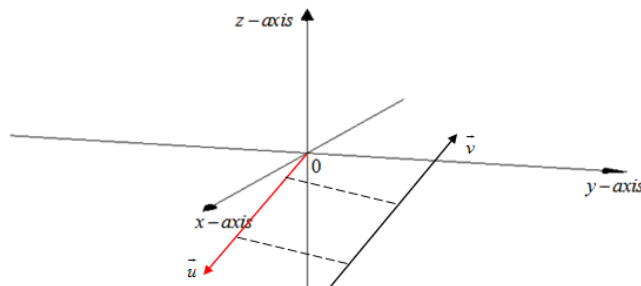
$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (2, 1, 0) - (1, 2, 3) = (1, -1, -3)$$

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = (1, 2, 3) - (2, 1, 0) = (-1, 1, 3)$$

Parallel vectors

Two vectors are parallel if and only if they have: The same direction, or opposite directions.

Thus, two vectors $\vec{u}$ and $\vec{v}$ are parallel if and only if one can be expressed as a scalar multiple of the other. i.e, if $\vec{u}$ is parallel to $\vec{v}$, then

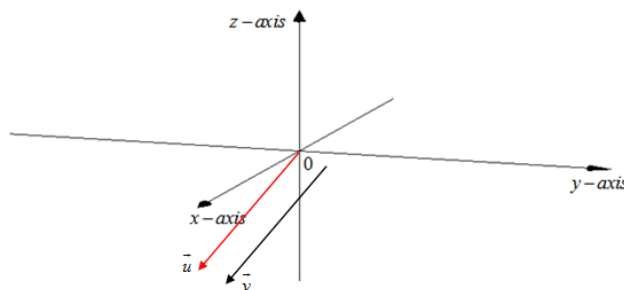


$\vec{u} = r\vec{v}$ or $\vec{v} = s\vec{u}$ for real numbers r and s . they point in the **same direction** if $r > 0$, they point in **opposite directions** if $r < 0$. Similar observation goes to s . In this case we write $\vec{u} \parallel \vec{v}$

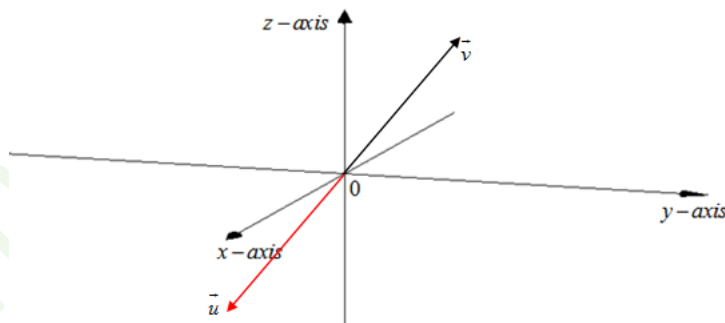
Example: For $\vec{v} = (1, 0, 3)$, the vector $\vec{u} = 2\vec{v} = 2(1, 0, 3) = (2, 0, 6)$ is parallel to $\vec{v}$.

Equal vectors

Two vectors are equal if they have the same length and the same direction. If $\vec{u}$ is equal to $\vec{v}$ we write $\vec{u} = \vec{v}$.



Opposite vectors: Two vectors are opposite if the coordinates of one vector are additive inverse of the coordinates of the other. That is, if $\vec{u}$ and $\vec{v}$ are opposite then $\vec{u} = -\vec{v}$.



Example: For $\vec{v} = (1, 0, 3)$, $\vec{u} = -\vec{v} = -(1, 0, 3) = (-1, 0, -3)$

Equality of two vectors: Two vectors are equal if and only if their corresponding components are equal. It means, if $\langle a_1, a_2, a_3 \rangle = \langle b_1, b_2, b_3 \rangle$, then: $a_1 = b_1$; $a_2 = b_2$; $a_3 = b_3$

Example: Given two vectors in $\mathbb{R}^3$: $\mathbf{u} = \langle 2x - 1, 4, z^2 \rangle$ and $\mathbf{v} = \langle 5, y + 2, 9 \rangle$.

If $\mathbf{u} = \mathbf{v}$, find the values of the scalars x, y , and z .

Solution: For the vectors u and v to be equal, the variables must be $x = 3$, $y = 2$, and $z = \pm 3$.



Application activity 7.1.1

Given points $A(6, 0, -3)$ and $B(3, -3, 0)$ and vectors $\vec{u} = (3, 4, 6)$, $\vec{v} = (1, 1, 1)$. Find and locate:

- a) Vector $\overrightarrow{AB}$
- b) Vector $\overrightarrow{BA}$
- c) Vector $-\overrightarrow{BA}$
- d) Vector $-\overrightarrow{AB}$
- e) Vector $-\vec{u}$
- f) Vector $-\vec{v}$

7.1.2 Addition and subtraction of vectors in 3 dimensions and properties



Activity 7.1.2

Given the vectors $\vec{a} = (4, -2, 1)$, $\vec{b} = (0, 3, 5)$ and $\vec{c} = (-1, 4, -3)$, calculate the following:

- 1) $\vec{a} + \vec{b}$ 2) $\vec{b} + \vec{c}$ 3) $\vec{a} - \vec{b}$ 4) $\vec{c} - \vec{b}$



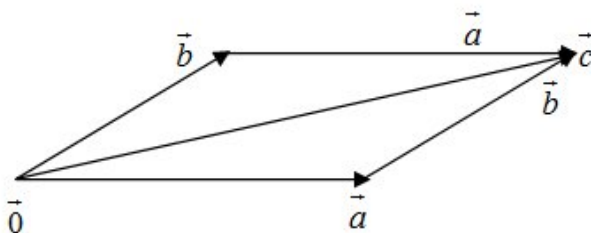
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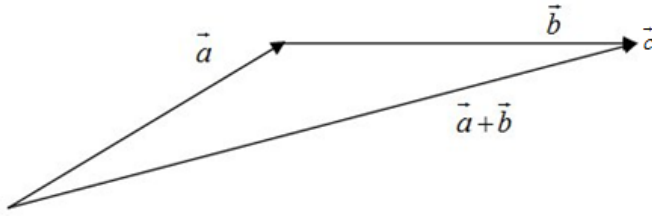
In three dimensions, vectors are treated as quantities with both magnitude and direction, represented by components along the x , y , and z axes (often using $\vec{i}$, $\vec{j}$ and $\vec{k}$ notation).

Addition, subtraction and scalar multiplication of vectors allow us to analyze quantities that have both magnitude and direction and are fundamental tools in mathematics, physics, and engineering.

Addition of two vectors

Graphical method: Draw the first vector, then place the tail of the second vector at the head (tip) of the first vector. The resultant vector is drawn from the tail of the first vector to the head of the second vector.





Analytical/algebraic method: Consider two vectors $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$.

Then, $\vec{a} + \vec{b} = (a_1, a_2, a_3) + (b_1, b_2, b_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$ is also in $\mathbb{R}^3$.

Example 1: If $\vec{a} = (1, 3, 4)$ and $\vec{b} = (3, 5, -7)$ in $\mathbb{R}^3$, then find $\vec{a} + \vec{b}$.

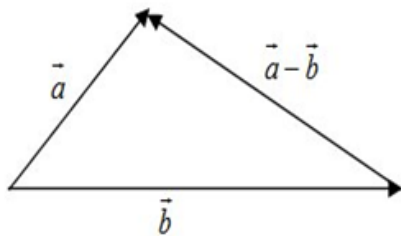
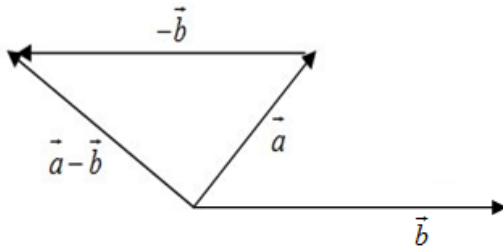
Solution: $\vec{a} + \vec{b} = (1, 3, 4) + (3, 5, -7) = (1 + 3, 3 + 5, 4 + (-7)) = (4, 8, -3)$.

Example 2: If $\vec{a} = \left(\frac{1}{2}, 3, \sqrt{2}\right)$ and $\vec{b} = (1, 0, 1)$ in $\mathbb{R}^3$, then find $\vec{a} + \vec{b}$.

Solution: $\vec{a} + \vec{b} = \left(\frac{1}{2}, 3, \sqrt{2}\right) + (1, 0, 1) = \left(\frac{1}{2} + 1, 3 + 0, \sqrt{2} + 1\right) = \left(\frac{3}{2}, 3, 1 + \sqrt{2}\right)$

Subtraction of two vectors

Graphically: The vector $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$



Algebraically: Consider two vectors $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$.

The subtraction $\vec{a} - \vec{b}$ is interpreted as adding the negative vector: $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$. This reverses the direction of $\vec{b}$ while keeping its magnitude the same.

Component-wise Operation: If $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$, then

$\vec{a} - \vec{b} = (a_1, a_2, a_3) + (-(b_1, b_2, b_3)) = (a_1, a_2, a_3) + (-b_1, -b_2, -b_3) = (a_1 - b_1, a_2 - b_2, a_3 - b_3)$ is also in $\mathbb{R}^3$.

Example 1: If $\vec{a} = (1, 3, 4)$ and $\vec{b} = (3, 5, -7)$ in $\mathbb{R}^3$, then find $\vec{a} - \vec{b}$.

Solution: $\vec{a} - \vec{b} = (1, 3, 4) - (3, 5, -7) = (1 - 3, 3 - 5, 4 - (-7)) = (-2, -2, 11)$.

Example 2: If $\vec{a} = \left(3, -\frac{3}{4}, 4\right)$ and $\vec{b} = \left(-2, \frac{1}{2}, -3\right)$ in $\mathbb{R}^3$, then find $\vec{a} - \vec{b}$.

Solution: $\vec{a} - \vec{b} = \left(3, -\frac{3}{4}, 4\right) - \left(-2, \frac{1}{2}, -3\right) = \left(3 - (-2), -\frac{3}{4} - \frac{1}{2}, 4 - (-3)\right) = \left(5, -\frac{5}{4}, 7\right)$.

Properties of addition of vectors in 3D

1. The addition defined above is defined in E_0 . That is, $\forall \vec{a}, \vec{b} \in E_0, \vec{a} + \vec{b} \in E_0$
2. It is commutative. That is, $\forall \vec{a}, \vec{b} \in E_0, \vec{a} + \vec{b} = \vec{b} + \vec{a}$
3. It is associative. That is, $\forall \vec{a}, \vec{b}, \vec{c} \in E_0, (\vec{a} + \vec{b}) + \vec{c} = (\vec{b} + \vec{c}) + \vec{a}$
4. The identity element is zero vector. That is, $\forall \vec{a} \in E_0, \exists \vec{0} \in E_0 : \vec{a} + \vec{0} = \vec{a} = \vec{0} + \vec{a}$
5. The symmetric element is the opposite of a vector.

That is, $\forall \vec{a} \in E_0, \exists (-\vec{a}) \in E_0 : \vec{a} + (-\vec{a}) = \vec{0}$

Properties of subtraction of vectors in 3D

❖ **It is non-commutative:** $\vec{a} - \vec{b} \neq \vec{b} - \vec{a}$. The order of vectors matters.

Specifically, $\vec{a} - \vec{b} = -(\vec{b} - \vec{a})$.

❖ **It is non-Associative:** $(\vec{a} - \vec{b}) - \vec{c} \neq \vec{a} - (\vec{b} - \vec{c})$ in general.

❖ **Zero Vector Property:** A vector subtracted from itself results in the zero vector: $\vec{a} - \vec{a} = \vec{0}$

Example 1. If $\vec{a} = (1, 3, 4)$ and $\vec{b} = (3, 5, -7)$ in $\mathbb{R}^3$, then compare $\vec{a} + \vec{b}$ and $\vec{b} + \vec{a}$.

Solution: $\vec{a} + \vec{b} = (1, 3, 4) + (3, 5, -7) = (4, 8, -3)$.

$$\vec{b} + \vec{a} = (3, 5, -7) + (1, 3, 4) = (4, 8, -3).$$

Hence $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, then the addition is commutative.



Application activity 7.1.2

Given points A(6,0,-3) and B(3,-3,0) and vectors $\vec{u} = (3, 4, 6)$, $\vec{v} = (1, 1, 1)$. Find

1. $\vec{AB}$
2. $\vec{AB} + \vec{u} - \vec{v}$
3. $\vec{AB} - 3\vec{u} + \vec{v}$
4. $4\vec{u} - \vec{AB} + 2\vec{v}$

7.1.3 Multiplication of a vector by a scalar



Activity 7.1.3

Locate the point P of coordinates (3,2,1) in the cartesian plan.

a) Draw the vector $\vec{OP} = \vec{P}$

b) Now calculate and physically create the double each component: Move 6 units in x-direction, Move 4 units in y-direction and Move 2 units in z-direction. Mark the end point as M. What are coordinate of the point M.

c) Draw the vector $\vec{OM} = \vec{M}$.

d) Compare the vector $\vec{P}$ and the vector $\vec{M}$.

Are they of the same length? Complete: $\vec{M} = \underline{\hspace{1cm}} \vec{P}$

e) Now, try to generalize, if r is a real number and $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ a vector of space,

Complete: $r \times \vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underline{\hspace{1cm}}$



Content summary

A scalar multiplication is an application from $\mathbb{R} \times \mathbb{R}^3$ to $\mathbb{R}^3$. Consider a scalar $\beta \in \mathbb{R}$ and a vector $\vec{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$. Then $\beta \times \vec{a} = \beta \vec{a} = \beta(a_1, a_2, a_3) = (\beta a_1, \beta a_2, \beta a_3)$ is also in $\mathbb{R}^3$.



Example 1: If $\vec{a} = (2, 3, -2)$ and $\beta = 2$. Then, find $\beta\vec{a}$.

Solution: $\beta\vec{a} = 2(2, 3, -2) = (2 \times 2, 2 \times 3, 2 \times (-2)) = (4, 6, -4)$.

Example 2. If $\vec{a} = (1, 3, 4)$ and $\vec{b} = (3, 5, -7)$ in $\mathbb{R}^3$ and $r = 2 \in \mathbb{R}$. then show that

$$2(\vec{a} + \vec{b}) = 2\vec{a} + 2\vec{b}.$$

Solution: $2(\vec{a} + \vec{b}) = 2[(1, 3, 4) + (3, 5, -7)] = 2(4, 8, -2) = (8, 16, -4)$.

$$2\vec{a} + 2\vec{b} = 2(1, 3, 4) + 2(3, 5, -7) = (2, 6, 8) + (6, 10, -14) = (8, 16, -6).$$

Hence $2(\vec{a} + \vec{b}) = 2\vec{a} + 2\vec{b}$.



Application activity 7.1.3

2) Carry out the following activities:

If the three forces act on an object which is at origin are $\vec{F}_1 = (5, 0, 2)$ and $\vec{F}_2 = (-2, 3, -1)$

Calculate three times the resultant force.

3) Given the vectors $\vec{a} = (4, -2, 1)$, $\vec{b} = (0, 3, 5)$ and $\vec{c} = (-1, 4, -3)$, calculate the following:

1. $2(\vec{a} + \vec{b})$
2. $-3(\vec{b} + \vec{c})$
3. $-2(3\vec{a} - \vec{b})$
4. $\frac{1}{2}(\vec{a} + 3\vec{b} - 4\vec{c})$.

1. Given $\vec{p} = (5, -3, 4)$. a) Find the vector $-\vec{p}$. b) Prove that $\vec{p} + (-\vec{p}) = \vec{0}$.

7.1.4 Sub vector space



Activity 7.1.4

1) Consider all vectors that lie on the vertical line passing through the origin. This includes vectors like $(0, 0, 5)$, $(0, 0, -2)$, and $(0, 0, 1)$.

- a. Is $(0, 0, 0)$ on this line? (Yes / No)
- b. Take $(0, 0, 2)$ and $(0, 0, -3)$. Add them: $(0, 0, -1)$. Is the result still on the line? (Yes / No)
- c. Take $(0, 0, 4)$. Multiply it by -10 to get $(0, 0, -40)$. Is this still on the line? (Yes / No)

2) Consider $V = \{(0, x, 3x) | x \in \mathbb{R}\}$

- a. What would be the value of x so that $(0,0,0) \in V$?
- b. Let $\vec{u} = (0, a, 3a), \vec{v} = (0, b, 3b)$ Show that for any real number α, β ; $\alpha\vec{u} + \beta\vec{v} \in V$



Content Summary

A **subspace** is simply a vector space that lives *inside* a larger vector space (like $\mathbb{R}^3$). It is a set of vectors that is "closed" under the operations of addition and scalar multiplication. In 3D space, subspaces are just **lines and planes** that pass through the **origin**.

Therefore, a sub vector space in $\mathbb{R}^3$ is any subset of vectors that contains the zero vector, closed under vector addition and it is closed under scalar multiplication.

Let us consider $V = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_n\}$ where $\vec{v}_i \in V$ such that $i = 1, 2, 3, \dots, n$.

The vector V is a **vector space** if the following 10 properties are satisfied :

For all vectors $u, v, w \in V$ and scalars $a, b \in F$

- 1) $\forall u, v \in V$, then $u + v \in V$ (closure under Addition)
- 2) $u + v = v + u$ (Commutative)
- 3) $(u + v) + w = u + (v + w)$ (Associative property of Addition)
- 4) $\exists 0 \in V : u + 0 = u, \forall u \in V$ (Existence of Additive Identity)
- 5) $\forall u \in V, \exists -u \in V : u + (-u) = 0$ (Additive Inverse)
- 6) $\forall a \in F, u \in V, au \in V$ (Closure under scalar multiplication)
- 7) $\forall a, b \in F, u \in V, a(bu) = (ab)u$ (Associative of scalar multiplication)
- 8) $\forall 1 \in F, u \in V, 1u = u$ (Identity under multiplication)
- 9) $\forall a \in F, u, v \in V, a(u + v) = au + av$ (Distributive over vector addition)
- 10) $\forall a, b \in F, u \in V, (a + b)u = au + bu$ (Distributive over scalar addition)

A subset W of $\mathbb{R}^3$ is called a **sub-vector space**, or just a **subspace**, of $\mathbb{R}^3$ if it has the following properties:

- The zero vector belongs to W , i.e $(0, 0, 0) \in W$,
- W is closed under vector addition, i.e if $\vec{u}, \vec{v} \in W$ then $\vec{u} + \vec{v} \in W$.
- W is closed under scalar multiplication, i.e if $\beta \in \mathbb{R}, \vec{u} \in W$, then $\beta\vec{u} \in W$.

Generally, if $(\mathbb{R}, F, +)$ is a subspace of $(\mathbb{R}, E, +)$, The following conditions are satisfied:

- a) $F \subset E$ b) $\vec{0} \in F$ c) $\vec{u}, \vec{v} \in F, \alpha, \beta \in \mathbb{R}; \alpha\vec{u} + \beta\vec{v} \in F$

Example 1. Consider $A = \{(x, 0, 2x), x \in \mathbb{R}\}$, we can show that $(\mathbb{R}, A, +)$ is a sub-vector space of $\mathbb{R}^3$.

Solution:

- $A \in \mathbb{R}^3$, If we take $x=0$, we see that $(0,0,0)$ belongs to A .
- Consider $\vec{k} = (k, 0, 2k), \vec{t} = (t, 0, 2t) \in A, \alpha, \beta \in \mathbb{R}$

$$\begin{aligned}\alpha\vec{k} + \beta\vec{t} &= \alpha(k, 0, 2k) + \beta(t, 0, 2t) = (\alpha k, 0, 2\alpha k) + (\beta t, 0, 2\beta t) \\ &= (\alpha k + \beta t, 0 + 0, 2\alpha k + 2\beta t) = (\alpha k + \beta t, 0 + 0, 2(\alpha k + \beta t)) \\ &= (y, 0, 2y) \text{ for } y = \alpha k + \beta t\end{aligned}$$

Then; $\alpha\vec{k} + \beta\vec{t} \in A$ therefore, A is a subspace of $\mathbb{R}^3$.

Example 2. Show that a plane β through the origin is a sub vector space of $\mathbb{R}^3$.

Solution:

- Zero: $\vec{0} \in \beta$ i.e $\vec{0} = (0, 0, 0) \in \beta \Rightarrow \beta \subset \mathbb{R}^3$.
- Consider two vectors $\vec{u}$ and $\vec{v}$ in plane β such that $\vec{u} = (x_1, y_1, z_1)$ and $\vec{v} = (x_2, y_2, z_2)$

$$\forall \vec{u}, \vec{v} \in \beta, \vec{u} + \vec{v} \in \beta : (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2) \in \beta \Rightarrow \beta \subset \mathbb{R}^3$$
- Consider a vector $\vec{u}$ in plane β and a scalar $\lambda \in \mathbb{R}$ such that $\vec{u} = (x_1, y_1, z_1)$.

$$\forall \vec{u} \in \beta, \text{ and } \lambda \in \mathbb{R} : \lambda(x_1, y_1, z_1) = (\lambda x_1, \lambda y_1, \lambda z_1) \in \beta \Rightarrow \beta \subset \mathbb{R}^3.$$

As we verify that a plane β through the origin is a sub vector space of $\mathbb{R}^3$, then we conclude that a 2D through the origin is a sub space of 3D.



Application activity 7.1.4

1) Show that the following are or are not sub-vector spaces of $\mathbb{R}^3$

- $F = \{(y, z, 0), y, z \in \mathbb{R}\}$
- $G = \{(2x, 3y, 0), x, y \in \mathbb{R}\}$

2) Consider the set $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

Work out the following:

- Verify whether the **zero vector** belongs to W .
- Check if the set is **closed under vector addition**.
- Check if the set is **closed under scalar multiplication**.

- d. Conclude whether W is a **subspace of $\mathbb{R}^3$ or not**.
- 3) Consider $A = \{(x, 0, 2x), x \in \mathbb{R}\}$. Show that $(\mathbb{R}, A, +, \cdot)$ is a sub-vector space of $\mathbb{R}^3$.

7.1.5 Linear combination of vectors



Activity 7.1.5

- 1) You go to the market; you buy 3 hen and 3 goats to be used in the ceremony. If H is the cost of a hen and G the cost of one goat. To write the expression of the total cost C , we combine the total cost for hens and the total cost for Goats. Complete that combination: $C = 2H + \dots\dots\dots$
- 2) To show that the vector $\vec{w} = (5, 3, -2)$ is a linear combination of vectors $\vec{u} = (1, -1, 0)$ and $\vec{v} = (1, 3, -1)$, we write $(5, 3, -2) = a(1, -1, 0) + b(1, 3, -1)$. Find the real numbers a and b .



Content Summary

A **linear combination** of vectors is simply a combination of vectors through two operations: **scalar multiplication** and **addition**. It explains how new vectors can be formed from other vectors using **scalar multiplication and vector addition**.

The vector $\vec{u}$ is called a linear combination of the vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3$ if there exists scalars c_1, c_2, c_3 such that $\vec{u} = c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\vec{u}_3$

Example 1. Show that every vector $\vec{v} = (x, y, z)$ is the linear combination of unit vectors $\vec{i} = (1, 0, 0)$, $\vec{j} = (0, 1, 0)$ and $\vec{k} = (0, 0, 1)$ of the vector space $\mathbb{R}^3$.

Solution: It is true because $(x, y, z) = x\vec{i} + y\vec{j} + z\vec{k} = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$.

Example 2: Write $\vec{v}$ as a combination of $\vec{u}_1, \vec{u}_2, \vec{u}_3$ where

$$\vec{v} = (3, 10, 7) \text{ and } \vec{u}_1 = (1, 3, -2), \vec{u}_2 = (1, 4, 2), \vec{u}_3 = (2, 8, 1)$$

Solution: A vector $\vec{v}$ is a linear combination of $\vec{u}_1, \vec{u}_2, \vec{u}_3$ if there exist scalars x, y and z such that $\vec{v} = x\vec{u}_1 + y\vec{u}_2 + z\vec{u}_3$

$$(3, 10, 7) = x(1, 3, -2) + y(1, 4, 2) + z(2, 8, 1)$$

$$\Leftrightarrow (3, 10, 7) = (x, 3x, -2x) + (y, 4y, 2y) + (2z, 8z, z)$$

$$\Leftrightarrow (3, 10, 7) = (x + y + 2z, 3x + 4y + 8z, -2x + 2y + z)$$

$$\Rightarrow \begin{cases} x + y + 2z = 3 \\ 3x + 4y + 8z = 10 \\ -2x + 2y + z = 7 \end{cases}$$

By elimination method (you can use any method), we get:

$$\Leftrightarrow \begin{cases} x + y + 2z = 3 \\ y + 2z = 1 \\ 4y + 5z = 13 \end{cases} \Leftrightarrow \begin{cases} x + y + 2z = 3 \\ y + 2z = 1 \\ -3z = 9 \end{cases} \Rightarrow x = 2, y = 7, z = -3$$

$$\text{Thus } \vec{v} = 2\vec{u}_1 + 7\vec{u}_2 - 3\vec{u}_3$$

Let $S = \{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ be a set of vectors in the vector space V . The set of all linear combinations of $\vec{u}_1, \vec{u}_2, \vec{u}_3$ is called the **span** of the set S , denoted by $\text{span}(S)$ or $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$.

Spanning set or a generating set for a vector space

A set of vectors $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ spans a vector space V if every vector $\vec{v} \in V$ can be written as a linear combination of vectors in S . That is $\vec{v} = a_1\vec{v}_1 + a_2\vec{v}_2 + a_3\vec{v}_3$

Example: We claim that the following vectors also form a spanning set of $\mathbb{R}^3$:

$$w_1 = (1, 0, 0), w_2 = (0, 1, 0), w_3 = (0, 0, 1)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbb{R}^3$, then $v = (a, b, c) = aw_1 + bw_2 + cw_3$.

$$\text{For example, } \vec{v} = -5\vec{w}_1 - 6\vec{w}_2 + 2\vec{w}_3$$

Linearly independent vectors:

The set of vectors $S = \{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ of a vector space V is said to be **linearly independent** provided that the equation $c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\vec{u}_3 = \vec{0}$ has only the trivial solution. $c_1 = c_2 = c_3 = 0$

A set of vectors is called **linearly dependent** if it is not linearly independent. Or if $c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\vec{u}_3 = \vec{0}$ for $c_1, c_2, c_3 \neq 0$

Example 1: Show that the vectors $\vec{u} = (1, -1, 0)$, $\vec{v} = (1, 3, -1)$ and $\vec{w} = (5, 3, -2)$ are linearly dependent.

Solution:

$$\text{Since } 3\vec{u} + 2\vec{v} - \vec{w} = \vec{0}$$

$$\text{or we can solve for } c_1(1, -1, 0) + c_2(1, 3, -1) + c_3(5, 3, -2) = \vec{0}$$

$$\begin{cases} c_1 + c_2 + 5c_3 = 0 & (1) \\ -c_1 + 3c_2 + 3c_3 = 0 & (2) \\ -c_2 - 2c_3 = 0 & (3) \end{cases}$$

From Equation (3) we have $c_2 = -2c_3$, substitute this into Equation (1) yields $c_1 + (-2c_3) + 5c_3 = 0$ implies that $c_1 = -3c_3$. Then put $c_1 = -3c_3$ and $c_2 = -2c_3$ into Equation (2) to get $-(-3c_3) + 3(-2c_3) + 2c_3 = 0$, implies that $3c_3 - 6c_3 + 2c_3 = 0 \Rightarrow c_3 = 0$. After substitution $c_1 = c_2 = c_3 = 0$

Therefore, the given three vectors are linearly Independent, not linearly dependent

Other Alternative,

$$\det \begin{pmatrix} 1 & 1 & 5 \\ -1 & 3 & 2 \\ 0 & -1 & -2 \end{pmatrix} = -1 \neq 0$$

Therefore, it is linearly Independent.

Example 2: Show that the vectors $\vec{u} = (1, 0, 0)$, $\vec{v} = (0, 1, 0)$ and $\vec{w} = (0, 0, 1)$ are linearly independent.

Solution:
$$\begin{cases} c_1 + 0c_2 + 0c_3 = 0 & (1) \\ 0c_1 + c_2 + 0c_3 = 0 & (2) \\ 0c_1 + 0c_2 + c_3 = 0 & (3) \end{cases}$$

This system has only trivial solution, and $c_3 = 0$. Therefore, the given three vectors are linearly independent.

Theorem 6.1: The three vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3$ in $\mathbb{R}^3$ are **linearly independent** if and only if the 3×3 matrix $A = [\vec{u}_1 \ \vec{u}_2 \ \vec{u}_3]$ with the vectors as columns have nonzero determinant.

Example 3: Show that the vectors $\vec{u} = (1, 2, 0)$, $\vec{v} = (0, -1, 3)$, $\vec{w} = (-1, 0, 2)$ are linearly independent.

Solution: The three vectors are linearly independent if

$$\begin{vmatrix} 1 & 0 & -1 \\ 2 & -1 & 0 \\ 0 & 3 & 2 \end{vmatrix} \stackrel{?}{=} 0 \Leftrightarrow 1 \begin{vmatrix} -1 & 0 \\ 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix} \stackrel{?}{=} 0 \Rightarrow -2 - 6 = -8 \neq 0 \quad c_1 = 0, \quad c_2 = 0$$

Thus, the given vectors are linearly independent.

Theorem 6.2: The three vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3$ in $\mathbb{R}^3$ are **linearly dependent** if and only if the 3×3 matrix $A = [\vec{u}_1 \ \vec{u}_2 \ \vec{u}_3]$ with the vectors as columns has **zero determinant**.

Example 4: Show that the vectors $\vec{u} = (1, 2, 3)$, $\vec{v} = (1, -1, -2)$, $\vec{w} = (2, 1, 1)$ are linearly dependent

Solution: The three vectors are linearly dependent if:

$$\begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 3 & -2 & 1 \end{vmatrix} \stackrel{?}{=} 0 \Leftrightarrow 1 \begin{vmatrix} -1 & 1 \\ -2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} \stackrel{?}{=} 0 \Rightarrow 1 + 1 - 2 = 0$$

Thus, the given vectors are linearly dependent.



Application activity 7.1.5

- 1) Write the vector $\vec{v} = (1, -2, 5)$ as a linear combination of the vectors $\vec{e}_1 = (1, 1, 1)$, $\vec{e}_2 = (1, 2, 3)$ and $\vec{e}_3 = (2, -1, 1)$.
- 2) Show that the vectors $\vec{u} = (1, 2, 3)$, $\vec{v} = (0, 1, 2)$ and $\vec{w} = (0, 0, 1)$ generate $\mathbb{R}^3$.
- 3) Which of the following collections of vectors are linearly independent in $\mathbb{R}^3$?
 - a) $(1, 1, 1), (1, 1, 0), (1, 0, 0)$
 - b) $(1, 0, 1), (0, 1, 0)$
 - c) $(1, 2, 4), (2, 1, 3), (4, -1, 1)$
- 4) Answer by true or False. The following vectors are linearly independent:
 - (a) the vectors $(1, 3, 2)$, $(2, 1, 3)$, and $(3, 2, 1)$.
 - (b) the vectors $(1, -3, 2)$, $(2, 1, -3)$, and $(-3, 2, 1)$

7.1.6 Basis and dimension of a vector space



Activity 7.1.6

Consider the set of 3 vectors $\{(1, 2, 0), (0, -1, 3), (-1, 0, 2)\}$ and the vector $\vec{u} = (3, 1, 4)$.

- a) Verify if the 3 vectors $\{(1, 2, 0), (0, -1, 3), (-1, 0, 2)\}$ are linearly independent.
- b) Find the value of a, b, c such that $(3, 1, 4) = a(1, 2, 0) + b(0, -1, 3) + c(-1, 0, 2)$.



Content Summary

Basis

A finite set S of vectors in a vector space V is called a **basis for V** if

- The vectors in S are linearly independent
- The vector in S span V (or S is a generating set of V).

A **basis** is simply a set of vectors that act as the "building blocks" or "coordinate system" for a vector space.

Example: The set of standard unit vectors in $\mathbb{R}^3$, $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is the **standard** (or **usual** or **canonical**) **basis** for $\mathbb{R}^3$.

Theorem 6.3: Let $S = \{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ be a basis for the vector space V . Then, any set of more than three vectors in V is linearly dependent.

Theorem 6.4: Any two bases of a vector space V consist of the same number of vectors.

Dimension

Let V be a vector space. If V has a basis consisting of n vectors we say that V has **dimension** n . The **dimension** is simply the **number of vectors** needed to form a basis. The sub space $\{\vec{0}\}$ of V is said to have dimension 0. V is said to be **finite dimensional** if there is a finite set of vectors that spans V , otherwise, we say that V is **infinite-dimensional**.

Theorem 6.5: If V is a vector space of dimension $n > 0$, then

- Any set of n linearly independent vectors spans V ;
- Any n vectors that span V are linearly independent.

Example: Given the vectors $\vec{u} = (3, -2, 4)$, $\vec{v} = (3, 2, -4)$, and $\vec{w} = (-6, 4, -8)$.

What is the dimension of $\text{span}(\vec{u}, \vec{v}, \vec{w})$?

Answer:

First, we verify that these are linearly independent as:

$$\begin{vmatrix} 3 & 3 & -6 \\ -2 & 2 & 4 \\ 4 & -4 & -8 \end{vmatrix} = 0. \text{ These results imply that } \dim \neq 3, \text{ but } \begin{vmatrix} 3 & 3 \\ -2 & 2 \end{vmatrix} \neq 0.$$

These results imply that $\dim$ of $\text{span}(\vec{u}, \vec{v}, \vec{w}) = 2$.

Theorem 6.6

A vector space V is called **finite dimensional** if it has a basis consisting of a finite number of vectors. The unique number of vectors in each basis for V is called the **dimension** of V and is denoted by $\dim(V)$. A vector space that is not finite dimensional is called **infinite dimensional**.

Example 1: Consider a vector space E of polynomials of degree 3 of x . Any element of E is of the form $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, $a_i \in \mathbb{R}$.

We verify that $(1, x, x^2, x^3)$ is a basis of E because $a_0 + a_1x + a_2x^2 + a_3x^3 = 0$ implies $a_0 = a_1 = a_2 = a_3 = 0$.

Moreover, $(1, x, x^2, x^3)$ is a generating set of E . This set has 4 elements, then $\dim(E) = 4$.

Example 2: We have seen that the set $\{\vec{e}_1 = (1, 0, 0), \vec{e}_2 = (0, 1, 0), \vec{e}_3 = (0, 0, 1)\}$ is the basis for $\mathbb{R}^3$.

Thus $\mathbb{R}^3$ has dimension 3.



Application activity 7.1.6

- 1) Show that $\{(1, 2, 3); (-2, 1, 0); (1, 0, 1)\}$ is a basis for $\mathbb{R}^3$.
- 2) Determine whether each of the following from a basis of $\mathbb{R}^3$:

a. $(1,1,1), (1,0,1);$

b. $(1,2,3), (1,3,5), (1,0,1), (2,3,0)$

3) Given the vectors $\vec{v}_1 = (3, -2, 4); \vec{v}_2 = (-3, 2, -4)$ and $\vec{v}_3 = (-6, 4, -8)$ what is the dimension of $\text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$?

4) Consider the vectors $\vec{u} = (1, -3, 2), \vec{v} = (2, -1, 1)$ and $\vec{w} = (1, 7, -4)$ of $\mathbb{R}^3$.

Verify if $\{\vec{u}, \vec{v}, \vec{w}\}$ is a basis of $\mathbb{R}^3$?

7.1.7 Coordinates of a vector relative to the given base



Activity 7.1.7

Consider the vector $\vec{u} = (2, 4, -1)$ and the basis $\{(3, 0, 0), (0, 3, 0), (0, 0, 3)\}$. Find the value of a, b, c such that $(2, 4, 1) = a(3, 0, 0) + b(0, 3, 0) + c(0, 0, 3)$. Can every vector of $\mathbb{R}^3$ be written as a linear combination of vectors of the given basis?



Content Summary

A coordinate vector is the set of "instructions" that tells you how to reach a specific point using a specific set of building blocks (a basis).

While a vector $\vec{v}$ is an absolute geometric object (an arrow in space), its **coordinate vector** depends on which basis you are using to describe it.

Suppose that $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis for a vector space V and that $\vec{u}$ is any vector from V . As $\vec{u}$ is a vector in V it can be expressed as a linear combination of the vectors from S as follows,

$$\vec{u} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3$$

The scalars c_1, c_2, c_3 are called the **coordinates** of $\vec{u}$ relative to the basis S . The **coordinate vector of $\vec{u}$ relative to S** is denoted by $[\vec{u}]_S$ and defined to be the following vector in $\mathbb{R}^3$.

$$[\vec{u}]_S = (c_1, c_2, c_3)$$
 The coordinate vector of vector $\vec{u}$ is unique.

Example 1: Determine the coordinate vector of $\vec{x} = (10, 5, 0)$ relative to the following bases.

a) The standard basis vectors for $\mathbb{R}^3$

b) The basis $A = \{\vec{e}_1 = (1, -1, 1), \vec{e}_2 = (0, 1, 2), \vec{e}_3 = (3, 0, -1)\}$

Solution: In each case we will need to determine how to write $\vec{x} = (10, 5, 0)$ as a linear combination of the given basis vectors.

c) The standard basis vectors for $\mathbb{R}^3$ is $\{\vec{e}_1 = (1, 0, 0), \vec{e}_2 = (0, 1, 0), \vec{e}_3 = (0, 0, 1)\}$

In this case the linear combination is simple to write down.

$$\vec{x} = (10, 5, 0) = 10(1, 0, 0) + 5(0, 1, 0) + 0(0, 0, 1)$$

And so, the coordinate vector for $\vec{x}$ relative to the standard basis vectors is $[\vec{x}]_S = (10, 5, 0)$. So, in the case of the standard basis vectors we have got that. This is, of course, what makes the standard basis vectors so nice to work with. The coordinate vectors relative to the standard basis vectors is just the vector itself.

d) The basis $A = \{\vec{e}_1 = (1, -1, 1), \vec{e}_2 = (0, 1, 2), \vec{e}_3 = (3, 0, -1)\}$

Now, in this case we will have a little work to do. We will first need to set up the following vector equation, $(10, 5, 0) = c_1(1, -1, 1) + c_2(0, 1, 2) + c_3(3, 0, -1)$ and we will need to determine the scalars c_1, c_2 and c_3 . We saw how to solve this kind of vector equation in both the section on Span and the section on Linear Independence. We need to set up the following system of equations:

$$\begin{cases} c_1 + 3c_3 = 10 \\ -c_1 + c_2 = 5 \\ c_1 + 2c_2 - c_3 = 0 \end{cases} \Rightarrow \begin{cases} c_1 = -2 \\ c_2 = 3 \\ c_3 = 4 \end{cases}$$

The coordinate vector for $\vec{x}$ relative to A is then, $[\vec{x}]_A = (-2, 3, 4)$

Example 2: Consider real space $\mathbb{R}^3$. The following vectors form a basis S of $\mathbb{R}^3$

$$\vec{u}_1 = (1, 3, -2), \vec{u}_2 = (1, 4, 2), \vec{u}_3 = (2, 8, 1).$$

Find the coordinate of $\vec{v} = (3, 10, 7)$ relative to the basis S .

Solution: These vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3$ form a basis S of $\mathbb{R}^3$ iff any are linearly independent and any vector of $\mathbb{R}^3$ is linearly combination of $\vec{u}_1, \vec{u}_2, \vec{u}_3$.

The coordinate of $\vec{v}$ is denoted as $[\vec{v}]_S = (x, y, z)$.

Find x, y and z are the points of coordinate of $\vec{v}$ in S , we have:

$$\begin{aligned} (3, 10, 7) &= x(1, 3, -2) + y(1, 4, 2) + z(2, 8, 1) \Leftrightarrow (3, 10, 7) = (x, 3x, -2x) + (y, 4y, 2y) + (2z, 8z, z) \\ &\Leftrightarrow (3, 10, 7) = (x + y + 2z, 3x + 4y + 8z, -2x + 2y + z) \Rightarrow \begin{cases} x + y + 2z = 3 \\ 3x + 4y + 8z = 10 \\ -2x + 2y + z = 7 \end{cases} \end{aligned}$$

By elimination method (you can use any method), we get:

$$\Rightarrow \begin{cases} x + y + 2z = 3 \\ y + 2z = 10 \\ 4y + 5z = 7 \end{cases} \Leftrightarrow \begin{cases} x + y + 2z = 3 \\ y + 2z = 10 \\ -3z = 9 \end{cases} \Rightarrow x = 2, y = 7 \text{ and } z = -3.$$

Thus, $\vec{v} = (2, 7, -3)$ in basis S .



Application activity 7.1.7

- Find the coordinate vector of the vector $\vec{u} = (3, 1, -4)$ and $\vec{v} = (3, -2, 1)$ relative to the basis $V = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$
- Find the coordinate vector of $\vec{v}$ relative to the basis $\{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ where
 - $\vec{v} = (4, -3, 2)$
 - $\vec{v} = (a, b, c)$
- Let V be the vector space of polynomials with degree less than or equal to 2: $V = \{at^2 + bt + c; a, b, c \in \mathbb{R}\}$. Find the coordinate vector of $\vec{v}$ relative to the basis $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ where $\vec{e}_1 = 1$, $\vec{e}_2 = t - 1$ and $\vec{e}_3 = (t - 1)^2$.

7.2 Euclidian vector space

7.2.1 Scalar or dot product of two vectors and properties



Activity 7.2.1

The **dot product** is a way to multiply two vectors to find a single number (a scalar).

Use the formula $(a, b, c) \cdot (d, e, f) = ad + be + cf$ to find:

a) $(2, 3, 4) \cdot (1, -2, 1)$

b) $(1, -2, 4) \cdot (1, 0, -1)$



Content Summary

The scalar product or dot product (or inner product) is an algebraic operation that takes two coordinate vectors and returns a single number. That is, the scalar product of two vectors of space is the application. $E_0 \times E_0 \rightarrow \mathbb{R}$

Algebraically, it is the sum of the products of the corresponding coordinates of the two vectors.

That is, the scalar product of vectors $\vec{u} = (a_1, a_2, a_3)$ and $\vec{v} = (b_1, b_2, b_3)$ of space is defined by

$$\vec{u} \cdot \vec{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Properties of scalar product

$$\forall \vec{u}, \vec{v} \in E_0$$

- a. If $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$, then $\vec{u} \cdot \vec{v} = 0$
- b. If $\vec{u} \parallel \vec{v}$ and $\vec{u}, \vec{v}$ have same direction, then $\vec{u} \cdot \vec{v} > 0$
- c. If $\vec{u} \parallel \vec{v}$ and $\vec{u}, \vec{v}$ have opposite direction, then $\vec{u} \cdot \vec{v} < 0$
- d. If $\vec{u} \perp \vec{v}$, then $\vec{u} \cdot \vec{v} = 0$
- e. Symmetry : $\forall \vec{u}, \vec{v} \in E_0, \vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- f. Bi-linearity : $\forall \vec{u}, \vec{v}, \vec{w} \in E_0, \vec{u} \cdot (a\vec{v} + b\vec{w}) = a\vec{v} \cdot \vec{u} + b\vec{w} \cdot \vec{u}, (\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$
- g. Positive definiteness : $\forall \vec{u} \in E_0 \setminus \{\vec{0}\}, \vec{u} \cdot \vec{u} > 0$

We define the square of $\vec{u}$ to be $\vec{u} \cdot \vec{u} = (\vec{u})^2$

Example: Find:

- a) The scalar product of $\vec{u} = (2, 3, 4)$ and $\vec{v} = (1, -2, 2)$
- b) The square of $\vec{u} = (2, 3, 4)$

Solution:

The scalar product of $\vec{u} = (2, 3, 4)$ and $\vec{v} = (1, -2, 2)$ is $\vec{u} \cdot \vec{v} = 2 - 6 + 8 = 4$

The square of $\vec{u} = (2, 3, 4)$ is $\vec{u} \cdot \vec{u} = 4 + 9 + 16 = 29$

Application of the scalar product: The following applications of the scalar product are going to be discussed.

1. The length of a vector $\mathbf{v}$ is the square root of its inner product by itself.
2. The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ is the length of their difference.
3. The cosine of the angle between two vectors equals their inner product divided by the product of their lengths.
4. Two vectors are perpendicular or orthogonal when their inner product equals zero.



Application activity 7.2.1

Find the scalar product $\vec{u} \cdot \vec{v}$

if

- a) $\vec{u} \cdot \vec{v} = (2, 3, 4)$ and $\vec{v} = (12, -3, 0)$
- b) $\vec{u} = (1, -2, -14)$ and $\vec{v} = (22, 0, 0)$
- c) $\vec{u} = (21, 4, -2)$ and $\vec{v} = (0, -1, 0)$
- d) $\vec{u} = (1, 0, 0)$ and $\vec{v} = (3, 3, 3)$

7.2.2 Magnitude (or normal or length) of a vector



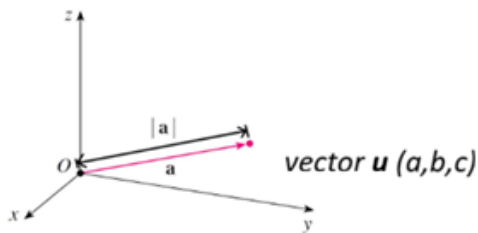
Activity 7.2.2

The magnitude of a vector $\vec{v} = (a, b, c)$ noted as $\|\vec{v}\| = \|(a, b, c)\|$ or simply $|\vec{v}| = |(a, b, c)|$ is a measure of its "size" or "length." It represents the distance from its starting point to its end point. Use the formula $\|(a, b, c)\| = \sqrt{a^2 + b^2 + c^2}$ or $|(a, b, c)| = \sqrt{a^2 + b^2 + c^2}$ to find a) $\|(2, 3, 4)\|$ b) $\|(1, -2, 1)\|$



Content Summary

$\forall \vec{u} \in E_0$ and $\vec{u} = (a, b, c)$, the magnitude of the vector $\vec{u}$ noted by $\|\vec{u}\|$ is defined as its length and is the square root of its square. $\forall \vec{u} \in E_0, \|\vec{u}\|^2 = \vec{u} \cdot \vec{u} = (\vec{u})^2$. Therefore, $\forall \vec{u} \in E_0, \|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}}$.



That is $\forall \vec{u} \in E_0, \|\vec{u}\| = \sqrt{(\vec{u})^2}$ or $\|\vec{u}\|^2 = (\vec{u})^2$ Given that $\vec{u} \cdot \vec{u} = (\vec{u})^2 = a^2 + b^2 + c^2$, $\|\vec{u}\| = \sqrt{a^2 + b^2 + c^2}$.

Note that the notation of absolute value $|\cdot|$ is also used for the magnitude of a vector. That is the magnitude of a vector $\vec{u}$ is also noted by $|\vec{u}|$.

Consequences

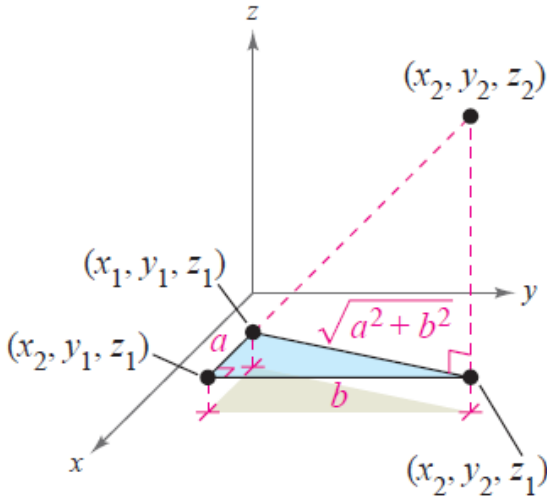
a) $\forall \vec{u} \in E_0$, if $\vec{u} = \vec{0}$ then $\|\vec{u}\| = 0$

b) $\forall \vec{u} \in E_0, k \in \mathbb{R}, \|k\vec{u}\| = |k| \|\vec{u}\|$

Vector joining two points: If A and B are two points, we can form a vector $\overrightarrow{AB}$ and the distance between these two points denoted $d(A, B)$ is given by $\|\overrightarrow{AB}\|$.

Thus, if $A(a_1, a_2, a_3)$ and $B(b_1, b_2, b_3)$ then

$$d(A, B) = \|\overrightarrow{AB}\| = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}$$



- Consider two vectors $\vec{u}$ and $\vec{v}$ on the same line:
If they have the same direction, then $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\|$
If they have the opposite direction, then $\vec{u} \cdot \vec{v} = -\|\vec{u}\| \|\vec{v}\|$
- Let θ be the angle between two vectors $\vec{u}$ and $\vec{v}$
If θ is an obtuse angle, then the scalar product $\vec{u} \cdot \vec{v}$ is negative
If θ is an acute angle, then the scalar product $\vec{u} \cdot \vec{v}$ is positive
- Unit vector:** A vector $\vec{u}$ is said to be **unit vector** if and only if its magnitude is 1. That is $\|\vec{u}\| = 1$
- Normalized vector:** The normalized vector of a vector is a vector in the same direction but with magnitude 1. It is also called the unit vector. Given a vector $\vec{v}$, the **normalized vector** parallel to $\vec{v}$ and with same direction is given by $\frac{\vec{v}}{\|\vec{v}\|}$.

Therefore, a **unit vector** $\vec{u}$ in the direction of $\vec{v}$ is $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$, $\vec{v} \neq \mathbf{0}$.

Example 1: The magnitude of $\vec{u} = (3, 2, 4)$ is $\|\vec{u}\| = \sqrt{9 + 4 + 16} = \sqrt{29}$

Example 2: The distance between $A(1, -1, 3)$ and $B(2, 4, 5)$ is

$$d(A, B) = \sqrt{(2-1)^2 + (4+1)^2 + (5-3)^2} = \sqrt{1 + 25 + 4} = \sqrt{30}$$

Example 3. The normalized vector parallel to $\vec{v} = (3, 2, 4)$ and with the same direction is given

$$\text{by } \vec{e} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\|\vec{v}\|} \vec{v} = \frac{1}{\sqrt{4+16+16}} \vec{v} = \frac{1}{6} (2, 4, 4) \text{ which is } \vec{e} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

Example 4: For $(2,2,1)$, the distance from the origin is $\sqrt{2^2 + 2^2 + 1^2} = \sqrt{9} = 3$ Length Unit (LU)

Example 5: Find the unit vector of this given vector $\vec{r} = (1, -2, 4)$.

Step1: Finding the magnitude of $\vec{r}$ which is $|\vec{r}| = \sqrt{1^2 + (-2)^2 + 4^2} = \sqrt{21}$ Length Unit

Step2: Finding the unit vector of $\vec{r}$ which is $\vec{u} = \frac{\vec{r}}{|\vec{r}|} = \frac{(1, -2, 4)}{\sqrt{21}} = \left(\frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}, \frac{4}{\sqrt{21}} \right)$.

Meaning of Euclidian vector space:

Let V be a vector space over the real numbers. V is called a Euclidean vector space if:

1. V is finite-dimensional and satisfies all the usual vector space axioms such as closure under addition and scalar multiplication, associativity, commutativity, and existence of zero vector.
2. There exists a function (scalar product) that takes two vectors and returns a real number, called the inner product, satisfying the properties seen here above: Bi-linearity, symmetry and positive definiteness.

Meaning of the Standard Euclidean Space $\mathbb{R}^n$

The most common example is $\mathbb{R}^n$ with the standard dot product:

For vectors $\vec{u} = (u_1, u_2, \dots, u_n)$ and $\vec{v} = (v_1, v_2, \dots, v_n)$, their dot product is:
 $\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$.



Application activity 7.2.2

Find the magnitude of

a) $\vec{u} = (21, 4, -2)$

b) $\vec{u} = (3, 3, 3)$

c) $\vec{u} = (22, 0, 0)$

d) $\vec{u} = (1, -2, -14)$

7.2.3 Angle between two vectors and the direction cosines



Activity 7.2.3

The angle θ between two vectors tells you how they are oriented relative to each other. It is derived directly from the dot product formula.

Consider two vectors $\vec{u} = (2, 2, 2)$ and $\vec{v} = (3, 3, 3)$

1) Find the scalar product $\vec{u} \cdot \vec{v}$

2) Find the product $\|\vec{u}\| \|\vec{v}\|$

3) Given that $\vec{u} \cdot \vec{v} = \|\vec{u}\| \cdot \|\vec{v}\| \cos(\vec{u}, \vec{v})$, find the angle θ between the two vectors $\vec{u}$ and $\vec{v}$



Content Summary

The angle θ between two non-zero vectors $\vec{u}$ and $\vec{v}$ in 3D space is the smaller angle ($0 \leq \theta \leq \pi$) formed when they are placed tail-to-tail.

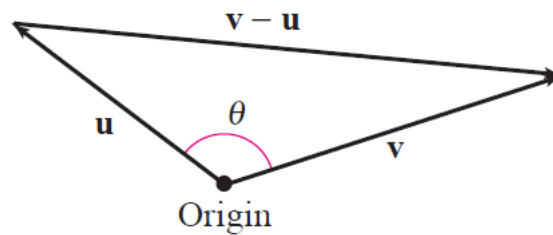
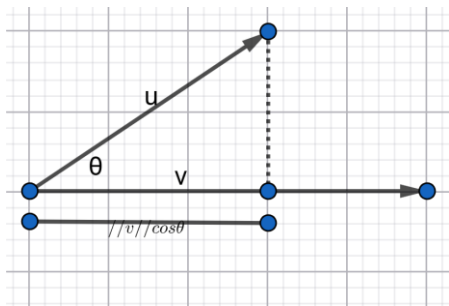
Direction cosines are the cosines of the angles that a vector makes with the positive coordinate axes. They completely specify the **direction** of a vector, independent of its magnitude.

Consider two non-zero vectors $\vec{u}$ and $\vec{v}$. Geometrically, the scalar product of $\vec{u}$ and $\vec{v}$ is .

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \cdot \|\vec{v}\| \cdot \cos(\vec{u}, \vec{v})$$

From this definition we can write. $\cos(\vec{u}, \vec{v}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|}$

Then the angle between vectors $\vec{u}$ and $\vec{v}$ is equal to $\cos^{-1}\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|}\right)$



Note that when we are calculating the angle between two vectors, we calculate the smallest positive angle.

Properties:

- If the two vectors are perpendicular their scalar product is zero means that the angle between them is $\frac{\pi}{2}$ (if the second is upward) or $\frac{-\pi}{2}$ (if the second is downward). Thus if $\vec{u} \perp \vec{v}$ then $\vec{u} \cdot \vec{v} = 0$
- If the two vectors are parallel then $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\|$ or, $\vec{u} \cdot \vec{v} = -\|\vec{u}\| \|\vec{v}\|$ means that the angle between them is 0 (if they have the same direction) or π (if they have the opposite direction).
- A vector is said to be normal vector or simply the normal to a surface if it is perpendicular to that surface. Often, the normal unit vector is desired, which is sometimes known as the unit normal.
- Cauchy-Schwarz Inequality: The absolute value of the inner product of two vectors is always less than or equal to the product of their lengths: $|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \cdot \|\vec{v}\|$

- Triangle Inequality: The length of the sum of two vectors is less than or equal to the sum of their lengths: $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$
- Pythagorean Theorem: If two vectors are perpendicular, the square of the length of their sum equals the sum of the squares of their length: If $\vec{u} \perp \vec{v}$, $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$.

Example 1: Consider the vector $\vec{u} = (3, 8, 1)$. What is the measure of the angle between this vector and z-axis of coordinates system?

Solution

$$\vec{u} = (3, 8, 1)$$

Take the normal vector on z-axis, $\vec{e} = (0, 0, 1)$

We need $\theta = \angle(\vec{u}, \vec{e})$

$$\cos \theta = \frac{\vec{u} \cdot \vec{e}}{\|\vec{u}\| \|\vec{e}\|} = \frac{1}{\sqrt{9+64+1} \cdot \sqrt{1}} = \frac{1}{\sqrt{74}}$$

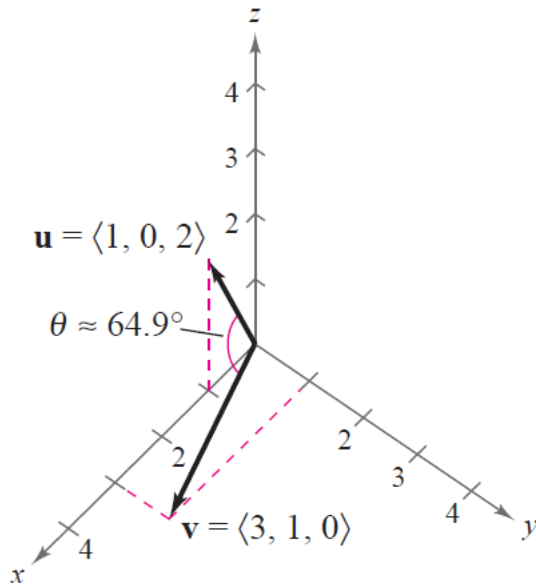
$$\cos \theta = \frac{1}{\sqrt{74}} \Leftrightarrow \theta = \arccos \frac{1}{\sqrt{74}} = 83.3^\circ$$

Example 2: Find angle between $u = (1, 0, 2)$ and $v = (3, 1, 0)$

Solution: $\|\mathbf{u}\| = \sqrt{1^2 + 0^2 + 2^2} = \sqrt{1+0+4} = \sqrt{5}$

$$\|\mathbf{v}\| = \sqrt{3^2 + 1^2 + 0^2} = \sqrt{9+1+0} = \sqrt{10}$$

$$\cos \theta = \frac{3}{\sqrt{5}\sqrt{10}} = \frac{3}{\sqrt{50}} = \frac{3}{5\sqrt{2}} \text{ and } \theta = \arccos\left(\frac{3}{\sqrt{50}}\right)$$

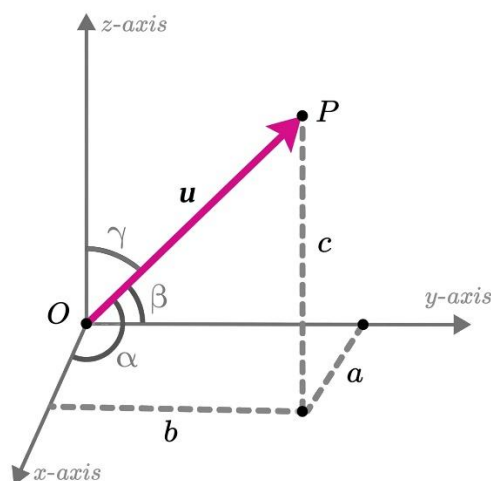
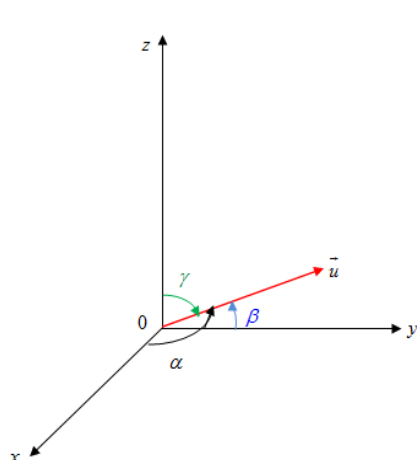


Direction cosine of a vector

Direction cosine (or directional cosine) of a vector is the angles between the vector and the three coordinates' axes. Or equivalently it is the component contributions of the basis to the unit vector.

The direction cosines of the vector $\vec{v} = (x, y, z)$ are:

$$\cos \alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}}, \cos \beta = \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \cos \gamma = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$



Note that the sum of squares of direction cosines of a vector is 1.

$$\text{In fact, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \left(\frac{x}{\|\vec{u}\|} \right)^2 + \left(\frac{y}{\|\vec{u}\|} \right)^2 + \left(\frac{z}{\|\vec{u}\|} \right)^2 = \frac{x^2 + y^2 + z^2}{\|\vec{u}\|^2} = \frac{\|\vec{u}\|^2}{\|\vec{u}\|^2} = 1$$

$$\text{Thus, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Example: Determine the direction cosines of the vector with components $(1, 2, -3)$.

Solution

$$\cos \alpha = \frac{1}{\sqrt{1^2 + 2^2 + (-3)^2}} = \frac{1}{\sqrt{14}}$$

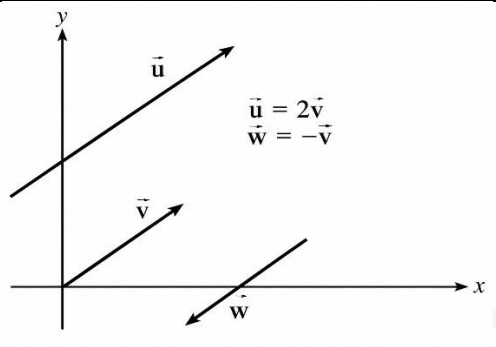
$$\cos \beta = \frac{2}{\sqrt{1^2 + 2^2 + (-3)^2}} = \frac{2}{\sqrt{14}}$$

$$\cos \gamma = \frac{-3}{\sqrt{1^2 + 2^2 + (-3)^2}} = \frac{-3}{\sqrt{14}}$$

Parallel vector:

Recall from the definition of scalar multiplication that positive scalar multiples of a nonzero vector v have the same direction as v , whereas negative multiples have the direction opposite that of v . In general, two nonzero vectors u and v are **parallel** if there is some scalar c such that $u = cv$

For example, the vectors u , v , and w are parallel because $u = 2v$ and $w = -v$



Vector w has initial point $(1, -2, 0)$ and terminal point $(3, 2, 1)$. Which of the following vectors is parallel to w ?

- a. $u = \langle 4, 8, 2 \rangle$ b. $v = \langle 4, 8, 4 \rangle$

Solution:

$$w = \langle 3 - 1, 2 - (-2), 1 - 0 \rangle = \langle 2, 4, 1 \rangle$$

$$u = \langle 4, 8, 2 \rangle = 2\langle 2, 4, 1 \rangle = 2w; \text{ You can conclude that } u \text{ is parallel to } w.$$

In this case, you need to find a scalar c such that $\langle 4, 8, 4 \rangle = c\langle 2, 4, 1 \rangle$. However, equating corresponding components produces $c = 2$ for the first two components and $c = 4$ for the third. So, the equation has no solution, and the vectors v and w are *not* parallel.



Application activity 7.2.3

1. Find the angle formed by the vectors

a) $\vec{u} = (2, 3, 4)$ and $\vec{v} = (12, -3, 0)$

b) $\vec{u} = (1, -2, -14)$ and $\vec{v} = (22, 0, 0)$

2. Find the direction cosines of the vector

a) $\vec{u} = (2, 3, 4)$

b) $\vec{v} = (12, -3, 0)$

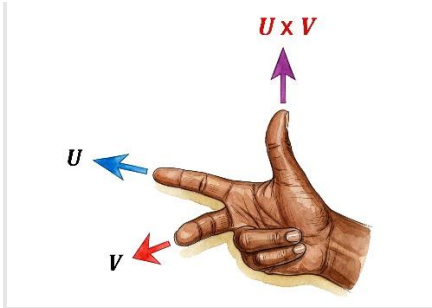
7.2.4 Vector product and its properties



Activity 7.2.4

The vector product (or cross product) is an operation that takes two vectors and produces a new vector that is perpendicular to both.

1) Consider vectors $\vec{u} = (2, -1, 3)$ and $\vec{v} = (1, 2, -1)$. Represent them in the cartesian plane and find the vector $\vec{w}$ that is perpendicular to both $\vec{u}$ and $\vec{v}$, we note it as $\vec{u} \times \vec{v}$:



2) Calculate the determinant

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 3 \\ 1 & 2 & -1 \end{vmatrix}$$

3) Comment on results in 1 and 2



Content Summary

The **vector product** (or **cross product** or **Gibbs vector product**) is a binary operation on two vectors in three-dimensional space. It results in a vector which is perpendicular to both of the vectors being multiplied and therefore normal to the plane containing them.

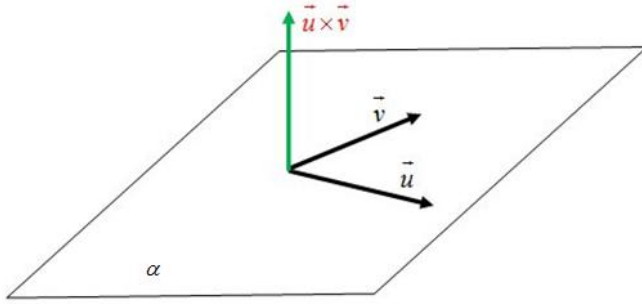
The direction of the resultant vector can be determined by the right-hand rule. The thumb and index finger held perpendicularly to one another represent the vectors and the middle finger held perpendicularly to the index and thumb indicates the direction of the cross vector.

Consider $\{\vec{i} = (1, 0, 0), \vec{j} = (0, 1, 0), \vec{k} = (0, 0, 1)\}$, a positive orthonormal basis of E_0 and two linearly independent vectors $\vec{u} = (a_1, a_2, a_3)$ and $\vec{v} = (b_1, b_2, b_3)$. The vector product of $\vec{u}$ and $\vec{v}$ is noted $\vec{u} \times \vec{v}$.

From activity 6.11,

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k} \quad \text{Or}$$

$$\vec{u} \times \vec{v} = \left(\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}, -\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}, \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \right)$$



Example: Find the vector product of $\vec{u} = (2, 3, 5)$ and $\vec{v} = (-2, 5, 6)$

Solution:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 5 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 5 & 6 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 5 \\ -2 & 6 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} \vec{k} = -7\vec{i} - 22\vec{j} + 16\vec{k}$$

Or

$$\vec{u} \times \vec{v} = (-7, -22, +16)$$

Properties of vector product

1. If $\vec{w}$ is vector product $\vec{u}$ and $\vec{v}$ then $\vec{w} \perp \vec{u}$ and $\vec{w} \perp \vec{v}$
2. The vector product is anti-symmetric:
 $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$.
3. If $\vec{u} = \vec{0}$ and $\vec{v} = \vec{0}$ then $\vec{u} \times \vec{v} = \vec{0}$
4. If two vectors are linearly dependent, then their vector product is a zero vector.
5. If $\vec{u} \times \vec{v} = \vec{0}$ then $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$
6. The vector product is bilinear:
 $\vec{u} \times (r\vec{v} + s\vec{w}) = r(\vec{u} \times \vec{v}) + s(\vec{u} \times \vec{w})$
 $(r\vec{u} + s\vec{v}) \times \vec{w} = r(\vec{u} \times \vec{w}) + s(\vec{v} \times \vec{w})$
7. The vector product is not associative:
 $(\vec{u} \times \vec{v}) \times \vec{w} \neq \vec{u} \times (\vec{v} \times \vec{w})$.

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in space and let c be a scalar.

1. $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$
2. $\vec{u} \times (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) + (\vec{u} \times \vec{w})$
3. $c \times (\vec{u} \times \vec{v}) = (c \times \vec{u}) \times \vec{v} = \vec{u} \times (c \times \vec{v})$
4. $\vec{u} \times \vec{0} = \vec{0} \times \vec{u} = \vec{0}$
5. $\vec{u} \times \vec{u} = \vec{0}$
6. $\vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$

Example:

Find a unit vector that is orthogonal to both $\mathbf{u} = 3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = -3\mathbf{i} + 6\mathbf{j}$.

Solution

The cross product $\mathbf{u} \times \mathbf{v}$, as shown in Figure 11.22, is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

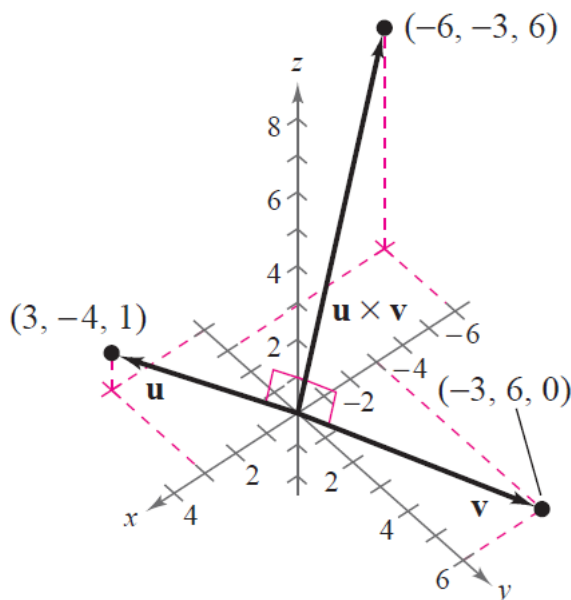
$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -4 & 1 \\ -3 & 6 & 0 \end{vmatrix} = -6\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$$

Because

$$\|\mathbf{u} \times \mathbf{v}\| = \sqrt{(-6)^2 + (-3)^2 + 6^2} = \sqrt{81} = 9$$

a unit vector orthogonal to both u and v is

$$\frac{\mathbf{u} \times \mathbf{v}}{\|\mathbf{u} \times \mathbf{v}\|} = -\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}.$$



...



Application activity 7.2.4

- If $\vec{a} = (2, 1, -1)$ and $\vec{b} = (-3, 4, 1)$ compute each of the following.
 - $\vec{a} \times \vec{b}$
 - $\vec{b} \times \vec{a}$
 - $\vec{a} \times \vec{a}$
 - $\vec{b} \times \vec{b}$
- Calculate the vector product $\vec{u} \times \vec{v}$ of the following vectors:
 - $\vec{u} = (1, 3, 1), \vec{v} = (-1, 2, 1)$
 - $\vec{u} = (3, 3, 2), \vec{v} = (5, 1, 0)$
 - $\vec{u} = (-3, 2, -1), \vec{v} = (0, 1, 1)$

7.2.5 Mixed product



Activity 7.2.5

- Find the determinant $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ where $\vec{i} = (1, 0, 0); \vec{j} = (0, 1, 0); \vec{k} = (0, 0, 1)$
- Find the scalar product of vector $\vec{u} = (c_1, c_2, c_3)$ and vector obtained in 1).



Content Summary

The mixed product (also called the **scalar triple product** or **box product** or **compound product**) of three vectors is a scalar which numerically equals the dot product of one vector by the cross product of the other two vectors.

Then the mixed product of the vectors $\vec{u} = (a_1, a_2, a_3)$, $\vec{v} = (b_1, b_2, b_3)$ and $\vec{w} = (c_1, c_2, c_3)$ is equals the dot product of the first vector by the vector product of the other two. It is denoted by $[\vec{u}, \vec{v}, \vec{w}]$

Thus, $[\vec{u}, \vec{v}, \vec{w}] = \vec{u} \cdot (\vec{v} \times \vec{w})$

From activity,

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

This product is equivalent to the development of a determinant whose columns are the coordinates of these vectors with respect to an orthonormal basis.

$$\text{That is, } \vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example: Calculate the mixed product $\vec{u} \cdot (\vec{v} \times \vec{w})$ of the following vectors: $\vec{u} = (2, -1, 3)$, $\vec{v} = (0, 2, -5)$ and $\vec{w} = (1, -1, -2)$

Solution:
$$\vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & -5 \\ 1 & -1 & -2 \end{vmatrix} = -9\vec{i} - 5\vec{j} - 2\vec{k} = (-9, -5, -2)$$

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = (2, -1, 3) \cdot (-9, -5, -2) = -18 + 5 - 6 = -19$$

Or

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} 2 & -1 & 3 \\ 0 & 2 & -5 \\ 1 & -1 & -2 \end{vmatrix} = -8 + 0 + 5 - 6 - 10 - 0 = -19$$

Mixed product properties

a) The mixed product does not change if the orders of its factors are circularly rotated, but changes sign if they are transposed. That is,

- $\vec{u} \cdot (\vec{v} \times \vec{w}) = \vec{w} \cdot (\vec{u} \times \vec{v}) = \vec{v} \cdot (\vec{w} \times \vec{u})$
- $\vec{u} \cdot (\vec{v} \times \vec{w}) = -\vec{w} \cdot (\vec{v} \times \vec{u}) = -\vec{v} \cdot (\vec{u} \times \vec{w})$

b) If three vectors are linearly dependent, the mixed product is zero.

Example: Calculate the mixed product $\vec{u} \cdot (\vec{v} \times \vec{w})$ of the following vectors: $\vec{u} = (2, -1, 3)$, $\vec{v} = (0, 2, -5)$ and $\vec{w} = (1, -1, -2)$

Solution:

$$\vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & -5 \\ 1 & -1 & -2 \end{vmatrix} = -9\vec{i} - 5\vec{j} - 2\vec{k} = (-9, -5, -2)$$

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = (2, -1, 3) \cdot (-9, -5, -2) = -18 + 5 - 6 = -19$$

$$\text{Or } \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 5 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 5 & 6 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 5 \\ -2 & 6 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} \vec{k} = -7\vec{i} - 22\vec{j} + 16\vec{k}$$



Application activity 7.2.5

Calculate the mixed product $\vec{u} \cdot (\vec{v} \times \vec{w})$ of the following vectors:

1. $\vec{u} = (1, 3, 1)$, $\vec{v} = (-1, 2, 1)$ and $\vec{w} = (1, 0, -1)$
2. $\vec{u} = (3, 3, 2)$, $\vec{v} = (5, 1, 0)$ and $\vec{w} = (2, 1, -3)$
3. $\vec{u} = (-3, 2, -1)$, $\vec{v} = (0, 1, 1)$ and $\vec{w} = (8, 0, 0)$

7.2.6 Applications of dot and cross vector product in real life



Activity 7.2.6

The dot and cross products are not just abstract mathematical tools; they are the fundamental building blocks for solving real-world problems involving work done by the force, area of a parallelogram or a triangle, Volume of a parallelepiped and tetrahedron.

1) Conduct research in the library or on internet and compile how the dot product and the cross product are used to solve some problems in real life.

2) From the definition of work done by a force on a body, if a constant force $\vec{F}$ acting on a particle displaces from A to B , express the work done in function of vectors $\vec{F}$ and $\overrightarrow{AB}$.

3) Constant forces $\vec{P} = 2\vec{i} - 5\vec{j} + 6\vec{k}$ and $\vec{Q} = -\vec{i} + 2\vec{j} - \vec{k}$ act on a particle. Determine the work done when the particle is displaced from A to B , the position vectors of A and B being $4\vec{i} - 3\vec{j} + 2\vec{k}$ and $6\vec{i} + \vec{j} - 3\vec{k}$ respectively.



Content Summary

1. Work done by the force

The **work done by a force** is an important application of the **scalar (dot) product**. It describes how much **energy a force transfers when moving an object through a displacement**.

Work is defined as the **scalar product (dot product)** of the **force vector** and the **displacement vector**. $W = \vec{F} \cdot \vec{AB}$

Where: W is work done, $\vec{F}$ is force vector, $\vec{d} = \vec{AB}$ is the displacement vector

Example 1: A force of magnitude 6 units acting parallel to $2\vec{i} - 2\vec{j} + \vec{k}$ displaces the point of application from $(1, 2, 3)$ to $(5, 3, 7)$. Find the work done.

Solution: The force of magnitude 6 units, acting parallel to $2\vec{i} - 2\vec{j} + \vec{k}$ is

$$6 \times \frac{(2\vec{i} - 2\vec{j} + \vec{k})}{\sqrt{2^2 + (-2)^2 + 1^2}} = 4\vec{i} - 4\vec{j} + 2\vec{k}$$

The displacement $= (5, 3, 7) - (1, 2, 3) = (4, 1, 4) = 4\vec{i} + \vec{j} + 4\vec{k}$

Work done $(4, -4, 2) \cdot (4, 1, 4) = 16 - 4 + 8 = 20$ units of work.

Example 2: Forces of magnitudes 5 and 3 units acting in the direction $6\vec{i} + 2\vec{j} + 3\vec{k}$ and $3\vec{i} - 2\vec{j} + 6\vec{k}$ respectively act on a particle which is displaced from the point $(2, 2, -1)$ to $(4, 3, 1)$. Find the work done by the forces.

Solution: First force of magnitude 5 units, acting in the direction $6\vec{i} + 2\vec{j} + 3\vec{k}$ is

$$5 \frac{6\vec{i} + 2\vec{j} + 3\vec{k}}{\sqrt{6^2 + 2^2 + 3^2}} = \frac{5}{7}(6\vec{i} + 2\vec{j} + 3\vec{k})$$

Second force of magnitude 3 units, acting in the direction $3\vec{i} - 2\vec{j} + 6\vec{k}$ is

$$3 \frac{3\vec{i} - 2\vec{j} + 6\vec{k}}{\sqrt{3^2 + (-2)^2 + 6^2}} = \frac{3}{7}(3\vec{i} - 2\vec{j} + 6\vec{k})$$

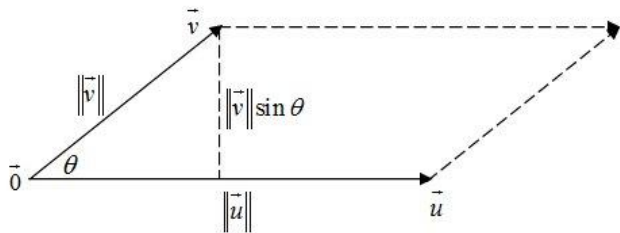
Resulting force is $\frac{5}{7}(6\vec{i} + 2\vec{j} + 3\vec{k}) + \frac{3}{7}(3\vec{i} - 2\vec{j} + 6\vec{k}) = \frac{1}{7}(39\vec{i} + 4\vec{j} + 33\vec{k})$

Displaced from the point $(2, 2, -1)$ to $(4, 3, 1)$ is $(4\vec{i} + 3\vec{j} + \vec{k}) - (2\vec{i} + 2\vec{j} - \vec{k}) = 2\vec{i} + \vec{j} + 2\vec{k}$

Work done: $\frac{1}{7}(39\vec{i} + 4\vec{j} + 33\vec{k}) \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = \frac{1}{7}(78 + 4 + 66) = \frac{148}{7}$.

2. Area of a parallelogram

Geometrically, the magnitude of the vector product of two vectors is the product of their magnitudes and the sine of the angle between them.



The area of a parallelogram with vectors $\vec{u}$ and $\vec{v}$ as two adjacent sides is given by $S = \|\vec{u} \times \vec{v}\|$

Thus, the magnitude of the vector product of two vectors $\vec{u}$ and $\vec{v}$ represents the area of a parallelogram with vectors $\vec{u}$ and $\vec{v}$ as two adjacent sides.

Example 1. Find the area of parallelogram with vectors $\vec{u} = (3, 0, 4)$ and $\vec{v} = (3, 2, 1)$ as two adjacent sides.

Solution:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & 4 \\ 3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} \vec{i} - \begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 3 & 0 \\ 3 & 2 \end{vmatrix} \vec{k} = -8\vec{i} + 9\vec{j} + 6\vec{k}$$

$$S = \|\vec{u} \times \vec{v}\| = \|-8\vec{i} + 9\vec{j} + 6\vec{k}\| = \sqrt{64 + 81 + 36} = \sqrt{181} \text{ area units}$$

Example 2. Find the area of the parallelogram formed by $\vec{a} = (2, -1, 1)$ and $\vec{b} = (1, 3, 2)$.

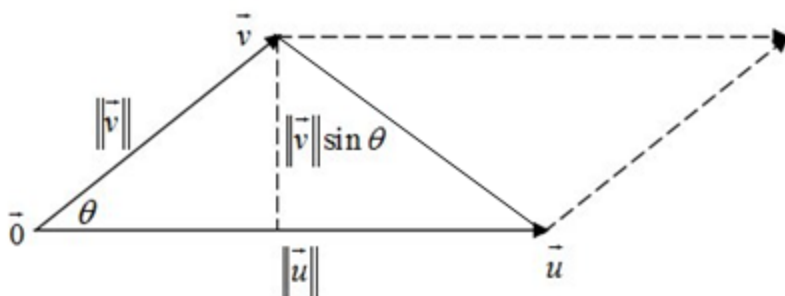
Solution: $\vec{a} \times \vec{b} = (2, -1, 1) \times (1, 3, 2) = (-5, -3, 7)$.

The area of that parallelogram is $\|\vec{a} \times \vec{b}\| = \sqrt{25 + 9 + 49} = \sqrt{83}$ Area unit.

Notice: Area of a triangle

Since the area of the parallelogram is twice the area of the triangle, we may use the vector product to find the area of triangle.

Thus, the area of triangle with vectors $\vec{u}$ and $\vec{v}$ as two sides is $S = \frac{1}{2} \|\vec{u} \times \vec{v}\|$



Consider the triangle ABC whose vertices are points $A(a_1, a_2, a_3)$, $B(b_1, b_2, b_3)$ and $C(c_1, c_2, c_3)$. Letting A to be the starting point, we can form two vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ and the area of this triangle is $S = \frac{1}{2} \|\overrightarrow{AB} \times \overrightarrow{AC}\|$

Example: Find the area of triangle with vectors $\vec{u} = (3, 0, 4)$ and $\vec{v} = (3, 2, 1)$ as two sides.

Solution:

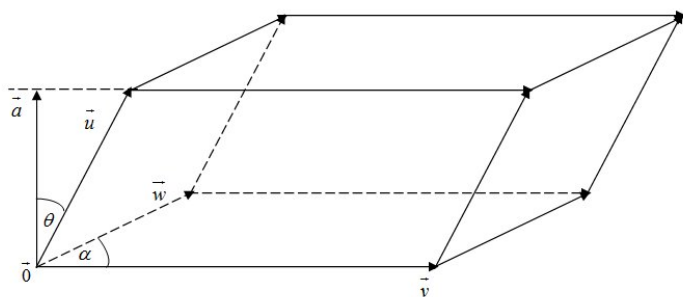
$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & 4 \\ 3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} \vec{i} - \begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 3 & 0 \\ 3 & 2 \end{vmatrix} \vec{k} = -8\vec{i} + 9\vec{j} + 6\vec{k}$$

$$S = \frac{1}{2} \|\vec{u} \times \vec{v}\| = \frac{1}{2} \|-8\vec{i} + 9\vec{j} + 6\vec{k}\| = \frac{1}{2} \sqrt{64 + 81 + 36} = \frac{1}{2} \sqrt{181} \text{ squared units.}$$

3. Volume of a parallelepiped and tetrahedron

Geometrically, the magnitude of the mixed product represents the volume of the parallelepiped whose edges are three vectors that meet in the same vertex.

Consider the following figure:



For a parallelepiped which has vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ as three concurrent edges, where $\vec{v}$ and $\vec{w}$ define its base, the volume is given by $V = \left| \vec{u} \cdot (\vec{v} \times \vec{w}) \right|$

Remember that if $\vec{u} = (a_1, a_2, a_3)$, $\vec{v} = (b_1, b_2, b_3)$ and $\vec{w} = (c_1, c_2, c_3)$ then.

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

If the parallelepiped is defined by four points, $A(a_1, a_2, a_3)$, $B(b_1, b_2, b_3)$, $W(c_1, c_2, c_3)$ and $D(d_1, d_2, d_3)$, its volume is $V = \left| \overrightarrow{AD} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) \right|$

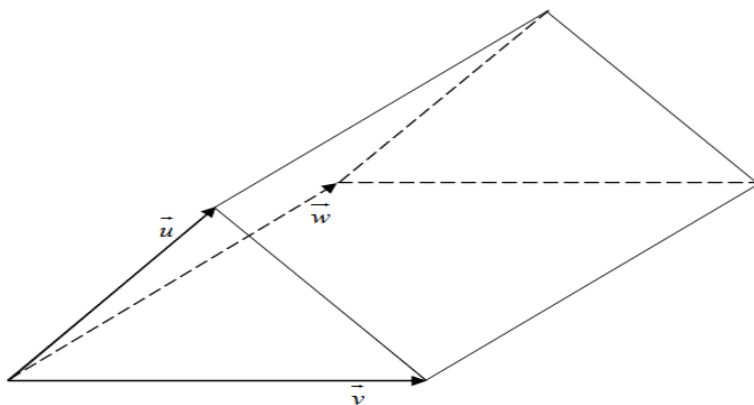
Note: A parallelepiped is a **prism (or polyhedron)** which has a **parallelogram** as its base.

Example 1: Find the volume of the parallelepiped formed by the vectors: $\vec{u} = (3, -2, 5)$
 $\vec{v} = (2, 2, -1)$ and $\vec{w} = (-4, 3, 2)$

Solution: $V = \vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} 3 & -2 & 5 \\ 2 & 2 & -1 \\ -4 & 3 & 2 \end{vmatrix} = 12 + 30 - 8 + 40 + 9 + 8 = 91$ cubic units.

Notice: Volume of a triangular prism

The parallelepiped can be split into 2 triangular prism of equal volume. Since the volume of a parallelepiped is the magnitude of the mixed product, then the volume of a triangular prism is equal to $\frac{1}{2}$ of the magnitude of the mixed product.



Thus, the volume of a triangular prism which has vectors $\vec{u} = (a_1, a_2, a_3)$, $\vec{v} = (b_1, b_2, b_3)$ and $\vec{w} = (c_1, c_2, c_3)$, as three concurrent edges, where $\vec{v}$ and $\vec{w}$ define its base, is given by

$$V = \frac{1}{2} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example: Find the volume of a triangular prism whose vertices are the points $A(1, 2, 1)$, $B(2, 4, 0)$, $C(-1, 2, 1)$ and $D(2, -2, 2)$.

Solution:

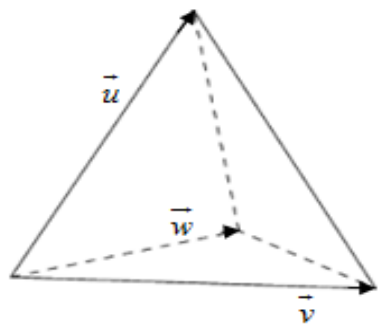
$$\overrightarrow{AB} = (1, 2, -1), \overrightarrow{AC} = (-2, 0, 0), \overrightarrow{AD} = (1, -4, 0)$$

The volume is: $V = \frac{1}{2} \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 0 \\ 1 & -4 & 1 \end{vmatrix} = \frac{1}{2} (0 - 8 + 0 - 0 - 0 + 4) = -2$

We need to take absolute value. Thus, the volume is $V = 2$ cubic units.

Volume of a tetrahedron

The parallelepiped can be split into 6 tetrahedra of equal volume. Since the volume of a parallelepiped is the magnitude of the mixed product, then the volume of a tetrahedron is equal to $\frac{1}{6}$ of the magnitude of the mixed product.



Thus, the volume of a tetrahedron which has vectors $\vec{u} = (a_1, a_2, a_3)$, $\vec{v} = (b_1, b_2, b_3)$ and $\vec{w} = (c_1, c_2, c_3)$, as three concurrent edges, where $\vec{u}$ and $\vec{w}$ define its base, is given by

$$V = \frac{1}{6} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Note: A tetrahedron is also called **triangular pyramid**.

Example: Find the volume of the tetrahedron whose vertices are the points $A(3, 2, 1)$, $B(1, 2, 4)$, $C(4, 0, 3)$ and, $D(1, 1, 7)$

Solution: $\vec{AB} = (-2, 0, 3)$, $\vec{AC} = (1, -2, 2)$, $\vec{AD} = (-2, -1, 6)$

The volume is: $V = \frac{1}{6} \begin{vmatrix} -2 & 0 & 3 \\ 1 & -2 & 2 \\ -2 & -1 & 6 \end{vmatrix} = \frac{1}{6} (24 - 3 + 0 - 12 - 4 - 0) = \frac{5}{6}$ cubic units



Application activity 7.2.6

1. A particle acted on by constant forces $2\vec{i} + \vec{j} - \vec{k}$, $\vec{i} - 2\vec{j} + 3\vec{k}$ and $3\vec{i} + \vec{j} + 5\vec{k}$ is displaced from the point $\vec{i} + 2\vec{j} + 3\vec{k}$ to the point $6\vec{i} + 3\vec{j} + \vec{k}$. Find the work done.

2. Constant forces $12\vec{i}-15\vec{j}+6\vec{k}$, $\vec{i}+2\vec{j}-2\vec{k}$ and $2\vec{i}+8\vec{j}+\vec{k}$ act on a point P which is displaced from the position $2\vec{i}-3\vec{j}+\vec{k}$ to the position $4\vec{i}+2\vec{j}+\vec{k}$. Find the total work done.
3. Find the area of a parallelogram with vectors
4. a) $\vec{u} = (1, -2, -14)$ and $\vec{v} = (22, 0, 0)$ as two adjacent sides
b) $\vec{u} = (21, 4, -2)$ and $\vec{v} = (0, -1, 0)$ as two adjacent sides
5. Find the area of triangle with vectors $\vec{u} = (1, 0, 0)$ and $\vec{v} = (3, 3, 3)$ as two sides
6. Find the area of a parallelogram whose sides are formed by the vectors $\vec{u} = (2, -3, 1)$ and $\vec{v} = (1, 4, 5)$.
7. Find the volume of the tetrahedron whose vertices are the points
a) $A(3, 1, 4)$, $B(1, 0, 0)$, $C(3, 4, 1)$ and $D(1, 0, 2)$.
b) $A(-1, -2, 1)$, $B(-5, 2, 3)$, $C(1, 1, 1)$ and $D(1, 1, 0)$.
8. Find the volume of the parallelepiped with adjacent sides $\vec{OA} = 3\vec{i} - \vec{j}$, $\vec{OB} = \vec{j} + 2\vec{k}$, $\vec{OC} = \vec{i} + 5\vec{j} + 4\vec{k}$ extending from origin of coordinates
9. Find the volume of the tetrahedron whose vertices are the points $A(2, -1, -3)$, $B(4, 1, 3)$, $C(3, 2, -1)$ and $D(1, 4, 2)$.



End of unit assessment

Assessing knowledge and understanding:

1. Comment by True or False:

In the Euclidean vector space $\mathbb{R}^3$, the result of the dot product of two vectors is always another vector that lies in the same plane as the original two.

2. Which of the following properties distinguishes the Euclidean vector space ($\mathbb{R}^n$ with the dot product) from a standard vector space ($\mathbb{R}^n$)?

- A) The closure property under vector addition.
- B) The existence of a zero vector.
- C) The ability to calculate the magnitude (or length) of a vector.
- D) The commutative property of vector addition.

3. Match the vector operation in Column A with its most accurate description or result in Column

B. Write the letter of the correct description next to the number of the operation.

Column A (Operation)	Column B (Description)
1. Scalar Multiplication	A. A fundamental operation in a Euclidean space that measures the similarity of direction of two vectors and results in a scalar.
2. Dot Product	B. An operation that changes the magnitude of a vector and, if the scalar is negative, its direction, but not its line of action.
3. Vector Subtraction	C. The process of finding the resultant vector from the tail of the first vector to the head of the second when they are placed head-to-tail.
4. Vector Addition	D. An operation geometrically equivalent to adding the negative of a vector, which helps find the relative position or displacement between two points represented by vectors.

Assessing skills

- Write the polynomial $s = t^2 + 4t - 3$ over $\mathbb{R}$ as a linear combination of the polynomials $p = t^2 - 2t + 5$, $q = 2t^2 - 3t$ and $r = t + 3$
- Show that the xy -plane $W = \{(a, b, 0)\}$ in $\mathbb{R}^3$ is generated by $\vec{u} = (1, 2, 0)$ and $\vec{v} = (0, 1, 0)$
- For two points $A(2, -1, 3)$ and $B(0, 2, 2)$, then find the coordinate of vector $\vec{v}$ from A to B and its magnitude.
- Consider the parallelepiped with sides determined by the following vectors:

$$\vec{a} = 3\vec{i} + 2\vec{j} + \vec{k}$$

$$\vec{b} = \vec{i} + \vec{j} + 2\vec{k}$$

$$\vec{c} = \vec{i} + 3\vec{j} + 3\vec{k}$$
 - Find the area of the face determined by the vector $\vec{a} = 3\vec{i} + 2\vec{j} + \vec{k}$, and $\vec{c} = \vec{i} + 3\vec{j} + 3\vec{k}$.
 - Find its volume and choose the correct answer in the following:
 - $V = 7$ cubic units
 - $V = 8$ cubic units
 - $V = 9$ cubic units
 - $V = 10$ cubic units
- Find the volume of the tetrahedron whose vertices are $(0, 1, 2), (3, 0, 1), (4, 3, 6), (2, 3, 2)$

Assessing attitudes and values

9. Comment by True or False

When solving a complex physics problem involving vector components, the most efficient and reliable path to the correct answer is to work completely alone to ensure no outside mistakes influence your final result.

10. In a Senior Five project, you use Euclidean vectors to model the forces on a bridge design. Your calculations show the bridge will hold. However, in the real world, similar bridges can sometimes fail even when these calculations are correct. What is the most important attitude to have in this situation? **Choose the correct answer.**

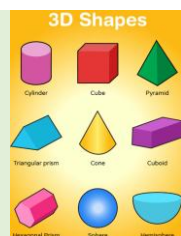
- A) Frustration, because the mathematical model is clearly useless if it can't predict reality perfectly.
- B) Acceptance, because engineers should expect failure and just build things stronger next time.
- C) Appreciation, because the vector model gives a strong foundation, but it's important to remember it is an idealization that doesn't account for every real-world variable (e.g., material fatigue, wind gusts).
- D) Scepticism, meaning you should abandon mathematical modelling entirely and rely only on physical testing.

11. A positive attitude towards learning mathematics involves several key attributes. **Match the challenge a student faces in Column A with the most constructive attitude or value from Column B that would help them overcome it.**

Column A (The Challenge)	Column B (The Attitude/Value)
1. You are trying to visualize a 3D vector problem (e.g., a plane's velocity relative to the wind) and are finding it difficult to draw or imagine.	A. Perseverance: Believing that the ability to solve complex problems grows with effort, and that being "stuck" is a normal part of the learning process.
2. You have attempted a vector problem multiple times and keep getting the wrong answer. It would be easy to give up and wait for the teacher to provide the solution.	B. Curiosity: Being motivated to understand the underlying principles of why the method works, which builds a more robust and adaptable knowledge.
3. You have learned the formula for calculating the dot product, but you are more interested in understanding <i>why</i> it	C. Creativity and resourcefulness: Actively seeking different strategies, such as using online 3D graphing tools, building a physical

produces a number related to the angle between vectors.	model with sticks, or trying a different coordinate system to better understand the problem.
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Unit 8: POINTS, STRAIGHT LINES AND SHAPES IN 3D

Key unit competence: Apply the meaning of coordinates in solving problems involving points, lines, planes and spheres in 3D space.

Learning objectives:

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain key concepts: Cartesian coordinates, vector/parametric equations, and geometric shapes (circle, sphere, plane).
- Apply the formulas for distance, midpoint, angles, and equations of lines, planes, and spheres.
- Identify the different forms of linear equations and their interrelationships.
- Describe the positional relationships between points, lines, planes and spheres.

Skills

- Represent points, lines, circles, and spheres using Cartesian, vector, and parametric forms. - Convert between different forms of linear equations (Cartesian, vector, parametric).
- Calculate distances, midpoints, angles, and centroids in 2D and 3D.
- Determine the position of points relative to lines, circles, planes, and spheres (e.g., inside/outside).
- Analyse the relationship between lines (parallel, intersecting, skew) and planes (parallel, intersecting).
- Solve complex geometric problems involving collinearity, coplanarity, and shortest distances. - Model real-world situations (e.g., navigation, architecture, physics) using coordinate geometry.

Attitudes and value

- Appreciate the power of coordinate systems in algebra and geometry.
- Value precision and accuracy in formulating equations and performing calculations.

- Develop spatial reasoning and visualization skills for 3D problems.
- Collaborate effectively to tackle multi-step geometric challenge.

Introduction

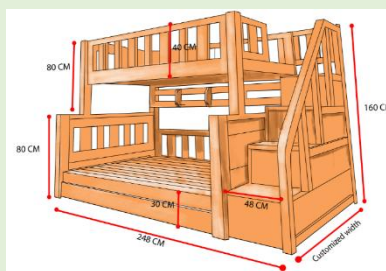
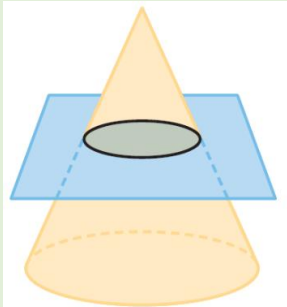
In the world of three-dimensional space, we extend the familiar two-dimensional Cartesian plane into a richer, more realistic framework by introducing the z-axis.

In this unit, we will explore how the simple concept of coordinates empowers us to precisely define and analyse geometric objects, moving from plotting individual points to deriving the equations of lines, planes, and spheres. You will learn to describe the positional relationships between these objects, determining if **lines are parallel, intersecting, or skew, and how planes intersect or form angles**. You will also master the techniques to calculate critical measurements like distances and midpoints. This foundational knowledge is not merely an abstract mathematical exercise; it is the language used in real life when finding the volume of some solids, and it is used in physics when analysing forces and motion in space, and navigation for plotting courses. This unit will give you the tools to model and solve complex, real-world spatial problems.



Introductory activity:

Observe handcraft objects, the cone and bed on the diagram below:



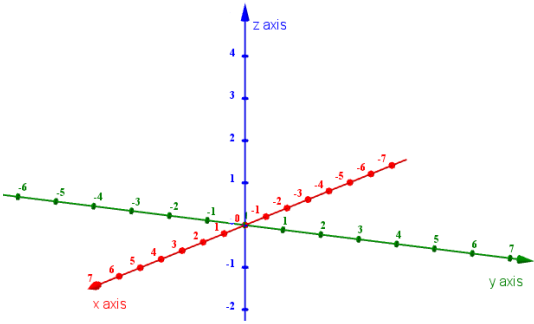
Write down what can illustrate **points, lines, angles, planes and shapes**. In each case explain your answer.

8.1. Points in 3D

8.1.1. Cartesian coordinates of a point and its representation in 3D

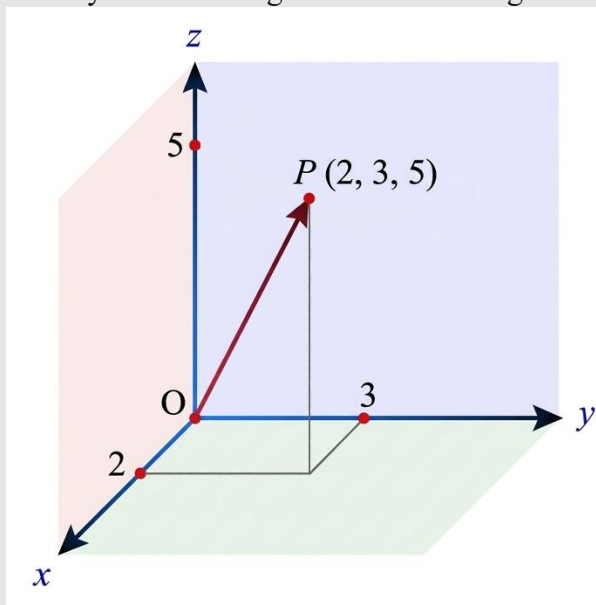
Activity 8.1.1

Consider the point $A(2, 3, 5)$. Copy the following figure on a piece of paper:



- Through the point 2 units from the origin, on the x -axis, draw the line parallel to the y -axis
- Through the point 3 units from the origin, on the y -axis, draw the line parallel to the x -axis and label P the intersection of the line in (a) and in (b)
- From point P draw the line parallel to the z -axis, at 5 units from P ; this is point A

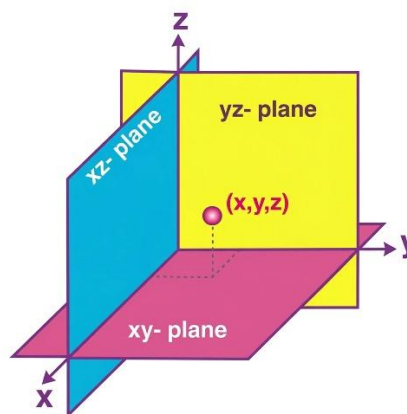
Check your answer against the following:



- Use the procedure above to plot point $B(-3, 1, 2)$.

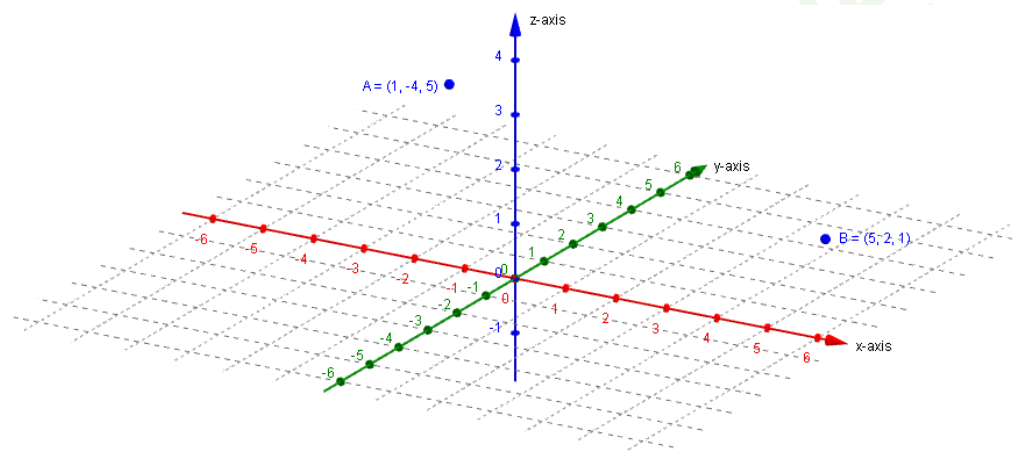
Content summary

Given a set of three axes (OX, OY and OZ) intersecting at the origin O, any two of the axes being perpendicular, any point M in the space can be characterised by an ordered triple of real numbers, called Cartesian coordinates of point M. They are denoted $M(x, y, z)$ are respectively the signed distances to move from the origin in the directions OX, OY and OZ, respectively.



Example: Represent on the Cartesian plane points $A(1, -4, 5)$ and $B(5, 2, 1)$

Solution



Application activity 8.1.1

Represent the following points in space

$A(1, 1, 1)$, $B(-1, 2, 3)$, $C(3, 4, 1)$

$D(-2, 1, 2)$, $E(3, 2, 1)$, $F(-2, 0, 1)$

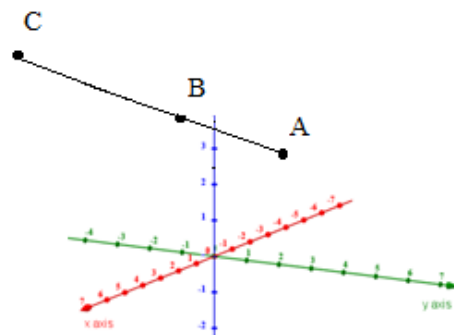
8.1.2. Distance between two points

Activity 8.1.2

Consider the following diagram.

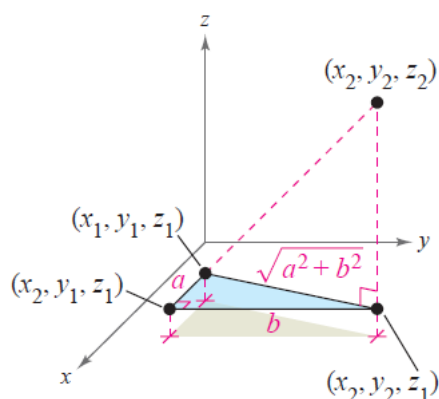
Compare the position of point B and the position of point C with respect to point A.

- Which one is near A?
- Can you find the distance $D(A,B)$ and $d(A,C)$ if $A(-1,-2,-3)$, $B(2,4,6)$ and $C(3,6,9)$?



Content summary

Given points $A(x_A, y_A, z_A)$ and $B(x_B, y_B, z_B)$ the distance between them is the positive real number denoted and defined by $d(A, B) = AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$



Example: Calculate the distance between points $A(1, -4, 5)$ and $B(5, 2, 1)$

Solution: $d = \sqrt{(5-1)^2 + (2+4)^2 + (1-5)^2} = \sqrt{68}$

Application activity 8.1.2

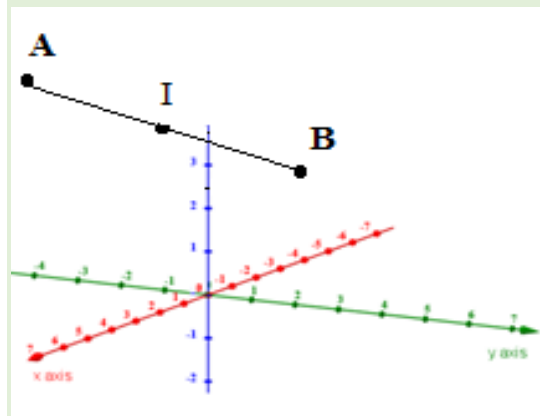
Consider the triangle with vertices at points $A(1, -1, 2)$; $B(3, -1, 3)$ and $C(1, 1, 1)$

- Find the lengths of sides AB and AC
- Hence, determine the type of triangle ABC.

8.1.3. Midpoint of a line segment and centroid of a geometric figure

Activity 8.1.3

Consider the diagram below where $A(2, -3, 5)$ and $B(0, 1, -1)$.



If I is such that $d(A, I) = d(I, B)$, describe the position of point, I with respect to points A and B .

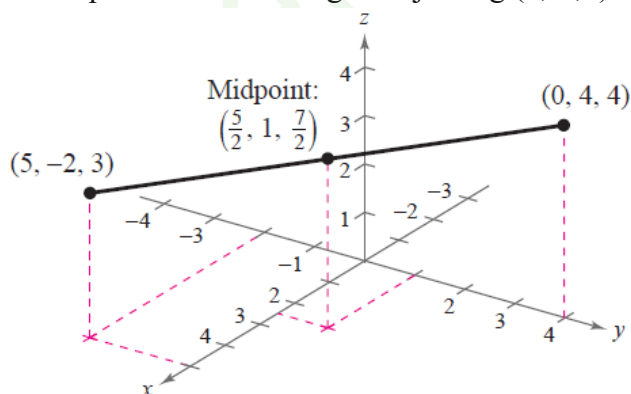
Content summary

Mid-point of a line segment:

Given that $A(x_A, y_A, z_A)$ and $B(x_B, y_B, z_B)$, to find the coordinates (x_I, y_I, z_I) in terms of the coordinates of A and B , we proceed as follows:

$$I\left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right)$$

The mid-point of the line segment joining $(5, -2, 3)$ and $(0, 4, 4)$ is:



Example: Find the coordinates of the midpoint of the line joining $A(1, 2, 3)$ and $B(3, 2, 1)$

Solution:

The coordinates of the midpoint of the line joining $(1, 2, 3)$ and $(3, 2, 1)$ is given by

$$\left(\frac{1+3}{2}, \frac{2+2}{2}, \frac{3+1}{2} \right) \text{ Which is } (2, 2, 2).$$

Centroid of a geometric figure

The **centroid** of **geometric figure** is the arithmetic mean (average) position of all points in the shape. In geometry, the synonym of centroid is **barycenter** or **geometric center**. In physics barycenter means the **physical center** of mass or the **center of gravity**.

Let $(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_n, y_n, z_n)$ be n points of space, their centroid is given by the formula:

$$\left(\frac{x_1 + x_2 + \dots + x_n}{n}, \frac{y_1 + y_2 + \dots + y_n}{n}, \frac{z_1 + z_2 + \dots + z_n}{n} \right)$$

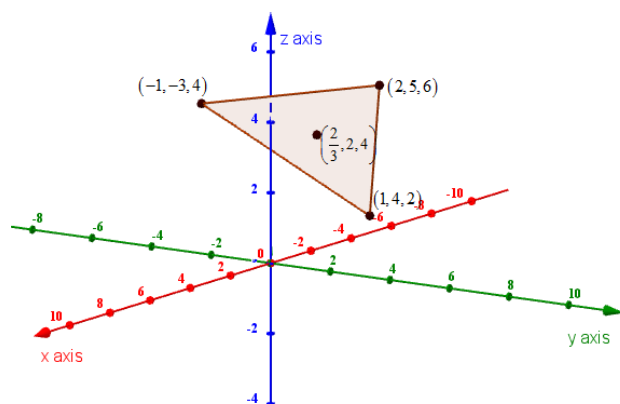
Example: Determine the centroid of the triangle built from the points

$(1, 4, 2), (-1, -3, 4)$ and $(2, 5, 6)$

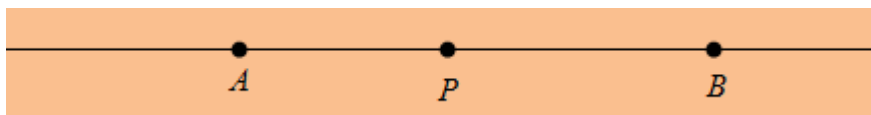
Solution:

The centroid of the triangle built from the points $(1, 4, 2), (-1, -3, 4)$ and $(2, 5, 6)$ is given by

$$\left(\frac{1-1+2}{3}, \frac{4-3+5}{3}, \frac{2+4+6}{3} \right) \text{ which is } \left(\frac{2}{3}, 2, 4 \right).$$

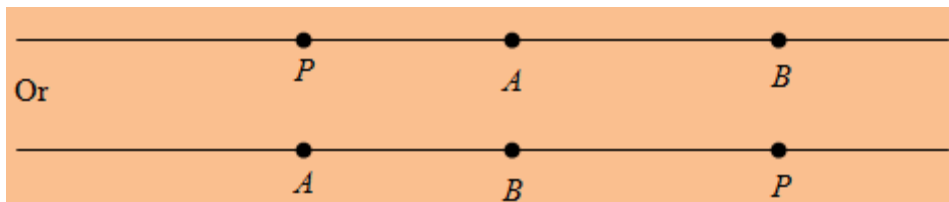
**The ratio formula**

Let P be a point on the line joining point A and point B . P divides this line internally (means that it lies between A and B) in the ratio $m : n$



We have $\overrightarrow{AP} = \frac{m}{n} \overrightarrow{PB}$. Then, $P = \frac{mB + nA}{m + n}$

Let point P divide the line externally (means that it does not lie between A and B) in the ratio $m : n$



We have $\overrightarrow{PA} = \frac{m}{n} \overrightarrow{PB}$ or $\overrightarrow{AP} = \frac{m}{n} \overrightarrow{BP}$.

Then, $P = \frac{mB - nA}{m - n}$.

Example: Find the position of the point P which divides $[AB]$

a) internally in the ratio 1:3

b) externally in the ratio 2:5

Solution

a) If P divides $[AB]$ internally in the ratio 1:3, we have $P = \frac{B + 3A}{4} = \frac{3}{4}A + \frac{1}{4}B$

b) If P divides $[AB]$ externally in the ratio 2:5, we have $P = \frac{2B - 5A}{-3} = \frac{5}{3}A - \frac{2}{3}B$.

Application activity 8.1.3

1) Find the coordinates of the midpoint of the line joining segment $[AB]$ if A and B are respectively:

a. (1, 3, 6) and (-1, 4, 5) b. (11, 2, 4) and (1, 3, -5) c. (-9, 8, 2) and (2, 3, 8)

2) Find the centroid of the figure built from the points

a) (6, 2, 0), (2, 4, 6) and (0, 0, 1).

b) (-5, 4, 10), (1, 3, 2), (0, 3, 7) and (2, 5, 9)

3) Find the coordinates of the point which divides the line joining (-2, 3, 5) to (1, -4, -6) in the ratio: a) 2:3 internally b) 2:3 externally.

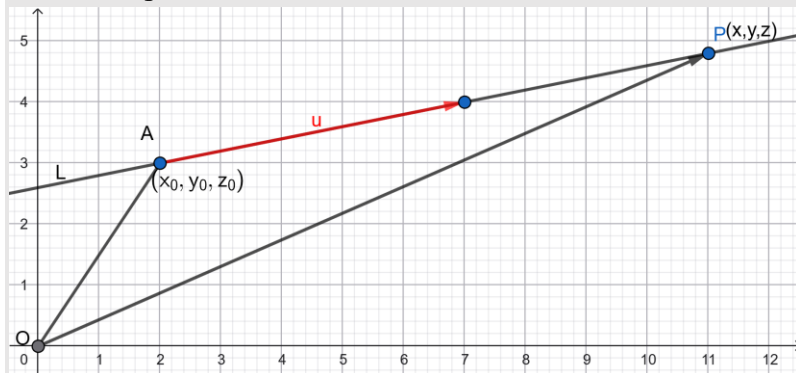
8.2. Straight lines in 3D

8.2.2. Equations of a straight line in 3D

Activity 8.2.2

a) How is a straight line determined in a three-dimensional space?

b) Line (L) passes through point A and is parallel to the vector $\vec{u}$, express, mathematically, the relationship between vectors $\vec{u}$ and $\overrightarrow{AP}$



a) Given that $P(x, y, z)$, $A(x_0, y_0, z_0)$ and $\vec{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ express the relationship found in part

b) In three different ways, use $\vec{i}, \vec{j}, \vec{k}$ as unit vectors in directions OX, OY and OZ , respectively.

Content summary

In a three-dimensional space, a straight line is determined by a point A through which it passes and a vector $\vec{u}$ parallel to the line. Vector $\vec{u}$ is the **direction** vector of the line.

Line (l) through point A and with direction vector $\vec{u}$ is the set of all points P in the space such that $\overrightarrow{AP}$ is a scalar multiple of $\vec{u}$, that is $\overrightarrow{AP} = t\vec{u}$, for some scalar t .

The way the equality $\overrightarrow{AP} = t\vec{u}$, is transformed gives rise to different equations:

Vectorequation: $\overrightarrow{OP} = \overrightarrow{OA} + t\vec{u}$ or $(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$, where t is parameter

the parametric equations:
$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$

the Cartesian equations: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ where $abc \neq 0$

Generally, we can find the Cartesian equation of line passing through the point $P(x_0, y_0, z_0)$ and parallel to the direction vector $\vec{v} = (a, b, c)$ in following manner:

$$\begin{vmatrix} x-x_0 & a \\ y-y_0 & b \\ z-z_0 & c \end{vmatrix} = \begin{vmatrix} x-x_0 & a \\ z-z_0 & c \end{vmatrix} = \begin{vmatrix} y-y_0 & b \\ z-z_0 & c \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} x & x_0 & a \\ y & y_0 & b \\ 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} x & x_0 & a \\ z & z_0 & c \\ 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} y & y_0 & b \\ z & z_0 & c \\ 1 & 1 & 0 \end{vmatrix} = 0$$

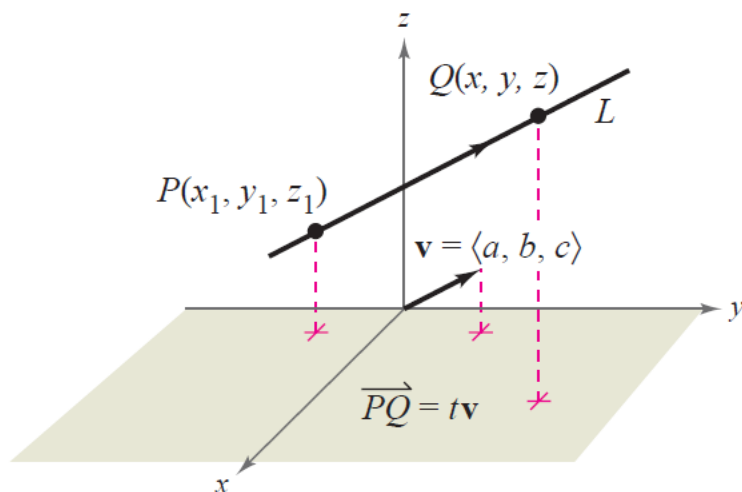
The vector $\mathbf{v}$ is the **direction vector** for the line L , and a , b , and c are the **direction numbers**. One way of describing the line L is to say that it consists of all points $Q(x, y, z)$ for which the vector PQ is parallel to $\mathbf{v}$.

This means that PQ is a scalar multiple of $\mathbf{v}$, and you can write

$PQ = t\mathbf{v}$, where t is a scalar.

$$PQ = \langle x - x_1, y - y_1, z - z_1 \rangle = \langle at, bt, ct \rangle = t\mathbf{v}$$

By equating corresponding components, you can obtain the **parametric equations of a line in space**.



Example:

- Find the vector, parametric and symmetric equations of the line (l) passing through the point $A(3, -2, 4)$ with direction vector $\vec{u} = (2, 3, 5)$.
- Find vector, parametric and symmetric equations of the line (r) passing through the points $A(3, -2, 5)$ and $B(1, 4, -2)$

Solution:

- Let $P(x, y, z)$ be any point on the line, then: equation, where t is a parameter

$$\text{Parametric equations : (l) : } \begin{cases} x = 3 + 2t \\ y = -2 + 3t \\ z = 4 + 5t \end{cases}$$

$$\text{Symmetric equations : (l) : } \overrightarrow{OP} = 3\vec{i} - 2\vec{j} + 4\vec{k} + t(2\vec{i} + 3\vec{j} + 5\vec{k}) \quad (l) : \frac{x-3}{2} = \frac{y+2}{3} = \frac{z-4}{5}$$

b) First we find the direction vector, which is $\overrightarrow{AB} = (-2, 6, -7)$

$$\text{Vector equation : } \overrightarrow{OP} = 3\vec{i} - 2\vec{j} + 5\vec{k} + t(-2\vec{i} + 6\vec{j} - 7\vec{k})$$

$$\text{Parametric equations : (r) : } \begin{cases} x = 3 - 2t \\ y = -2 + 6t \\ z = 5 - 7t \end{cases}$$

$$\text{Symmetric equations : (r) : } \frac{x-3}{-2} = \frac{y+2}{6} = \frac{z-5}{-7}$$

Line defined by two position vectors:

A line passing through points $P(x_0, y_0, z_0)$ and $Q(x_1, y_1, z_1)$ and $V(x, y, z)$ any point on the line has: Vector equation $\overrightarrow{PV} = r\overrightarrow{PQ}$, where r is a parameter.

- Parametric equations:
$$\begin{cases} x = x_0 + r(x_1 - x_0) \\ y = y_0 + r(y_1 - y_0) \\ z = z_0 + r(z_1 - z_0) \end{cases}$$
- The cartesian equations:
$$\frac{x-x_0}{x_1-x_0} = \frac{y-y_0}{y_1-y_0} = \frac{z-z_0}{z_1-z_0}$$

Or we can take two of them

$$\begin{cases} \frac{x-x_0}{x_1-x_0} = \frac{y-y_0}{y_1-y_0} \\ \frac{x-x_0}{x_1-x_0} = \frac{z-z_0}{z_1-z_0} \\ \frac{y-y_0}{y_1-y_0} = \frac{z-z_0}{z_1-z_0} \end{cases} \Leftrightarrow \begin{cases} (x-x_0)(y_1-y_0) = (y-y_0)(x_1-x_0) \\ (x-x_0)(z_1-z_0) = (z-z_0)(x_1-x_0) \\ (y-y_0)(z_1-z_0) = (z-z_0)(y_1-y_0) \end{cases}$$

Taking two of them, we have:

$$\begin{cases} (x-x_0)(y_1-y_0) = (y-y_0)(x_1-x_0) \\ (x-x_0)(z_1-z_0) = (z-z_0)(x_1-x_0) \end{cases}$$

These equations can be found using determinants:

$$\begin{vmatrix} x-x_0 & x_1-x_0 \\ y-y_0 & y_1-y_0 \end{vmatrix} = \begin{vmatrix} x-x_0 & x_1-x_0 \\ z-z_0 & z_1-z_0 \end{vmatrix} = \begin{vmatrix} y-y_0 & y_1-y_0 \\ z-z_0 & z_1-z_0 \end{vmatrix} = 0$$

Or

$$\begin{vmatrix} x & x_0 & x_1 \\ y & y_0 & y_1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} x & x_0 & x_1 \\ z & z_0 & z_1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} y & y_0 & y_1 \\ z & z_0 & z_1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Example:

Find vector, parametric and symmetric equations of the line M passing through the points $A(3, -2, 5)$ and $B(1, 4, -2)$

Solution:

First, we find the direction vector, which is $\overrightarrow{AB} = (-2, 6, -7)$

Vector equation: $M = \overrightarrow{AV} = r\overrightarrow{AB}$, with $V(x, y, z)$ and r is a parameter.

$$\text{Parametric equations: } M \equiv \begin{cases} x = 3 - 2r \\ y = -2 + 6r \\ z = 5 - 7r \end{cases}$$

$$\text{Cartesian equations (eliminating parameter } r): M \equiv \frac{-x+3}{2} = \frac{y+2}{6} = \frac{-z+5}{7}$$

Or we can take two of them

$$\begin{cases} \frac{-x+3}{2} = \frac{y+2}{6} \\ \frac{-x+3}{2} = \frac{-z+5}{7} \\ \frac{y+2}{6} = \frac{-z+5}{7} \end{cases} \Leftrightarrow \begin{cases} -6x+18 = 2y+4 \\ -7x+21 = -2z+10 \\ 7y+14 = -6z+30 \end{cases} \Leftrightarrow \begin{cases} -6x-2y+14 = 0 \\ -7x+2z+11 = 0 \\ 7y+6z-16 = 0 \end{cases} \Leftrightarrow \begin{cases} 3x+y-7 = 0 \\ 7x-2z-11 = 0 \\ 7y+6z-16 = 0 \end{cases}$$

$$\text{Taking two of them, we have: } \begin{cases} 3x+y-7 = 0 \\ 7x-2z-11 = 0 \end{cases}$$

$$\text{Or we can use determinants: } \begin{vmatrix} x-3 & -2 \\ y+2 & 6 \end{vmatrix} = \begin{vmatrix} x-3 & -2 \\ z-5 & -7 \end{vmatrix} = \begin{vmatrix} y+2 & 6 \\ z-5 & -7 \end{vmatrix} = 0$$

Taking two of them, we have: $\begin{vmatrix} x-3 & -2 \\ y+2 & 6 \end{vmatrix} = 0$ and $\begin{vmatrix} x-3 & -2 \\ z-5 & -7 \end{vmatrix} = 0$

$$\Leftrightarrow \begin{cases} 6x - 18 + 2y + 4 = 0 \\ -7x + 21 + 2z - 10 = 0 \end{cases}$$

$$M \equiv \begin{cases} 3x + y - 7 = 0 \\ 7x - 2z - 11 = 0 \end{cases}$$

OR

$$\begin{vmatrix} x & 3 & 1 \\ y & -2 & 4 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} x & 3 & 1 \\ z & 5 & -2 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} y & -2 & 4 \\ z & 5 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Taking two of them, we have

$$\begin{vmatrix} x & 3 & 1 \\ y & -2 & 4 \\ 1 & 1 & 1 \end{vmatrix} = 0 \text{ and } \begin{vmatrix} x & 3 & 1 \\ z & 5 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Leftrightarrow \begin{cases} -2x + y + 12 + 2 - 4x - 3y = 0 \\ 5x + z - 6 - 5 + 2x - 3z = 0 \end{cases} \Leftrightarrow \begin{cases} -6x - 2y + 14 = 0 \\ 7x - 2z - 11 = 0 \end{cases} \text{ And}$$

$$\text{finally, } M \equiv \begin{cases} 3x + y - 7 = 0 \\ 7x - 2z - 11 = 0 \end{cases}$$

Application activity 8.2.2

- 1) Find vector, parametric and symmetric equations of the line (l) passing through points A and B in each of the following cases:
 - a) $A(2,1,4)$ and $B(3,1,1)$. b) $A(1,1,3)$ and $B(2,5,4)$
 - c) $A(2,1,4)$ and $B(6,3,2)$. d) $A(1,1,1)$ and $B(4,5,6)$.
- 2) Find vector, parametric and symmetric equations of the line M passing through the points $A(2,1,4)$ and $B(3,1,1)$.

8.2.3 Condition of co-linearity of three points

Activity 8.2.3

From the method of finding equation of a line passing through two given points, determine which of the following sets of points lie on the same line

$$A = \{(1, 2, 3), (1, -4, 3), (-1, 0, 5)\}$$

$$B = \{(2, 1, -3), (1, -7, 6), (-4, 4, 0)\}$$

Content summary

Establish a condition for which three given points may satisfy to lie on the same line

The three points $A(a_1, a_2, a_3)$, $B(b_1, b_2, b_3)$ and $C(c_1, c_2, c_3)$ are collinear (means that they lie on the same line) if the following conditions are satisfied

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Alternative method:

Three points A, B and C are collinear if the vectors formed from these points are linearly dependent. That is, $\overrightarrow{AB} = k\overrightarrow{AC}$, $k \in \mathbb{R}_0$

Example: Prove that points $A(2, -1, 3)$, $B(4, 3, 5)$ and $C(6, 7, 7)$ are collinear.

Solution: Remember that the three points (a_1, a_2, a_3) ; (b_1, b_2, b_3) and (c_1, c_2, c_3) are collinear if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Or

$$\begin{vmatrix} 2 & 4 & 6 \\ -1 & 3 & 7 \\ 1 & 1 & 1 \end{vmatrix} = 6 - 6 + 28 - 18 - 14 + 4 = 0$$

$$\begin{vmatrix} 2 & 4 & 6 \\ 3 & 5 & 7 \\ 1 & 1 & 1 \end{vmatrix} = 10 + 18 + 28 - 30 - 14 - 12 = 0$$

Thus, the three points are collinear.

Alternative method: $\overrightarrow{AB} = (2, 4, 2)$, $\overrightarrow{AC} = (4, 8, 4)$.

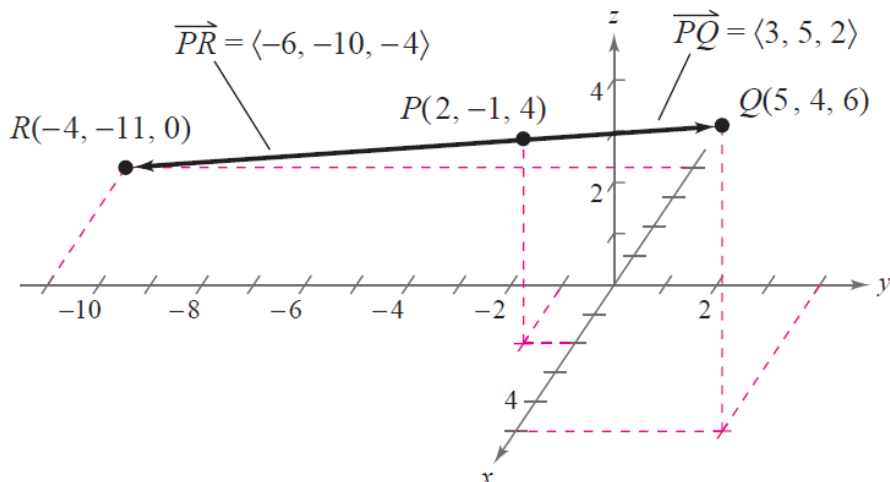
We see that $\overrightarrow{AC} = 2\overrightarrow{AB}$. Thus, the three points are collinear.

Example: Determine whether the points $P(2, -1, 4)$, $Q(5, 4, 6)$, and $R(-4, -11, 0)$ are collinear.

Solution

The component forms of PQ and PR are

$\overrightarrow{PQ} = \langle 5-2, 4-(-1), 6-4 \rangle = \langle 3, 5, 2 \rangle$ and $\overrightarrow{PR} = \langle -4-2, -11-(-1), 0-4 \rangle = \langle -6, -10, -4 \rangle$. Because $\overrightarrow{PR} = -2\overrightarrow{PQ}$, you can conclude that they are parallel. Therefore, the points P , Q , and R lie on the same line, as shown



Finding the terminal point of a vector

The initial point of the vector $v = \langle 4, 2, -1 \rangle$ is $P(3, -1, 6)$. What is the terminal point of this vector?

Solution

Using the component form of the vector whose initial point is $P(3, -1, 6)$ and whose terminal point is $Q(q_1, q_2, q_3)$, you can write

$$\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2, q_3 - p_3 \rangle = \langle q_1 - 3, q_2 + 1, q_3 - 6 \rangle = \langle 4, 2, -1 \rangle$$

This implies that $q_1 - 3 = 4$, $q_2 + 1 = 2$, and $q_3 - 6 = -1$. The solutions of these three equations are $q_1 = 7$, $q_2 = 1$, and $q_3 = 5$. So, the terminal point is $Q(7, 1, 5)$.

Application activity 8.2.3

- 1) Verify if the points $A(2, 3, 1)$, $B(5, 4, 3)$, $C(2, 1, 2)$ are collinear
- 2) Show that the point $A(1, 1, 1)$, $B(3, 2, 4)$, $C(-1, 0, -2)$ are collinear and find the equations, in parametric form, of the line they lie on.

8.2.4 Relationships between Lines

Activity 8.2.4

In each of the following pair of lines, after verifying if they are parallel, determine whether they are coincident or different. If not parallel, find intersection point if any.

$$1) L_1 : \vec{r} = (2 - \lambda)\vec{i} + 2(1 + \lambda)\vec{j} + (1 + 3\lambda)\vec{k}, \quad L_2 : 6(1 - x) = 3(y - 1) = 2(z - 1),$$

$$2) L_1 : x = \frac{y+2}{2} = 5 - z, \quad L_2 : \frac{x-1}{-1} = \frac{-3-y}{3} = z - 4$$

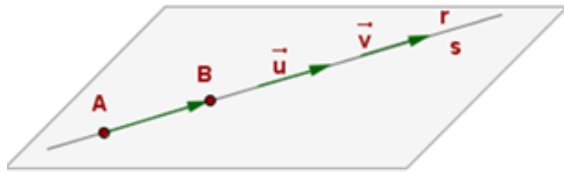
$$3) L_1 : \vec{r} = 5\vec{i} + 4\vec{j} + 5\vec{k} + \lambda(2\vec{i} + \vec{j} + \vec{k}), L_2 : x - 1 = 2(y - 2) = 2(z - 3)$$

Content summary

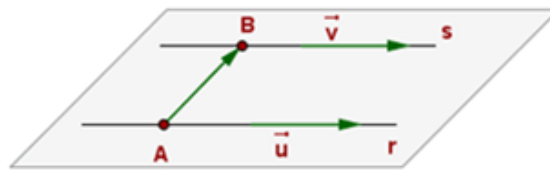
Consider two lines $L_1 : r = a + td_1$, $L_2 : r = b + sd_2$, where: $\vec{a}$ and $\vec{b}$ are position vectors of points on each line, $\vec{d}_1; \vec{d}_2$ are direction vectors; t, s are real parameters.

Case 1: Parallel lines:

- Two lines are **parallel** when their direction vectors are proportional: $\vec{d}_1 = k\vec{d}_2$.
- Lines are **coincident** if they are parallel **and** the vector between points on each line is also parallel to the direction vector: $\vec{b} - \vec{a} = \lambda\vec{d}_1$.



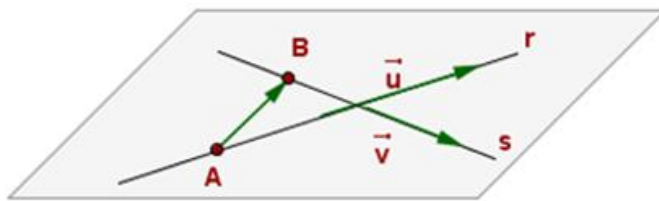
Lines r and s are coincident



Lines r and s are parallel and distinct

Case 2: Intersecting lines:

Two lines **intersect** if there exist real parameters t , and s satisfying: $\vec{a} + t\vec{d}_1 = \vec{b} + s\vec{d}_2$



Lines r intersect line s

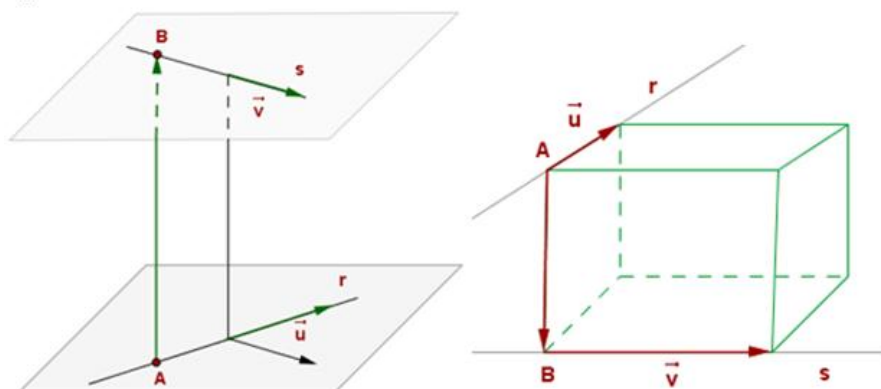
A solution (t, s) (exists implies that the lines intersect). Direction vectors must **not** be parallel:

$$\vec{d}_1 \nparallel \vec{d}_2$$

Case 3: Skewed lines:

Two lines are skewed (lines lie in **different planes**) if they are **not parallel** and **do not intersect**.

In this case, $\vec{d}_1 \nparallel \vec{d}_2$ and $\vec{a} + t\vec{d}_1 \neq \vec{b} + s\vec{d}_2$ for all t and s



Lines r and s are skew

Example 1:

Determine if the lines are parallel, intersect, skew or identical

$$M \equiv \begin{cases} x = 3 - 2r \\ y = -2 + 6r \\ z = 5 - 7r \end{cases}, L_2 \equiv \begin{cases} x = 5 - 2t \\ y = -2 + 4t \\ z = 1 - 2t \end{cases}$$

Solution:

First, we see that the direction vectors are $\vec{u} = (1, -2, 1)$ and $\vec{v} = (-2, 4, -2)$ (coefficients of parameter).

From two direction vectors we see that they are scalar multiple since $\vec{v} = -2\vec{u}$, means that the two lines are parallel, then show if they are identical. You can even determine if they pass through the same points or not. Then, determine if the two sets of parametric equations produce the same points for different values of t .

Let $t = 0$ for L_1 , the point produced is $(3, 2, 4)$.

Set the x from L_2 equal to the x -coordinate produced by L_1 and solve for t .

$$3 = 5 - 2t \Leftrightarrow -2 = -2t \Rightarrow t = 1$$

Now $t = 1$ for L_2 and the point $(3, 2, -1)$ is produced. Since the z -coordinates are not equal, the lines are not identical.

So, they are parallel and distinct.

Example 2:

Determine if the line 1 and line 2 intersect. If so, find the point of intersection.

$$\text{Line 1: } \begin{cases} x = 3 + 2t \\ y = -2t \\ z = 4 - t \end{cases} \quad \text{Line 2: } \begin{cases} x = 4 - s \\ y = 3 + 5s \\ z = 2 - s \end{cases}$$

Solution

Direction vectors: $\vec{v}_1 = (2, -2, -1)$, $\vec{v}_2 = (-1, 5, -1)$

Since $\vec{v}_2 \neq k \cdot \vec{v}_1$, the lines are not parallel. Thus, they either intersect or they are skew lines.

Keep in mind that the lines may have a point of intersection or a common point, but not necessarily for the same value of t . So, equate each coordinate, but replace the t in Line 2 with an s .

$$3 + 2t = 4 - s$$

$$-2t = 3 + 5s$$

$$4 - t = 2 - s$$

System of 3 equations with 2 unknowns – Solve the first 2 and check with the 3rd equation.

Solving the system, we get $t = 1$ and $s = -1$.

Line 1: $t = 1$ produces the point $(5, -2, 3)$

Line 2: $s = -1$ produces the point $(5, -2, 3)$

The lines intersect at this point.

Example 3:

Determine whether the following lines are identical or parallel and distinct

$$L_1 \equiv \begin{cases} x = 2 + r \\ y = 1 + 2r \\ z = 3 + 3r \end{cases} \quad L_2 \equiv \begin{cases} x = 3 - t \\ y = 3 - 2t \\ z = 6 - 3t \end{cases}$$

Solution

Two direction vectors for two lines are scalar multiples, means that the two lines are parallel. To determine if they are distinct or identical we need to check if there is a common point for two lines.

To check if there is a common point for two lines, let $r = 0$ for first line we have the point $(2, 1, 3)$. Using x -coordinate of this point in second line to obtain $2 = 3 - t \Rightarrow t = 1$. For this value of t , second line produce the point $(2, 1, 3)$. Then, this is one of the common points.

Or

To check if there is common points for two lines: we solve the following system

$$2 + r = 3 - t$$

$$1 + 2r = 3 - 2t$$

$$3 + 3r = 6 - 3t$$

Taking the first two equations, we have

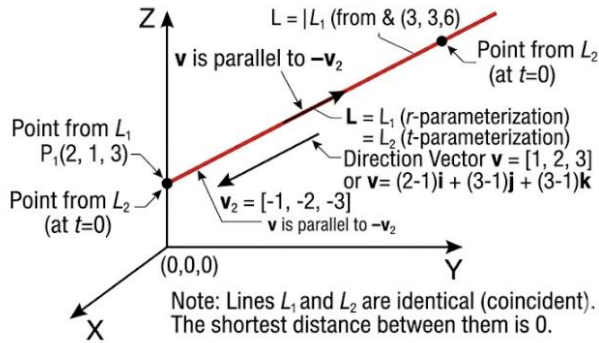
$$\begin{cases} 2 + r = 3 - t \\ 1 + 2r = 3 - 2t \end{cases} \Rightarrow \begin{cases} -4 - 2r = -6 + 2t \\ \underline{1 + 2r = 3 - 2t} \\ -3 = -3 \end{cases}$$

The obtained statement is true, means that there are two points on two lines with same x and y components.

Now, we check for z component. Taking the first and the last equations, we have

$$\begin{cases} 2+r=3-t \\ 1+2r=3-2t \end{cases} \Rightarrow \begin{cases} -4-2r=-6+2t \\ \underline{1+2r=3-2t} \\ -3=-3 \end{cases}$$

Again, the obtained statement is true. Then there are two points on two lines with same x , y and z components. Thus, there is a common point.



Hence the two lines are identical.

Example 4:

Prove that the lines $x-1=2-y=\frac{z+5}{2}$ and $\vec{r}=2\mu\vec{i}-3\vec{j}+(\mu-2)\vec{k}$ are skew.

Solution

Direction vector of first line is $\vec{u}=(1,-1,2)$ and direction vector of second line is $\vec{v}=(2,0,1)$.
 $\vec{u} \neq k\vec{v}$, then the two line are not parallel.

Parametric equations of first line:
$$\begin{cases} x=1+t \\ y=2-t \\ z=2t-5 \end{cases}$$

Parametric equations of second line:
$$\begin{cases} x=2\mu \\ y=-3 \\ z=\mu-2 \end{cases}$$

These two lines meets when
$$\begin{cases} 1+t=2\mu \\ 2-t=-3 \\ 2t-5=\mu-2 \end{cases}$$

From first two equations $t=5$ and $\mu=3$ but these values do not verify the third equation since $2(5)-5=5$ and $3-2=1$.

Therefore, the lines are skew.

Application activity 8.2.4:

In each of the following decide whether the given lines are skew or they intersect. If they intersect, find the coordinates of their common point.

- 1) $4x = 4y = z + 3$ and $\frac{x-7}{2} = y-5 = \frac{z-12}{6}$
- 2) $\vec{r} = -2\vec{i} - 3\vec{j} - 13\vec{k} + \lambda(2\vec{i} + 2\vec{j} + 3\vec{k})$ and $\vec{r} = -\vec{i} + 3\vec{j} - 5\vec{k} + \mu(3\vec{i} - 2\vec{j} - 2\vec{k})$
- 3) $\frac{x-1}{2} = y-1 = z-2$ and $x-4 = \frac{y+2}{3} = \frac{z+1}{-2}$

8.2.5 Angle between two lines

Activity 8.2.5

Consider two lines and $L_1 : \vec{r} = 5\vec{i} + 4\vec{j} + 5\vec{k} + \lambda(\vec{i} + \vec{j} - \vec{k})$ and $L_2 : x-1 = \frac{y+2}{5} = 3-z$

- 1) Find their direction vectors and then, the angle between those vectors.
- 2) Try and find the angle between these two lines.

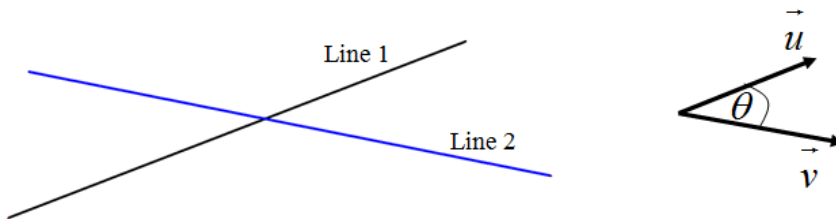
Remember this: depending on the position of observer, between two non-perpendicular lines, there are two angles (acute and obtuse).

Content summary

Let the lines be $L_1 : \vec{r} = \vec{a} + t\vec{u}$, and $L_2 : \vec{r} = \vec{b} + s\vec{v}$ where $\vec{u} = (a_1, b_1, c_1)$ and $\vec{v} = (a_2, b_2, c_2)$ are the **direction vectors**, where: $\vec{a}$ and $\vec{b}$ are position vectors of points on each line t and s are real parameters. We define the angle between two lines to be the acute angle (angle which lies

between 0 and 90 degrees) between their direction vectors, $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \times \|\vec{v}\|}$

Note that this definition works equally well if the lines do not actually cut each other since we then just slide the two direction vectors together until their tails meet.



The angle θ between the lines is the angle between their direction vectors.

The angle between the lines is found by working out the dot product of $\vec{u}$ and $\vec{v}$.

We have $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$

From this, knowing $\vec{u}$ and $\vec{v}$, θ is given by $\theta = \arccos \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|} \right)$

Example 1:

Calculate the angle between the lines AB and AC for $A(1,2,3)$, $B(4,5,6)$ and $C(3,2,0)$.

Solution

Line AB has direction vector $\vec{u} = (3, 3, 3)$

Line AC has direction vector $\vec{v} = (2, 0, -3)$

Let θ be the angle between two lines, then $\cos \theta = \frac{6+0-9}{\sqrt{27}\sqrt{13}} = \frac{-3}{\sqrt{351}}$

$$\theta = \arccos\left(\frac{-3}{\sqrt{351}}\right) = 99.2^\circ$$

The acute angle is $180 - 99.2 = 80.8$

Thus, between the lines AB and AC is 80.8 degrees.

Application activity 8.2.5

Find the angle between the lines

$$1) \quad L_1 \equiv \begin{cases} x = 1 + t \\ y = 4 + t \\ z = 8 \end{cases} \quad L_2 \equiv \begin{cases} x = 1 + 2t \\ y = 1 + t \\ z = -1 - 2t \end{cases}$$

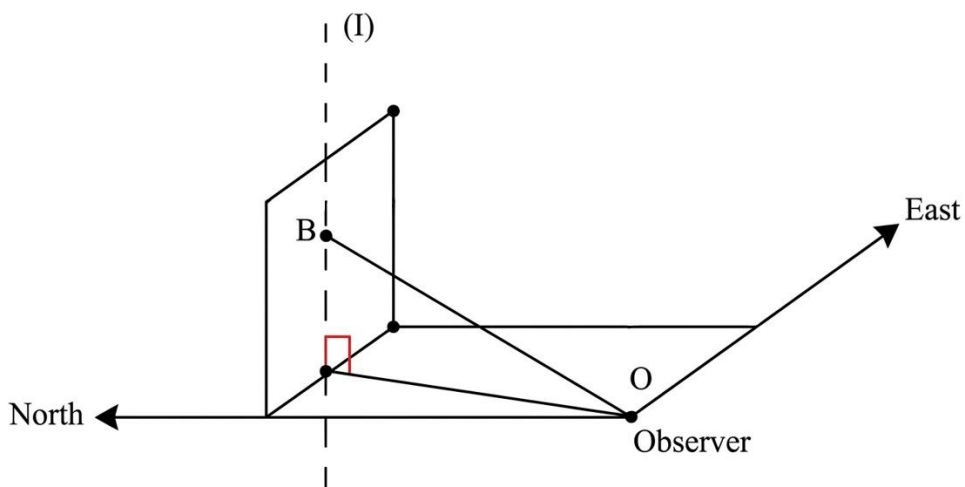
$$2) \quad L_1 \equiv \begin{cases} x = 2 + 2t \\ y = 1 + 2t \\ z = -2 + 2t \end{cases} \quad L_2 \equiv \begin{cases} x = 2 + 2t \\ y = 4 - 2t \\ z = -1 - t \end{cases}$$

to the nearest hundredth of a radian

8.2.6. Distance from a point to a line

Activity 8.2.6

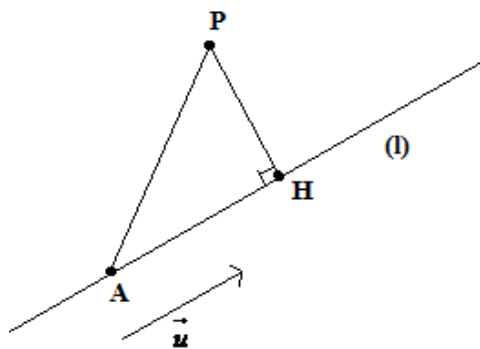
Consider the diagram below to answer the questions:



- a) Label A the intersection point of the plane through point O and perpendicular to line (l) and plot it on the diagram?
- b) What is the distance from point O to line (l) ?
- c) Write down a direction vector of line (l)
- d) In the right angled triangle OAB , let θ be the angle between the direction vector of line (l) and $\overrightarrow{BO}$, express the distance from point O to line (l) in terms of $\sin \theta$. Multiply and divide the side containing $\sin \theta$ by the magnitude of the direction vector. Hence, write down the formula for the distance from a point to a line.

Content summary

Let (l) be the straight line through point A and with direction vector $\vec{u}$.



To find the distance d from a point $P(x_0, y_0, z_0)$ to a line l in 3D, we use the properties of the **cross product**. The distance from point P to line (l) is the distance from P to H , where H is the foot of the perpendicular to (l) through P .

You need:

- **A point on the line:** Let's call it $A(x_1, y_1, z_1)$
- **A direction vector:** Let's call it $u(u_1, u_2, u_3)$
- vector $\overrightarrow{AP}$ that starts at the point A on the line and ends at the point P in space:

$$\overrightarrow{AP} = (x_0 - x_1, y_0 - y_1, z_0 - z_1)$$

The shortest distance d is the magnitude of the component of vector $\overrightarrow{AP}$ perpendicular to the line. Using the cross product, the formula is:

$$\|\overrightarrow{PH}\| = \frac{\|\overrightarrow{AP} \wedge \vec{u}\|}{\|\vec{u}\|}$$

Example

Find the distance from the point $Q(1, 3, -2)$ to the line given by the parametric equations:

$$\begin{cases} x = 2 + t \\ y = -1 - t \\ z = 3 + 2t \end{cases}$$

Solution

From the parametric equations we know the direction vector, $\vec{u} = (1, -1, 2)$ and if we let $t = 0$ a point A on the line is $A(2, -1, 3)$

$$\text{Thus } \overrightarrow{AQ} = \begin{pmatrix} 2-1 \\ -1-3 \\ 3+2 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 5 \end{pmatrix}$$

Find the cross product is $\vec{u} \wedge \overrightarrow{AQ} = -3\vec{i} + 3\vec{j} + 3\vec{k}$; its magnitude is $\|\vec{u} \wedge \overrightarrow{AQ}\| = 3\sqrt{3}$

The distance is $\frac{\|\vec{u} \wedge \overrightarrow{AQ}\|}{\|\vec{u}\|} = \frac{3\sqrt{3}}{\sqrt{6}} = \frac{3\sqrt{2}}{2}$ units

Application activity 8.2.6

Find the distance from the point A to the line (l) , in each of the following cases:

a) $A(0, 0, 12)$; $(l): \begin{cases} x = 4t \\ y = -2t \\ z = 2t \end{cases}$

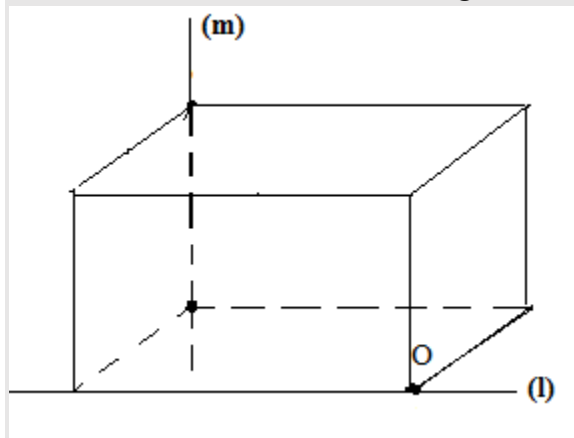
b) $A(2, 1, 3)$; $(l): \vec{r} = 2\vec{i} + \vec{j} + 3\vec{k} + \lambda(\vec{i} + 3\vec{j})$

c) $A(3, -1, 4)$; $(l): \begin{cases} x = 4 - t \\ y = 3 + 2t \\ z = -5 + 3t \end{cases}$

8.2.7 Distance between two skew lines

Activity 8.2.7

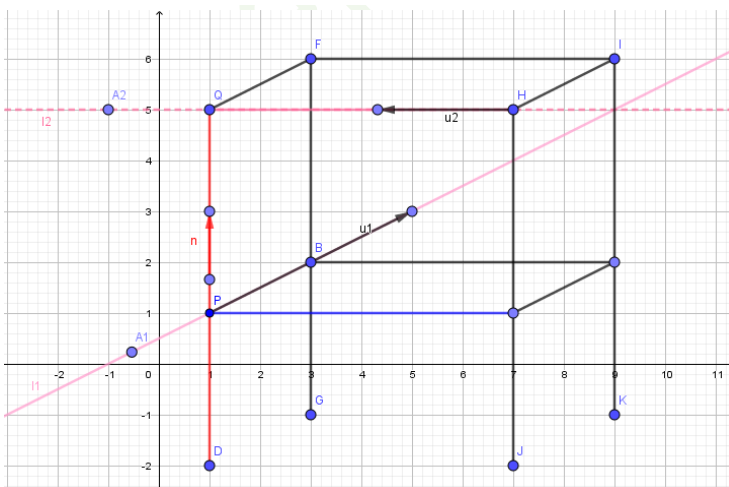
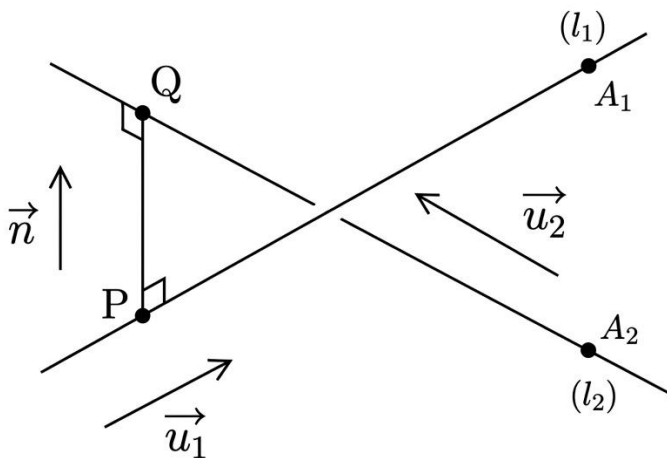
Use a real box and consider the diagram below to answer the questions:



- Copy the cuboid and shade the face illustrating the plane containing line (l) and perpendicular to line (m) .
- Plot the intersection point of line (m) and the plane shaded in part (a) and label it A
- Shade on the cuboid (in different color) the face illustrating the plane containing line (m) and perpendicular to line (l)
- Plot the intersection point of line (l) and the plane shaded in part (c) and label it B
- What is the distance between lines (l) and (m) ?

Content summary

Let (l_1) be the line through point A_1 and with direction vector $\vec{u}_1$, and (l_2) the line through point A_2 and with direction vector $\vec{u}_2$.



If P and Q are points, one on each line, which are the closest together, then the line (PQ) is perpendicular to both lines (l_1) and (l_2) , and hence parallel to the normal $\vec{n} = \vec{u}_1 \wedge \vec{u}_2$

The shortest distance lies along a vector that is perpendicular to both lines. We find this using the **cross product** of their direction vectors: $\vec{n} = \vec{u}_1 \wedge \vec{u}_2$

The distance is obtained in this way: we take a vector connecting any two points on the lines, $\overrightarrow{A_1A_2}$ and project it onto this normal vector $\vec{n} = \vec{u}_1 \wedge \vec{u}_2$. The formula for the shortest distance D

$$\text{is } \|\overrightarrow{PQ}\| = \frac{|\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2)|}{\|\vec{u}_1 \wedge \vec{u}_2\|}$$

In particular, if lines (l_1) and (l_2) are coincident or secant the distance between them is zero.

If lines (l_1) and (l_2) are parallel and distinct, then the distance between them is the distance from any point on any of the two lines to the other line.

Example

Find the distance between the lines (l_1) and (l_2) in each of the following cases:

$$a) (l_1): \vec{r} = 5\vec{i} + 3\vec{j} + t(2\vec{i} - \vec{j}) \quad \text{and} \quad (l_2): \vec{R} = 2\vec{i} + 9\vec{k} + s(\vec{j} - \vec{k})$$

$$b) (l_1): \frac{x-5}{3} = \frac{y-7}{-16} = \frac{z-3}{7} \quad \text{and} \quad (l_2): \frac{x-9}{3}, \frac{y-13}{8}, \frac{z-15}{-5}$$

Solution

$$a) (l_1) \text{ passes through point } A_1(5,3,0) \text{ and has direction vector } \vec{u}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$(l_2) \text{ passes through point } A_2(2,0,9) \text{ and has direction vector } \vec{u}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\vec{n} = \vec{u}_1 \wedge \vec{u}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \vec{k} = \vec{i} + 2\vec{j} + 2\vec{k}$$

The distance between $\vec{n} = \vec{u}_1 \wedge \vec{u}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \vec{k} = \vec{i} + 2\vec{j} + 2\vec{k}$ these lines is,

$$\|\vec{PQ}\| = \frac{|\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2)|}{\|\vec{u}_1 \wedge \vec{u}_2\|} \quad \text{where} \quad \overrightarrow{A_1A_2} = \begin{pmatrix} -3 \\ -3 \\ 9 \end{pmatrix}; \quad \overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2) = -3(2) - 3(2) + 9(2) = 9 \quad \text{and}$$

$$\|\vec{u}_1 \wedge \vec{u}_2\| = \sqrt{1^2 + 2^2 + 2^2} = 3$$

It follows that the distance between the two lines is $\|\vec{PQ}\| = \frac{|\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2)|}{\|\vec{u}_1 \wedge \vec{u}_2\|} = \frac{|9|}{3} = 3$

b) (l_1) passes through point $A_1(5, 7, 3)$ and has direction vector $\vec{u}_1 = \begin{pmatrix} 3 \\ -16 \\ 7 \end{pmatrix}$

(l_2) passes through point $A_2(9, 13, 15)$ and has direction vector $\vec{u}_2 = \begin{pmatrix} 3 \\ 8 \\ -5 \end{pmatrix}$

The distance between these lines is, $\|\vec{PQ}\| = \frac{|\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2)|}{\|\vec{u}_1 \wedge \vec{u}_2\|}$ where $\overrightarrow{A_1A_2} = \begin{pmatrix} 4 \\ 6 \\ 12 \end{pmatrix}$,

$$\vec{u}_1 \wedge \vec{u}_2 = \begin{vmatrix} -16 & 7 \\ 8 & -5 \end{vmatrix} \vec{i} - \begin{vmatrix} 3 & 7 \\ 3 & -5 \end{vmatrix} \vec{j} + \begin{vmatrix} 3 & -16 \\ 3 & 8 \end{vmatrix} \vec{k} = 14\vec{i} + 36\vec{j} + 72\vec{k}$$

$$\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2) = 4(14) + 6(36) + 12(72) = 1136$$

$$\text{and } \|\vec{u}_1 \wedge \vec{u}_2\| = \sqrt{14^2 + 36^2 + 72^2} = \sqrt{6676}$$

It follows that the distance between the two lines is $\|\vec{PQ}\| = \frac{|\overrightarrow{A_1A_2} \cdot (\vec{u}_1 \wedge \vec{u}_2)|}{\|\vec{u}_1 \wedge \vec{u}_2\|}$

$$\|\vec{PQ}\| = \frac{|1136|}{\sqrt{6676}} = 13.9$$

Application activity 8.2.7

Find the distance between the lines

$$1) \quad L_1 \equiv \begin{cases} x = 1 + 4t \\ y = 5 - 4t \\ z = -1 + 5t \end{cases} \quad L_2 \equiv \begin{cases} x = 2 + 8t \\ y = 4 - 3t \\ z = 5 + t \end{cases}$$

$$2) L_1 \equiv \begin{cases} x = 3 + 2t \\ y = -2t \\ z = 4 - t \end{cases} \quad L_2 \equiv \begin{cases} x = 4 - t \\ y = 3 + 5t \\ z = 2 - t \end{cases}$$

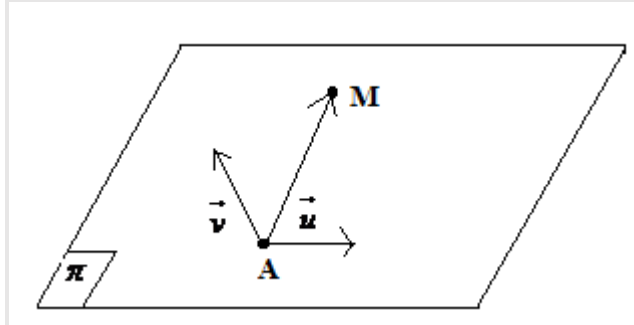
8.3. Planes in 3D

8.3.1. Equations of a plane in 3D

Activity 8.3.1

a) How is a plane determined

b) Plane π passes through point A and is parallel to both vectors $\vec{u}$ and $\vec{v}$ of different directions.



Express, mathematically, the relationship between vectors $\vec{u}, \vec{v}$ and $\overrightarrow{AM}$

c) Given that $M(x, y, z), A(x_0, y_0, z_0), \vec{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} a' \\ b' \\ c' \end{pmatrix}$, express the relationship found in

part (b) in three different ways, use $\vec{i}, \vec{j}, \vec{k}$ as unit vectors in the directions OX, OY and OZ , respectively

Content summary

In a three-dimensional space, a plane π is determined in six different ways that can be brought to one:

By a point A through which π passes and two vectors $\vec{u}$ and $\vec{v}$ of different directions, both parallel to the plane. Vectors $\vec{u}$ and $\vec{v}$ is the **direction** vectors of the plane.

Plane π through point A and with direction vectors $\vec{u}$ and $\vec{v}$ is the set of all points M in the space such that, $\overrightarrow{AM} = t\vec{u} + s\vec{v}$ for some scalars t and s .

The way the equality $\overrightarrow{AM} = t\vec{u} + s\vec{v}$ is transformed gives rise to different equations:

- **Vectorequation:** $\overrightarrow{OM} = \overrightarrow{OA} + t\vec{u} + s\vec{v}$ or,
 $(x, y, z) = (x_0, y_0, z_0) + t(a, b, c) + s(a', b', c')$ where t and s are parameters.

- The **parametric equations**:
$$\begin{cases} x = x_0 + at + a's \\ y = y_0 + bt + b's \\ z = z_0 + ct + c's \end{cases}$$
- The **Cartesian equation**: $n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0$ where
$$\vec{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \vec{u} \wedge \vec{v}$$

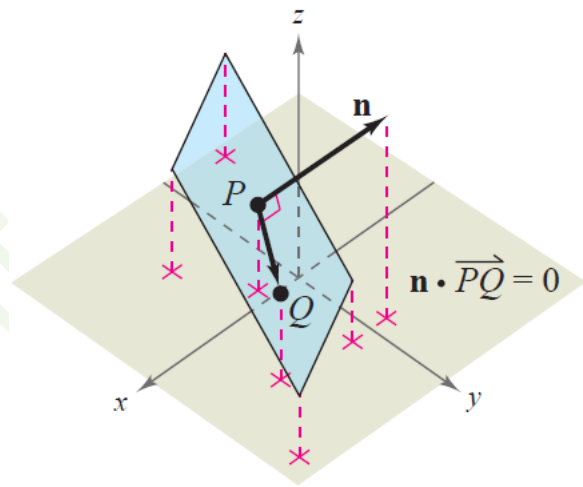
In general: Consider the plane containing the point $P(x_1, y_1, z_1)$ having a nonzero normal vector $\mathbf{n} = \langle a, b, c \rangle$, as shown

This plane consists of all points $Q(x, y, z)$ for which the vector PQ is orthogonal to $\mathbf{n}$. Using the dot product, you can write the following $\mathbf{n} \cdot \overrightarrow{PQ} = 0$

$$\begin{aligned} \langle a, b, c \rangle \cdot \langle x - x_1, y - y_1, z - z_1 \rangle &= 0 \\ a(x - x_1) + b(y - y_1) + c(z - z_1) &= 0 \end{aligned}$$

The plane containing the point (x_1, y_1, z_1) and having normal vector $\mathbf{n} = \langle a, b, c \rangle$ can be represented by the **standard form of the equation of a plane**:

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$



Example

- Find the vector, parametric and symmetric equations of the plane, α , passing through the point $A(2, 7, -1)$ with direction vectors $\vec{u} = (3, 1, 1)$ and $\vec{v} = (-1, -2, -3)$
- Find the vector, parametric and Cartesian equation of plane β containing points $A(3, -2, -1)$ and $B(4, 2, 7)$ with direction vector $\vec{u} = (1, 1, 3)$
- Find vector, parametric and Cartesian equation of plane β passing through points $A(1, 3, 5)$, $B(-2, 5, 4)$ and $C(3, -6, -5)$

Solution

a) Let $P(x, y, z)$ represents any point of plane α , t and s be the parameters.

The vector equation: $x\vec{i} + y\vec{j} + z\vec{k} = 2\vec{i} + 7\vec{j} - \vec{k} + t(3\vec{i} + \vec{j} + \vec{k}) + s(-\vec{i} - 2\vec{j} - 3\vec{k})$

Parametric equations:
$$\begin{cases} x = 2 + 3t - s \\ y = 7 + t - 2s \\ z = -1 + t - 3s \end{cases}$$

Cartesian equation:

$$\vec{u} \wedge \vec{v} = \begin{vmatrix} 1 & 1 \\ -2 & -3 \end{vmatrix} \vec{i} - \begin{vmatrix} 3 & 1 \\ -1 & -3 \end{vmatrix} \vec{j} + \begin{vmatrix} 3 & 1 \\ -1 & -2 \end{vmatrix} \vec{k} = -\vec{i} + 8\vec{j} - 5\vec{k}$$

The Cartesian equation is $-1(x - 2) + 8(y - 7) - 5(z + 1) = 0$

Which is equivalent to $\alpha \equiv x - 8y + 5z + 59 = 0$

b) Let $X(x, y, z)$ be any point on the plane

Direction vectors are $\overrightarrow{AB} = (1, 4, 8)$ and $\vec{u} = (1, 1, 3)$

The vector equation: $\vec{r} = 3\vec{i} - 2\vec{j} - \vec{k} + t(\vec{i} + 4\vec{j} + 8\vec{k}) + s(\vec{i} + \vec{j} + 3\vec{k})$, where t and s are parameters

Parametric equations:
$$\begin{cases} x = 3 + t + s \\ y = -2 + 4t + s \\ z = -1 + 8t + 3s \end{cases}$$

Cartesian equation: $\vec{u} \wedge \vec{v} = \begin{vmatrix} 4 & 8 \\ 1 & 3 \end{vmatrix} \vec{i} - \begin{vmatrix} 1 & 8 \\ 1 & 3 \end{vmatrix} \vec{j} + \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} \vec{k} = 4\vec{i} + 5\vec{j} - 3\vec{k}$

The Cartesian equation is $4(x - 3) + 5(y + 2) - 3(z + 1) = 0$

Which is equivalent to $4x + 5y - 3z - 5 = 0$

c. Let A be the starting point. Then the two direction vectors are $\overrightarrow{AB} = (-3, 2, -1)$ and $\overrightarrow{AC} = (2, -9, -10)$

Let $X(x, y, z)$ represents any point on this plane, then

Vector equation, $\vec{r} = \vec{i} + 3\vec{j} + 5\vec{k} + t(-3\vec{i} + 2\vec{j} - \vec{k}) + s(2\vec{i} - 9\vec{j} - 10\vec{k})$ where t and s are parameters.

Parametric equations:
$$\begin{cases} x = 1 - 3t + 2s \\ y = 3 + 2t - 9s \\ z = 5 - t - 10s \end{cases}$$

Cartesian equation: $29x + 32y - 23z - 10 = 0$

Application activity 8.3.1

Find vector, parametric and Cartesian equation of plane β passing through points

- a) $A(2, 4, 1), B(1, 3, -1)$ and $C(2, 1, 3)$
- b) $A(1, 1, 1), B(4, -2, 1)$ and $C(-2, 4, 3)$
- c) $A(3, 6, 0), B(1, 0, 1)$ and $C(5, 1, 7)$
- d) $A(4, 3, 8), B(-4, 1, 1)$ and $C(-2, 8, 6)$

8.3.2 Condition of co-planarity of four points

Activity 8.3.2

From the equation of plane passing through the three given points, determine which of the following sets of points lie on the same plane

- 1) $A = \{(1, 2, -1), (2, 3, 1), (3, -1, 0)\}$ and $\{(1, 2, 1)\}$
- 2) $B = \{(-2, 1, 1), (0, 2, 3), (1, 0, -1)\}$ and $\{(2, 1, -1)\}$
- 3) $C = \{(1, 0, -1), (0, 2, 3), (-2, 1, 1)\}$ and $\{(4, 2, 3)\}$

Is there any shortcut to verify if the given four points lie in the same plane?

If there is, indicate any.

Content summary

Suppose you have four points: A, B, C and D such that their coordinates are respectively.

$(a_1, a_2, a_3); (b_1, b_2, b_3); (c_1, c_2, c_3)$ and (d_1, d_2, d_3)

To check if they are coplanar, create three vectors starting from a common point (e.g., point A):

$$\begin{aligned}\{AB\} &= B - A = (b_1 - a_1, b_2 - a_2, b_3 - a_3) \\ \bullet \quad \{AC\} &= C - A = (c_1 - a_1, c_2 - a_2, c_3 - a_3) \\ \{AD\} &= D - A = (d_1 - a_1, d_2 - a_2, d_3 - a_3)\end{aligned}$$

The points are coplanar if their **scalar triple product** is zero:

$$AB \cdot (AC \times AD) = 0$$

If the points are coplanar, the 3 vectors are linearly dependent, meaning one can be written as a combination of the others.

It means that they lie on the same plane if the following condition is satisfied.

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} a_1 - d_1 & b_1 - d_1 & c_1 - d_1 \\ a_2 - d_2 & b_2 - d_2 & c_2 - d_2 \\ a_3 - d_3 & b_3 - d_3 & c_3 - d_3 \end{vmatrix} = 0$$

or

The use of the transpose matrix is also correct.

Example 1: Show that the points $(4,0,0)$, $(0,6,0)$, $(0,0,3)$ and $\left(1, \frac{9}{2}, 0\right)$ are coplanar.

Solution:
$$\begin{vmatrix} 4 & 0 & 0 & 1 \\ 0 & 6 & 0 & \frac{9}{2} \\ 0 & 0 & 3 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0$$

Let the third row be fixed:

$$\begin{vmatrix} 4 & 0 & 0 & 1 \\ 0 & 6 & 0 & \frac{9}{2} \\ 0 & 0 & 3 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0 \begin{vmatrix} 0 & 0 & 1 \\ 6 & 0 & \frac{9}{2} \\ 1 & 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 4 & 0 & 1 \\ 0 & 0 & \frac{9}{2} \\ 1 & 1 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 0 & 1 \\ 0 & 6 & \frac{9}{2} \\ 1 & 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 4 & 0 & 0 \\ 0 & 6 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 0 - 0 + 3(24 + 0 + 0 - 6 - 18) = 0$$

Thus, the given points are coplanar.

Note: Three distinct points are always coplanar, but a fourth point or more added in space can exist in another plane. So, if points are in a line, it is possible to put them on a plane, but there may be points on the plane that are not all in a straight line, thus collinear points are coplanar but coplanar points need not to be collinear.

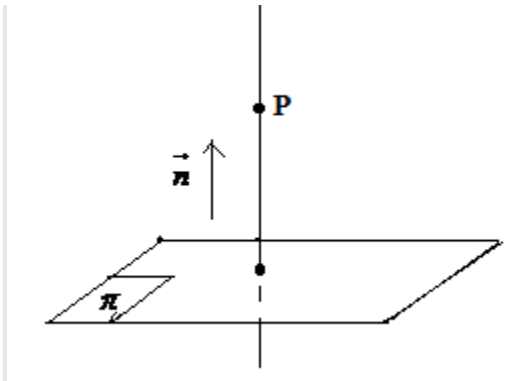
Application activity 8.3.2

1. Determine if the points $A(1,2,3)$, $B(4,7,8)$, $C(3,5,5)$, $D(-1,-2,-3)$ and $E(2,2,2)$ are coplanar.
2. Calculate the value of x for the coplanar set of points $A(0,0,1)$, $B(0,1,2)$, $C(-2,1,3)$ and $D(x, x-1, 2)$.
3. What is the condition for a, b and c so that the points $A(1,0,1)$, $B(1,1,0)$, $C(0,1,1)$ and $D(a, b, c)$ are coplanar?
4. Calculate the value of a for the points $(a, 0, 1)$, $(0, 1, 2)$, $(1, 2, 3)$ and $(7, 2, 1)$ so that they are coplanar. Also, calculate the equation of the plane that contains them.

8.3.3. Distance from a point to a plane

Activity 8.3.3

Consider the diagram below to answer the questions:



- Copy the diagram and plot the intersection point of the line (l) perpendicular to plane π through point P and label it H.
- What is the distance from point P to plane π ?
- Express mathematically the relationship between the direction vector $\vec{n}$ of line (l) and vector $\overrightarrow{HP}$. Hence, find the parametric equations of line (l) through point $P(x_0, y_0, z_0)$ and with direction vector $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$
- Use the values of x , y and z from parametric equations of line (l) in the Cartesian equation of plane $\pi : ax + by + cz + d = 0$, to find the value of the parameter.
- Find the expression of the distance from point $P(x_0, y_0, z_0)$ to plane $\pi : ax + by + cz + d = 0$

Content summary

To find the distance from the plane to the point P,

- Create vector QP where the point Q is on the plane such that the vector QP is perpendicular to the plane. This vector goes from the plane to your point.

- Project vector QP onto vector normal of the plane: The scalar projection of vector PQ in the direction of the normal vector gives the perpendicular height. Its magnitude is the distance requested:

$$d = \frac{|\overrightarrow{QP} \cdot \vec{n}|}{|\vec{n}|} = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

In particular:

- If point P lies in the plane π , then the distance from P to the plane is zero.
- If P lies in the plane such that the two planes are parallel (their Cartesian equations can be expressed as $ax + by + cz + d = 0$; $ax + by + cz + d' = 0$) then to find the distance between

them, it is sufficient to find the distance from a point in one of the two planes to the other plane.

- If a straight line is parallel to a plane, the direction vector $\vec{w}$ of the line can be expressed as $\vec{w} = t\vec{u} + s\vec{v}$ for some real values t and s , where $\vec{u}$ and $\vec{v}$ are direction vectors of the plane.

Alternatively, the dot product of $\vec{w} \cdot (\vec{u} \wedge \vec{v}) = \vec{w} \cdot \vec{n} = 0$. In this case, to find the distance from the line to the plane, it is sufficient to find the distance from a point on the line to the plane.

Example

- Find the distance from the point $P(2, 4, 7)$ to the plane. $\alpha \equiv 3x + 5y - 6z = 18$
- Show that planes $\pi_1 : 3x - y + 2z - 6 = 0$ and $\pi_2 : 6x - 2y + 4z + 4 = 0$ are parallel and find the distance between them
- Show that line $(l) : \frac{x-2}{3} = \frac{y-1}{-2} = \frac{z-1}{5}$ is parallel to plane $\pi : x - y - z + 2 = 0$ and find the distance from the line to the plane

Solution

a) The distance is $\frac{|3(2) + 5(4) - 6(7) - 18|}{\sqrt{3^2 + 5^2 + (-6)^2}} = \frac{34}{\sqrt{70}}$

b) The equation $6x - 2y + 4z + 4 = 0$ can be re organized as $3x - y + 2z + 2 = 0$ which shows that the two planes are parallel. A point in plane $6x - 2y + 4z + 4 = 0$ is $p(0, 0, 2)$. The distance between the planes is $\frac{|3(0) - 0 + 2(2) - 6|}{\sqrt{9 + 1 + 4}} = \frac{6}{\sqrt{14}}$

c) The direction vector of the line is $\vec{u} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$ and a vector normal to the plane is $\vec{n} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$

We have $\vec{u} \cdot \vec{n} = 3(1) - 2(-1) + 5(-1) = 0$ Therefore; the line is parallel to the plane.

The distance from the line to the plane is $\frac{|2 - 1 - 1 + 2|}{\sqrt{1 + 1 + 1}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

Application activity 2.3.3

- Find the distance from the point $(2, -3, 4)$ to the plane $x + 2y + 2z = 13$
- Find the distance from the point $(0, 1, 1)$ to the plane $4y + 3z = -12$
- Find the distance from the point $(0, -1, 0)$ to the plane $2x + y + 2z = 4$
- Find the shortest distance between the planes $x + 2y + 6z = 1$ and $x + 2y + 6z = 10$
- Find the shortest distance between the planes $-2x + y + z = 0$ and $6x - 3y - 3z - 5 = 0$

8.3.4 Position of a line and a plane

Activity 8.3.4

- 1) Let $\vec{u} = (a, b, c)$ be the direction vector of the line L perpendicular to the plane passing through the points $A(a_1, a_2, a_3)$ and $X(x, y, z)$.
 - a) From two points $A(a_1, a_2, a_3)$ and $X(x, y, z)$, find the direction vector $\overrightarrow{AX}$ of plane α
 - b) Find the scalar product of the vector $\vec{u}$ and the vector $\overrightarrow{AX}$ obtained in 1 and equate the result to zero since the two vectors are perpendicular. Expand the obtained equation. What can you say about the expanded equation?
- 2) In each of the following pairs of line and plane, after verifying if they are parallel or not, determine whether the given line lies in the plane or they are strictly parallel or if the line pierces the plane.
 - a) $\vec{r} = 4\vec{i} + \vec{j} + 2\vec{k} + \lambda(3\vec{i} + \vec{j} + 2\vec{k})$ and $\vec{r} \cdot (\vec{i} + \vec{j} + \vec{k}) = 4$, $[\vec{r} = (x, y, z)]$
 - b) $x - 1 = 2 - y = \frac{3 - z}{4}$ and $\vec{r} \cdot (2\vec{i} - 2\vec{j} + \vec{k}) = 1$, $[\vec{r} = (x, y, z)]$

Content summary

A line L is perpendicular to plane α if and only if each direction vector of L is perpendicular to each direction vector of α or the scalar product of direction vector of the line and the direction vector of the plane is zero.

In this case the direction vector of the line is perpendicular to the plane and is said to be the **normal** or **orthogonal** vector of the plane.

Note that the normal vector of the plane can be found by finding the vector product of its two non proportional direction vectors.

The Cartesian equation of plane passing through the point (a_1, a_2, a_3) with orthogonal vector (a, b, c) is $a(x - a_1) + b(y - a_2) + c(z - a_3) = 0$

A line and a plane are parallel if the direction vector of the line is perpendicular to the normal vector of the plane. There are two possibilities: they are strictly parallel or the line lies in the plane. To see this do the following: Find any point of the line and check if this point lies on the plane. If this point lies on the plane, then the line is in the plane otherwise the plane and the line are strictly parallel.

In summary: For a line in parametric equation $l: \vec{r} = \vec{r}_0 + t\vec{d}$ where $\vec{r}_0$ is a point on the line. $\vec{d}$ is the direction vector.

A plane $\pi: ax + by + cz + d = 0$ and $\vec{n} = (a, b, c)$ is the normal to the plane. The position of the line and the plane is described:

Condition	Meaning
$\vec{d} \cdot \vec{n} = 0$	Line $\parallel$ plane
$\vec{d} \cdot \vec{n} = 0$ and point satisfies plane	Line lies in plane
$\vec{d} \cdot \vec{n} \neq 0$	Line intersects plane
$\vec{d} = k\vec{n}$	Line $\perp$ plane

Example 1: Find the Cartesian equation of plane α passing through the point $(2, -3, 4)$ and perpendicular to the line defined by the points $A(1, 5, 7)$ and $B(-2, 2, 3)$.

Solution: The direction vector of the line is $\vec{AB} = (-3, -3, -4)$.

This is the normal vector of the plane α

$$\text{Thus, } \alpha \equiv -3(x - 2) - 3(y + 3) - 4(z - 4) = 0$$

$$\text{Or } \alpha \equiv 3x + 3y + 4z - 13 = 0$$

Example 2: Find the equation of plane β passing through the point $P(6, -1, 9)$ and

$$\text{perpendicular to the line } \begin{cases} 4x - 3y + 5z = 13 \\ 6x + 7y - 4z = 23 \end{cases}$$

Solution

The direction vector of the line is

$$\left(\begin{vmatrix} -3 & 5 \\ 7 & -4 \end{vmatrix}, -\begin{vmatrix} 4 & 5 \\ 6 & -4 \end{vmatrix}, \begin{vmatrix} 4 & -3 \\ 6 & 7 \end{vmatrix} \right) = (12 - 35, 16 + 30, 28 + 18) = (-23, 46, 46)$$

This vector can be written as $(-23, 46, 46) = -23(1, -2, -2)$ Thus, we can take the direction vector to be $(1, -2, -2)$.

Remember that to find the direction vector of the line we can equate the right-hand sides to zero,

$$\text{i.e., } \begin{cases} 4x - 3y + 5z = 0 \\ 6x + 7y - 4z = 0 \end{cases}$$

Next, replace any variable in the equation by any chosen value and find values of other remaining variables. Here, let $x = 1$, we have

$$\begin{cases} 4-3y+5z=0 \\ 6+7y-4z=0 \end{cases} \Leftrightarrow \begin{cases} -3y+5z=-4 \\ 7y-4z=-6 \end{cases} \Leftrightarrow \begin{cases} -12y+20z=-16 \\ 35y-20z=-30 \\ \hline 23y=-46 \Rightarrow y=-2, z=-2 \end{cases}$$

Then the equation of plane is

$$\beta \equiv (x-6) - 2(y+1) - 2(z-9) = 0$$

Or

$$\beta \equiv x - 2y - 2z + 10 = 0$$

Example 3: Find the equation of plane passing through the point $A(2, 3, -6)$ with vectors $\vec{u} = (-1, 5, 3)$ and $\vec{v} = (4, -4, 1)$ as direction vectors.

Solution

The normal $\vec{n}$ vector of this plane is given by the vector product of its two direction vectors.

$$\vec{n} = \vec{u} \times \vec{v} = \left(\begin{vmatrix} 5 & 3 \\ -4 & 1 \end{vmatrix}, -\begin{vmatrix} -1 & 3 \\ 4 & 1 \end{vmatrix}, \begin{vmatrix} -1 & 5 \\ 4 & -4 \end{vmatrix} \right) = (17, 13, -16)$$

The equation of plane is

$$17(x - 2) + 13(y - 3) - 16(z + 6) = 0 \text{ or } 17x + 13y - 16z = 169$$

Example 4: Show that the plane $2x - y - 3z = 4$ is parallel to the line

$$\begin{cases} x = -2 + 2t \\ y = -1 + 4t \\ z = 4 \end{cases}$$

Solution

First, we check if the normal vector of the plane and the direction vector of the line are perpendicular.

$\vec{n} = (2, -1, -3)$ is normal to the plane. $\vec{u} = (2, 4, 0)$ is the direction vector of the line.

$\vec{n} \cdot \vec{u} = 4 - 4 + 0 = 0$. Then, the given line and plane are parallel.

One point of the line is $(-2, -1, 4)$ for $t = 0$. We must check whether this point lie or does not lie

on the plane. That is, $2(-2) - 1(-1) - 3(4) \stackrel{?}{=} 4 \Rightarrow -15 \neq 4$

Then, this point does not lie on the plane.

Hence, the line and the plane are parallel and distinct.

Application activity 8.3.4

3) Consider the plane $2x + y - 4z = 4$ and the line

$$\begin{cases} x = t \\ y = 2 + 3t \\ z = t \end{cases}$$

Find all points of intersection.

4) Consider the plane $2x + y - 4z = 4$ and the line

$$\begin{cases} x = 1 + t \\ y = 4 + 2t \\ z = t \end{cases}$$

Find all points of intersection.

3. Consider the plane $2x + y - 4z = 4$ and the line

$$\begin{cases} x = t \\ y = 4 + 2t \\ z = t \end{cases}$$

Find all points of intersection.

8.3.5 Angles of lines and planes

Activity 8.3.5

- When analyzing the relationship between a line, its projection, and the normal vector, what geometric shape is formed that allows us to use trigonometric ratios
- Explain why we use the sine of the angle θ rather than the cosine when calculating the relationship between a line's direction vector $\vec{u}$ and a plane's normal vector $\vec{n}$
- How do we express the sine of the angle between a line and a plane in terms of the Cartesian components of the direction vector $\vec{a} = (a, b, c)$ and the normal vector $\vec{n} = (n_1, n_2, n_3)$?

Content Summary

To find the angle θ between different spatial elements, use the following formulas:

A. Angle Between Two Lines

Given direction vectors $\vec{u}$ and $\vec{v}$

$$\cos(\theta) = \frac{|\vec{u} \cdot \vec{v}|}{\|\vec{u}\| \|\vec{v}\|}$$

B. Angle Between Two Planes

Two distinct planes in three-space either are parallel or intersect in a line. If they intersect, you can determine the angle $\theta: 0 \leq \theta \leq 90^\circ$ between them from the angle between their normal vectors.

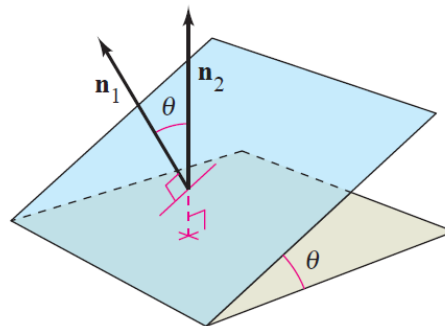
Specifically, if vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are normal to two intersecting planes, the angle θ between the normal vectors is equal to the **angle between the**

two planes and is given by $\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}$

Consequently, two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are

- *perpendicular* if $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$.

- *parallel* if vector $\mathbf{n}_1$ is a scalar multiple of $\mathbf{n}_2$.



C) Angle Between a Line and a Plane

Given a direction vector $\vec{u}$ for the line and a normal vector $\vec{n}$ for the plane:

$$\sin(\theta) = \frac{|\vec{u} \cdot \vec{n}|}{\|\vec{u}\| \|\vec{n}\|}$$

Example:

Find the angle between the line with direction vector $\vec{u} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$, the plane with normal vector

$$\vec{n} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

Solution: Calculate the scalar product: $\vec{u} \cdot \vec{n} = (1)(2) + (-1)(1) + (1)(2) = 2 - 1 + 2 = 3$

Calculate the magnitudes: $\|\vec{u}\| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$

$$\|\vec{n}\| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

Use the Sine formula:

$$\sin(\theta) = \frac{3}{3\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\theta = \arcsin\left(\frac{1}{\sqrt{3}}\right) \approx 35.26^\circ$$

Note: We use sin here because the angle between the line and the plane is the complement of the angle between the line and the normal vector.

Application activity 8.3.5

1) Determine the angle α between the direction vector of a line $\vec{u} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$ and the normal vector

of a plane $\vec{n} = \begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix}$

2) Find the angle between the line with direction vector $\vec{u} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and the plane $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

3) Calculate the angle between the line $\frac{x}{1} = \frac{y}{\sqrt{3}} = \frac{z}{0}$ and the plane $z = 0$

4) Calculate the obtuse angle between two intersecting lines whose direction vectors are

$$\vec{a} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} \text{ and } \vec{b} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

5) A light ray follows the direction vector $\vec{v} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Determine the angle θ it makes with the plane

$$x + y + \sqrt{2}z = 4$$

6) Find the angle between the line $\vec{r} = \lambda \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix}$ and the plane $y + 5z = 1$

7) Determine the angle between the line defined by the parametric equations $x = t, y = t, z = t$ and the plane $x + z = 0$

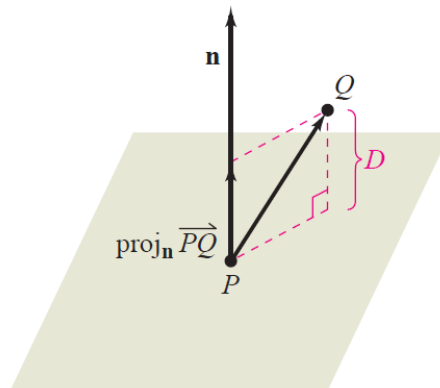
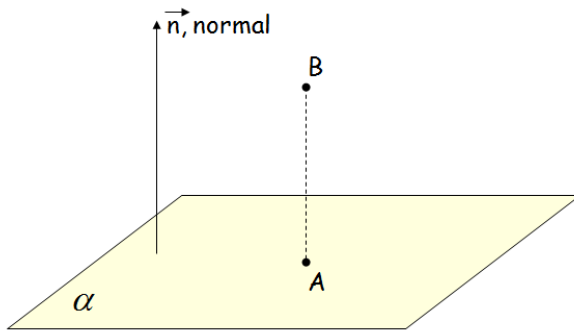
8.3.6 Shortest distance from a point to a plane

Activity 8.3.6

Consider plane $\alpha \equiv ax + by + cz = d$ passing through point $A(a_1, a_2, a_3)$. Consider another point $B(a_1, a_2, a_3)$ which does not lie on plane α . Let $\vec{e}$ be the normalized normal vector of plane α . The shortest distance from point B to plane α can be written as $\|\vec{AB}\| = \|\vec{e}\| \|\vec{AB}\| = \|\vec{e} \cdot \vec{AB}\|$. Develop the expression $\|\vec{e} \cdot \vec{AB}\|$ to find the expression for $\|\vec{AB}\|$.

Content summary:

The distance from point B to plane α is the shortest distance given by the length of perpendicular from that point to the plane.



$$d(B, \alpha) = d(A, B) = \|\vec{AB}\|$$

The **distance between a plane and a point** Q (not in the plane) is

$$D = \|\text{proj}_{\mathbf{n}} \vec{PQ}\| = \frac{|\vec{PQ} \cdot \mathbf{n}|}{\|\mathbf{n}\|} \text{ where } P \text{ is a point in the plane and } \mathbf{n} \text{ is normal to the plane.}$$

From activity, the distance from point $B(b_1, b_2, b_3)$ to plane $\alpha \equiv ax + by + cz = d$ is given by

$$d(B, \alpha) = \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{\|\mathbf{n}\|} = \frac{|ab_1 + bb_2 + cb_3 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

From this result we deduce the **normal equation** of a plane: $lx + my + nz + t = 0$ is the equation of the plane. The vector (l, m, n) is normal to the plane. If $l.l + m.m + n.n = 1$ that normal vector is a unit vector. In that case we say that $lx + my + nz + t = 0$ is a normal equation of the plane.

Example:

$$0.5x - 0.5y + \frac{\sqrt{2}}{2}z + 5 = 0 \text{ is the normal equation of the plane}$$

Transform an equation of a plane to a normal equation

If $\alpha \equiv ax + by + cz = d$ then, normal equation of this plane is $\alpha \equiv \frac{ax + by + cz - d}{\pm\sqrt{a^2 + b^2 + c^2}} = 0$

General rule for calculating the shortest distance between a point and a plane

To find the distance between a point and plane,

- Find the normal equation of the plane, which is $\alpha \equiv \frac{ax + by + cz - d}{\pm\sqrt{a^2 + b^2 + c^2}} = 0$
- In the expression $\frac{ax + by + cz - d}{\sqrt{a^2 + b^2 + c^2}}$ replace the variables by the corresponding coordinates of the point.

Example 2: Find the distance from the point $p(2, 4, 7)$ to the plane $\alpha \equiv 3x + 5y - 6z = 18$

Solution: The normal equation of this plane is $\alpha \equiv \frac{3x + 5y - 6z - 18}{\pm\sqrt{9 + 25 + 36}} = 0$ Or

$$\alpha \equiv \frac{3x + 5y - 6z - 18}{\pm\sqrt{70}} = 0$$

Now, in the expression $\frac{3x + 5y - 6z - 18}{\sqrt{70}}$, replacing the variables by the corresponding coordinates of the point, we have

$$d(P, \alpha) = \frac{|3 \times 2 + 5 \times 4 - 6 \times 7 - 18|}{\sqrt{70}} \text{ Or } d(P, \alpha) = \frac{34}{\sqrt{70}}$$

Note:

1. Shortest distance between two planes

When calculating the distance between two planes, first check if the planes are parallel. If they are not, they intersect and the distance is zero. If they are parallel, find an arbitrary point in one of the planes and calculate its distance to the other plane. If two planes are parallel, they never meet, and the distance between them is constant at every point. To find the distance D between two parallel planes given by the equations:

- Plane 1: $ax + by + cz + d_1 = 0$

2. Plane 2: $ax + by + cz + d_2 = 0$

(Note: Since they are parallel, they share the same normal vector $\mathbf{n} = \langle a, b, c \rangle$. If the coefficients are multiples, you must simplify them so a , b , and c are identical in both equations before using this formula.)

The distance formula is: $D = \frac{|d_2 - d_1|}{\sqrt{a^2 + b^2 + c^2}}$

Note that if two planes coincide (identical) the shortest distance is zero.

If you do not want to memorize the specific "plane-to-plane" formula, you can use the **point-to-plane** method:

1. Pick any arbitrary point $P(x_1, y_1, z_1)$ that lies on the first plane.
2. Calculate the distance from that point P to the second plane ($ax + by + cz + d_2 = 0$)

using the standard distance formula: $D = \frac{|ax_1 + by_1 + cz_1 + d_2|}{\sqrt{a^2 + b^2 + c^2}}$

Example: Calculate the distance between the two planes given below:

$$2x - 3y + 3z = 12 \text{ and } -6x + 9y - 9z = 27$$

Solution: First, let us check to see if they are parallel.

Their normal vectors are $\vec{n} = (2, -3, 3)$ and $\vec{m} = (-6, 9, -9)$. We see that $\vec{m} = -3\vec{n}$, then the two planes are parallel. Now, pick a point in the second plane and calculate the distance to the first plane. Let $x = y = 0$. Then $z = -3$. A point in the second plane is $P(0, 0, -3)$. Use the distance formula to calculate the distance from point P to the first plane. The first plane is:

$$2x - 3y + 3z = 12$$

$$\Leftrightarrow 2x - 3y + 3z - 12 = 0$$

Distance is $\frac{|2 \times 0 - 3 \times 0 - 3 \times 3 - 12|}{\sqrt{2^2 + (-3)^2 + 3^2}} = \frac{21}{\sqrt{22}}$

2. Shortest distance between a line and a plane

When calculating the distance between a line and a plane, first check if the line is parallel to the plane. If not, they intersect and the distance is zero. If they are parallel, find a point on the line and calculate its distance to the plane.

Example: Find the shortest distance between the plane $2x - y - 3z = 4$ and the line

$$\begin{cases} x = -2 + 2t \\ y = -1 + 4t \\ z = 4 \end{cases}$$

Solution: First, we check if the line is parallel to the plane. We need to know that the normal vector of the plane and the direction vector of the line are perpendicular.

$\vec{n} = (2, -1, -3)$ is normal to the plane $\vec{u} = (2, 4, 0)$ is the direction vector of the line $\vec{n} \cdot \vec{u} = 4 - 4 + 0 = 0$. Then, the given line and the given plane are parallel. One point of the line is $(-2, -1, 4)$ for $t = 0$. The distance from this point to the plane is

$$d = \frac{|2(-2) - (-1) - 3(4)|}{\sqrt{4 + 1 + 9}} = \frac{|-4 + 1 - 12 - 4|}{\sqrt{14}} = \frac{19}{\sqrt{14}} \text{ units}$$

Application activity 8.3.6

- 1) Find the distance from the point $(2, -3, 4)$ to the plane $x + 2y + 2z = 13$
- 2) Find the shortest distance between the planes $x + 2y + 6z = 1$ and $x + 2y + 6z = 10$
- 3) Find the shortest distance between the planes $-2x + y + z = 0$ and $6x - 3y - 3z - 5 = 0$

8.3.7 Position of two planes

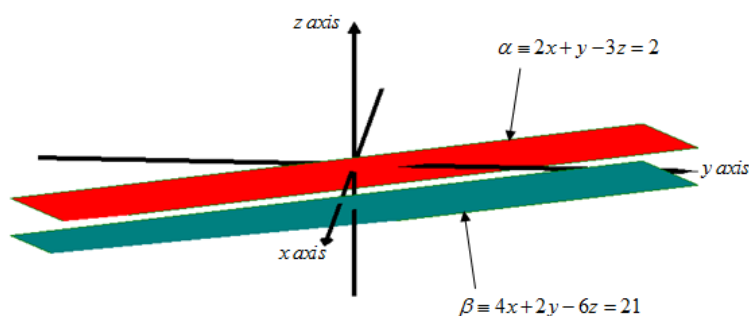
Activity 8.3.7

The equations of two planes are $\alpha \equiv 2x + y - 3z = 2$ and $\beta \equiv 4x + 2y - 6z = 21$

- a) Find the normal vector for each plane
- b) What is the relationship between the two normal vectors?
- c) Consider the relationship of normal vector and the independent constant and conclude about the position of the two planes: are they coinciding or parallel?

Since $\vec{v} = 2\vec{u}$, $\vec{v} = 2\vec{u}$ the two planes are parallel. We need to know if they are distinct or not.

$(2, 1, -3, 2)$ and $(4, 2, -6, 21)$ are not proportional, thus the two planes are parallel and distinct and hence no intersection between them.



Content summary

Consider planes of equations

$$\alpha \equiv a_1x + b_1y + c_1z = d_1$$

$$\beta \equiv a_2x + b_2y + c_2z = d_2$$

We need $S = \alpha \cap \beta$ that is

$$S = \{(x, y, z) \in \mathbb{R}^3 : a_1x + b_1y + c_1z = d_1 \text{ and } a_2x + b_2y + c_2z = d_2\}.$$

Two cases occur:

Case 1. Normal vectors are proportional:

$$(a_1, b_1, c_1) = k(a_2, b_2, c_2), \quad k \in \mathbb{R}_0 \Rightarrow \alpha \parallel \beta$$

- If $(a_1, b_1, c_1, d_1) = k(a_2, b_2, c_2, d_2), \quad k \in \mathbb{R}_0$

The two planes coincide. That is, $\alpha = k\beta, k \in \mathbb{R}_0$. So $S = \alpha$ or $S = \beta$

Normal vectors are parallel (collinear) and the distance from origin satisfies the same proportion.

The planes are identical; every point on P_1 lies on P_2 .

- If $(a_1, b_1, c_1, d_1) \neq k(a_2, b_2, c_2, d_2), \quad k \in \mathbb{R}_0$

The two planes are parallel and distinct and hence no intersection. Thus, $S = \emptyset$.

Normal vectors are parallel, but the constants are **not** in the same proportion. The planes never intersect; they are parallel with a constant distance between them.

Case 2. Normal vectors are not proportional:

$$(a_1, b_1, c_1) \neq k(a_2, b_2, c_2), \quad k \in \mathbb{R}_0 \Rightarrow \alpha \nparallel \beta$$

The two planes intersect and their intersection is a line defined by the equations of the two planes taken together. The planes intersect in a **straight line**.

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \end{cases}$$

Note that the general equation of a straight line in space is

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \end{cases}$$

The direction vector of this line is

$$\left(\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}, -\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}, \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \right)$$

Or to find the direction vector of the line we can equate the right-hand sides of the general equations to zero.

$$\text{i.e., } \begin{cases} a_1x + b_1y + c_1z = 0 \\ a_2x + b_2y + c_2z = 0 \end{cases}$$

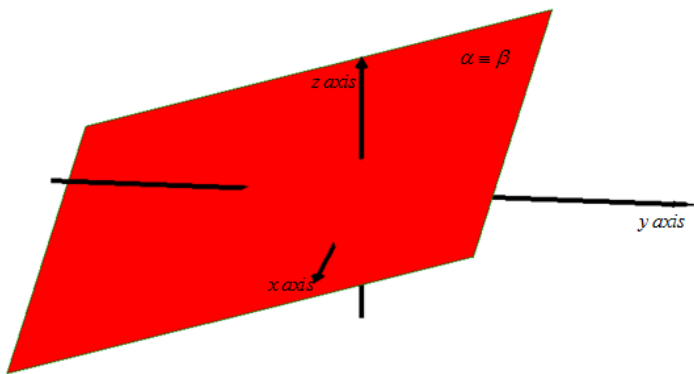
We can replace any variable in the equation by any chosen value and find values of other remaining variables.

Example 1: Find the intersection of $\alpha \equiv 6x - 10y + 14z = 38$ and $\beta \equiv 3x - 5y + 7z = 19$

Solution

We see that $\alpha = 2(3x - 5y + 7z = 19) = 2\beta$

So, the two planes coincide and the intersection is any one of them.



Example 2:

Find intersection of $\alpha \equiv 4x - 3y + 7z = -3$ and $\beta \equiv 5x + 2y - 6z = 25$

Solution

The normal vectors are $\vec{n}_1 = (4, -3, 7)$ and $\vec{n}_2 = (5, 2, -6)$

$\vec{n}_1$ and $\vec{n}_2$ are not proportional, thus the two planes intersect.

Solving the simultaneous equations, $4x - 3y + 7z = 0$ and $5x + 2y - 6z = 0$

and $5x + 2y - 6z = 0$ you get one of the direction vectors of intersection line.

$$\left(\begin{vmatrix} -3 & 7 \\ 2 & -6 \end{vmatrix}, -\begin{vmatrix} 4 & 7 \\ 5 & -6 \end{vmatrix}, \begin{vmatrix} 4 & -3 \\ 5 & 2 \end{vmatrix} \right) = (18 - 14, 24 + 35, 8 + 15) \\ = (4, 59, 23)$$

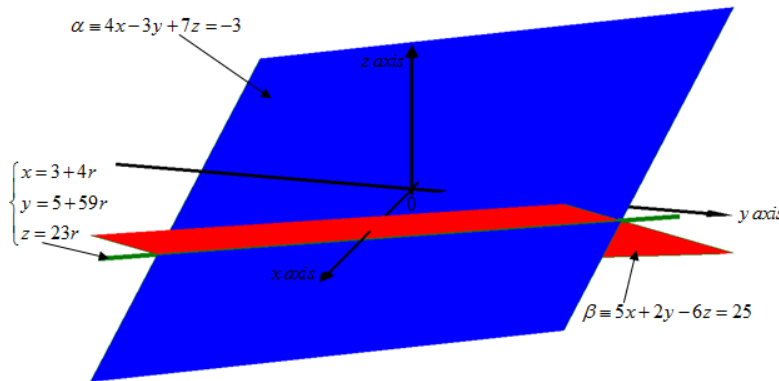
The two planes are secant. So, there is a line of intersection.

The direction vector of the line is $(4, 59, 23)$. To find a common point of the two planes, let one variable to be zero and solve for others. Let $z = 0$, solving for others gives $x = 3, y = 5$. Thus, the point on the line is $(3, 5, 0)$.

The equations of the line of intersection are $\begin{cases} x = 3 + 4r \\ y = 5 + 59r \\ z = 23r \end{cases}$, where r is a parameter.

Or the line of intersection can be defined by the equations

$$\begin{cases} 4x - 3y + 7z = -3 \\ 5x + 2y - 6z = 25 \end{cases}$$



Example 2: Given the Cartesian equations of the line

$$, L \equiv \begin{cases} 3x - 7y = 4 \\ 5x + 2z = 1 \end{cases} \text{ find its direction vector.}$$

Solution

The direction vector is

$$\left(\begin{vmatrix} -7 & 0 \\ 0 & 2 \end{vmatrix}, -\begin{vmatrix} 3 & 0 \\ 5 & 2 \end{vmatrix}, \begin{vmatrix} 3 & -7 \\ 5 & 0 \end{vmatrix} \right) = (-14, -6, 35)$$

Note that any other non-zero scalar multiple of this vector is also a direction vector for the given line. Or To find the direction vector, let $x = 1$ and replace this value in the system

$$\begin{cases} 3x - 7y = 0 \\ 5x + 2z = 0 \end{cases}$$

We obtain the vector $\left(1, \frac{3}{7}, -\frac{5}{2}\right)$. Since any other non zero scalar multiple of this vector is also a direction vector for the given line, we can multiply this vector by -14 to obtain $(-14, -6, 35)$.

Example 3: Find the equation of the intersection line of planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ in standard form.

Solution

Direction vector of intersection line is

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = \vec{i} \begin{vmatrix} -6 & -2 \\ 1 & -2 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & -2 \\ 2 & -2 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & -6 \\ 2 & 1 \end{vmatrix} = 14\vec{i} + 2\vec{j} + 15\vec{k}$$

Any nonzero scalar multiple of this vector will do as well.

We find a common point by assigning a value to one equation and calculating the other two from the given equations. For instance, letting $z = 0$ in the two equations and solving for y and x simultaneously yields $(3, -1, 0)$, so is one point on the line.

Thus, the line has standard form equation $\frac{x-3}{14} = \frac{y+1}{2} = \frac{z}{15}$

Notice: Given that plane β passes through points $A(a_1, a_2, a_3)$ and $B(b_1, b_2, b_3)$ is perpendicular to another plane $\alpha \equiv c_1x + c_2y + c_3z = d$, the mixed product can easily help us to find the equation of plane β .

In fact, the normal vector of plane α , which is $\vec{n} = (c_1, c_2, c_3)$, is the direction vector of the plane β and another direction vector of β is vector $\overrightarrow{AB}$. The normal vector of β is now $\vec{m} = \vec{n} \times \overrightarrow{AB}$ and the equation of β is $\vec{m} \cdot \overrightarrow{AX}$ where $X(x, y, z)$ represents any point on plane β . We can write $\beta \equiv \vec{m} \cdot \overrightarrow{AX}$

This can be written in determinant form as follows

$$\beta \equiv \begin{vmatrix} x-a_1 & y-a_2 & z-a_3 \\ b_1-a_1 & b_2-a_2 & b_3-a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

Application activity 8.3.7

- 1) Find the equation of plane γ passing through the point $A(3, 5, 2)$ and perpendicular to the plane $\beta \equiv -4x - y + z = 4$, $\gamma \equiv x - 2y + 2z + 3 = 0$
- 2) Find the equation of plane β passing through the point $P(5, 8, 1)$ and parallel to the plane $\alpha \equiv 3x - 5y - 7z = 12$
 $\beta \equiv 3x - 5y - 7z + 32 = 0$
- 3) Find the equation of plane passing through the points $A(3, 2, -1)$ and $B(0, 5, -3)$ and perpendicular to the plane $3x - 4y + 6z = 13$

8.4 Shapes in 3D

Activity 8.4

You are familiar with 3D shapes in your everyday life. In fact, you are most likely inside of a 3D shape right now (i.e. a room that is most likely in the shape of a cube or a rectangular prism). Other everyday examples include balls (spheres), cubes (boxes), cylinders (cans), and cones.

Conduct research in the library or on internet to explore at least 6 Three D shapes, and how they look like in space: for each one, find image example of similar real life object, properties, and features that set it apart from all of the other 3D shapes such as number of sides, angles, curves, and faces.

Content summary

The main 3D shapes are the following:

Cone  Flat Face: 1 Curved Face: 1 Edges: 1 Vertices: 1	Sphere  Curved Face: 1 Edges: 0 Vertices: 0	Cube  Faces: 6 Edges: 12 Vertices: 8	Rectangular Prism  Faces: 6 Rectangular bases: 2 Rectangular faces: 6 Edges: 12 Vertices: 8
Cylinder  Flat Face: 2 Curved Face: 1 Curved Edges: 2 Vertices: 0	Triangular Prism  Faces: 5 Triangular bases: 2 Rectangular faces: 3 Edges: 9 Vertices: 6	Square Pyramid  Faces: 5 Square bases: 1 Triangular faces: 4 Edges: 8 Vertices: 5	Triangular Pyramid  Faces: 4 Triangular bases: 1 Triangular faces: 3 Edges: 6 Vertices: 5

Properties that make (3D) shapes different from 2D shapes:

- **Surface Area:** The total area covered by all the outer surfaces of a 3D shape.
- **Volume:** The space occupied by a 3D shape. It is measured in cubic units.
- **Dimensions:** Every 3D shape is defined by three dimensions: length, width, and height, which determine its overall size and form.

3D Shapes formulas

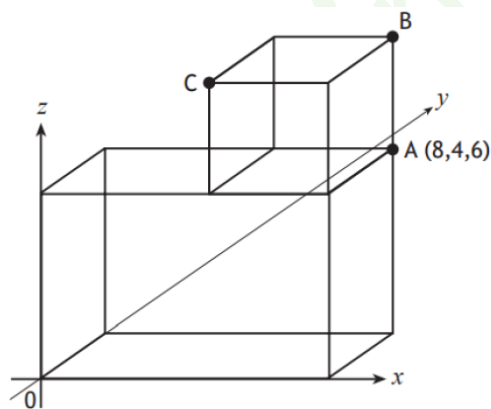
Here is the list of formulas for 3D shapes, including surface area and volume:

3D shape	Surface Area	Volume
Cube	$6a^2$	a^3
Cuboid	$2(lb + bh + hl)$	$2h(l + b)$
Cone	$\pi r(r + l)$ (here l is slant height)	$\frac{1}{3}\pi r^3$

3D shape	Surface Area	Volume
Cylinder	$2\pi r(r + h)$	$\pi r^2 h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Square Pyramid	$s^2 + 2sl$ (here s is the base area)	$\frac{1}{3}s^2 h$
Triangular Prism	$(b \times h) + (s_1 + s_2 + s_3)$	$\frac{1}{2}(bhl)$
Hexagonal Pyramid	$3ab + 3bs$	abh

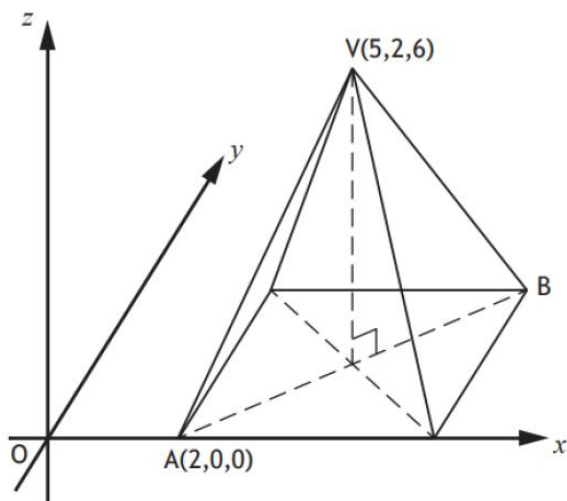
Examples:

1) The diagram shows the cube placed on top of a cuboid, relative to the coordinate's axis. A is the point $(8, 4, 6)$. Write the coordinates of B and C.



Solution: $B(8, 4, 10)$ and $C(4, 0, 10)$

2) The diagram shows a rectangular based pyramid, relative to the coordinate axes. A is the point $(2, 0, 0)$; V is the point $(5, 2, 6)$.



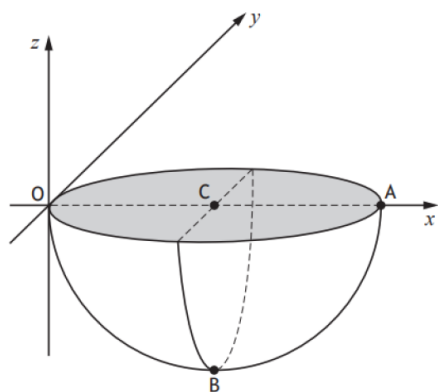
- Write the coordinates of B.
- Calculate the length of the edge AV of the pyramid.

Solution:

- $B(8, 4, 0)$
- 7 units of length.

3) the diagram shows a hemisphere relative to the coordinates axes:

A is the point $(6, 0, 0)$. C is the midpoint of diameter OA and B is vertically below C.



- State the coordinates of B
- Calculate the volume of the hemisphere in function of π .

Solution:

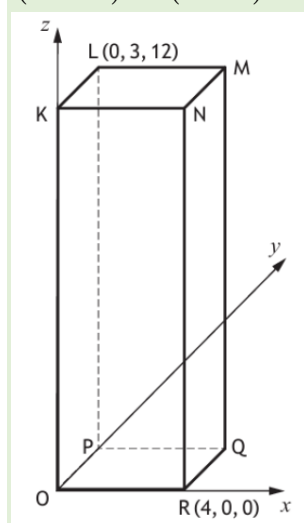
- $B(3, 0, -3)$
- $V = 18\pi$.

Application activity 8.4

1) Provide the shape similar to the following real objects and its characteristics.



2) The diagram shows a cuboid, KLMNOPQR, relative to the coordinates axes. L has coordinates $(0, 3, 12)$; $R(4, 0, 0)$.



- Write the coordinates of M.
- Calculate the length of the diagonal space OM.

8.5 Sphere

8.5.1 Equation of a sphere in 3 dimensions

Sphere is the locus of a point in space which remains at a constant distance called the **radius** from a fixed point called the **centre** of the sphere.



Activity 8.5.1

- Develop the equation $(x - k)^2 + (y - l)^2 + (z - m)^2 = r^2$
- Compare the equation $x^2 + y^2 + z^2 + ax + by + cz + d = 0$ and the one obtained in 1) and find the value of k , l , m and r in function of a , b , c , and d .



Content Summary

The equation of a sphere of centre (k, l, m) and radius r is given by

$$S \equiv (x-k)^2 + (y-l)^2 + (z-m)^2 = r^2$$

From activity 8.22,

The general equation of a sphere is: $x^2 + y^2 + z^2 + ax + by + cz + d = 0$

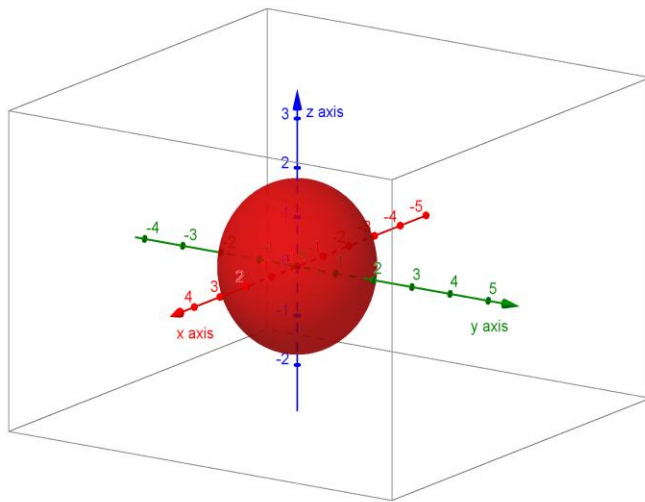
In this equation:

The center, is $\Omega = \left(-\frac{a}{2}, -\frac{b}{2}, -\frac{c}{2} \right)$

and the radius is given by

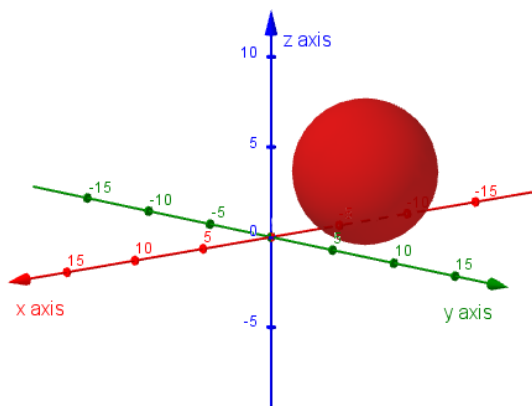
$$r = \frac{1}{2} \sqrt{a^2 + b^2 + c^2 - 4d}, \text{ provided that } a^2 + b^2 + c^2 - 4d > 0$$

From this general equation, as it contains four constants a, b, c and d , if we are given that a sphere passes through four points; each of which gives one independent equation in the constants. We can find the values of four constants a, b, c and d by solving the four equations.



Example 1: Find the equation of the sphere whose centre is $(-6, 1, 3)$ and radius 4.

Solution: The required equation is given by $x^2 + y^2 + z^2 + 12x - 2y - 6z + 36 + 1 + 9 = 16$ which is $x^2 + y^2 + z^2 + 12x - 2y - 6z + 30 = 0$



Example 2. Find the coordinates of centre and radius of the sphere $4x^2 + 4y^2 + 4z^2 - 8x + 16y + 20z + 9 = 0$.

Solution: First, we put this equation in general form by dividing both sides by 4. That is,

$$x^2 + y^2 + z^2 - 2x + 4y + 5z + \frac{9}{4} = 0$$

Now the centre is

$$\left(-\frac{-2}{2}, -\frac{4}{2}, -\frac{5}{2}\right) = \left(1, -2, -\frac{5}{2}\right)$$

$$\text{The radius is } \frac{1}{2} \sqrt{(-2)^2 + 4^2 + 5^2 - 4 \times \frac{9}{4}} = \frac{1}{2} \sqrt{4 + 16 + 25 - 9} = 3$$

Thus, the centre is $\left(1, -2, -\frac{5}{2}\right)$ and the radius is 3.

Alternative method

We could get the centre and radius by completing the squares. That is

$$4x^2 + 4y^2 + 4z^2 - 8x + 16y + 20z + 9 = 0 \Rightarrow x^2 + y^2 + z^2 - 2x + 4y + 5z = -\frac{9}{4}$$

$$\Rightarrow x^2 - 2x + y^2 + 4y + z^2 + 5z = -\frac{9}{4}$$

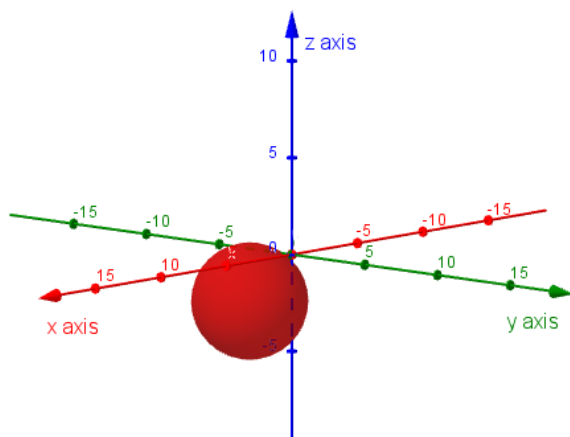
$$\Rightarrow (x-1)^2 - 1 + (y+2)^2 - 4 + \left(z + \frac{5}{2}\right)^2 - \frac{25}{4} = -\frac{9}{4}$$

$$\Rightarrow (x-1)^2 + (y+2)^2 + \left(z + \frac{5}{2}\right)^2 = -\frac{9}{4} + 1 + 4 + \frac{25}{4}$$

$$\Rightarrow (x-1)^2 + (y+2)^2 + \left(z + \frac{5}{2}\right)^2 = \frac{-9 + 4 + 16 + 25}{4}$$

$$\Rightarrow (x-1)^2 + (y+2)^2 + \left(z + \frac{5}{2}\right)^2 = 9 \Rightarrow (x-1)^2 + (y+2)^2 + \left(z + \frac{5}{2}\right)^2 = 3^2$$

Thus, the centre is $\left(1, -2, -\frac{5}{2}\right)$ and the radius is 3.



Example 3: Find the equation of sphere which passes through the points $(1, 2, 3), (0, -2, 4), (4, -4, 2)$ and $(3, 1, 4)$.

Solution: Let the equation of sphere be $x^2 + y^2 + z^2 + ax + by + cz + d = 0$

Substituting these four points into this equation gives:

$$\begin{cases} 1 + 4 + 9 + a + 2b + 3c + d = 0 \\ 0 + 4 + 16 + 0 - 2b + 4c + d = 0 \\ 16 + 16 + 4 + 4a - 4b + 2c + d = 0 \\ 9 + 1 + 16 + 3a + b + 4c + d = 0 \end{cases} \Leftrightarrow \begin{cases} a + 2b + 3c + d = -14 \\ -2b + 4c + d = -20 \\ 4a - 4b + 2c + d = -36 \\ 3a + b + 4c + d = -26 \end{cases}$$

Solving gives $\begin{cases} a = -4 \\ b = 2 \\ c = -2 \\ d = -8 \end{cases}$

Thus, the equation is $x^2 + y^2 + z^2 - 4x + 2y - 2z - 8 = 0$

Example 4: Prove that the equation of sphere described on the line segment joining the points $(2, -1, 4)$ and $(-2, 2, -2)$ as diameter, is $x^2 + y^2 + z^2 - y - 2z - 14 = 0$

Solution: The midpoint of the points $(2, -1, 4)$ and $(-2, 2, -2)$, which is the centre of the sphere,

is $\frac{1}{2}(2 - 2, -1 + 2, 4 - 2) = \left(0, \frac{1}{2}, 1\right)$.

The radius of the sphere is a half the distance between the given two points:

$$r = \frac{1}{2}\sqrt{(-2 - 2)^2 + (2 + 1)^2 + (-2 - 4)^2} = \frac{1}{2}\sqrt{16 + 9 + 36} = \frac{1}{2}\sqrt{61}$$

The equation of sphere is

$$(x - 0)^2 + \left(y - \frac{1}{2}\right)^2 + (z - 1)^2 = \left(\frac{1}{2}\sqrt{61}\right)^2$$

Or

$$x^2 + y^2 + z^2 - y - 2z - 14 = 0 \text{ as required.}$$



Application activity 8.5.1

- Find the equation of the sphere with
 - Centre $(1, 2, 3)$ and radius 4
 - Centre $(3, -1, 1)$ and radius $\sqrt{3}$
 - Centre $(4, 0, -1)$ and radius 7
- Find the centre and radius of the sphere
 - $x^2 + y^2 + z^2 - 22x - 6y + 66 = 0$
 - $x^2 + y^2 + z^2 + 8x - 16y - 14z + 93 = 0$
 - $3x^2 + 3y^2 + 3z^2 - 54y - 18z - 318 = 0$
- Describe the sets of points in space whose coordinates satisfy the given inequalities
 - $x^2 + y^2 + z^2 < 4$
 - $x^2 + y^2 + z^2 + 4x - 6y + 8z + 25 \leq 0$

8.5.2 Position of a sphere and other objects



Activity 8.5.2

In each of the following cases, find the distance between the given point P and the centre of the given sphere S . Deduce if the point lies inside the sphere, outside the sphere or on the sphere.

- $S \equiv (x-1)^2 + (y+2)^2 + (z-2)^2 = 4, \quad P(2, 4, 3)$
- $S \equiv (x-3)^2 + (y-2)^2 + (z-1)^2 = 6, \quad P(1, 1, 0)$
- $S \equiv (x+2)^2 + y^2 + (z-1)^2 = 37, \quad P(-1, -2, 1)$



Content Summary

Position of a point and a sphere

Consider a sphere S with radius r and centre $c(a, b, c)$ and any point $P(a_1, a_2, a_3)$

- If $d(c, P) < r$, the point lies inside the sphere S
- If $d(c, P) = r$, the point lies on the sphere S
- If $d(c, P) > r$, the point lies outside the sphere S

In all cases $d(c, P)$ is the distance between point P and centre c of sphere S

Example 1: Find the position of point $A(4,5,6)$ and the sphere $(x-2)^2 + (y-1)^2 + (z+1)^2 - 37 = 0$

Solution: Centre of sphere is $(2,1,-1)$ and its radius is $r = \sqrt{37}$

The distance between the centre of the sphere and the given point is $d = \sqrt{2^2 + 4^2 + 7^2} = \sqrt{69}$
Here $d > r$. Thus, the point lies outside the sphere.

Example 2: Describe the position of point $A(1,-2,1)$ and the sphere $(x+1)^2 + (y+2)^2 + (z-1)^2 = 56$

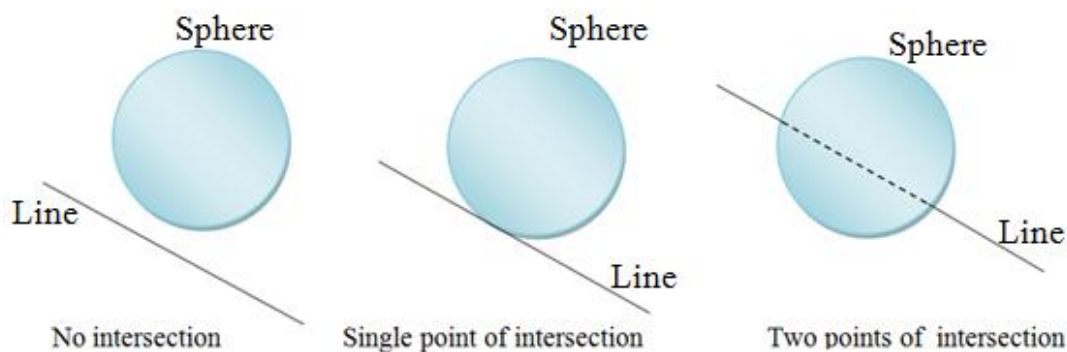
Solution: Centre of sphere is $(-1,-2,1)$ and its radius is $r = \sqrt{56}$

The distance between the centre of the sphere and the given point is $d = 2$
Here $d < r$. Thus, the point lies inside the sphere.

Position of a sphere and a line

There are three possible line-sphere intersections:

- One point of intersection,
- Two points of intersection and
- No intersection.



If the line passes through the centre of a sphere, there are two points of intersection and those points are called **antipodal** points.

Consider a sphere S with radius r and centre $c(a,b,c)$ and a line L

- If $d(c,L) < r$, there are two points of intersection
- If $d(c,L) = r$, there is a single point of intersection

- If $d(c, L) > r$, there is no intersection

In all cases $d(\Omega, L)$ is the shortest distance from the centre c of sphere S to the line L

We are interested in the case where the line-plane intersection is a point. In this case the line is tangent to the sphere and there is one point of intersection. At this point there are many lines tangent to the sphere and they are included in the plane tangent to the sphere at this point.

Let the point of intersection be $P(a, b, c)$ and centre of sphere be $\Omega = (k, l, m)$.

The vector $\overrightarrow{\Omega P} = (a - k, b - l, c - m)$ is orthogonal to the line. From this vector we can find the direction vector of the line.

Example 8.1

Consider the sphere S passing through the point $P(2, -1, 3)$ and with centre $C(1, 2, -3)$. Find the equations of the line D tangents to the sphere S at point P .

Solution

There are many possible answers. First, we need the vector from point P to the centre of the sphere, i.e. $\overrightarrow{CP} = (1, -3, 6)$. Next, we need the vector $\vec{u}$ which is perpendicular to the vector $\overrightarrow{CP}$ and that vector will be the direction vector of line D . Let take $\vec{u} = (6, 4, 1)$ $\vec{u} = (6, 4, 1)$.

So, the line D passing through the point $P(2, -1, 3)$ and with direction vector $\vec{u} = (6, 4, 1)$ has

$$\text{equations } D \equiv \begin{cases} x = 2 + 6r \\ y = -1 + 4r \\ z = 3 + r \end{cases}$$

Consider the sphere $S \equiv x^2 + y^2 + z^2 = 16$ and the line D passing through the points $P(1, 2, -4)$ and $P(-2, 1, 3)$. Find the common points.

Solution

The direction vector of the line is $\overrightarrow{PQ} = (-3, -1, 7)$ and the parametric equations are

$$\begin{cases} x = 1 - 3r \\ y = 2 - r \\ z = -4 + 7r \end{cases}$$

Putting these values into the equation of the sphere, we have $(1 - 3r)^2 + (2 - r)^2 + (-4 + 7r)^2 = 16$

$$\Leftrightarrow 1 - 6r + 9r^2 + 4 - 4r + r^2 + 16 - 56r + 49r^2 = 16$$

$$\Leftrightarrow 59r^2 - 66r + 5 = 0 \Rightarrow r = \frac{33 + \sqrt{794}}{59} \text{ or } r = \frac{33 - \sqrt{794}}{59}$$

If $r = \frac{33 + \sqrt{794}}{59}$, we have

$$\begin{cases} x = 1 - 3 \left(\frac{33 + \sqrt{794}}{59} \right) = \frac{-40 - 3\sqrt{794}}{59} \\ y = 2 - \frac{33 + \sqrt{794}}{59} = \frac{85 - \sqrt{794}}{59} \\ z = -4 + 7 \left(\frac{33 + \sqrt{794}}{59} \right) = \frac{-5 + 7\sqrt{794}}{59} \end{cases}$$

If $r = \frac{33 - \sqrt{794}}{59}$, we have

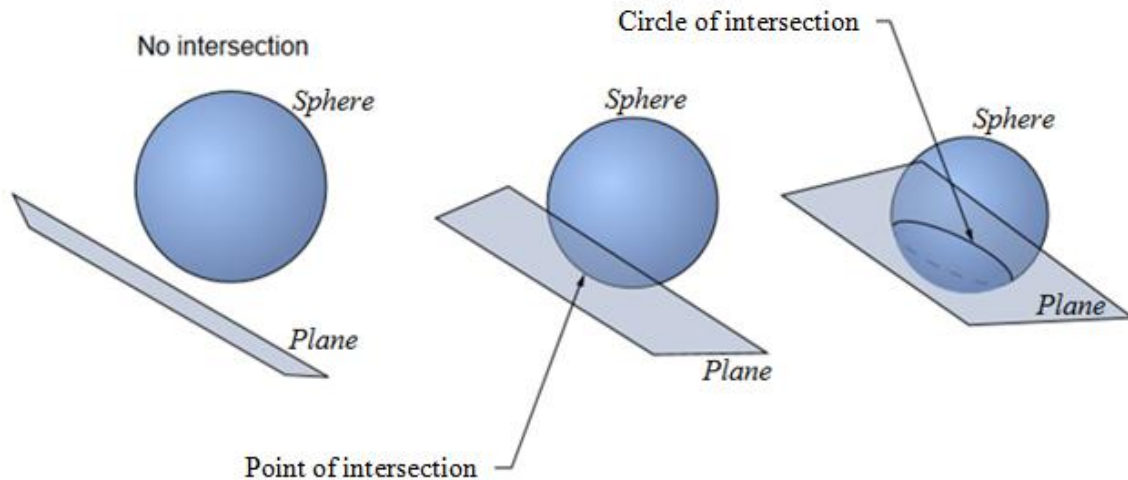
$$\begin{cases} x = 1 - 3 \left(\frac{33 - \sqrt{794}}{59} \right) = \frac{-40 + 3\sqrt{794}}{59} \\ y = 2 - \frac{33 - \sqrt{794}}{59} = \frac{85 + \sqrt{794}}{59} \\ z = -4 + 7 \left(\frac{33 - \sqrt{794}}{59} \right) = \frac{-5 - 7\sqrt{794}}{59} \end{cases}$$

Then there are two points of intersection:

$$\left(\frac{-40 - 3\sqrt{794}}{59}, \frac{85 - \sqrt{794}}{59}, \frac{-5 + 7\sqrt{794}}{59} \right) \text{ and } \left(\frac{-40 + 3\sqrt{794}}{59}, \frac{85 + \sqrt{794}}{59}, \frac{-5 - 7\sqrt{794}}{59} \right)$$

Position of a sphere and a plane

Consider a sphere $S \equiv x^2 + y^2 + z^2 + ax + by + cz + d = 0$ with centre $\Omega = (k, l, m)$ and radius r and plane $\alpha \equiv hx + ny + pz = q$, their position appears in three cases.



- 1) If $d(\Omega, \alpha) < r$, the plane cuts the sphere, and the intersection is a circle whose centre is on the line through the centre of the sphere and perpendicular to the plane. When the plane cuts the sphere, we call it **plane section of a sphere**.

The normal vector of the plane is the direction vector of this perpendicular line. Since this perpendicular line passes through the centre of the sphere, we can find its parametric equations (having the point and the direction vector). To find the intersection of this line with the plane we will plug in the values of x, y , and z from the parametric equation of the perpendicular line, into the equation of the sphere to find the value of the parameter and then that value of the parameter into the parametric equations of the perpendicular line to find the point of intersection which will be the centre of the circle of intersection.

The radius can be found using Pythagorean rule. Since the radius of the sphere and the distance from the centre of the sphere to the plane can be found, if P is the centre of this circle, Ω is the centre of the sphere and Q is any point on the circle which is also a point on the sphere then $d(\Omega, Q)$ is the radius of the sphere, $d(\Omega, P)$ is the distance from the centre of the sphere to the plane and $d(P, Q)$ is the radius of the circle.

$$\text{Then } d(P, Q) = \sqrt{[d(\Omega, Q)]^2 - [d(\Omega, P)]^2}.$$

Remarks

- a. Two equations, one of a sphere and the other of the plane, together represent a circle.

$$\text{Thus, } \begin{cases} x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases} \text{ is a circle.}$$

Hence the circle of intersection of the sphere $S \equiv x^2 + y^2 + z^2 + ax + by + cz + d = 0$ and

$$\text{the plane } \alpha \equiv hx + ny + pz = q, \text{ is given by } \begin{cases} x^2 + y^2 + z^2 + ax + by + cz + d = 0 \\ hx + ny + pz = q \end{cases}$$

- b. If we add a scalar multiple of the second equation using a constant k ,
i.e. $x^2 + y^2 + z^2 + ax + by + cz + d + k(hx + ny + pz - q) = 0$ we will have a sphere.

- c. **Sphere through a given circle**

Given the circle

$$\begin{cases} x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases}$$

The equation of sphere through this circle is
 $x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 + k(a_2x + b_2y + c_2z + d_2) = 0.$

- 2) If $d(\Omega, \alpha) > r$, there is no intersection.

3) If $d(\Omega, \alpha) = r$, the plane is tangent to the sphere, and the intersection is the point. To find this point of intersection we proceed in the same way as in case 1 above.

The **tangent plane** at point $P(a_1, a_2, a_3)$ on the sphere $x^2 + y^2 + 2kx + 2ly + 2mz + d = 0$ is $a_1x + a_2y + a_3z + k(x + a_1) + l(y + a_2) + m(z + a_3) + d = 0$

In all cases $d(\Omega, \alpha)$ is the distance from the center $\Omega = (k, l, m)$ of the sphere $S \equiv x^2 + y^2 + z^2 + ax + by + cz + d = 0$ with radius r to the plane $\alpha \equiv hx + ny + pz = q$. It is given by $d(\Omega, \alpha) = \frac{|hk + nl + pm - q|}{\sqrt{h^2 + n^2 + p^2}}$

Note that the section of a sphere by a plane through its centre is known as **great circle**.

The centre and radius of a great circle are the same as those of the sphere.

Example 1. Consider the sphere $S \equiv x^2 + y^2 + z^2 - 2x - 15 = 0$ and the plane $\alpha \equiv 3x - 2y + 5z = 6$. Find their intersection.

Solution: The centre of the sphere is $\Omega\left(-\frac{-2}{2}, 0, 0\right) = \Omega(1, 0, 0)$

The radius of the sphere is

$$\frac{1}{2}\sqrt{(-2)^2 + 0 + 0 + 4 \times 15} = \frac{1}{2}\sqrt{64} = 4$$

The distance between the sphere and the plane is

$$\frac{|3 + 0 + 0 - 6|}{\sqrt{9 + 4 + 25}} = \frac{3}{\sqrt{38}} = \frac{3\sqrt{38}}{38}$$

Since this distance is less than the radius of the sphere there is a circle of intersection.

Then the radius of the circle of intersection is

$$\sqrt{4^2 - \left(\frac{3}{\sqrt{38}}\right)^2} = \sqrt{16 - \frac{9}{38}} = \sqrt{\frac{599}{38}}$$

The normal vector of the plane which is also the direction vector of the perpendicular line of the plane through the centre $(1, 0, 0)$ of the sphere is $(3, -2, 5)$.

Then the parametric equations of this perpendicular line are
$$\begin{cases} x = 1 + 3r \\ y = -2r \\ z = 5r \end{cases}$$

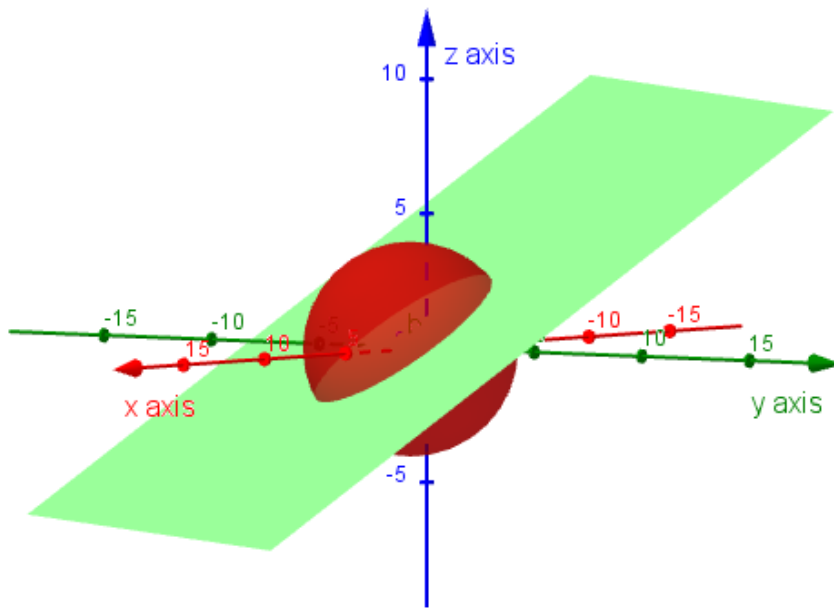
Putting them into the equation of the plane to find the value of the parameter r gives

$$3(1+3r) - (-2r) + 5(5r) = 6 \Leftrightarrow 3 + 38r = 6 \Leftrightarrow r = \frac{3}{38}$$

Putting this value of r into the parametric equations of the perpendicular line of the plane through the centre of the sphere to find the centre of the circle of intersection gives

$$\begin{cases} x = 1 + 3\frac{3}{38} \\ y = -2\frac{3}{38} \\ z = 5\frac{3}{38} \end{cases} \Leftrightarrow \begin{cases} x = \frac{47}{38} \\ y = \frac{-6}{38} = \frac{-3}{19} \\ z = \frac{15}{38} \end{cases}$$

Then the centre of the circle of the intersection is $\left(\frac{47}{38}, \frac{-3}{19}, \frac{15}{38}\right)$



Example 2: Consider the plane $\alpha \equiv 2x + y - z = 7$ and the sphere $S \equiv x^2 + y^2 + z^2 - 4x + 6y - 15 = 0$

Find

- the Cartesian equations of two planes β and γ tangent to the sphere S and parallel to the plane α .
- points of intersection of sphere S with planes β and γ .

Solution:

The centre of the sphere is $(2, -3, 0)$ and the radius is $\frac{1}{2}\sqrt{16+36+60} = \frac{1}{2}\sqrt{112} = 2\sqrt{7}$. The equations of the planes β and γ tangent to the sphere S and parallel to the plane α have the form $2x + y - z = k$. The distance from the centre of the sphere to each plane β or γ is equal to the radius of the sphere. That is $\frac{|4-3+0-k|}{\sqrt{4+1+1}} = 2\sqrt{7} \Leftrightarrow \frac{|1-k|}{\sqrt{6}} = 2\sqrt{7} \Leftrightarrow |1-k| = 2\sqrt{42}$
 $\Leftrightarrow 1-k = 2\sqrt{42}$ or $1-k = -2\sqrt{42} \Leftrightarrow k = 1+2\sqrt{42}$ or $k = 1-2\sqrt{42}$. Thus, the two planes are $\beta \equiv 2x + y - z = 1+2\sqrt{42}$ and $\gamma \equiv 2x + y - z = 1-2\sqrt{42}$

These two points lie on the perpendicular line of planes β and γ passing through the centre of the sphere $(2, -3, 0)$ and the direction vector of this perpendicular is the normal vector for the two planes which is $(2, 1, -1)$.

The parametric equations for this line are
$$\begin{cases} x = 2 + 2r \\ y = -3 + r \\ z = -r \end{cases}$$

Putting these values of x, y , and z into $\beta \equiv 2x + y - z = 1+2\sqrt{42}$ gives

$$4 + 4r - 3 + r + r = 1 + 2\sqrt{42} \Rightarrow r = \frac{\sqrt{42}}{3} \text{ and putting them into } \gamma \equiv 2x + y - z = 1 - 2\sqrt{42}$$

gives $4 + 4r - 3 + r + r = 1 - 2\sqrt{42} \Rightarrow r = -\frac{\sqrt{42}}{3}$ Putting these two values of the parameter into the parametric equations of the perpendicular line gives:

$$\begin{cases} x = 2 + \frac{2\sqrt{42}}{3} \\ y = -3 + \frac{\sqrt{42}}{3} \\ z = -\frac{\sqrt{42}}{3} \end{cases} \quad \begin{cases} x = \frac{6 + 2\sqrt{42}}{3} \\ y = \frac{-9 + \sqrt{42}}{3} \\ z = -\frac{\sqrt{42}}{3} \end{cases}$$

and

$$\begin{cases} x = 2 - \frac{2\sqrt{42}}{3} \\ y = -3 - \frac{\sqrt{42}}{3} \\ z = \frac{\sqrt{42}}{3} \end{cases} \quad \begin{cases} x = \frac{6-2\sqrt{42}}{3} \\ y = \frac{-9-\sqrt{42}}{3} \\ z = \frac{\sqrt{42}}{3} \end{cases}$$

Thus, the two points of intersection are

$$\left(\frac{6+2\sqrt{42}}{3}, \frac{-9+\sqrt{42}}{3}, -\frac{\sqrt{42}}{3} \right) \text{ and } \left(\frac{6-2\sqrt{42}}{3}, \frac{-9-\sqrt{42}}{3}, \frac{\sqrt{42}}{3} \right)$$

Example 4. Find the centre and the radius of the circle

$$\begin{cases} x^2 + y^2 + z^2 - 2y - 4z = 11 \\ x + 2y + 2z = 15 \end{cases}$$

Solution: The given circle is the intersection of sphere $S \equiv x^2 + y^2 + z^2 - 2y - 4z = 11$ and plane $\alpha \equiv x + 2y + 2z = 15$.

The centre of this sphere is $\Omega = (0, 1, 2)$ and its radius is $r = 4$.

Parametric equation of the line through the centre of the sphere and perpendicular to the plane

$$\text{are } \begin{cases} x = t \\ y = 1 + 2t \\ z = 2 + 2t \end{cases}$$

Putting these values into the equation of the sphere we will have the centre of the circle: $t + 2 + 4t + 4 + 4t = 15$ or $t = 1$. Substituting this value of t in equation of line gives the centre of the circle which is $(1, 3, 4)$.

The radius of the circle:

$$\text{The distance of the centre of the sphere from the plane is } d = \frac{|0 + 2 + 4 - 15|}{\sqrt{9}} = \frac{9}{3} = 3$$

Then the radius of the circle is $\sqrt{4^2 - 3^2} = \sqrt{7}$.

Example 5. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 + 2x + 4y - 6z - 6 = 0$ at $(1, 2, 3)$

Solution: We see that $k = 1, l = 2, m = -3, a_1 = 1, a_2 = 2, a_3 = 3$.

$$\text{Therefore, } x + 2y + 3z + (x+1) + 2(y+2) - 3(z+3) - 6 = 0$$

$$\Leftrightarrow x + 2y + 3z + x + 1 + 2y + 4 - 3z - 9 - 6 = 0$$

$\Rightarrow 2x + 4y - 10 = 0$ Which is the required equation.

Alternative method

The vector formed by the point $(1, 2, 3)$ and the centre of sphere is the normal vector of the needed plane.

Centre of sphere is $(-1, -2, 3)$. Normal vector is $\vec{n} = (-2, -4, 0)$

Since the plane passes through the point $(1, 2, 3)$ then the required equation is

$$-2(x-1) - 4(y-2) + 0(z-3) = 0$$

$$\Leftrightarrow -2x + 2 - 4y + 8 = 0$$

$$\Leftrightarrow -2x - 4y + 10 = 0$$

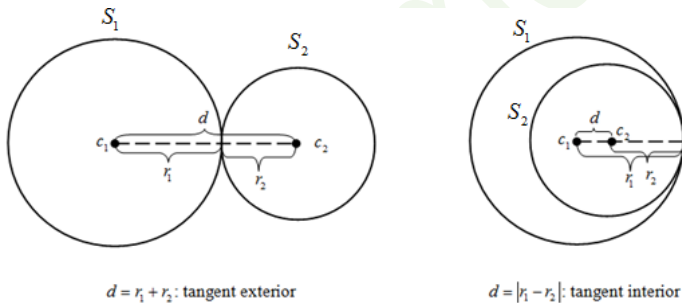
Or

$$2x + 4y - 10 = 0$$

Position of two spheres

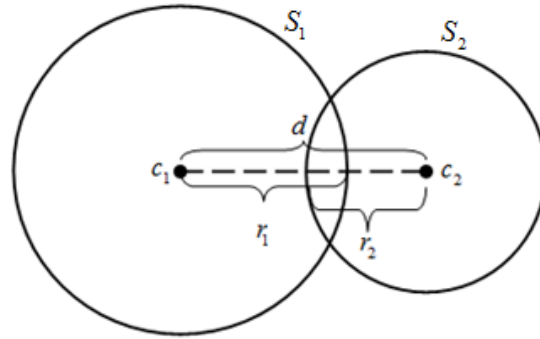
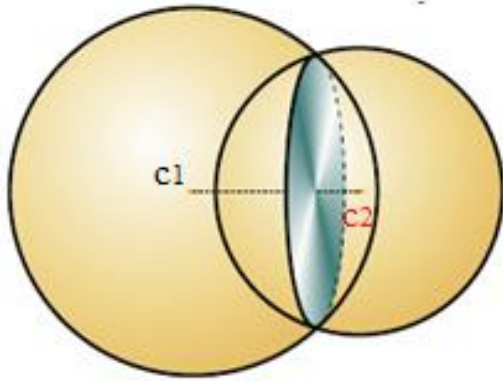
Consider two spheres with centers c_1 and c_2 ; radii r_1 and r_2 . The position of these two spheres depends on the distance between their center $d(c_1, c_2)$

Case 1: $d = r_1 + r_2$: Tangent exterior, $d = |r_1 - r_2|$: tangent interior



The intersection is a point

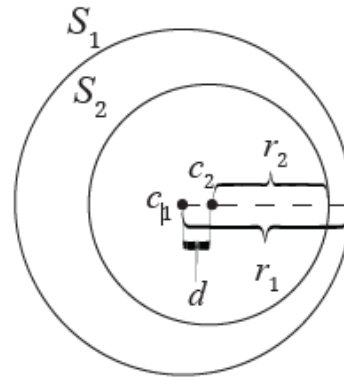
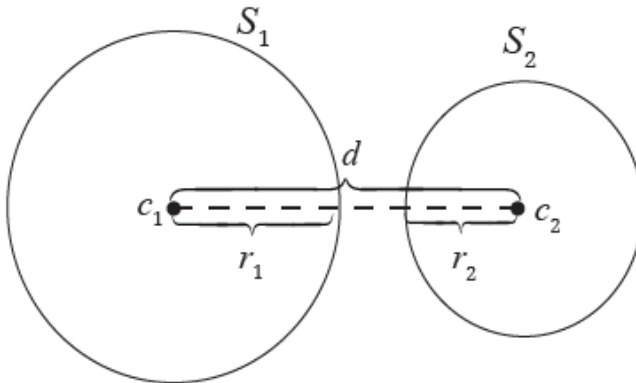
Case 2: $|r_1 - r_2| < d < r_1 + r_2$



The intersection is a circle

Case 3: $d > r_1 + r_2$: Exterior

$d < r_1 + r_2$: Interior



No intersection

- If $d > r_1 + r_2$: Two spheres are exterior and hence no intersection.
- If $d < r_1 + r_2$: Two spheres are interior and hence no intersection.
- If $d = r_1 + r_2$: Two spheres are tangent exterior and hence there is a point of intersection.

To find this point of intersection:

We can find, by writing the parametric equations, the line through the two centers of two spheres. Intersect that line with the two spheres we obtain four points of intersection (two for each sphere). Among these four points, the common point will be the point of intersection of two spheres.

Or We subtract one equation of the sphere from the other to obtain a plane tangent to both spheres at the point of intersection. The intersection of this plane with one of the two spheres is the intersection of two spheres (we are on the case of sphere-plane intersection where the plane is tangent to the sphere).

- If $d = |r_1 - r_2|$: Two spheres are tangent interior and hence there is a point of intersection.

To find this point of intersection use the same method as in the case above.

- If $|r_1 - r_2| < d < r_1 + r_2$: One sphere cuts another. The intersection is a circle.

Given two spheres we subtract one from the other to obtain a plane. In this plane lies a circle that is the circle where the two spheres intersect. The task here is to somehow parameterize a curve that will generate explicitly the points on this circle. Call the centre of the circle of intersection and its radius, say C and R respectively. Now we are on the case of sphere-plane intersection.

We can find by writing the parametric equations of the line through the two centers of two spheres. Intersect that line with the plane of intersection and obtain the point which will be the centre of the circle of intersection.

Likewise, we can determine the radius of the circle by using the Pythagorean Theorem in the following way: determine the distance of the centre of one of the spheres from the obtained plane.

This is one leg of a right triangle. The desired radius is the other leg. The radius of the sphere is the hypotenuse.

Note:

- Sphere through the intersection of two spheres:

Given two spheres $x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 = 0$ and

$$x^2 + y^2 + z^2 + a_2x + b_2y + c_2z + d_2 = 0$$

The equation of sphere through the intersection of these two spheres is

$$x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 + k(x^2 + y^2 + z^2 + a_2x + b_2y + c_2z + d_2) = 0$$

- Angle of intersection of two spheres

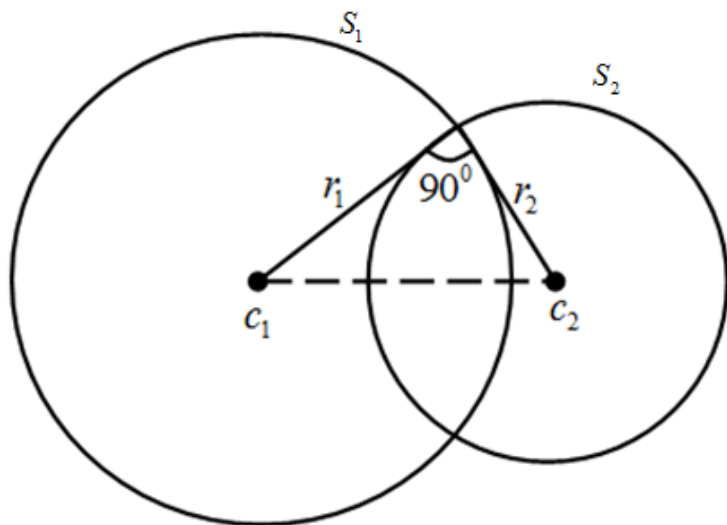
Angle of intersection of two spheres at a common point is the angle between the tangent planes to them at that point, and is, therefore, also equal to the angle between the radii of the spheres to the common point; since the radii being perpendicular to the respective tangent planes at the point.

- Condition of orthogonality of two spheres

Given two spheres $x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 = 0$ and

$$x^2 + y^2 + z^2 + a_2x + b_2y + c_2z + d_2 = 0$$

If the angle of intersection is a right angle the two spheres are said to be orthogonal.



The condition is $r_1^2 + r_2^2 = (C_1C_2)^2$

Example : Sphere S_1 has centre $C_1(3, -2, 5)$ and radius 4. Find the equation of sphere S_2 with centre $C_2(7, 6, -8)$ tangents to S_1 exterior.

Solution: First, we calculate the distance between the centers:

$$d(C_1, C_2) = \sqrt{4^2 + 8^2 + (-13)^2} = \sqrt{249}$$

Since one sphere is tangent to another exterior the radius of S_2 is $\sqrt{249} - 4$.

The equation of S_2 is $(x-7)^2 + (y-6)^2 + (z+8)^2 = (\sqrt{249} - 4)^2$

$$\text{Or } x^2 + y^2 + z^2 - 14x - 12y + 16z + \sqrt{249} - 116 = 0$$

Example 2: Find the intersection between sphere $S_1 \equiv x^2 + y^2 + z^2 - 2y = 8$ and $S_2 \equiv x^2 + y^2 + z^2 - 12x - 2y = -33$

Solution: For $S_1 \equiv x^2 + y^2 + z^2 - 2y = 8$,

$$\text{Centre is } C_1 = \left(\frac{0}{2}, \frac{2}{2}, \frac{0}{2} \right) = (0, 1, 0),$$

$$\text{Radius is } R_1 = \frac{1}{2} \sqrt{0+4+0+32} = 3$$

$$\text{For } S_2 \equiv x^2 + y^2 + z^2 - 12x - 2y = -33:$$

$$\text{Centre is } C_2 = \left(\frac{12}{2}, \frac{2}{2}, \frac{0}{2} \right) = (6, 1, 0),$$

Radius is $R_2 = \frac{1}{2}\sqrt{144+4+0-132} = 2$, $d(C_1, C_2) = \sqrt{6^2+0+0} = 6$, $R_1 + R_2 = 3+2 = 5$,

$$d(C_1, C_2) > R_1 + R_2$$

Then the two spheres are exterior and hence no intersection between them.

Example 3: Find the intersection between sphere $S_1 \equiv x^2 + y^2 + z^2 + 2x - 6y + 1 = 0$ and $S_2 \equiv 4x^2 + 4y^2 + 4z^2 + 10x - 25y - 2z = 0$

Solution: For $S_1 \equiv x^2 + y^2 + z^2 + 2x - 6y + 1 = 0$:

$$\text{Centre is } C_1 = \left(-\frac{2}{2}, \frac{6}{2}, \frac{0}{2}\right) = (-1, 3, 0),$$

$$\text{Radius is } R_1 = \frac{1}{2}\sqrt{4+36+0-4} = 3$$

$$\text{For } S_2 \equiv 4x^2 + 4y^2 + 4z^2 + 10x - 25y - 2z = 0:$$

$$\text{Centre is } C_2 = \left(-\frac{5}{4}, \frac{25}{8}, \frac{1}{4}\right),$$

$$\text{Radius is } R_2 = \frac{1}{2}\sqrt{\frac{25}{4} + \frac{625}{16} + \frac{1}{4} - 0} = \frac{1}{2} \times \frac{27}{4} = \frac{27}{8}$$

$$d(C_1, C_2) = \sqrt{\left(\frac{-1}{4}\right)^2 + \left(\frac{1}{8}\right)^2 + \left(\frac{1}{4}\right)^2} = \frac{3}{8}$$

$$R_1 + R_2 = 3 + \frac{27}{8} = \frac{51}{8},$$

$$|R_1 - R_2| = \left|3 - \frac{27}{8}\right| = \left|-\frac{3}{8}\right| = \frac{3}{8}$$

Then, $d(C_1, C_2) = |R_1 - R_2|$, means that the two spheres are tangent interior. There is a point of intersection.

The plane, say α , through the point of intersection is given by $S_1 - S_2$. i.e., $\alpha \equiv -2x + y + 2z + 4 = 0$.

The normal vector of this plane is the direction vector of its perpendicular line which passes through the centers of the two spheres (the vector formed by the two centers is also a direction vector of this line).

The parametric equations of this line (taking the centre of the first sphere to be the point on the

line) are
$$\begin{cases} x = -1 - 2t \\ y = 3 + t \\ z = 2t \end{cases}$$

Intersecting this line and the plane will give us a point of intersection of two spheres:

Putting the equations of the line into the equation of the plane gives $2 + 4t + 3 + t + 4t = -4$

Or $t = -1$ and then
$$\begin{cases} x = 1 \\ y = 2 \\ z = -2 \end{cases}$$

Hence, the point of intersection for the given spheres is $(1, 2, -2)$.

Example 4: Find the intersection between sphere $S_1 \equiv x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$ and $S_2 \equiv x^2 + y^2 + z^2 + 6z - 4 = 0$

Solution: For $S_1 \equiv x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$, $c_1 = (0, -5, 2)$ and $r_1 = \sqrt{37}$

For $S_2 \equiv x^2 + y^2 + z^2 + 6z - 4 = 0$,

$c_2 = (0, 0, -3)$ and $r_2 = \sqrt{13}$

Distance between centers: $c_1c_2 = \sqrt{25 + 25} = 5\sqrt{2}$, $r_1 + r_2 = \sqrt{37} + \sqrt{13}$

and then $d < r_1 + r_2$

$|r_1 - r_2| = |\sqrt{37} - \sqrt{13}|$ and then $|r_1 - r_2| < d$.

We see that $|r_1 - r_2| < d < r_1 + r_2$.

Thus, there is a circle of intersection.

The plane through the intersection is $\alpha \equiv S_1 - S_2$, that is $\alpha \equiv 5y - 5z - 2 = 0$.

Taking $S_1 \equiv x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$ and $\alpha \equiv 5y - 5z - 2 = 0$, the circle of intersection is

given by
$$\begin{cases} x^2 + y^2 + z^2 + 10y - 4z - 8 = 0 \\ 5y - 5z - 2 = 0 \end{cases}$$

To find the centre of this circle we need a perpendicular line through the centre of S_1 . The normal vector of plane α is the direction vector of this perpendicular line. Then this perpendicular line

has equations:
$$\begin{cases} x = 0 \\ y = -5 + 5t \\ z = 2 - 5t \end{cases}$$

The centre of the circle is the intersection of this line and plane α .

Putting these parametric equations of the perpendicular line into the equation of the plane α

$$\text{gives: } 5(-5+5t)-5(2-5t)-2=0 \Rightarrow t = \frac{37}{50} \text{ and then } \begin{cases} x=0 \\ y=-5+5 \times \frac{37}{50} = -\frac{13}{10} \\ z=2-5 \times \frac{37}{50} = -\frac{17}{10} \end{cases}$$

Thus, the centre of the circle is $\left(0, -\frac{13}{10}, -\frac{17}{10}\right)$.

To find the radius of the circle of intersection we need the length from the centre of sphere S_1 to the plane α .

$$\text{This length is } d_1(c_1, \alpha) = \frac{|-25-10-2|}{\sqrt{5+25}} = \frac{37}{\sqrt{50}}$$

Or

The length from the centre of sphere S_1 to the plane α is given by the distance between the centre of sphere S_1 and the centre of the circle of intersection since this circle lies on plane α

That is

$$d_1(c_1, \alpha) = \sqrt{0 + \left(-5 + \frac{13}{10}\right)^2 + \left(2 + \frac{17}{10}\right)^2} = \sqrt{\frac{1369}{100} + \frac{369}{100}} = \frac{37}{\sqrt{50}}$$

$$\text{Now, the radius of the circle of intersection is } \sqrt{r_1^2 - d_1^2} = \sqrt{37 - \frac{369}{50}} = \sqrt{\frac{1481}{50}} = \frac{1}{5} \sqrt{\frac{1481}{2}}.$$

Note that the circle of intersection can also be defined by the equation of sphere S_2 and plane

$$\alpha. \text{ That is, circle of intersection is given by } \begin{cases} x^2 + y^2 + z^2 + 6z - 4 = 0 \\ 5y - 5z - 2 = 0 \end{cases}$$

Using the same method, the centre and the radius will be the same.



Application activity 8.5.2

1) Describe the position of point $P(1, 2, 3)$ and the sphere $x^2 + y^2 + z^2 - 4x + 2y - 2z - 8 = 0$

2) Find the equation to a sphere which passes through the circle $\begin{cases} x^2 + y^2 + z^2 - 2x + 2y + 4z - 3 = 0 \\ 2x + y + z = 4 \end{cases}$ and through the point $(1, 2, -1)$.

- 3) Find the intersection between sphere $S_1 \equiv x^2 + y^2 + z^2 - 2x - 2y = 34$ and $S_2 \equiv x^2 + y^2 + z^2 - 8x - 2y = -13$
- 4) Find the equation of the sphere which passes through the circle $x^2 + y^2 + z^2 - 4x - y + 3z + 12 = 0$, $2x + 3y - 7z = 10$ and touch the plane $x - 2y + 2z = 1$.



END OF UNIT ASSESSMENT

Assessing knowledge and understanding:

1) For two distinct lines in 3D space, if they are not parallel and do not intersect, they are classified as skew lines.

- A. True
B. False

2) Which of the following equations represents a sphere with a center at $C(1, -2, 3)$ and a radius of 5?

A: $(x-1)^2 + (y+2)^2 + (z-3)^2 = 25$

B: $(x+1)^2 + (y-2)^2 + (z+3)^2 = 25$

C: $(x-1)^2 + (y+2)^2 + (z-3)^2 = 5$

D: $(x+1)^2 + (y-2)^2 + (z+3)^2 = 5$

3) Match each equation in Column A with the correct geometric description in Column B.

Column A (Equation)	Column B (Description)
1. $\vec{r} = \vec{a} + t\vec{b}$	A. Cartesian equation of a plane
2. $(x-h)^2 + (y-k)^2 = r^2$	B. Vector equation of a line
3. $Ax + By + Cz + D = 0$	C. Equation of a circle in 2D

4) Determine the vector equation of the line which is parallel to the vector $\vec{u} = 2\vec{i} + 3\vec{j} - \vec{k}$ and which passes through the point $A(1, 1, 1)$.

5) Find the vector equation of the line which passes through the point $A(-1, 2, 1)$ and which is parallel to the vector $\vec{u} = (1, 2, 3)$.

6 Given points $A(2, -1, 1)$ and $B(5, 2, -2)$.

a) Find $\overrightarrow{AB}$ b) The vector equation of the line that passes through A and B .

6. Calculate the angle between the planes $2x + 3y + z = 10$ and $x - 2y - 3z + 12 = 0$

8. Find the angle between the lines:

a. $\frac{x}{2} = -(y+1) = \frac{z+1}{-2}$ and $x - 6 = y = \frac{1}{2}(z - 1)$

b. $\vec{r} = \vec{i} - \vec{j} + t(\vec{i} + \vec{j} - \vec{k})$ and $\begin{cases} x = -2 + 3t \\ y = 6t \\ z = 1 + 2t \end{cases}$

9. Find the distance from point $P(1, 5, -4)$ to plane $\pi: 3x - y + 2z - 6 = 0$

10) Find the equation of the sphere through the circle $x^2 + y^2 + z^2 = 4, \quad z = 0$ cutting the sphere $x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$ orthogonally.

1.1 Comment by True or False:

When modeling the flight path of a drone delivering a package in a city, using a 3D coordinate system allows for precise navigation by incorporating altitude as a third dimension, demonstrating the power of coordinate systems to represent real-world complexity.

A. True B. False

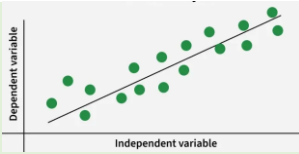
12. You and a group are working on a multi-step problem to find the shortest distance between two skew lines representing proposed pipelines. One group member makes a sign error in the cross-product calculation, leading to an incorrect distance. What is the most effective way to demonstrate the value of precision and collaboration in this scenario?

- A. Quickly point out the mistake, state the correct answer, and move to the next problem to save time.
- B. Blame the group member for the error to ensure they are more careful next time.
- C. Review the vector calculation step-by-step with the group, explain where the sign error occurred, and discuss why careful arithmetic is critical for accurate modeling in engineering contexts.

- D. Assume the error is negligible and submit the answer as is, since the pipeline project is just a hypothetical exercise.

13. Match each scenario on the left with the attitude or value it most directly reflects on the right.

Column A: Scenario	Column B: Attitude/Value
1. A student sketches a 3D coordinate system and rotates it mentally to determine whether two lines appear to be intersecting or skew before performing calculations.	A. Appreciation for the power of coordinate systems
2. A team divides the steps of finding the intersection between a line and a plane, then cross-checks each other's algebra to ensure the final point satisfies both equations.	B. Spatial reasoning and visualization
3. A student remarks, "It's fascinating how plotting points in 3D lets us solve geometric problems that would be nearly impossible with just shapes and no coordinates."	C. Collaboration and precision



UNIT 9: BIVARIATE STATISTICS

Key unit Competence: You will be able to extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

Learning objectives

Knowledge and understanding:

- Explain two variable data and mention examples
- Explain correlation and list its different types.
- State the formula for covariance and its role in measuring the direction of a linear relationship.
- Calculate and interpret correlation coefficients to quantify relationships.
- Recall the equation of a straight line ($y = mx + c$) and its relation to the concept of a regression line.

Skills:

- Construct and visualize bivariate data: Create accurate scatter diagrams to visually represent the relationship between two variables.
- Execute statistical calculations: Compute covariance, standard deviation, and Spearman's Rank Correlation Coefficient from raw datasets.
- Derive and apply regression models: Determine the equation of a regression line and use it to perform predictions (estimating y for a given x).
- Manipulate output to interpret and evaluate relationships: Analyze statistical results (scatter plots, coefficients, equations) to draw conclusions and assess the strength and nature of variable relationships.

Attitudes and values:

- Develop a critical attitude towards statistical results (especially regarding correlation vs. causation) and recognize the ethical responsibility to present findings transparently without misleading conclusions.
- Appreciate the importance of accuracy and meticulousness in statistical work and demonstrate a willingness to use statistical evidence to support arguments and inform decision-making.
- Foster a collaborative spirit when working on data analysis projects with peers and appreciate the importance of statistical methods for describing and understanding real-world relationships.



Introductory activity:

In Kabeza village, after her 9 observations about farming, UMULISA saw that in every house observed, where there are X cows there are also Y domestic ducks. She got the following results in form of ordered pairs (x, y) :

$(1, 4), (2, 8), (3, 4), (4, 12), (5, 10)$

$(6, 14), (7, 16), (8, 6), (9, 18)$

a) Represent this information graphically in (x, y) – coordinates

b) Find the equation of a straight line joining many points of the graph and guess the name of this line.

c) According to your observation from (a), explain in your own words if there is any relationship between the number of Cows (X) and the number of domestic ducks (Y).



9.1 Bivariate data and scatter diagram



Activity 9.1

Consider the situation in which the mass y (g) of a chemical is related to the time, x minutes, for which the chemical reaction has been taking place. Data were given in the following table:

Time, x min	5	7	12	16	20
Mass, y g	4	12	18	21	24

a) Plot the above information in (x, y) coordinates.

b) Explain in your own words the relationship between the variation of x and y .



Content Summary

In statistics, **bivariate** or **double series** includes technique of analysing data in two variables, the focus on the relationship between a dependent variable y and an independent variable x .

Example: Relationship or dependency between: - Age and weight; - weight and height; - Years of education and salary; - Amount of daily exercise and cholesterol level.

As with data for a single variable, we can describe bivariate data both graphically and numerically. In both cases, we will be primarily concerned with determining whether there is a *linear* relationship between the two variables under consideration or not.

Note: Keep in mind that a statistical relationship between two variables does not necessarily imply a *causal* relationship between them. For example, a strong relationship between weight and height does not imply that either variable causes the other.

Bivariate analysis

It is a statistical method used to explore the relationship between **two variables** in order to understand whether and how the two variables are related and then, describe the **nature**, **strength**, and **direction** of that relationship.

Bivariate analysis is quantitative, it examines two variables denoted as X and Y and it aims to understand to what extent one variable (possibly independent) predicts the value of another (possibly dependent). It helps uncover **correlations** and **associations** between different factors in data analysis.

Types of bivariate analysis

The type of bivariate analysis used depends on the **nature of the variables** involved — whether they are **numerical**, **categorical**, or **ordinal**. The choice of statistical technique is guided by how these variables interact.

a) Numerical and numerical:

In this type, both independent and dependent variables are numerical.

Example:

- **Height and Weight** – Do taller people tend to weigh more?
- **Study Hours and Test Scores** – Is there a relationship between time spent studying and performance?

b) Categorical and categorical: Here, both variables are categorical. The goal is often to test for association or independence.

Example: Gender and product preference – Do preferences vary between male and female customers?

3) Numerical and categorical:

In this scenario, one variable is numerical (usually dependent) and the other is categorical (often the independent).

Example: Education Level and income – How does education level impact salary?

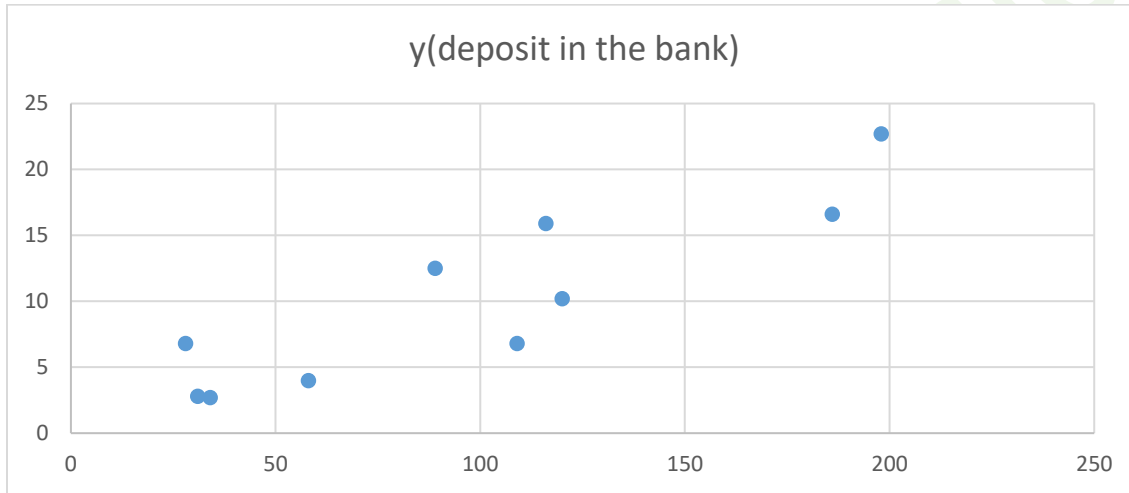
Scatter plots or Scatter diagram

A scatter diagram (or scatter plot) is a graphical tool that displays the relationship between two numerical variables using Cartesian coordinates (X and Y axes) to plot data points.

Consider the following data which relate x as the respective number of branches that 10 different banks have in a given common market, with y , the corresponding market share of total deposits held by the banks:

x	198	186	116	89	120	109	28	58	34	31
y	22.7	16.6	15.9	12.5	10.2	6.8	6.8	4.0	2.7	2.8

If each point (x, y) of the data is plotted in an x, y coordinate plane, **the scatter plot or Scatter diagram** is obtained.



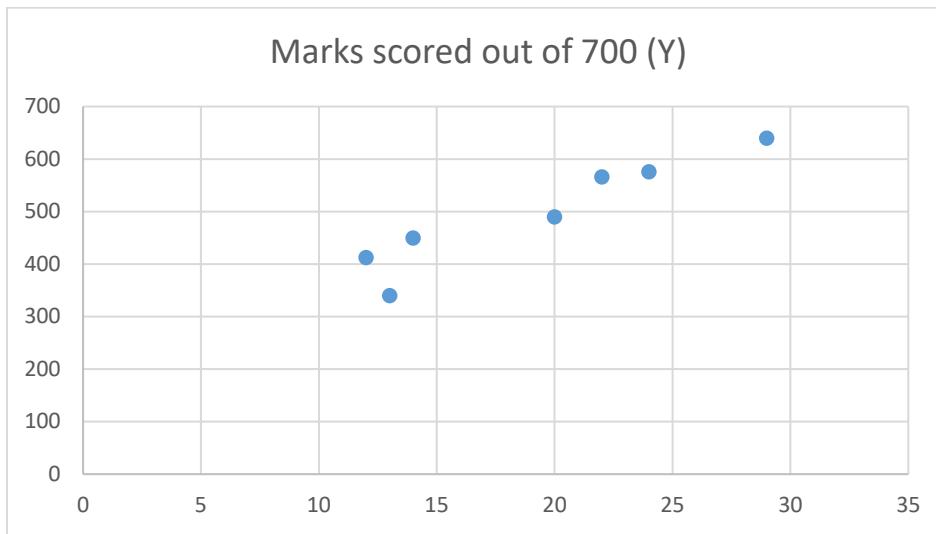
For instance, consider a study examining the relationship between exercise duration (X) and heart rate (Y) during physical activity. By analyzing this bivariate data, we can determine if there's a correlation between the duration of exercise (independent) and heart rate (dependent).

Example:

A teacher collected data of total hours studied by students and total marks scored by them:

Total Hours Studied in a Week (X)	14	12	20	22	24	29	13
Marks scored out of 700 (Y)	450	413	490	566	576	640	340

Scatter diagram



The teacher can determine the correlation between the two variables, to show that the marks scored by a student in a test is directly correlated to number of hours studied by him/her.

When do we use a scatter diagram:

- When you have paired numerical data
- When your dependent variable may have multiple values for each value of your independent variable
- When trying to determine whether the two variables are related, such as:
 - When trying to identify potential root causes of problems
 - After brainstorming causes and effects using a fishbone diagram to determine objectively whether a particular cause and effect are related
 - When determining whether two effects that appear to be related both occur with the same cause
 - When testing for autocorrelation before constructing a control chart.



Application activity 9.1

One measure of personal fitness is the time taken for an individual's pulse rate to return to normal after strenuous exercise, the greater the fitness, the shorter the time. Following a short program of strenuous exercise Norman recorded his pulse rates P at time t minutes after he had stopped exercising. Norman's results are given in the table below.

t	0.5	1.0	1.5	2.0	3.0	4.0	5.0
P	125	113	102	94	81	83	71

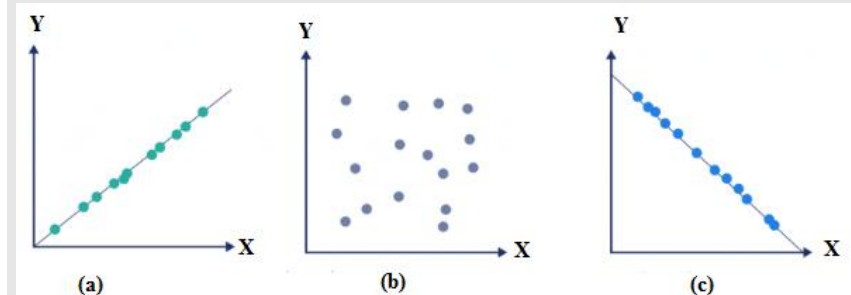
Draw a scatter diagram to represent this information in (x, y) coordinates .

9.2 Correlation and types of correlation



Activity 9.2

Observe the diagram and the graphs plotted by joining some (x,y) points of the function $y = f(x)$:



- At which diagram do we have many points on a straight line?
- If we consider the linear function $y = f(x)$, which line has a positive slope, which one has a negative slope?
- Consider each case; Can you say that there is a relationship between the variation of x and the variation of y ?



Content summary

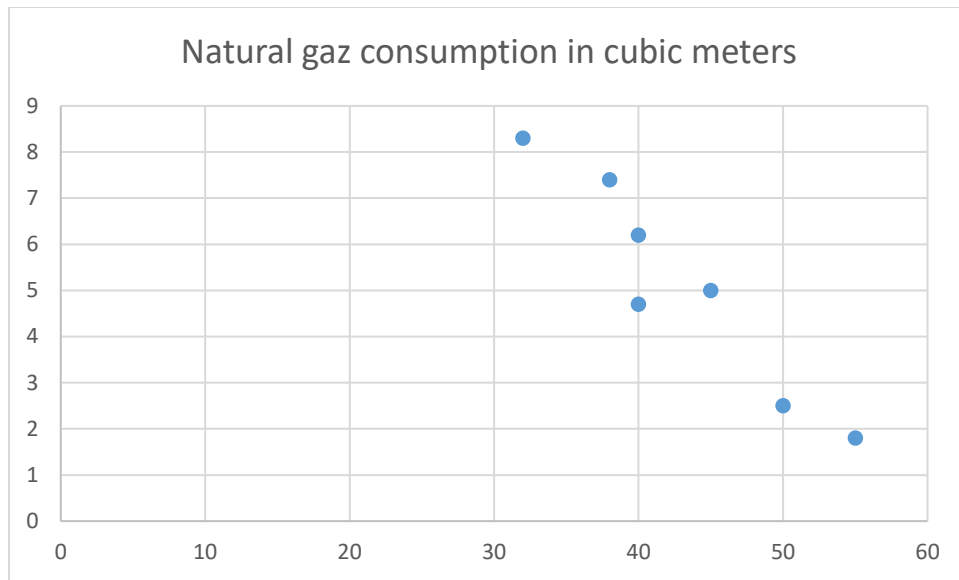
Correlation: Correlation refers to the relationship or association between two numerical variables.

There are several characteristics we consider when describing the correlation between two variables: direction, linearity, strength, outliers, and causation.

The scatter plot or scatter diagram (in the figure above) indicates that, roughly speaking, the market share increases as the number of branches increases. We say that x and y have a **positive correlation**.

On the other hand, consider the data below, which relate average daily temperature x , in degrees Fahrenheit, and daily natural gas consumption y , in cubic metre.

$x, ^\circ F$	50	45	40	38	32	40	55
y, m^3	2.5	5.0	6.2	7.4	8.3	4.7	1.8



We see that y tends to decrease as x *increases*. Here, x and y have a ***negative correlation***

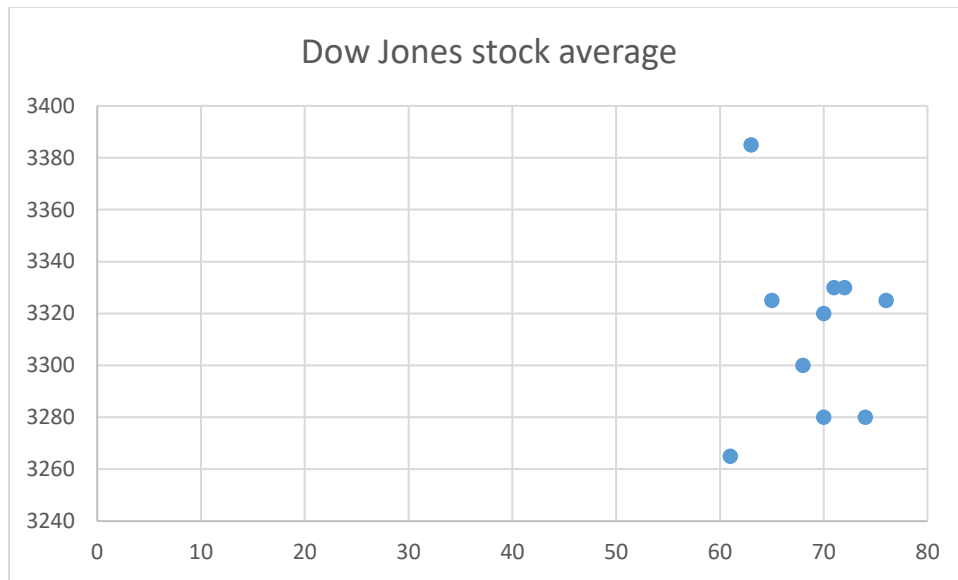
Finally, consider the data items (x, y) below, which relate daily temperature x over a 10-day period to the Dow Jones stock average y .

(68, 3300); (61, 3265); (63, 3385); (72, 3330); (76, 3325); (70, 3320); (71, 3330); (65, 3325); (70, 3280); (74, 3280) ;

Here's the data organized in a table:

X	63	72	76	70	71	65	70	74	68	61
Y	338	333	332	332	333	332	328	328	330	326

On the scatter diagram:



There is no apparent relationship between x and y (*no correlation* or *Weak correlation*). Note that the correlation is a mutual relationship between two or more things.

Examples of correlation in real life include:

- As students' study time increases, the tests average increases too.
- As the number of trees cut down increases, soil erosion increases too.
- The more you exercise your muscles, the stronger they get.
- As a child grows, so does the clothing size.
- The more one smokes, the fewer years he will have to live.

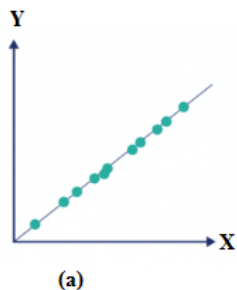
Types of correlation

There are several characteristics we consider when describing the correlation between two variables: **Direction, linearity, strength, outliers, and causation.**

Direction:

Positive correlation:

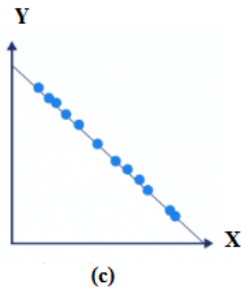
For a generally *upward* trend, we say that the correlation is **positive**. An increase in the independent variable generally results in an increase in the dependent variable.



- 1) The more times people have unprotected sex with different partners, the more the rates of HIV is in a society.
- 2) As students study time increases, the tests average increases too.

Negative correlation:

For a generally *downward* trend, we say that the correlation is **negative**. An increase in the independent variable generally results in a decrease in the dependent variable.

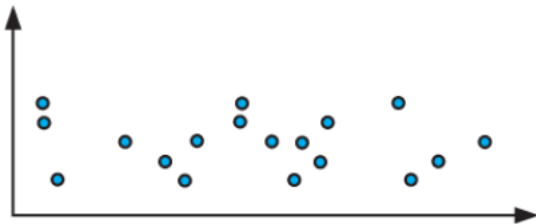


The correlation is negative because the increase of the variable x contributes to the decrease of the variable y .

Examples: The more alcohol one consumes, the less the judgment one has.

No correlation:

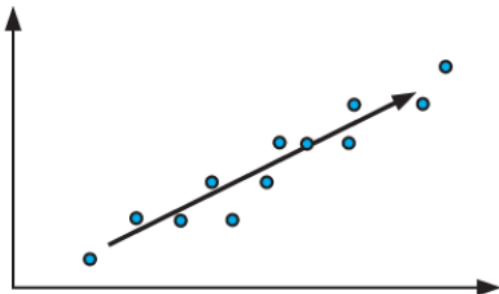
For *randomly scattered* points, with no upward or downward trend, we say there is **no correlation**.



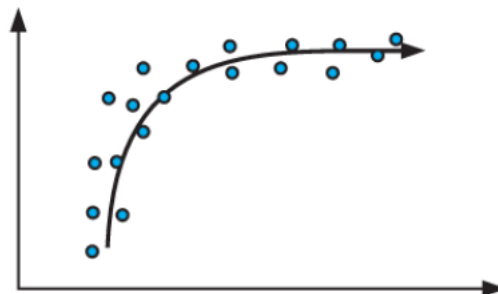
LINEARITY

When a trend exists, if the points approximately form a straight line, we say the trend is linear.

These points are roughly linear.

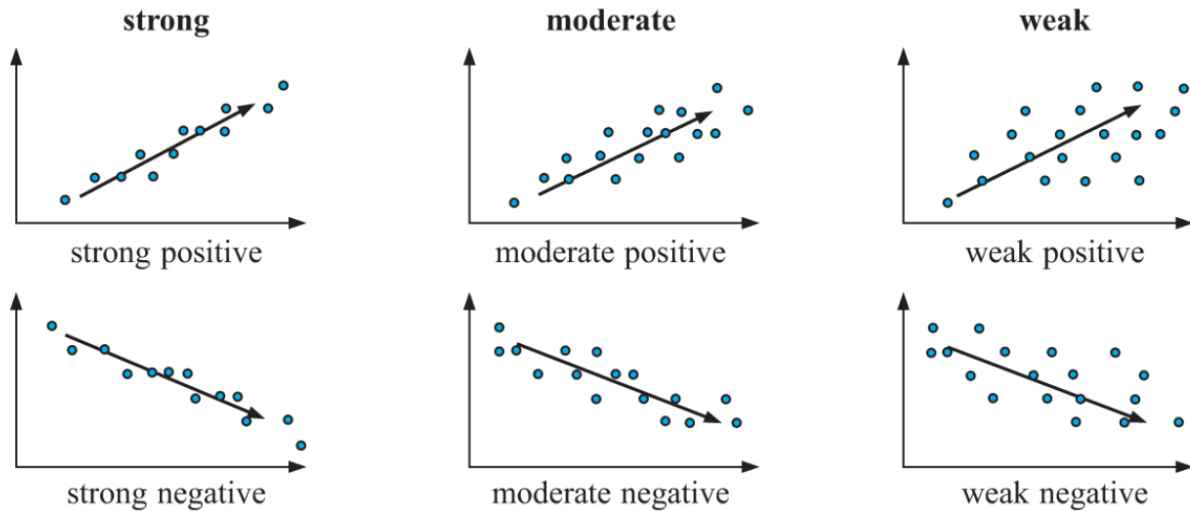


These points do not follow a linear trend.



STRENGTH

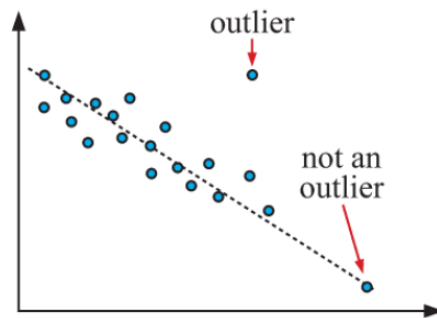
To describe how closely the data follows a pattern or trend, we talk about the strength of correlation. It is usually described as either strong, moderate, or weak.



OUTLIERS

Outliers are isolated points which do not follow the trend outlier formed by the main body of data.

If an outlier is the result of a recording or graphing error, it should be discarded. However, if the outlier is a genuine piece of data, it should be kept.



CAUSALITY

Correlation between two variables does not necessarily mean that one variable causes the other. If a change in one variable causes a change in the other variable, then we say that a causal relationship exists between them. In these cases, we can say that the independent variable explains the dependent variable.

In cases where a causal relationship is not apparent, we cannot conclude that a causal relationship exists based on high correlation alone.

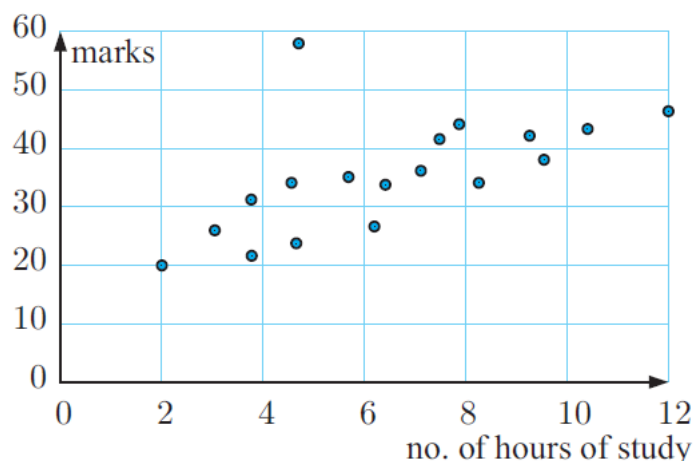
Example:

The scatter diagram alongside shows the marks obtained by students in a test out of 50 marks, plotted against the number of hours each student studied for the test.

a) Describe the correlation between the variables.

b) How should the outlier be treated?

Explain your answer.



Solution:

a) Description of the Correlation

-The correlation between the number of hours of study and the marks obtained is **positive**. This indicates that as the independent variable (hours of study) increases, the dependent variable (marks) generally results in an increase as well.

Visually, the data follows a generally **upward trend**.

b) Treatment of the Outlier

The outlier is the data point located at approximately (4.5 hours, 58 marks). It should be treated as follows:

- **Identification of error:** You should investigate this point for a potential recording or entry error. The test was out of **50 marks**, yet this student has a score of approximately **58**, which is an impossible value for the given scale. As the value is mathematically impossible within the context of a 50-mark test, the outlier should be **discarded** from the data set or **corrected** if the original test paper can be verified.
- **Impact on analysis:** Removing this impossible value will allow for a more accurate calculation of the correlation coefficient and a better analysis of other parameters.



Application activity 9.2:

One measure of personal fitness is the time taken for an individual's pulse rate to return to normal after strenuous exercise, the greater the fitness, the shorter the time. Following a short program of strenuous exercise Norman recorded his pulse rates P at time t minutes after he had stopped exercising. Norman's results are given in the table below.

t	0.5	1.0	1.5	2.0	3.0	4.0	5.0
P	125	113	102	94	81	83	71

Explain the relationship between Norman's pulse P and time t .

9.3 Covariance



Activity 9.3

Complete the following table

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	3	6			
2	5	9			
3	7	12			
4	3	10			
5	2	7			
6	6	8			
	$\sum_{i=1}^6 x_i = \dots$	$\sum_{i=1}^6 y_i = \dots$			$\sum_{i=1}^6 (x_i - \bar{x})(y_i - \bar{y}) = \dots$
	$\bar{x} = \dots$	$\bar{y} = \dots$			

What can you get from the following expressions?

- 1) $\sum_{i=1}^k (x_i - \bar{x})(x_i - \bar{x})$
- 2) $\sum_{i=1}^k (x_i - \bar{x})(y_i - \bar{y})$



Content summary

In case of two variables, say x and y , there is another important result called **covariance of x and y** , denoted $\text{cov}(x, y)$

The **covariance of variables x and y** is a measure of how these two variables change together. If the greater values of one variable mainly correspond with the greater values of the other variable, and the same holds for the smaller values, i.e. the variables tend to show similar behaviour, the covariance is positive. In the opposite case, when the greater values of one variable mainly correspond to the smaller values of the other, i.e. the variables tend to show opposite behaviour, the covariance is negative. If covariance is zero the variables are said to be **uncorrelated**, it means that there is no linear relationship between them.

Therefore, the **sign of covariance** shows the tendency in the linear relationship between the variables. The magnitude of covariance is not easy to interpret.

Covariance of variables x and y , where the summation of frequencies $\sum_{i=1}^k f_i = n$ are equal for both variables, is defined to be

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^k f_i (x_i - \bar{x})(y_i - \bar{y})$$

Developing this formula, we find that the covariance is also given by

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^k f_i x_i y_i - \bar{x} \bar{y}.$$

Examples

1) Find the covariance of x and y in following data sets

x	3	5	6	8	9	11
y	2	3	4	6	5	8

Solution

We have

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
3	2	-4	-2.6	10.4
5	3	-2	-1.6	3.2
6	4	-1	-0.6	0.6
8	6	1	1.4	1.4
9	5	2	0.4	0.8
11	8	4	3.4	13.6
$\sum_{i=1}^6 x_i = 42$	$\sum_{i=1}^6 y_i = 28$			$\sum_{i=1}^6 (x_i - \bar{x})(y_i - \bar{y}) = 30$
$\bar{x} = 7$	$\bar{y} = 4.6$			

Thus,

$$\text{cov}(x, y) = \frac{1}{6} \sum_{i=1}^6 (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{6}(30) = 5$$

2) Find the covariance of the following distribution where some pairs (x_i, y_i) are repeated with the frequency f_i .

x	0	2	4
y			

1	2	1	3
2	1	4	2
3	2	5	0

Solution

Convert the double entry into a simple table and compute the arithmetic means

x_i	y_i	f_i	$x_i f_i$	$y_i f_i$	$x_i y_i f_i$
0	1	2	0	2	0
0	2	1	0	2	0
0	3	2	0	6	0
2	1	1	2	1	2
2	2	4	8	8	16
2	3	5	10	15	30
4	1	3	12	3	12
4	2	2	8	4	16
4	3	0	0	0	0
		$\sum_{i=1}^9 f_i = 20$	$\sum_{i=1}^9 x_i f_i = 40$	$\sum_{i=1}^9 y_i f_i = 41$	$\sum_{i=1}^9 x_i y_i f_i = 76$

$$\bar{x} = \frac{40}{20} = 2, \quad \bar{y} = \frac{41}{20} = 2.05$$

$$\text{cov}(x, y) = \frac{76}{20} - 2 \times 2.05 = -0.3$$

Alternative method

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^k f_i x_i y_i - \bar{x} \bar{y}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k x_i f_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^k y_i f_i$$

$x \backslash y$	0	2	4	Total
1	2	1	3	6
2	1	4	2	7
3	2	5	0	7
Total	5	10	5	20

$$\text{aligned}\bar{x} = \frac{1}{20}(0 \times 5 + 2 \times 10 + 4 \times 5) = \frac{40}{20} = 2$$

$$\bar{y} = \frac{1}{20}(1 \times 6 + 2 \times 7 + 3 \times 7)$$

$$= \frac{41}{20} = 2.05$$

$$\begin{aligned} \text{cov}(x, y) &= \frac{1}{20}(0 \times 1 \times 2 + 0 \times 2 \times 1 + 0 \times 3 \times 2 + 2 \times 1 \times 1 + 2 \times 2 \times 4 + 2 \times 3 \times 5 + 4 \times 1 \times 3 + 4 \times 2 \times 2 + 4 \times 3 \times 0) \\ &\quad - 2 \times 2.05 \\ &= \frac{1}{20}(0 + 0 + 0 + 2 + 16 + 30 + 12 + 16 + 0) - 4.1 \\ &= \frac{76}{20} - 4.1 = -0.33 \end{aligned}$$



Application activity 9.3:

1. The scores of 12 students in their mathematics and physics classes are

Mathematics	2	3	4	4	5	6	6	7	7	8	10	10
Physics	1	3	2	4	4	4	6	4	6	7	9	10

Find the covariance of the distribution

2. The values of two variables x and y are distributed according to the following table

$x \backslash y$	100	50	25
14	1	1	0
18	2	3	0
22	0	1	2

Calculate the covariance

9.4 Coefficient of correlation and its properties



Activity 9.4

Consider the following table

x	y
3	6
5	9
7	12
3	10
2	7
6	8

1. Find the variance of x and its square root named as the standard variation σ_x . Remember that the variance of x is $\sigma_x^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$
2. Find the variance of y and its square root which is the standard variation σ_y .
3. Find $\text{cov}(x, y)$
4. Calculate the ratio $\frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$



Content summary

The **Pearson's coefficient of correlation** (or **Product moment coefficient of correlation** or simply **coefficient of correlation**), denoted by r , is a measure of the strength of linear correlation between two variables.

The coefficient of correlation between two variables x and y is given by: $r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

Where, $\text{cov}(x, y)$ is covariance of x and y ;

σ_x is the standard deviation for x ;

σ_y is the standard deviation for y .

Properties of the coefficient of correlation

- a) The coefficient of correlation does not change the measurement scale. That is, if the height is expressed in meters or feet, the coefficient of correlation does not change.
- b) The sign of the coefficient of correlation is the same as the covariance.

- c) The square of the coefficient of correlation is equal to the product of the gradient of the regression line of y on x , and the gradient of the regression line of x on y .

In fact, $r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

Squaring both sides gives

$$\begin{aligned} r^2 &= \left[\frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \right]^2 \\ &= \frac{\text{cov}^2(x, y)}{\sigma_x^2 \sigma_y^2} \\ &= \frac{\text{cov}(x, y)}{\sigma_x^2} \times \frac{\text{cov}(x, y)}{\sigma_y^2} = ac, \end{aligned}$$

- d) If the coefficient of correlation is known, it can be used to find the gradients or slopes of two regression lines.

We know that the gradient of the regression line is $\frac{\text{cov}(x, y)}{\sigma_x^2}$.

From this, we have $\frac{\text{cov}(x, y)}{\sigma_x^2} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_x} \times \frac{\sigma_y}{\sigma_y} = r \frac{\sigma_y}{\sigma_x}$

Thus, the **gradient of the regression line** of y on x is given by $r \frac{\sigma_y}{\sigma_x}$ and the gradient of the regression line of x on y is given by $r \frac{\sigma_x}{\sigma_y}$.

e) **Cauchy Inequality:** $\text{cov}^2(x, y) \leq \sigma_x^2 \sigma_y^2$

f) The coefficient of correlation takes value ranging between -1 and +1. That is $-1 \leq r \leq 1$

g) If the linear coefficient of correlation takes values closer to **-1**, the **correlation is strong and negative** and will become stronger the closer r approaches -1.

h) If the linear coefficient of correlation takes values close to **1** the **correlation is strong and positive**, and will become stronger the closer r approaches 1

i) If the linear coefficient of correlation takes values close to **0**, the **correlation is weak**.

j) If $r = 1$ or $r = -1$, there is **perfect correlation** and the line on the scatter plot is increasing or decreasing respectively.

k) If $r = 0$, there is **no linear correlation**.

Example

A test is made over 200 families on number of children (x) and number of beds y per family.

Results are collected in the table below

$y \backslash x$	0	1	2	3	4	5	6	7	8	9	10
1	0	2	7	5	2	0	0	0	0	0	0
2	2	2	10	8	15	1	0	0	0	0	0
3	1	3	5	6	8	6	1	0	0	0	0
4	0	2	8	2	6	12	10	8	0	0	0
5	0	1	0	2	5	6	10	5	7	3	3
6	0	0	0	2	4	5	5	2	3	3	2

- a) What is the average number for children and beds per a family?
b) Find the covariance
c) Can we confirm that there is a high linear correlation between the number of children and number of beds per family?

Solution

- a) Average number of children per family:

Contingency table:

$y \backslash x$	0	1	2	3	4	5	6	7	8	9	10	Total
1	0	2	7	5	2	0	0	0	0	0	0	16
2	2	2	10	8	15	1	0	0	0	0	0	38
3	1	3	5	6	8	6	1	0	0	0	0	30
4	0	2	8	2	6	12	10	8	0	0	0	48
5	0	1	0	2	5	6	10	5	7	3	3	42
6	0	0	0	2	4	5	5	2	3	3	2	26
Total	3	10	30	25	40	30	26	15	10	6	5	200

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k f_i x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^k f_i y_i$$

Marginal series:

x_i	0	1	2	3	4	5	6	7	8	9	10	Total

f_i	3	1	3	2	40	30	26	15	1	6	5	$\sum f_i = 200$
		0	0	5					0			
$f_i x_i$	0	1	6	7	16	15	15	10	8	5	5	$\sum f_i x_i = 900$
		0	0	5	0	0	6	5	0	4	0	

y_i	1	2	3	4	5	6	Total
f_i	16	38	30	48	42	26	$\sum f_i = 200$
$f_i y_i$	16	76	90	192	210	156	$\sum f_i y_i = 740$

The means are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{11} f_i x_i = \frac{900}{200} = 4.5 \text{ and}$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^6 f_i y_i = \frac{740}{200} = 3.7$$

There are about 5 children per family and about 4 beds per family.

b) The covariance is calculated as follow:

$$Cov(x, y) = \left(\frac{1}{n} \sum_{(i,j)=(1,1)}^{(p,q)=(11,6)} f_{i,j} x_i y_j \right) - \bar{x} \bar{y}, \text{ where } i \text{ assumes values from } 1 \text{ to } p = 11, \text{ and } j \text{ assumes values from } 1 \text{ to } q = 6, \text{ or}$$

$$\text{cov}(x, y) = \frac{1}{200} \sum_{i=1}^{66} f_i x_i y_i - \bar{x}\bar{y}$$

Where $\bar{y} = 3.7$ and $\bar{x} = 4.5$

$$\begin{aligned}
 &= \frac{1}{200} \left(\begin{aligned} &0 + 2 + 14 + 15 + 8 + 0 + 4 + 40 + 48 + 120 + 10 + 0 \\ &9 + 30 + 54 + 96 + 90 + 18 + 0 + 8 + 64 + 24 + 96 \\ &+ 240 + 240 + 224 + 0 + 5 + 0 + 30 + 100 + 150 \\ &+ 300 + 175 + 280 + 135 + 150 + 0 + 36 + 96 + 150 \\ &+ 180 + 84 + 144 + 162 + 120 \end{aligned} \right) - 4.5 \times 3.7 \\
 &= \frac{3751}{200} - 16.65 \\
 &= 18.7555 - 16.65 \\
 &= 2.105
 \end{aligned}$$

c) The correlation coefficient is given by $\frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

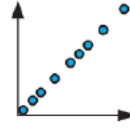
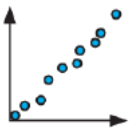
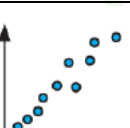
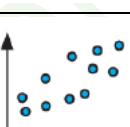
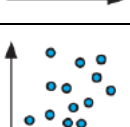
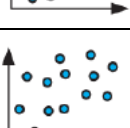
$$\begin{aligned}
 \sigma_y^2 &= \frac{1}{200} \sum_{i=1}^6 f_i y_i^2 - (\bar{y})^2 \\
 &= \frac{1}{200} (16 + 38 \times 4 + 30 \times 9 + 48 \times 16 + 42 \times 25 + 26 \times 36) - (3.7)^2 \\
 &= 15.96 - 13.69 \\
 &= 2.27 \\
 \sigma_x^2 &= \frac{1}{200} \sum_{i=1}^{11} f_i x_i^2 - (\bar{x})^2 = \frac{1}{200} \left(\begin{aligned} &0 + 10 + 4 \times 30 + 9 \times 25 + 40 \times 16 + 30 \times 25 + \\ &36 \times 26 + 49 \times 15 + 64 \times 10 + 81 \times 6 + 100 \times 5 \end{aligned} \right) - (4.5)^2 \\
 &= 25.21 - 20.25 = 4.96
 \end{aligned}$$

Therefore, the correlation coefficient is

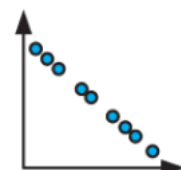
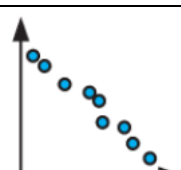
$$r = \frac{2.105}{\sqrt{4.96} \sqrt{2.27}} \approx 0.63$$

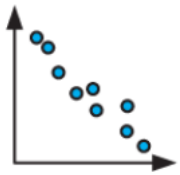
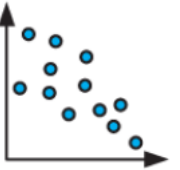
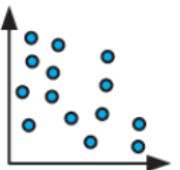
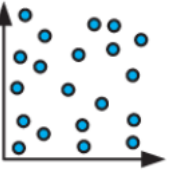
- The values of r range from -1 to +1.
 - A positive value for r indicates the variables are **positively correlated**. An increase in one of the variables will result in an increase in the other.
 - A negative value for r indicates the variables are **negatively correlated**. An increase in one of the variables will result in a decrease in the other.
 - The **size** of r indicates the **strength** of the correlation. A value of r close to +1 or -1 indicates strong correlation between the variables.
- A value of r close to zero indicates weak correlation between the variables.

Strength of positive linear correlation r

r Value	Description	Scatter diagram
$r=1$	perfect positive correlation	
$0.95 \leq r < 1$	very strong positive correlation	
$0.87 \leq r < 0.95$	strong positive correlation	
$0.7 \leq r < 0.87$	moderate positive correlation	
$0.5 \leq r < 0.7$	weak positive correlation	
$0 < r < 0.5$	very weak positive correlation	

Strength of negative linear correlation r

Correlation Coefficient (r)	Description	Scatter diagram
$r=-1$	perfect negative correlation	
$-1 < r \leq -0.95$	very strong negative correlation	

$-0.95 < r \leq -0.87$	strong negative correlation	
$-0.87 < r \leq -0.7$	moderate negative correlation	
$-0.7 < r \leq -0.5$	weak negative correlation	
$-0.5 < r < 0$	very weak negative correlation	

Note: Discuss how to use a calculator or the excel sheet to find the coefficient of correlation

The coefficient of determination r^2

To help describe the *strength* of correlation we calculate the **coefficient of determination r^2** .

$$r^2 = \left[\frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \right]^2 = \frac{\text{cov}^2(x, y)}{\sigma_x^2 \sigma_y^2} = \frac{\text{cov}(x, y)}{\sigma_x^2} \times \frac{\text{cov}(x, y)}{\sigma_y^2} \text{ or}$$

$$r^2 = \frac{(\sum xy - n\bar{x}\bar{y})^2}{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}$$

This is simply the square of Pearson's correlation coefficient r , and as such the direction of correlation is eliminated.

Here is the data from the image organized into an editable table:

value	strength of correlation
$r^2 = 0$	No correlation
$0 < r^2 < 0.25$	Very weak correlation

$0.25 \leq r^2 < 0.50$	Weak correlation
$0.50 \leq r^2 < 0.75$	moderate correlation
$0.75 \leq r^2 < 0.90$	strong correlation
$0.90 \leq r^2 < 1$	Very strong correlation
$r^2 = 1$	Perfect correlation

The table above is a guide for describing the strength of linear correlation using the coefficient of determination.



Application activity 9.4:

1) The scores of 12 students in their mathematics and physics classes are

Mathematics	2	3	4	4	5	6	6	7	7	8	10	10
Physics	1	3	2	4	4	4	6	4	6	7	9	10

Find the correlation coefficient distribution and interpret it.

2) The values of the two variables X and Y are distributed according to the following table:

X \ Y	0	2	4
1	2	1	3
2	1	4	2
3	2	5	0

Calculate the correlation coefficient.

9.5a Line of the best fit



Activity 9.5a

At a junior tournament, a group of young athletes throw a discus. The *age* and *distance thrown* are recorded for each athlete.

Athlete	A	B	C	D	E	F	G	H	I	J	K	L
Age (years)	12	16	16	18	13	19	11	10	20	17	15	13

Distance thrown (m)	20	35	23	38	27	47	18	15	50	33	22	20
---------------------	----	----	----	----	----	----	----	----	----	----	----	----

1) Calculate the mean of the X values $\bar{x}$, and the mean of the Y values $\bar{y}$.

2) Mark the mean point $(\bar{x}, \bar{y})$ on the scatter diagram.

3) Draw a line through the mean point such that about the same number of data points are above the line as below it.



Content summary

The line of the best fit is a straight line drawn through a scatter diagram that best represents the relationship between two variables.

It serves as a visual and mathematical tool to express the general trend of the data and can be used to make predictions for values within the data range.

In general, it is only worth drawing a line of best fit if the correlation between the variables is strong. There is no fixed rule, but we suggest that a line of best fit is not appropriate if $[r] < 0.85$. If we draw the line just by observing the points, we call it a line of best fit by eye. This line will vary from person to person.

We draw the line through the mean point so it follows the trend of the data and there are about the same number of points above the line as below the line.

For the **Opening Problem**, the mean point is (15, 29). So, we draw our line of best fit through (15, 29). We can use the line of best fit to estimate the value of y for any given value of x , and vice versa.



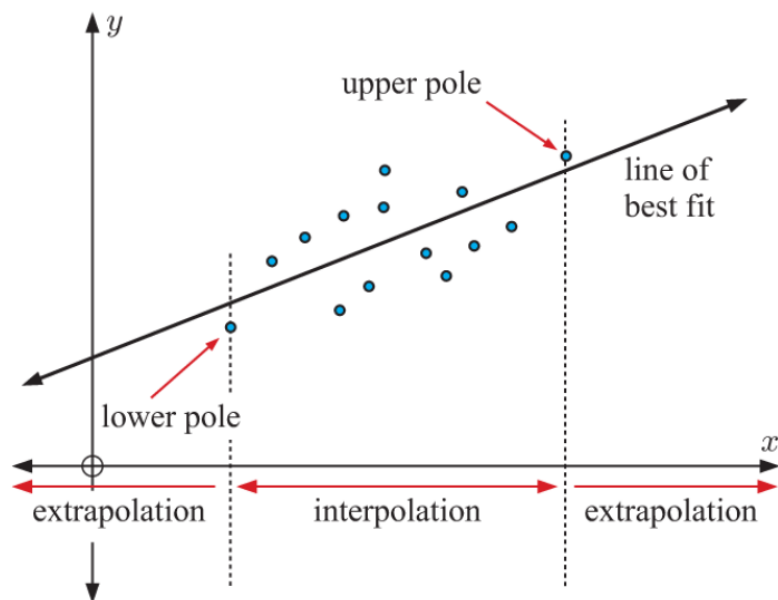
Consider the data in the scatter diagram alongside. The data with the highest and lowest values are called the **poles**.

The line of best fit for the data is also drawn on the scatter diagram. We can use this line

to predict the value of one variable for a given value of the other.

- If we predict a y value for an x value in between the poles, we say we are interpolating in between the poles.

- If we predict a y value for an x value outside the poles, we say we are extrapolating outside the poles.



Example:

On a hot day, six cars were left in the sun for various lengths of time. The length of time each car was left in the sun was recorded, as well as the temperature inside the car at the end of the period.

Car	A	B	C	D	E	F
Time (x min)	50	5	25	40	15	45
Temperature (y °C)	47	28	36	42	34	41

a) Calculate $\bar{x}$ and $\bar{y}$.

b) Draw a scatter diagram for the data.

c) Plot the mean point $(\bar{x}, \bar{y})$ on the scatter diagram and draw a line of best fit through the mean point.

d) Predict the temperature of a car which has been left in the sun for:

i) 35 minutes

ii) 75 minutes.

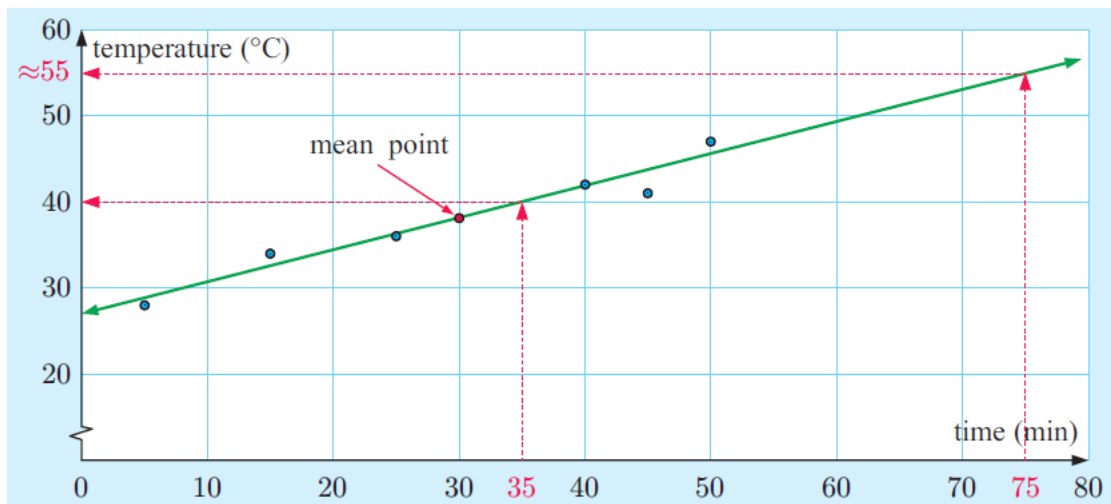
e) Comment on the reliability of your predictions in d.

Solution:

$$\text{a) } \bar{x} = \frac{50 + 5 + 25 + 40 + 15 + 45}{6} = 30,$$

$$\bar{y} = \frac{47 + 28 + 36 + 42 + 34 + 41}{6} = 38$$

(b and c)



d) i) When $x = 35$, $y \approx 40$. So, the temperature of a car left in the sun for 35 minutes will be approximately 40°C .

ii) When $x = 75$, $y \approx 55$. So, the temperature of a car left in the sun for 75 minutes will be approximately 55°C .

e) The prediction in **d i** is reliable, as the data appears linear, and this is an interpolation.

The prediction in **d ii** may be unreliable, as it is an extrapolation, and the linear trend displayed by the data may not apply over 75 minutes.



Application activity 9.5a:

Fifteen students were weighed, and their pulse rates were measured:

Weight (x kg)	61	52	47	72	62	79	57	45	67	71	80	58	51	43	55
Pulse rate (y beats per min)	65	59	54	74	69	87	61	59	70	69	75	60	56	53	58

a) Draw a scatter diagram for the data.

b) Calculate r and r^2 .

c) Describe the relationship between *weight* and *pulse rate*.

d) Calculate the mean point $(\bar{x}, \bar{y})$.

e) Plot the mean point on the scatter diagram, and draw a line of best fit through the mean point.

f) Estimate the pulse rate of a student who weighs 65 kg. Comment on the reliability of your estimate.

9.5b Regression lines



Activity 9.5b

Given the data in the table below:

x	5	7	12	16	20
y	4	12	18	21	24

Determine:

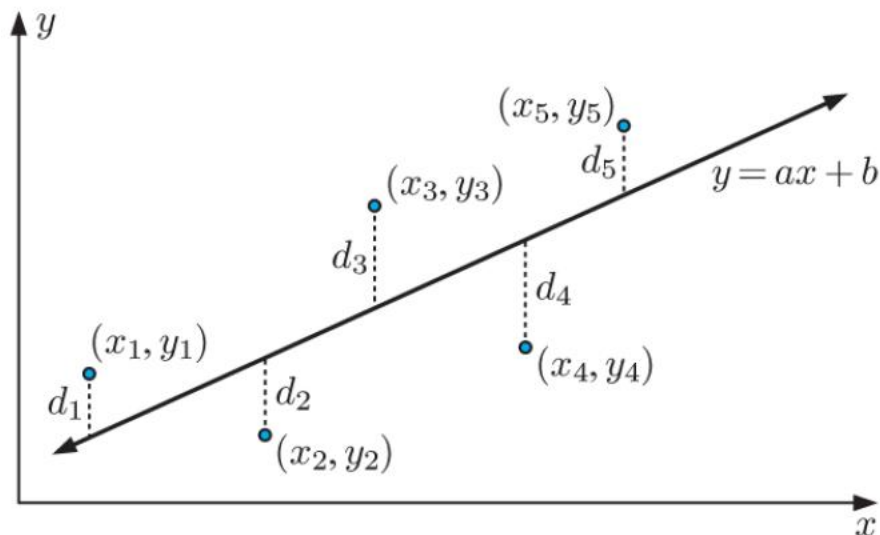
- Variance of x
- Variance of y
- Covariance of (x, y)
- The value a given by $a = \frac{\text{cov}(x, y)}{\text{var } x} = \frac{S_{x,y}}{S_{x,x}}$
- The value b given by $b = \bar{Y} - a\bar{X}$
- Establish the equation of the line $y = ax + b$
- In the Cartesian plane, represent the data (x, y) in the table above and the line $y = ax + b$ found in (f).
- Discuss the position of the points of coordinate (x, y) with respect to the line $y = ax + b$



Content summary

The problem with drawing a line of best fit by eye is that the line drawn will vary from one person to another. For consistency, we use a method known as linear regression to find the equation of the line which best fits the data. The most common method is the method of “least squares”.

In least squares linear regression, we minimise the sum of the squares of the vertical distances between each data point and the regression line.



For any line we draw to model the points, we can find the vertical distances $d_1, d_2, d_3, \dots$ between each point and the line.

We can then square each of these distances, and find their sum $d_1^2 + d_2^2 + d_3^2 + \dots$

If the line is a good fit for the data, most of the distances will be small, and so the sum of their squares will also be small.

The **least squares regression line** is the line which makes this sum as small as possible.

Equation of regression line

The equation $y = ax + b$ is the equation of the regression line of y on x , and $x = cy + d$ is the equation of the regression line of x on y .

We use the regression line of y on x to **predict** a value of y for any given value of x and vice versa, we use the regression line of x on y , to predict a value of x for a given value of y .

The “best” line would make the best predictions: the observed y -values should stray as little as possible from the line. This straight line is the regression line from which we can adjust its algebraic expressions, and it is written as $y = ax + b$, where a is the gradient and b is the y -intercept.

The regression line y on x is written as $y = \frac{\text{cov}(x, y)}{\sigma_x^2} x + \left(\bar{y} - \frac{\text{cov}(x, y)}{\sigma_x^2} \bar{x} \right)$

The above regression line can be re-written as

$$L_{y/x} \equiv y - \bar{y} = \frac{\text{cov}(x, y)}{\sigma_x^2} (x - \bar{x})$$

Note that **the regression line x on y** is $x = cy + d$, where c is the gradient of the line and

d is the x -intercept, it is given by $x = \frac{\text{cov}(x, y)}{\sigma_y^2} y + \left(\bar{x} - \frac{\text{cov}(x, y)}{\sigma_y^2} \bar{y} \right)$

This line is written as $L_{x/y} \equiv x - \bar{x} = \frac{\text{cov}(x, y)}{\sigma_y^2} (y - \bar{y})$

Shortcut method of finding regression line

To abbreviate the calculations, the two regression lines can be determined as follow:

a) The equation of the regression line of y on x is $L_{y/x} \equiv y = ax + b$ and the values of a and b are found by solving the simultaneous equations:

$$\begin{cases} \sum_{i=1}^k f_i y_i = a \sum_{i=1}^k f_i x_i + b n \\ \sum_{i=1}^k f_i x_i y_i = a \sum_{i=1}^k f_i x_i^2 + b \sum_{i=1}^k f_i x_i \end{cases}$$

These equations are called the **normal equations** for y on x .

b) The equation of the regression line of x on y is $L_{x/y} \equiv x = cy + d$ and the values of c and d are found by solving the simultaneous equations:

$$\begin{cases} \sum_{i=1}^k f_i x_i = c \sum_{i=1}^k f_i y_i + d n \\ \sum_{i=1}^k f_i x_i y_i = c \sum_{i=1}^k f_i y_i^2 + d \sum_{i=1}^k f_i y_i \end{cases}$$

These equations are called the **normal equations** for x on y .

Examples:

Let us consider the following bivariate data for this situation. A farmer sold hens in different periods and received interest in thousands of Rwandan francs. The following table shows the number x of hens sold and the corresponding interest y in thousands of Rwandan francs earned in different periods:

x (number of hens)	3	5	6	8	9	11
y (thousands Frw)	2	3	4	6	5	8

Find the equation of the regression line of y on x , and the equation of the regression line of x on y for the following data and estimate the value of y for $x = 4, x = 7, x = 16$ and the value of x for $y = 7, y = 9, y = 16$

Solution

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
3	2	-4	-2.6	16	6.76	10.4

5	3	-2	-1.6	4	2.56	3.2
6	4	-1	-0.6	1	0.36	0.6
8	6	1	1.4	1	1.96	1.4
9	5	2	0.4	4	0.16	0.8
11	8	4	3.4	16	11.56	13.6
$\sum_{i=1}^6 x_i = 42$	$\sum_{i=1}^6 y_i = 28$			$\sum_{i=1}^6 (x_i - \bar{x})^2 = 42$	$\sum_{i=1}^6 (y_i - \bar{y})^2 = 23.36$	$\sum_{i=1}^6 (x_i - \bar{x})(y_i - \bar{y}) = 30$

$$\bar{x} = \frac{42}{6} = 7, \bar{y} = \frac{28}{6} = 4.7$$

$$\text{cov}(x, y) = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y}) = \frac{30}{6} = 5$$

$$\sigma_x^2 = \frac{42}{6} = 7, \sigma_y^2 = \frac{23.36}{6} = 3.89$$

$$L_{y/x} \equiv y - \bar{y} = \frac{\text{cov}(x, y)}{\sigma_x^2} (x - \bar{x})$$

$$L_{y/x} \equiv y - 4.7 = \frac{5}{7} (x - 7)$$

The equation of the regression line of y on x is generally $L_{y/x} \equiv y - \bar{y} = \frac{\text{Cov}(x, y)}{\sigma_x^2} (x - \bar{x})$

$$\text{where } \text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^6 (x_i - \bar{x})(y_i - \bar{y}) = \frac{30}{6} = 5$$

In this case it becomes:

$$L_{y/x} \equiv y = \frac{5}{7} x - 0.3.$$

Using this equation, we can predict the interest to be earned when the farmer sells 4 hens, 7 hens or 16 hens.

x	4	7	16
y	2.5	4.7	11.1

By joining the obtained points (number of hens, interest), we can get the following figure: Since the data appears to be linearly related, we can find a straight-line model that fits the data better than all other possible straight-line models.

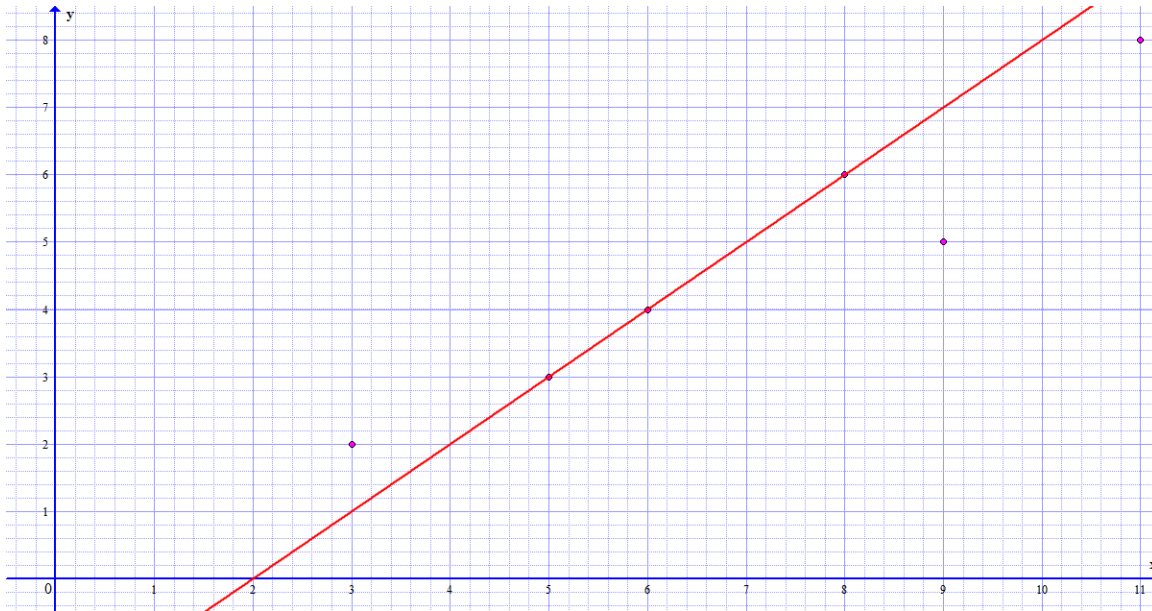


Figure: Scatter diagram

And

$$L_{x/y} \equiv x - \bar{x} = \frac{\text{cov}(x, y)}{\sigma_y^2} (y - \bar{y})$$

$$L_{x/y} \equiv x - 7 = \frac{5}{3.89} (y - 4.7)$$

Finally, the equation of the regression line of x on y is $L_{x/y} \equiv x = 1.3y + 1$

On the other hand, when we have the interest y , we can predict the number of hens sold using the equation of the regression line of x on y . This is $L_{x/y} \equiv x - \bar{x} = \frac{\text{Cov}(x, y)}{\sigma_y^2} (y - \bar{y})$.

Hence, $L_{x/y} \equiv x = 1.3y + 1$.

Using this equation, we can predict the number of hens to be sold if the farmer received the interest of 7000Frw, 9000Frw or 16000Frw

y	7	9	16
x	10.1	12.7	21.8

Note: As x represents the number of hens, it should be an integer.

Therefore, the interest of 7000Frw, 9000Frw and 16000Frw correspond to 10 hens, 13 hens and 22 hens respectively.

Alternative method

x	y	x^2	y^2	xy
3	2	9	4	6
5	3	25	9	15
6	4	36	16	24
8	6	64	36	48
9	5	81	25	45
11	8	121	64	88
$\sum_{i=1}^6 x_i = 42$	$\sum_{i=1}^6 y_i = 28$	$\sum_{i=1}^6 x_i^2 = 336$	$\sum_{i=1}^6 y_i^2 = 154$	$\sum_{i=1}^6 x_i y_i = 226$

$$L_{y/x} \equiv y = ax + b$$

$$\begin{cases} \sum_{i=1}^k f_i y_i = a \sum_{i=1}^k f_i x_i + b n \\ \sum_{i=1}^k f_i x_i y_i = a \sum_{i=1}^k f_i x_i^2 + b \sum_{i=1}^k f_i x_i \end{cases}$$

$$\begin{cases} 28 = 42a + 6b \\ 226 = 336a + 42b \end{cases} \Leftrightarrow \begin{cases} a = \frac{5}{7} \\ b = -0.3 \end{cases}$$

Thus, the line of y on x is $L_{y/x} \equiv y = \frac{5}{7}x - 0.3$

$$x = 4 \Rightarrow y = 2.5$$

$$x = 7 \Rightarrow y = 4.7$$

$$x = 16 \Rightarrow y = 11.1$$

$$L_{x/y} \equiv x = cy + d$$

$$\begin{cases} \sum_{i=1}^k f_i x_i = c \sum_{i=1}^k f_i y_i + d n \\ \sum_{i=1}^k f_i x_i y_i = c \sum_{i=1}^k f_i y_i^2 + d \sum_{i=1}^k f_i y_i \end{cases}$$

$$\begin{cases} 42 = 28c + 6d \\ 226 = 154c + 28d \end{cases} \Leftrightarrow \begin{cases} c = 1.3 \\ d = 1 \end{cases}$$

Thus, the line of x on y is

$$L_{x/y} \equiv x = 1.3y + 1$$

$$y = 7 \Rightarrow x = 10.1$$

$$y = 9 \Rightarrow x = 12.7$$

$$y = 16 \Rightarrow x = 21.8$$



Application activity 9.5:

1. Consider the following table

x	y
60	3.1
61	3.6
62	3.8
63	4
65	4.1

- Find the regression line of y on x
- Calculate the approximate y value for the variable $x = 64$

2. The values of two variables x and y are distributed according to the following table

$x \backslash y$	100	50	25
14	1	1	0
18	2	3	0
22	0	1	2

Find the regression lines.

9.6 Interpretation of statistical data



Activity 9.6

Explain in your own words how statistics, especially bivariate statistics, can be used in our daily life.



Content summary

Bivariate statistics can help in prediction of a value for one variable if we know the value of the other.

Examples

1) One measure of personal fitness is the time taken for an individual's pulse rate to return to normal after strenuous exercise, the greater the fitness, the shorter the time. Following a short program of strenuous exercise Norman recorded his pulse rates P at time t minutes after he had stopped exercising. Norman's results are given in the table below.

t	0.5	1.0	1.5	2.0	3.0	4.0	5.0
P	125	113	102	94	81	83	71

Estimate Norman's pulse rate 2.5 minutes after stopping the exercise program.

Solution

t	P	t^2	P^2	tP
0.5	125	0.25	15625	62.5
1	113	1	12769	113
1.5	102	2.25	10404	153
2	94	4	8836	188
3	81	9	6561	243
4	83	16	6889	332
5	71	25	5041	355
$\sum_{i=1}^7 t_i = 17$	$\sum_{i=1}^7 P_i = 669$	$\sum_{i=1}^7 t_i^2 = 57.5$	$\sum_{i=1}^7 P_i^2 = 66125$	$\sum_{i=1}^7 t_i P_i = 1446.5$

We need the line $P = at + b$

Use the formula

$$\begin{cases} \sum_{i=1}^7 P_i = a \sum_{i=1}^7 t_i + bn \\ \sum_{i=1}^7 t_i P_i = a \sum_{i=1}^7 t_i^2 + b \sum_{i=1}^7 t_i \end{cases}$$

We have:
$$\begin{cases} 669 = 17a + 7b \\ 1446.5 = 57.5a + 17b \end{cases}$$

Solving we have
$$\begin{cases} a = -11 \\ b = 122.3 \end{cases}$$

Then $P = -11t + 122.3$

So, the Norman's pulse rate 2.5 minutes after stopping the exercise program is estimated to be $P = -11(2.5) + 122.3$ or 94.8.

2) Collect data on the mass and the height of 10 people. If X represents the mass and Y represents the height.

a) Organise the data in the table

b) Plot the scatter diagram

c) Calculate the mean of x , the mean of y , the variance of x and the variance of y

d) Calculate the $\text{cov}(x, y)$, the correlation coefficient (r) then interpret the relationship between the mass and the height of these people

e) Establish the equation of the regression line y in function of x .



Application activity 9.6:

An old film is treated with a chemical in order to improve the contrast. Preliminary tests on nine samples drawn from a segment of the film produced the following results.

Sample	A	B	C	D	E	F	G	H	I
x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
y	49	60	66	62	72	64	89	90	90

The quantity x is a measure of the amount of chemical applied, and y is the contrast index, which takes values between 0 (no contrasts) and 100 (maximum contrast).

- Plot a scatter diagram to illustrate the data.
- It is subsequently discovered that one of the samples of film was damaged and produced an incorrect result. State which sample you think this was.

In all subsequent calculations, this incorrect sample was ignored. The remaining data can be summarized as follows: $\sum x = 23.5$, $\sum y = 584$, $\sum x^2 = 83.75$, $\sum y^2 = 44622$, $\sum xy = 1883$, $n = 8$.

- Calculate the product moment correlation coefficient,
- State with a reason whether it is sensible to conclude from your answer to part (iii) that x and y are linearly related.
- The line of regression of y on x has equation $y = a + bx$. Calculate the value of a and b , each correct to three significant figures.
- Use your regression line to estimate what the contrast index corresponding to the damaged piece of film would have been if the piece has been undamaged.

Vii) State with a reason, whether it would be sensible to use your regression equation to estimate the contrast index when the quantity of chemical applied to the film is zero.



END OF UNIT ASSESSMENT

A. Assessing knowledge and understanding:

1) Answer by true or false:

In a set of paired data, the independent variable is the one that is plotted on the vertical axis (y -axis), and its value is presumed to be affected by the other variable.

2) A researcher is studying the relationship between the number of hours a student studies (X) and their final exam score (Y). After collecting data, she calculates a correlation coefficient of $r = 0.92$. Which of the following is the most appropriate conclusion?

- A) Studying few hours causes higher exam scores.
- B) There is a strong, positive linear relationship between hours studied and exam scores.
- C) 92% of the variation in exam scores is explained by few hours studied.
- D) The data points are randomly scattered with no discernible pattern.

3) Match the term on the left with its correct description on the right.

Column A (Term)	Column B (Description)
1. Correlation Coefficient (r)	A. A measure of the direction and strength of a linear relationship.
2. Extrapolation	B. The variable that is plotted on the x-axis in an experiment.
3. Independent Variable	C. Using a regression model to make a prediction for an X-value far outside the range of the collected data. The action of concluding something by assuming that existing trends will remain applicable
4. Interpolation	D. a process of approximating the value of a given function at a given set of discrete points.

B. Assessing skills

4) A line of best fit is drawn for the data and passes through two specific points: (63, 3385) and (74, 3280).

- a) Calculate the slope of this line.
- b) Assuming this linear trend continues, use this line to predict the Y value when $X = 80$.

5) The scores awarded by two judges at an ice-skating competition are shown in the table.

Competitor	A	B	C	D	E	F	G	H	I	J
Judge A	5	6.5	8	9	4	2.5	7	5	6	3
Judge b	6	7	8.5	9	5	4	7.5	5	7	4.5

- Construct a scatter diagram for this data with Judge A's scores on the horizontal axis and Judge B's scores on the vertical axis.
- Copy and complete the following comments about the scatter diagram: There appears to be _____ correlation between Judge A's scores and Judge B's scores. This means that as Judge A's scores increase, Judge B's scores _____

6) The following results were obtained from line-ups in Mathematics and Physics examinations:

	Mathematics (x)	Physics (y)
Mean	47.5	39.5
Standard deviation	16.8	10.8

The coefficient of correlation $r = 0.95$.

Find both equations of the regression lines. Also estimate the value of y for $x=30$.

7) For a set of 20 pairs of observations of the variables x and y , it is known that $\sum_{i=1}^k f_i x_i = 250$, $\sum_{i=1}^k f_i y_i = 140$, and that the regression line of y on x passes through $(15,10)$.

Find the equation of that regression line and use it to estimate y when $x = 10$.

C. Assessing attitudes and values

8) Read the statement below and indicate your level of agreement on a scale of 1 to 5, where:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Statement: "Understanding the relationship between two variables (like the one in our dataset) is a useful skill for making informed decisions in my everyday life and future career." Provide a brief explanation for your rating.

Sample Answer (for a rating of 4 or 5):

Rating: 5 (Strongly Agree)

Explanation: I believe this skill is very important. In daily life, we are constantly bombarded

with claims about what causes what (e.g., "this product leads to better health"). Knowing how to interpret correlations helps me be a more critical consumer of information. In my future career in business, understanding relationships between variables like advertising spending and sales revenue would be essential for making smart choices.

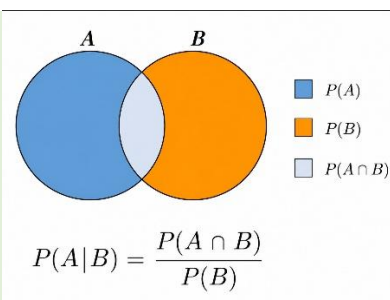
9) A company wants to prove that its tutoring program is highly effective. They collect data on the number of tutoring sessions students attended (X) and their final grades (Y). The correlation is positive but moderate ($r = 0.4$). In their advertising, the company publishes a graph that only shows the data for the students who attended the most sessions and uses a truncated y-axis (starting at 80% instead of 0%) to make the increase in grades look much steeper than it is.

Question: What is your opinion on the company's actions? Do you think this is an acceptable way to present data? Why or why not?

Sample Answer:

I think the company's actions are **unacceptable and unethical**. While they are using real data, they are deliberately manipulating the way it is presented to deceive the public.

- **Excluding data** (only showing the most successful students) hides the full picture and gives a false impression of the program's overall effectiveness.
- **Manipulating the scale** of the graph exaggerates the small improvement, making it look far more significant than it really is.
This is a misuse of statistics because its goal is to mislead rather than to inform. Ethical data presentation should be transparent, show the full range of data, and use appropriate scales so people can draw their own, accurate conclusions.
- own, accurate conclusions.



UNIT 10: ELEMENTARY AND CONDITIONAL PROBABILITY

Key unit competence: Apply counting techniques and probability rules in determining and interpreting the likelihood of events under equally likely assumptions.

Learning objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain key terms: experiment, outcome, sample space, event (simple and compound).
- Identify and state the three basic axioms of probability.
- Explain mutually exclusive (disjoint) and independent events.
- Explain the difference between independent and dependent events.
- Identify the concept of conditional probability as a change in the sample space.
- Apply the formulas for the addition rule, multiplication rule, and conditional probability.
- Identify the purpose and application of Bayes' Theorem.

Skills:

- Calculate the probability of an event using the classical (theoretical) approach.
 - Apply the addition rule for mutually exclusive and non-mutually exclusive (inclusive) events.
 - Apply the multiplication rule for independent and dependent events.
 - Construct and use tree diagrams to model multi-stage experiments and calculate probabilities.
 - Calculate conditional probabilities using the formula.
 - Use two-way tables and Venn diagrams to organize information and solve probability problems.
 - Apply Bayes' Theorem to solve practical problems involving conditional probabilities.
- Interpret probability values in context, assessing likelihood and risk

Attitudes and values

- Appreciate the role of probability in quantifying uncertainty and making informed decisions in daily life.
- Cultivate a critical attitude towards probabilistic claims in media, gambling, and risk assessment.
- Foster collaborative skills through group problem-solving activities.
- Value accuracy in calculation and clarity in the interpretation of result.

Introduction

This unit covers the basic concepts and principles of probability, which is the study of chance and uncertainty. You will explore the meaning of probability and its importance in everyday life situations such as games, weather prediction, and decision making. The unit introduces key concepts including experiments, sample space, and events, and explains how to determine the probability of simple events. You will study basic probability rules, including the probability of mutually exclusive (incompatible) events and inclusive events using the addition rule.

The unit also introduces the concept of independent events and how to calculate their probabilities using the multiplication rule. In addition, you will be introduced to tree diagrams as a tool for representing possible outcomes and solving probability problems. Emphasis is placed on applying counting techniques and probability rules to determine and interpret the likelihood of events under equally likely assumptions.



Introductory activity:

The teacher brings a bag containing colored balls: 3 red balls, 2 blue balls and 1 green ball and asks one student to pick one ball without looking.

1. What are the possible colors the student can pick?
2. Which color do you think is most likely to be picked? Why?
3. Which color is least likely?
4. Can you list all possible outcomes of this experiment?
5. How can we measure the “chance” of picking a red ball?

10.1 Concept of probability and examples



Activity 10.1

Consider the deck of 52 playing cards.

	Ace	2	3	4	5	6	7	9	10	Jack	Queen	King
Clubs:												
Diamonds:												
Hearts:												
Spades:												

Suppose that you are choosing one card.

- a) How many possibilities do you have for the cards to be chosen? What is the chance of choosing one card?
- b) How many possibilities do you have for the kings to be chosen? What is the related chance?
- c) How many possibilities do you have for the ace of hearts to be chosen? What is the related chance?



Content Summary

Probability is the possibility (or chance) that something will happen. The concept of probability can be illustrated in the context of a game of 52 playing cards composed of four suits of 13 cards each. If a player selects a card at random (by simple random sampling), then each card has the same chance or same probability of being selected. The chance of selecting one card is $1/52$.

We cannot say beforehand whether the card will be red or black. That depends on chance. Such phenomena are called probabilistic. The theory of probability is concerned with this type of phenomena.

Probability is a concept which numerically measures the degree of uncertainty and therefore of certainty of occurrence of events.

Example 1

If a dotted die is rolled and the number of dots that show up is noted, what is the probability of getting 5 dots?



Solution: If a die is rolled until a “6” is obtained, and the number rolled before getting first “6” is counted, then we have that the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, $n(\Omega) = 6$. There are 6 possible outcomes. If A is the event: getting 5, $A = \{5\}$, and $n(A) = 1$. The chance of getting 5 is the probability of getting 5. That is $p(A) = \frac{n(A)}{n(\Omega)} = \frac{1}{6}$

Example 2

Consider the experiment that consists in rolling a die and recording the number that shows up. Let A be the event “the even number is shown” and B be the event “the odd number less than 5 is shown”. Define the events A and B. and find their probability

Solution

We have the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$.

$A = \{2, 4, 6\}$, $n(A) = 3$. $B = \{1, 3\}$ and $n(B) = 2$

$$p(A) = \frac{n(A)}{n(\Omega)} = \frac{3}{6} = 0.5 \text{ and } p(B) = \frac{n(B)}{n(\Omega)} = \frac{2}{6} = \frac{1}{3}$$

a) Experiment: An action or process that leads to one or more possible outcomes.

Example: Tossing a coin, Rolling a die, Picking a card from a deck.

In all these cases there are a number of possible results (outcomes) which can occur but there is an uncertainty as to which one of them will actually occur. Each performance in a random experiment is called a **trial**. The result of a trial in a random experiment is called an **outcome**, an elementary event, or a sample point.

c) Sample Space (S)

The set of all possible outcomes of an experiment.

Example: Tossing a coin:

$S = \{\text{Head, Tail}\}$.

A sample space may be discrete or continuous.

Discrete sample space:

- It can be finite: The number of possible outcomes is **finite**.
- It can be infinite: The number of possible outcomes is **countable infinite**, which means that there is an infinite number of possible outcomes, but the outcomes can be put in a one-to-one correspondence with the positive integers.

Example

If a die is rolled and the number that show up is noted, then $S = \{1, 2, 3, \dots, 6\}$.



If a die is rolled until a “6” is obtained, and the number of rolls made before getting first “6” is **counted**, then we have that $S = \{0, 1, 2, 3, \dots\}$.

Continuous sample space: If the sample space contains one or more intervals, the sample space is then **uncountable infinite**.

Example

A die is rolled until a “6” is obtained and the time needed to get this first “6” is recorded. In this case, we have that $S = \{t \in \mathbb{R}^2 : t > 0\} = (0, \infty)$.

Sample Space S : All possible amounts of rainfall tomorrow in millimetres (e.g., from 0 to 10 mm).

Event H : "It rains more than 1mm". $H = \{x \in S : x > 1\}$

Event I : "It does not rain". $I = \{\phi\}$

More special events:

- **Sure Event / Certain Event:** The event that is equal to the entire sample space S . It is guaranteed to occur.
- **Impossible event:** The empty set ϕ . It cannot occur.
- **Simple event:** An event that consists of exactly one outcome.
- **Compound event:** An event that consists of two or more outcomes.

d) Event (I) : An event is a set of outcomes of an experiment; it is a subset of the sample space.

If S is the **sample space** of a random experiment, then an **event E** is a subset of S . Mathematically: $E \subseteq S$.

We say that an event **occurs** if the outcome of the experiment is an element of E .

The null set ϕ is thus an event known as the **impossible event**. The sample space S corresponds to the **sure event**.

In particular, every elementary outcome is an event, and so, is the sample space itself.

Note:

- An elementary outcome is sometimes called a **simple event**, whereas a **compound event** is made up of at least two elementary outcomes.
- To be precise we should distinguish between the elementary outcome w , which is an element of S and the elementary event $\{w\} \subset S$
- The events are denoted by capital letters such as A, B, C and so on.

Examples:

1. Toss a coin: Event “getting Head” $\rightarrow \{\text{Head}\}$
2. Roll a die: Event “getting an even number” $\rightarrow \{2, 4, 6\}$
3. Pick a card: Event “drawing a Queen” $\rightarrow \{\text{Queen of Hearts, Queen of Diamonds, Queen of Clubs, Queen of Spades}\}$.



Application activity 10.1

- 1) **Comment by True or False**
 - a) Probability can never be greater than 1.
 - b) The sample space contains all possible outcomes of an experiment.
 - c) Picking a red ticket is an impossible even.
- 2) Define the following terms:
 - a) **Experiment**
 - b) **Sample Space**
 - c) **Event**
- 3) If two students pick tickets one after another without replacing the first ticket, find the probability that both pick a blue ticket.
- 4) Organize the sample space for tossing a coin twice. Then find the probability of getting exactly one Head.

10.2 Probability of an Event



Activity 10.2

During a school raffle, the teacher has a box containing **5 red tickets, 3 blue tickets, and 2 green tickets**. The students are asked to pick one ticket without looking.

Dialogue:

- **Alice:** “I hope I pick a red ticket!”
- **Eric:** “But you cannot be sure. Maybe you will pick blue or green.”
- **Teacher:** “Exactly! But we can calculate the **chance or probability** of picking each color. That is what we study in probability.”

Questions

1. How many possible outcomes are there when picking one ticket?
2. Which outcomes are favorable if you want a red ticket?
3. Can you predict the exact color of the ticket before picking it? Why?



Content Summary

The probability of an event $A \subset \Omega$, is a real number obtained by applying to A the function P defined by:

$$P(A) = \frac{\text{Number of outcomes in } A}{\text{Total number of outcomes in the sample space}} = \frac{n(A)}{n(\Omega)}$$

This is the formula for the classical probability; It uses the **sample space** Ω .

Probability can be expressed as a fraction, decimal or percentage.

Example

1) For a card drawn from an ordinary deck, find the probability of getting a queen.

Solution: There are 52 cards in a deck and there are 4 queens,

$$P(\text{queen}) = \frac{n(\text{queen})}{n(\Omega)} = \frac{4}{52} = \frac{1}{13} = 0.076 = \frac{7.6}{100}$$

2) A letter is chosen from the letters of the word “**MATHEMATICS**”. What is the probability that the letter chosen is an “A”?

Solution

Since two of the eleven letters are A's, we have two favourable outcomes.

There are eleven letters, so we have 11 possible outcomes.

Thus, the probability of choosing a letter A is

3) A survey involving 120 people about their preferred breakfast showed that;
55 eat eggs for breakfast, 40 drink juice for breakfast and 25 eat both eggs and drink juice for breakfast.

(a) Represent the information on a Venn diagram.

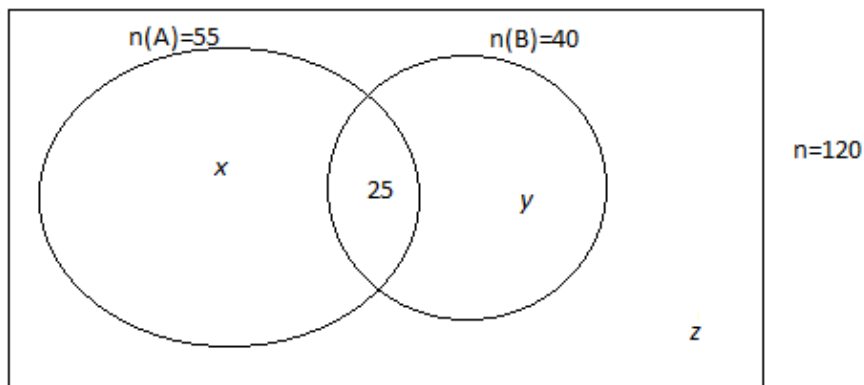
(b) Calculate the following probabilities.

(i) A person selected at random takes only one type for breakfast.

(ii) A person selected at random takes neither eggs nor juice for breakfast.

Solution:

a) Let x = those who eat eggs, y = those who take juice and z represents those who did not take anything.



Here, we can now solve for the number of people who take eggs only:

$$x = 55 - 25 = 30.$$

So, 30 people took Eggs only.

$$\text{Also, } y = 40 - 25 = 15.$$

So, 15 people took Juice only.

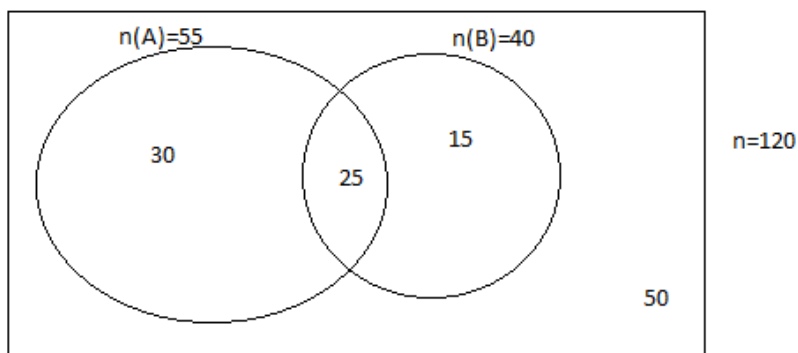
$$\text{Hence } 30 + 25 + 15 + z = 120$$

$$z = 120 - (30 + 25 + 15)$$

$$z = 120 - 70 = 50.$$

The number of people who did not take anything for breakfast is 50.

Therefore,



b) i) Let E be the event “A person takes one type of breakfast”, $P(E) = \frac{45}{120}$

ii) $A \cup B$ is the event “A person takes eggs or juice for breakfast”

then $(A \cup B)'$ is the event “A person takes neither eggs nor juice for breakfast”

for which the probability is $P[(A \cup B)'] = \frac{50}{120}$.

Properties of probabilities

Referring to rules mentioned above, probabilities assigned to events on a sample space Ω can be summarized in the following properties:

a) $P(E) \geq 0$ for every $E \subset \Omega$

b) $P(\Omega) = 1$

c) If $E \subset F \subset \Omega$, then $P(E) \leq P(F)$

d) If A and B are disjoint subsets of Ω , then $P(A \cup B) = P(A) + P(B)$

e) $P(A') = 1 - P(A)$ for every $A \subset \Omega$.

Formula for empirical (classical) probability

Given a frequency distribution, the probability of an event being in a given class is

$$P(A) = \frac{\text{frequency for the class}}{\text{Total frequency in the distribution}} = \frac{f}{n}$$

This probability is called empirical probability and is based on observation.

Examples

1) A researcher asked 25 staff of an institution if they liked the way their breakfast is prepared. The responses were classified as “Yes”, “No”, and “Undecided”. The results were categorized in a frequency distribution as follows:

Response	Frequency
----------	-----------

Yes	15
No	8
Undecided	2
Total	25

- What is the probability of selecting a person who disliked the way the breakfast is prepared?
- What is the probability of selecting a person who liked the way the breakfast is prepared?
- What is the probability of selecting a person who neither like nor disliked the way the breakfast is prepared?

Solution

- $P(\text{non}) = \frac{f}{n} = \frac{8}{25}$
- $P(\text{yes}) = \frac{f}{n} = \frac{15}{25}$
- $P(\text{undecided}) = \frac{f}{n} = \frac{2}{25}$



Application activity 10.2

- True or False:
 - The probability of an impossible event is 1.
 - Probability of an event ranges from 0 to 1.
 - All outcomes in an experiment have the same probability.
- A bag contains 4 red, 3 blue, and 5 yellow marbles. One marble is drawn at random. Find the probability of picking:
 - A red marble.
 - A yellow marble.
- A die is rolled. Find the probability of:
 - Getting a number less than 4
 - Getting a number greater than 4

10.3 Basic probability rules



Activity 10.3

At the school snack shop, there is a basket containing **10 chocolates, 6 biscuits, and 4 candies**. The teacher asks a student to pick **one snack without looking**.

Dialogue

- Lina:** "I hope I get a chocolate!"
- Sam:** "But we don't know which snack we will pick. Maybe a biscuit or candy."

- **Teacher:** “Exactly! But we can use **probability rules** to figure out how likely it is to pick each snack.”

Question

1. What is the probability of picking a chocolate?
2. What is the probability of picking a candy?
3. What is the probability of **not picking a biscuit**?



Content Summary

Probability rules help us **calculate the likelihood of events** in a consistent way. They are used for **single events** or **combined events** like “and,” “or,” and “not” scenarios.

These rules are important for **predicting outcomes** in games, experiments, and real-life situations.

Basic Rules

Rule 1: Probability Range

$$0 \leq P(E) \leq 1$$

- ❖ Probability of any event is **never negative** and **never greater than 1**.
- ❖ $P(E) = 0 \rightarrow$ impossible event
- ❖ $P(E) = 1 \rightarrow$ certain event

Example:

- Rolling a die: Event “getting 7 ” $\rightarrow P(E) = 0$
- Tossing a coin: Event “getting Head” $\rightarrow P(E) = \frac{1}{2}$

The event that cannot occur is an impossible event $A = \emptyset$, and if $A = \emptyset$ then $P(A) = 0$

Example

When a single die is rolled, find the probability of getting a 9. $A = \emptyset$

Solution

$\Omega = \{1, 2, 3, 4, 5, 6\}$, it is impossible to get a 9. $P(\text{getting a 9}) = 0$.

Rule 2: Probability of the Sample Space

$$P(S) = 1$$

The sum of probabilities of all possible outcomes in the sample space = 1.

Example:

Toss a coin: $S = \{H, T\}$

$$P(H) + P(T) = \frac{1}{2} + \frac{1}{2} = 1$$

Rule 3: Complementary Rule

When E and E' are complementary events, $P(E) = 1 - P(E')$

Consider two different events, A and B , which may occur when an experiment is performed.

- The event $A \cup B$ is the event which occurs if A or B or both A and B occur, i.e., at least one of A and B occurs.
- The event $A \cap B$ is the event which occurs if A and B occur.
- The event $A - B$ is the event which occurs when A occurs and B does not occur.
- The event A' is the event which occurs when A does not occur $P(\text{not } E) = 1 - P(E)$

The probability that an event **does not occur** is 1 minus the probability that it occurs.

Example:

Rolling a die: Event E = “getting 3”

$$P(\text{not } 3) = 1 - \frac{1}{6} = \frac{5}{6}$$

Example

1. When a die is rolled, the event E of getting odd number is that $E = \{1, 3, 5\}$ and $P(E) = \frac{3}{6} = \frac{1}{2}$

The event F of not getting an odd number is a complement of E . $F = E' = \{2, 4, 6\}$. As

$$P(\Omega) = 1, P(F) = 1 - \frac{1}{2} = \frac{1}{2}.$$

2. If the probability that a person lives in an industrialized country of the world is $\frac{1}{4}$, find the probability that a person does not live in an industrialized country.

Solution

$P(\text{not living in an industrialized country}) = 1 - P(\text{living in an industrialized country})$

$$P = 1 - \frac{1}{5} = \frac{4}{5}.$$

Rule 4: Addition Rule (Mutually Exclusive Events)

If events A and B **cannot occur together** (mutually exclusive):

$$P(A \text{ or } B) = P(A) + P(B)$$

Example: Roll a die: Event $A = \{2\}$, Event $B = \{5\} \rightarrow$ mutually exclusive

$$P(2 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

Example: When you roll a fair die, there are six possible outcomes. Each outcome in the sample space has probability of $\frac{1}{6}$. Hence, the sum of the probabilities of all outcomes is given by

$$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1$$

Rule 5: Multiplication Rule (Independent Events)

If events A and B are **independent**:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example:

- Toss a coin and roll a die:
 - Event $A = \text{Head} \rightarrow P(A) = \frac{1}{2}$
 - Event $B = 4 \rightarrow P(B) = \frac{1}{6}$
 - $P(\text{Head and } 4) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$



Application activity 10.3

1) True or False:

- a) Probability of any event cannot be greater than 1
- b) Events A and B can be independent and mutually exclusive at the same time
- c) The sum of probabilities in a sample space is 1

2) A coin is tossed twice. Find:

- a) Probability of two Heads
- b) Probability of at least one Tail

3) A bag has 5 red and 3 blue balls. One ball is drawn, replaced, then another is drawn. Find:

- a) Red then Blue
- b) Two Reds

10.4 Probability of mutually exclusive (incompatible) events



Activity 10.4

Given a deck of 52 playing cards. If a card is drawn from that pack, Find

- (a) the probability $P(A)$ that the card is a club
- (b) the probability $P(B)$ that the card is a diamond

- (c) the probability P that the card is a club or a diamond.
 (d) Compare the probability P to the $P(A)$ and $P(B)$.



Content Summary

If two events A and B are such that $A \cup B = \Omega$ and $P(A \cup B) = 1$, these two events are said to be **exhaustive**.

- When $A \cap B = \phi$, A and B are **mutually exclusive** or disjoint and $P(A \cup B) = P(A) + P(B)$

This is called addition **rule** for mutually exclusive events.

- When two events A and B are **not mutually exclusive** $A \cap B \neq \phi$.

Generally, given a finite sample space, say $\Omega = \{a_1, a_2, a_3, \dots, a_n\}$, we can find a finite probability by assigning to each point $a_i \in \Omega$ a real number P_i , called the probability of a_i , satisfying the following:

a) $P_i \geq 0$ for all integers i , $1 \leq i \leq n$;

b) $\sum_{i=1}^n P_i = 1$

If E is an event, then the probability $P(E)$ is defined to be the sum of the probabilities of the sample points in E .

Example

2) A coin is weighted so that heads is three times as likely to appear as tails.

Find $P(H)$ and $P(T)$

Solution

Let $P(T) = p_1$, then $P(H) = 3p_1$.

But $P(H) + P(T) = 1$

Therefore $3p_1 + p_1 = 1 \Leftrightarrow 4p_1 = 1 \Rightarrow p_1 = \frac{1}{4}$

Thus, $P(H) = \frac{3}{4}$ and $P(T) = \frac{1}{4}$.

3) Events A and B are such that they are both mutually exclusive and exhaustive. Find the relation between these two events.

Solution:

If A and B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$

If A and B are mutually exclusive then $P(A \cup B) = 1$,

therefore, $P(A) + P(B) = 1$ and $P(B) = 1 - P(A)$

But, $P(A') = 1 - P(A)$, therefore, $P(B) = P(A')$ i.e. $B = A'$. Similarly, $A = B'$

Thus, if events A and B are such that they are both mutually exclusive and exhaustive, then they are complementary.

4) A pen is drawn from a basket containing 10 pens of which 5 are red and 3 are black. If A is the event: “a pen is red” and B is the event: “a pen is black”, find $P(A), P(B), P(A \cup B)$.

Solution

There are 5 red pens, then $P(A) = \frac{5}{10} = \frac{1}{2}$

There are 3 black pens, then $P(B) = \frac{3}{10}$

Since the pen cannot be red and black at the same time, then $A \cap B = \emptyset$ and two events are mutually exclusive so

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= \frac{1}{2} + \frac{3}{10} = \frac{4}{5} \end{aligned}$$

Note that if $A = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$, where $A_1, A_2, A_3, \dots, A_n$ are incompatible events,

then we may write that $P(A) = \sum_{i=1}^n P(A_i)$ for $n = 2, 3, \dots$

Therefore, **the addition rule for exclusive events can be extended to any number of exclusive events and be written as follows:**

$$P(A_1 \text{ or } A_2 \text{ or } A_3 \dots A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n)$$

Example

In race in which there are no dead hearts, the probability that John wins is 0.3, the probability that Paul wins is 0.2 and the probability that Marks wins is 0.4. Find the probability that

- John or Mark wins
- John or Paul or Mark wins
- Someone else wins.

Solution

Since only one person wins, the events are mutually exclusive;

$$\text{a) } P(\text{John or Mark wins}) = P(\text{John wins}) + P(\text{Mark wins}) = 0.3 + 0.4 = 0.7$$

$$\begin{aligned} P(\text{John or Paul or Mark wins}) &= P(\text{John wins}) + P(\text{Paul wins}) + P(\text{Mark wins}) \\ &= 0.3 + 0.2 + 0.4 = 0.9 \end{aligned}$$

$$\text{b) } P(\text{someone else wins}) = 1 - 0.9 = 0.1$$



Application activity 10.4

- 1) True or False:
 - a) If two events are mutually exclusive, they can occur at the same time.
 - b) If two events are exhaustive, their probabilities add up to 1.
 - c) Mutually exclusive and exhaustive events are complementary.
- 2) If events A and B are mutually exclusive, then:
 - a) $P(A \cap B) = 1$
 - b) $P(A \cap B) = 0$
 - c) $P(A \cup B) = 0$
 - d) $P(A) = P(B)$

3) A die is thrown once.

Let A = “number less than 5” and B = “number greater than 3

Find:

- a) $P(A)$
- b) $P(B)$
- c) $P(A \cap B)$
- d) $P(A \cup B)$

10.5 Probability non-mutually exclusive events and additive rule



Activity 10.5

- 1) A die is rolled once.
 - a) What is the probability of getting an even number?
 - b) What is the probability of getting a number greater than 3?
 - c) Can a number be both even and greater than 3? Give an example.
 - d) How would you find the probability of getting an even number or a number greater than 3?
- 2) In a class, some students play football and some play basketball.
 - a) Can a student play both games?
 - b) If yes, how do you calculate the probability of selecting a student who plays football or basketball?



Content Summary

Events are called **non-mutually exclusive** if they **can occur at the same time**.

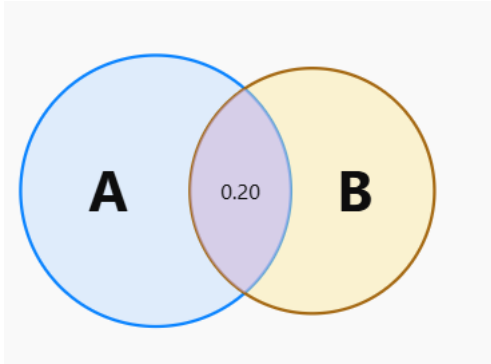
- These events **overlap**.

- Some outcomes belong to both events.

Examples:

- Getting a number that is **even and greater than 3**
- A student who plays **both football and basketball**

For any two events A and B : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Where:

- $P(A)$: Probability of event A
- $P(B)$: Probability of event B
- $P(A \cap B)$: Probability that **both A and B occur**

The intersection is subtracted to **avoid double counting**.

Important Observations

- If events **overlap**, use the full formula.
- If events are **mutually exclusive**, then: $P(A \cap B) = 0$

So, the formula becomes: $P(A \cup B) = P(A) + P(B)$

Example 1

A card is drawn from a deck of 52 playing cards. If A is an event of drawing an ace and B is an event of drawing a spade. Find $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$

Solution

There are 4 aces, then $P(A) = \frac{4}{52} = \frac{1}{13}$

There are 13 spades, then $P(B) = \frac{13}{52} = \frac{1}{4}$

There is 1 ace of spades, then $P(A \cap B) = \frac{1}{52}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{13} + \frac{1}{4} - \frac{1}{52} = \frac{4}{13}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{4}{13}$$

Example

- A die is thrown once. Let A be the event: “the number obtained is less than 5” and B be the event: “the number obtained is greater than 3”. Find probability of $A \cup B$.

• Solution:

Here $A = \{1, 2, 3, 4\}$ and $B = \{4, 5, 6\}$, then $A \cup B = \{1, 2, 3, 4, 5, 6\}$ and then

$$P(A \cup B) = P(\Omega) = 1[10pt]$$

$$\text{Or } P(A) = \frac{4}{6}, P(B) = \frac{3}{6}, A \cap B = \{4\}, \text{ then } P(A \cap B) = \frac{1}{6}$$

$$\text{Therefore, } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{4}{6} + \frac{3}{6} - \frac{1}{6} = 1$$



Application activity 10.5

1) True or False

- a) Two inclusive events cannot happen at the same time.
- b) $P(A \cap B)$ represents the probability that both events occur.
- c) In inclusive events, we subtract the intersection to avoid double counting.
- d) $P(A \cup B) = P(A) + P(B)$ always holds for inclusive events.
- e) If two events overlap, they are inclusive.

2) Which formula represents inclusive events?

- a) $P(A \cup B) = P(A) - P(B)$
- b) $P(A \cup B) = P(A) + P(B)$
- c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- d) $P(A \cap B) = P(A) + P(B)$

3) If two events are inclusive, then:

- a. They cannot occur together
- b. They are mutually exclusive
- c. They can occur together
- d. Their probabilities must be equal

4) Given that $P(A) = 0.7$, $P(B) = 0.6$, and $P(A \cap B) = 0.4$

- (a) Find $P(A \cup B)$
- (b) Interpret your answer

10.6 Probability of independent events and examples



Activity 10.6

A coin is tossed and a die is rolled at the same time.

1. List the possible outcomes of this experiment.
2. Find:
 - (a) The probability of getting a Head
 - (b) The probability of getting a 4
3. Find the probability of getting a Head and a 4.
4. Does getting a Head affect the number obtained on the die? Explain.
5. What can you conclude about events that do not influence each other?



Content Summary

In probability, two events A and B are **independent** if the occurrence (or non-occurrence) of one event does **not** affect the probability of the other event occurring.

In this case, the probability of A given B is the same as the probability of A alone. It is written as $P(A|B) = P(A)$.

Probability of independent events

$P(A \cap B)$ is the probability that both A and B happen. When two events are independent, the probability of both occurring is $P(A \text{ and } B) = P(A) \times P(B)$ or $P(A \cap B) = P(A) \times P(B)$

This is the **multiplication rule** of independent events or **and rule** for independent events.

Example

A fair die is thrown twice. Find the probability that two fives are thrown.

Solution

On one throw, $P(5) = \frac{1}{6}$

On two throws,

$$P(5_1 \text{ and } 5_2) = P(5_1 \times 5_2) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

$$P(\text{two fives are thrown}) = \frac{1}{36}$$

Example

A factory runs two machines. The first machine operates for 80% of the time while the second machine operates for 60% of the time and at least one machine operates for 92% of the time. Do these two machines operate independently?

Solution:

Let the first machine be M_1 and the second machine be M_2 ,

$$P(M_1) = 80\% = 0.8, P(M_2) = 60\% = 0.6 \text{ and } P(M_1 \cup M_2) = 92\% = 0.92$$

$$\text{Now, } P(M_1 \cup M_2) = P(M_1) + P(M_2) - P(M_1 \cap M_2)$$

$$P(M_1 \cap M_2) = P(M_1) + P(M_2) - P(M_1 \cup M_2)$$

$$= 0.8 + 0.6 - 0.92 = 0.48 = 0.8 \times 0.6 = P(M_1) \times P(M_2)$$

Thus, the two machines operate independently.

Example

A die is thrown twice. Find the probability of obtaining a 4 on the first throw and an odd number on the second throw.

Solution: Let A be the event: “a 4 is obtained on the first throw”, then $P(A) = \frac{1}{6}$.

That is $A = \{4\}$,

B be the event: “an odd number is obtained on the second throw”. That is $B = \{1, 3, 5\}$

Since the result on the second throw is not affected by the result on the first throw, A and B are independent events.

There are 3 odd numbers, then

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

$$\text{Therefore, } P(A \cap B) = P(A) \times P(B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

Example: A coin is weighted so that heads is three times as likely to appear as tails. Find $P(H)$ and $P(T)$.

Solution:

∴ Let $P(T) = p_1$, then $P(H) = 3p_1$ But $P(H) + P(T) = 1$

Thus, $P(H) = \frac{3}{4}$ and $P(T) = \frac{1}{4}$.

Therefore $p_1 + 3p_1 = 1 \Leftrightarrow 4p_1 = 1 \Rightarrow p_1 = \frac{1}{4}$

Example

Suppose a day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Suppose two parts are selected from the batch, but the first part is replaced before the second part is selected. What is the probability that the second part is defective (denoted as B) given that the first part is defective (denoted as A)? The probability needed can be expressed as $P(B \setminus A)$ Because the first part is replaced prior to selecting the second part, the batch still contains 850 parts, of which 50 are defective. Therefore, the probability of B does not depend on whether or not the first part was defective. That is, Also, the probability that both parts are defective.

Solution: $P(B \setminus A) = \frac{50}{850}$

Also, the probability that both parts are defectives is

$$P(A \cap B) = P(B \setminus A)P(A) = \left(\frac{50}{850}\right) \cdot \left(\frac{50}{850}\right) = 0.0035.$$

**Application activity 10.6**

1) True or False

- Independent events affect each other.
- For independent events, probabilities are multiplied.
- Tossing a coin twice gives independent events.
- Drawing without replacement gives independent events.
- If $P(A \cap B) = P(A) \times P(B)$, then A and B are independent.

2) Which of the following represents independent events?

- Drawing two cards without replacement
- Tossing a coin and rolling a die

3. Selecting students without replacement
4. Picking balls without replacement

3) A die is rolled and a coin is tossed. Find the probability of:

- (a) Getting an even number and a Head
- (b) Getting a number less than 4 and a Tail

10.7 Multiplication rule



Activity 10.7

A bag contains 3 red balls and 2 blue balls. One ball is drawn, replaced, and then another ball is drawn.

Tasks:

1. Find the probability of drawing:
 - (a) a red ball on the first draw
 - (b) a red ball on the second draw
2. Find the probability of drawing two red balls.
3. Does the first draw affect the second draw? Explain.

Question:

How can we determine the probability of two events occurring together?



Content Summary

The **Multiplication Rule** is used to find the probability that **two or more events occur together**.

(a) Independent Events

For two independent events A and B (one event does **not affect** the other):

$$P(A \cap B) = P(A) \times P(B)$$

- $P(A)$: Probability of event A
- $P(B)$: Probability of event B
- $P(A \cap B)$: Probability that both events occur

The multiplication rule can be extended to any number of independent events:

$$P(A_1 \text{ and } A_2 \text{ and } A_3 \dots A_n) = P(A_1) \times P(A_2) \times P(A_3) \times \dots P(A_n)$$

Example

1) Events A, B and C are independent. If the $P(A) = \frac{1}{3}$; $P(B) = \frac{1}{4}$ and $P(C) = \frac{2}{5}$. Find the probability of A, B, and C.

Solution

If events A, B and C are independent, then the $P(A \text{ and } B \text{ and } C) =$

$$P(A) \times P(B) \times P(C) = P\left(\frac{1}{3}\right) \times P\left(\frac{1}{4}\right) \times P\left(\frac{2}{5}\right) = \frac{2}{60} = \frac{1}{30}.$$

(b) Dependent Events

Dependent Events: Two events A and B are dependent if the occurrence of A affects the probability of B.

Rule: $P(A \cap B) = P(A) \times P(B \setminus A)$

Where $P(B \setminus A)$ is the probability of B given that A has already occurred.

Example:

A bag contains 5 balls (3 red, 2 blue). Two balls are drawn **without replacement**. Find the probability of drawing **two red balls**.

Solution

- First Draw: $P(R_1) = \frac{3}{5}$
- Second draw: $P(R_2 \setminus R_1) = \frac{2}{4}$

$$P(R_1 \cap R_2) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$



Application activity 10.7

1) A coin is tossed twice. Find the probability of getting:

- (a) Two Heads
- (b) One Head and one Tail

2) A bag contains 4 red balls and 3 blue balls. Two balls are drawn without replacement. Find the probability of drawing two blue balls.

3) True or False

1. The multiplication rule is used to find the probability of two events occurring together.
2. For independent events, probabilities are added.
3. For dependent events, conditional probability is used.
4. Replacement makes events independent.

10.8 Multiplication rule using tree diagrams



Activity 10.8

A coin is tossed two times.

1. Draw all possible outcomes.
2. Find the probability of getting:
 - (a) Two Heads
 - (b) One Head and one Tail

3. How can a tree diagram help you find the probability of combined events?

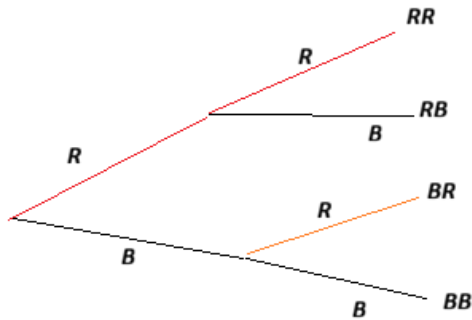


Content Summary

Multiplication rule for independent events (since replacement makes draws independent):

$$P(A \cap B) = P(A) \times P(B)$$

Tree diagram and probability:



- $R \rightarrow R: \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$
- $R \rightarrow B: \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$
- $B \rightarrow R: \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$
- $B \rightarrow B: \frac{2}{5} \times \frac{2}{5} = \frac{4}{25}$

(a) $P(\text{both red}) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

(b) $P(\text{one red, one blue}) = P(RB \text{ or } BR) = \frac{6}{25} + \frac{6}{25} = \frac{12}{25}$

Example

Same bag: 3 red, 2 blue. Two balls are drawn without replacement. Find Probabilities

(a) $P(\text{both red})$

(b) $P(\text{second ball red})$.

Sample Space $S = \{RR, RB, BR, BB\}$

Tree diagrams

A **tree diagram** is a means which can be used to show the probabilities of certain **outcomes** occurring when two or more **trials** take place in succession.

The **outcome** is written at the end of the branch and the fraction on the branch gives the probability of the outcome occurring.

For each **trial** the number of branches is equal to the number of possible outcomes of that trial.

In the diagram there are two possible outcomes, A and B , of each trial.

First draw: $P(R) = \frac{3}{5}, P(B) = \frac{2}{5}$

Second draw probabilities depend on first:

- If first was P : (2R, 2B left $\rightarrow$ total 4)

$$P(R \setminus R) = \frac{2}{4}, P(B \setminus R) = \frac{1}{4}$$

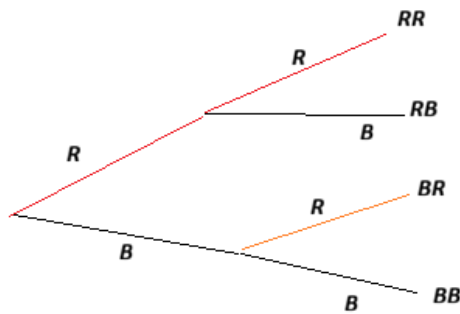
- If first was B : (3R, 1B left $\rightarrow$ total 4)

$$P(R \setminus B) = \frac{3}{4}, P(B \setminus B) = \frac{1}{4}$$

Formula Used: Multiplication rule for dependent events:

$$P(A \cap B) = P(A) \times P(B)$$

Tree diagram and probability:



- $R \rightarrow R: \frac{3}{5} \times \frac{3}{4} = \frac{9}{20} = \frac{3}{10}$
- $R \rightarrow B: \frac{3}{5} \times \frac{2}{5} = \frac{6}{25} = \frac{3}{10}$
- $B \rightarrow R: \frac{2}{5} \times \frac{3}{4} = \frac{6}{20} = \frac{3}{10}$
- $B \rightarrow B: \frac{2}{5} \times \frac{1}{4} = \frac{2}{20} = \frac{1}{10}$

Find Probabilities

(a) $P(\text{both red}) = P(RR) = \frac{3}{10}$

(b) $P(\text{second ball red}) = P(2\text{nd red}) = P(RR) + P(BR) = \frac{3}{10} + \frac{3}{10} = \frac{6}{10} = \frac{3}{5}$

Example

Probability of rain $P(R)=0.3$, no rain $P(N)=0.7$. If rain, probability of traffic jam $P(J \setminus R)=0.8$. If no rain, probability of traffic jam $P(J \setminus N)=0.2$. Find

(a) $P(\text{rain and jam})$

(b) $P(\text{jam})$

Solution

Sample Space

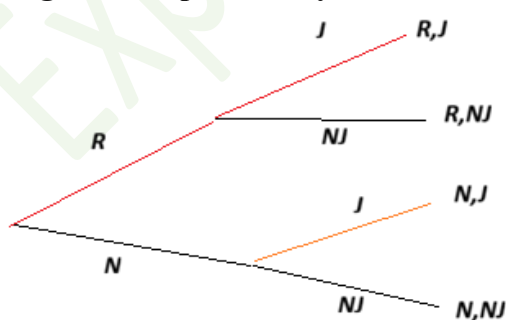
Let R = Rain, N = No rain, J = Jam, NJ = No jam.

$$S = \{(R, J), (R, NJ), (N, J), (N, NJ)\}$$

Formula used multiplication rules for dependent events:

$P(A \cap B) = P(A) \times P(B \setminus A)$ because B depends on A (traffic jam probability changes with rain).

Tree diagram and probability



- $R \rightarrow J: 0.3 \times 0.8 = 0.24$
- $R \rightarrow NJ: 0.3 \times 0.2 = 0.06$
- $N \rightarrow J: 0.7 \times 0.2 = 0.14$
- $N \rightarrow NJ: 0.7 \times 0.8 = 0.56$

(a) $P(\text{rain and jam}) = P(R \cap J) = 0.24$

(b) $P(\text{jam}) = P(J) = P(R \cap J) + P(N \cap J) = 0.24 + 0.14 = 0.38$

Example

A bag contains 8 balls of which 3 are red and 5 are green. One ball is drawn at random, its color is noted and the ball replaced in the bag. A ball is again drawn from the bag and its color is noted. Find the probability the ball drawn will be

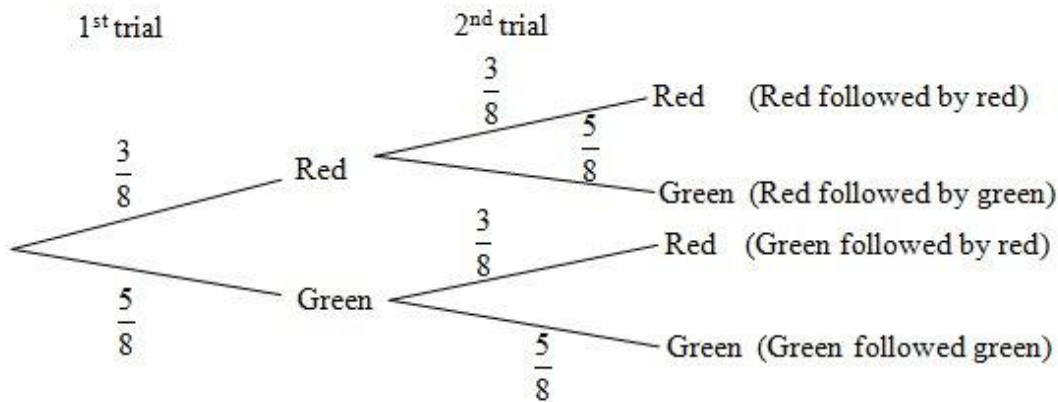
- a) red followed by green,
- b) red and green in any order,
- c) of the same colour.

Solution

Since there are 3 red balls and 5 green balls, for the 1st trial, the probability of choosing a red ball is $\frac{3}{8}$ and probability of choosing a green ball is $\frac{5}{8}$ and since after the 1st trial the ball is

replaced in the bag, for the second trial the probabilities are the same as in the first trial.

Draw a tree diagram showing the probabilities of each outcome of the two trials.



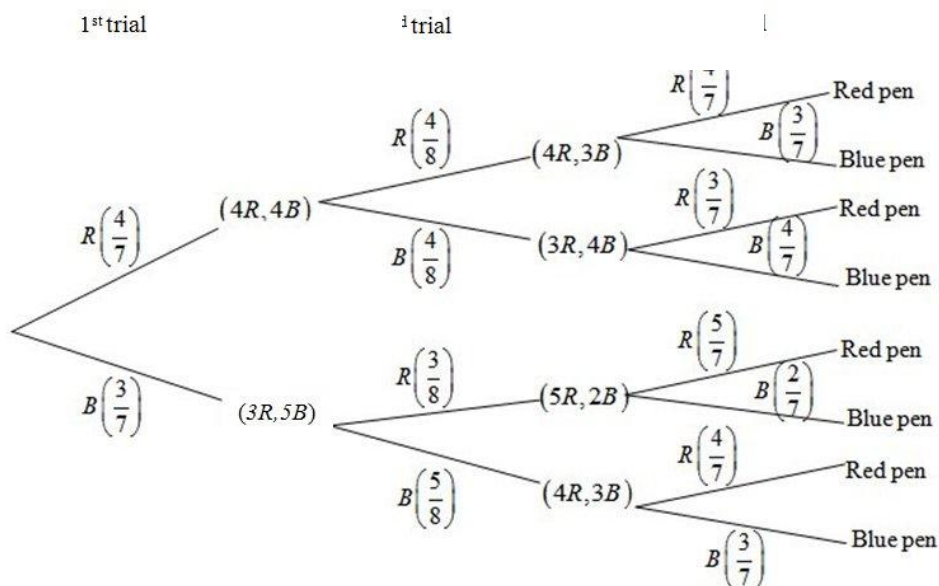
- a) $P(\text{Red followed by green}) = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$
- b) $P(\text{Red and green in any order}) = \frac{3}{8} \times \frac{5}{8} + \frac{5}{8} \times \frac{3}{8} = \frac{15}{32}$
- c) $P(\text{both of the same colors}) = \frac{3}{8} \times \frac{3}{8} + \frac{5}{8} \times \frac{5}{8} = \frac{17}{32}$

Example

A bag (1) contains 4 red pens and 3 blue pens. Another bag (2) contains 3 red pens and 4 blue pens. A pen is taken from the first bag (1) and placed into the second bag (2). The second bag (2) is shaken and a pen is taken from it and placed in the first bag (1). If now a pen is taken from the first bag, use the tree diagram to find the probability that it is a red pen.

Solution

Tree diagram is given below:



From tree diagram, the probability to have a red pen is

$$P(R) = \frac{4}{7} \times \frac{4}{8} \times \frac{3}{7} + \frac{4}{7} \times \frac{4}{8} \times \frac{3}{7} + \frac{3}{7} \times \frac{3}{8} \times \frac{5}{7} + \frac{3}{7} \times \frac{5}{8} \times \frac{4}{7}$$

$$= \frac{64}{392} + \frac{48}{392} + \frac{45}{392} + \frac{60}{392} = \frac{31}{56}$$



Application activity 10.8

1) True or False

- The probability of two independent events A and B occurring together is found by adding their probabilities.
- In a tree diagram, probabilities along a branch are multiplied to find joint probability.
- The multiplication rule can be used for both dependent and independent events.

2) A bag contains 3 red balls and 2 blue balls. Two balls are drawn **without replacement**.

What is the probability that both balls are red?

- a) $\frac{3}{5} \times \frac{3}{5}$ b) $\frac{3}{5} \times \frac{2}{4}$ c) $\frac{2}{5} \times \frac{3}{5}$ d) $\frac{3}{5} + \frac{2}{4}$

3) A box contains 2 green balls and 3 yellow balls. Two balls are drawn one after the other without replacement.

- Draw a tree diagram to represent the situation.
- Use the multiplication rule to find the probability of drawing two green balls.

10.9 Introduction to dependent events



Activity 10.9

"A teacher has a box containing 5 red balls and 3 blue balls. She asks a student to pick one ball, note its color, and then pick another ball without putting the first one back. What do you think happens to the chances of picking a red ball the second time?"

1. What is the probability of picking a red ball first?
2. If the first ball picked was red, do you think the probability of picking a red ball second will change? Why?
3. What would happen if the first ball was put back before picking the second one?



Content Summary

Dependent Events: Two events A and B are dependent if the occurrence of A affects the probability of B .

Rule: $P(A \cap B) = P(A) \times P(B \setminus A)$

Where $P(B \setminus A)$ is the probability of B given that A has already occurred.

Example 1:

In a class of **20 students, 12 are girls and 8 are boys**. Two students are selected **one after another without replacement**.

Find the probability that **both students selected are girls**.

Solution

Let: A = First girl, B = Second girl

$$\text{Then: } P(A) = \frac{12}{20}$$

After selecting one girl, 19 students remain and 11 girls remain.

$$P(B \setminus A) = \frac{11}{19}$$

$$\text{Thus } P(A \cap B) = P(A) \times P(B \setminus A), \quad P(A \cap B) = \frac{12}{20} \times \frac{11}{19} = \frac{132}{380} = \frac{33}{95}$$

Therefore: The events are dependent because selecting the first student changes the number of students remaining.

Example 2:

A basket contains **6 oranges and 4 mangoes**. Two fruits are picked **one after another without replacement**.

Find the probability that **both fruits are oranges**.

Solution

$$P(\text{First orange}) = \frac{6}{10}$$

After picking one orange, 9 fruits remain, including 5 oranges.

$$P(\text{second orange} \setminus \text{first orange}) = \frac{5}{9}$$

$$P(\text{Both oranges}) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$$

Therefore: The probability of the second event depends on the result of the first event, so these are dependent events.



Application activity 10.9

1) True or False

1. When two events are dependent, the occurrence of the first event affects the probability of the second event.
2. When items are selected **without replacement**, the events are usually independent.

2) Choose the correct answer

1. A bag contains **5 red balls and 3 blue balls**. Two balls are drawn **without replacement**. What is the probability that the **first ball is red and the second ball is also red**?
 a) $\frac{5}{8} \times \frac{4}{7}$ b) $\frac{5}{8} \times \frac{5}{8}$ c) $\frac{4}{7}$ d) $\frac{3}{8}$
2. Which of the following situations represents **dependent events**?
 A. Tossing a coin twice
 B. Rolling a die two times
 C. Choosing two students from a class **without replacement**
 D. Spinning a spinner twice

10.10 Conditional probability: Probability of event B occurring when event A has already taken place



Activity 10.10

In a class of 25 students, 15 play football, 10 play basketball, and 6 play both football and basketball.

The teacher selects a student who plays football.

1. How many students play football?
2. Among the students who play football, how many also play basketball?
3. If a student is known to play football, what is the probability that the student also plays basketball?



Content Summary

The probability of an event B given that event A has occurred is called the **conditional probability** of B given A and is written $P(A/B)$

In this case $P(B/A)$ is the probability that B occurs considering A as the sample space, and since the subset of A in which B occurs is $A \cap B$ then

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

From this result, we have general statement of the multiplication law:

$$P(B \cap A) = P(A) \times P(B | A)$$

This shows us that the probability that two events will both occur is the product of the probability that one will occur and the conditional probability that the other will occur given that the first has occurred. We can also write $P(B \cap A) = P(A) \times P(B | A)$. Since A and B are interchangeable. If A and B are **independent**, then the probability of B is not affected by the occurrence of A and so, $P(B | A) = P(B)$ giving $P(B \cap A) = P(A) \times P(B)$

Example 1

A die is tossed. Find the probability that the number obtained is a 4 given that the number is greater than 2.

Solution: Let A be the event: “the number is a 4”, then $A = \{4\}$

B be the event: “the number is greater than 2”, then $B = \{3, 4, 5, 6\}$ and $P(B) = \frac{4}{6} = \frac{2}{3}$

But $A \cap B = \{4\}$ and $P(A \cap B) = \frac{1}{6}$

Therefore, $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{6}}{\frac{1}{6}} = 1$

Example 2: At a middle school, 18% of all students play football and basketball, and 32% of all students play football. What is the probability that a student who plays football also plays basketball?

Solution: Let A be a set of students who play football and B a set of students who play basketball then the set of students who play both games is $A \cap B$. We have, $P(A) = 32\% = 0.32$.

$P(A \cap B) = 18\% = 0.18$ We need the probability of B known that A has occurred.

Therefore, $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.32} = 0.562 = 56\%$

Notice: Contingency table

Contingency table (or **Two-Way table**) provides a different way of calculating probabilities. The table helps in determining conditional probabilities quite easily. The table displays sample values in relation to two different variables that may be dependent or contingent on one another.

Below, the contingency table shows the favourite leisure activities for 50 adults, 20 men and 30 women. Because entries in the table are frequency counts, the table is a **frequency table**.

	Dance	Sports	TV	Total
Men	2	10	8	20
Women	16	6	8	30
Total	18	16	16	50

Entries in the total row and total column are called **marginal frequencies** or the **marginal distribution**. Entries in the body of the table are called **joint frequencies**.

Example 3

Suppose a study of speeding violations and drivers who use car phones produced the following fictional data:

	Speeding violation in the last year	No speeding violation in the last year	Total
Car phone user	25	280	305
Not a car phone user	45	405	450
Total	70	685	755

Calculate the following probabilities using the table:

- P(person is a car phone user)
- P(person had no violation in the last year)
- P(person had no violation in the last year AND was a car phone user)
- P(person is a car phone user OR person had no violation in the last year)
- P(person is a car phone user GIVEN person had a violation in the last year)
- P(person had no violation last year GIVEN person was not a car phone user)

Solution

$$\text{a) } P(\text{person is a car phone user}) = \frac{\text{number of car phone users}}{\text{total number in study}} = \frac{305}{755}$$

$$\text{b) } P(\text{person had no violation in the last year}) = \frac{\text{number that had no violation}}{\text{total number in study}} = \frac{685}{755}$$

$$\text{c) } P(\text{person had no violation in the last year AND person is a car phone user}) = \frac{280}{755}$$

$$\text{d) } P(\text{person is a car phone user OR person had no violation in the last year}) = \left(\frac{305}{755} + \frac{685}{755} \right) - \frac{280}{755} = \frac{710}{755}$$

e) The sample space is reduced to the number of persons who had a violation. Then

$$P(\text{person is a car phone user GIVEN person had a violation in the last year}) = \frac{25}{70}.$$

f) The sample space is reduced to the number of persons who were not car phone users. Then

$$P(\text{person had no violation last year GIVEN person was not a car phone user}) = \frac{405}{450}$$



Application activity 10.10

1) True or False

- Conditional probability is the probability of an event occurring after another event has already occurred.
- $P(B \setminus A) = \frac{P(A \cap B)}{P(A)}$
- If two events are independent, then $P(B \setminus A) = P(B)$
- Conditional probability uses the original sample space without any change.

2) Choose the Correct Answer

- In a class, 15 students play football and 6 play both football and basketball. What is the probability that a student who plays football also plays basketball?
 - $\frac{6}{25}$
 - $\frac{6}{15}$
 - $\frac{15}{25}$
 - $\frac{10}{25}$
- A card is drawn from a deck. If the card is known to be red, what is the probability that it is a Queen?
 - $\frac{1}{26}$
 - $\frac{2}{52}$
 - $\frac{2}{26}$
 - $\frac{1}{13}$
- Which formula represents conditional probability?

$$\text{a) } P(A \cup B) \quad \text{b) } P(A \cap B) \quad \text{c) } P(B \setminus A) = \frac{P(A \cap B)}{P(A)} \quad \text{d) } P(A) + P(A)$$

3) A bag contains five discs, three of which are red. A box contains six discs, four of which are red. A card is selected at random from a normal pack of 52 cards, if the card is a club a disc is removed from the bag and if the card is not a club a disc is removed from the box. Find the probability that, if the removed disc is red it came from the bag.

10.11 Basic formulae and properties of conditional probability



Activity 10.11

In a school, there are **40 students**:

- 25 students like Mathematics
- 18 students like Physics
- 10 students like both Mathematics and Physics

The teacher randomly selects a student **who likes Mathematics**.

- How many students like Mathematics?

2. Among those who like Mathematics, how many also like Physics?
3. If a student already likes Mathematics, what is the chance that they also like Physics?
4. Does knowing that a student likes Mathematics change the probability of liking Physics?



Content Summary

a) Basic formulae

Conditional probability is the probability of an event B occurring **given that event A has already occurred**.

We use this key formula:

$$P(B \setminus A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

When we multiply $P(A)$ on both sides we get:

$$P(B \setminus A) \times P(A) = \frac{P(A \cap B)}{P(A)} \times P(A) \quad \rightarrow \quad P(B \setminus A) \times P(A) = P(A \cap B)$$

We obtain: $P(A \cap B) = P(A) \times P(B \setminus A)$ This is the Multiplication Rule

Where:

- $P(B \setminus A)$: Probability of B given A
- $P(A \cap B)$: Probability that both A and B occur
- $P(A)$: Probability of A

Multiplication Rule:

$$P(A \cap B) = P(A) \times P(B \setminus A)$$

Example 1: A bag contains: 3 Red balls and 2 Blue balls.

A student draws **two balls one after another WITHOUT replacement**.

What is the probability that:

First ball is **Red**

Second ball is also **Red**

Solution

Let: Event A : First ball is Red. Event B : Second ball is Red

Let us find $P(A) = ?$

Here we have: Total balls=5, Red balls=3

Then $P(A) = \frac{3}{5}$

Let us also find $P(B \setminus A)$

After taking one red ball: Remaining balls =4 and remaining red balls =2

$$P(B \setminus A) = \frac{2}{4} = \frac{1}{2}$$

By applying Multiplication Rule, we get $P(A \cap B) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$

Therefore, the probability of getting two red balls is $\frac{3}{10}$

This means: The probability of both events happening equals the probability of one event times the conditional probability of the other.

Conditional Probability in reverse: $P(A \cap B) = P(B) \times P(A \setminus B)$

b) properties of conditional probability

1. Probability range

$$0 \leq P(B \setminus A) \leq 1$$

A conditional probability is always between 0 and 1.

2. Certain Event Property

$$P(A \setminus A) = 1 \text{ (if } P(A) > 0\text{)}$$

If event A has already happened, the probability that A occurs is **1 (certain)**.

3. Impossible Event Property

$$P(\emptyset \setminus A) = 0$$

The probability of an impossible event remains **0**, even if A has occurred.

4. Complement rule

$$P(B^c \setminus A) = 1 - P(B \setminus A)$$

The probability that B does **not** occur given A

5. Chain rule (Extended Multiplication Rule)

For three events: $P(A \cap B \cap C) = P(A) \times P(B \setminus A) \times P(C \setminus A \cap B)$

Used for **multiple dependent events**



Application activity 10.11

1. True or False

- a) Conditional probability $P(B \setminus A)$ is always between 0 and 1.
- b) If A has already occurred, $P(A \setminus A) = 1$.

2. Choose the Correct Answer

A bag contains 3 red balls and 2 blue balls. Two balls are drawn **without replacement**. What is the probability that both balls are red

- a) $\frac{3}{5}$ b) $\frac{1}{2}$ c) $\frac{3}{10}$ d) $\frac{2}{5}$

3. There are in a bag: 4 green balls, 3 yellow balls and 3 red balls

A student draws three balls one after another without replacement.

- Let A = first ball is green
- B = second ball is yellow
- C = third ball is red

Find $P(A \cap B \cap C)$ using the **Chain Rule**.

10.12 Tree diagrams and probability problems



Activity 10.12

A school organizes a quiz competition. A student answers two questions one after another. For each question, the student can either answer **Correct (C)** or **Wrong (W)**.

Questions:

1. What are the possible outcomes after answering two questions?

2. Can you list all possible combinations?
3. How can we represent these outcomes in a structured way?



Content Summary

1) Tree Diagram

A **tree diagram** is simply a way of representing a sequence of events. Tree diagrams are particularly useful in probability since they record all possible outcomes in a clear and uncomplicated manner.

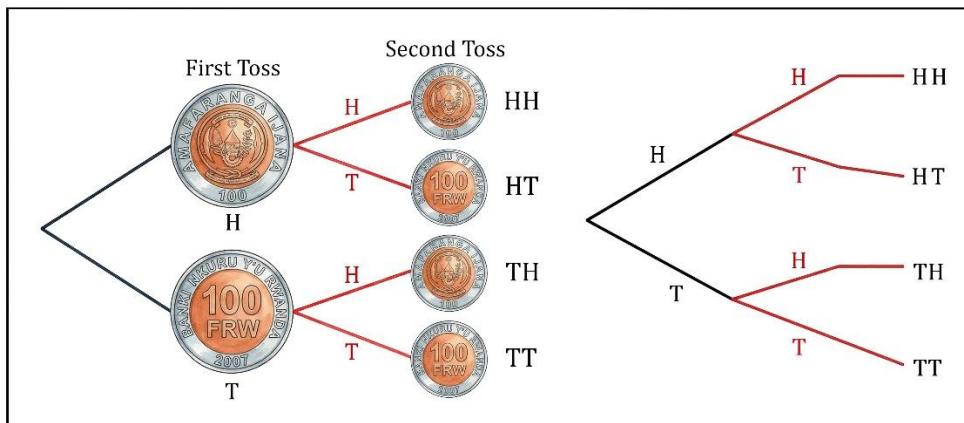
It has branches and sub-branches which help us to see the sequence of events and all the possible outcomes at each stage. **How to Construct a Tree Diagram**

1. Start with a single point (root).
2. Draw branches for the first event.
3. From each branch, draw branches for the second event.
4. Continue for all stages.
5. Assign probabilities along each branch.
6. Multiply probabilities along a path to find the probability of an outcome.

Example 1:

Using a tree diagram, determine all the possible outcomes that can be obtained when a coin is tossed twice

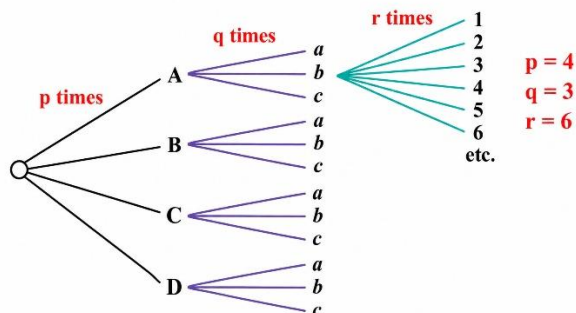
Solution: In the first toss, we get either a head (H) or a tail (T). On getting a H in the first toss, we can get a H or T in the second toss. Likewise, after getting a T in the first toss, we can get a H or T in the second toss



$$S = \{HH, HT, TH, TT\}$$

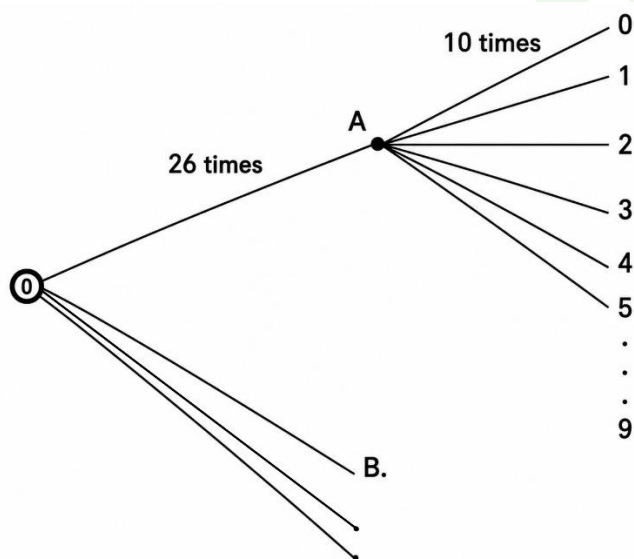
We see that the number of possible outcomes is 4. There are 2 selections of the first choice and 2 selections of the second. This is $2 \times 2 = 4$.

Multiplication principle: If a task consists of a sequence of choices in which there are p selections for the first choice, q selections for the second choice, r selections for the third choice, and so on, then the task of making these selections can be done in $n = p \times q \times r$.



Example 1: How many two-symbol code words can be formed if the first symbol is a letter (uppercase) and the second symbol is a digit?

Solution: It sometimes helps to begin by listing some of the possibilities. The code consists of a letter (uppercase) followed by a digit, so some possibilities are $A_0, A_1, A_2, \dots, B_0, B_1, B_2, B_3, \dots, B_9$.



The task consists of making two selections.

The first selection requires choosing an uppercase letter (26 choices) and the second task requires choosing a digit (10 choices).

By the Multiplication Principle, there are:

$n = 26 \times 10 = 260$ different code words of the type described.

2) Probability Problems

Probability is a number that shows **how likely an event is to happen**. It tells us the chance of something occurring, and it is always **between 0 and 1**, where 0 means the event **cannot happen** and 1 means the event **must happen**.

To find probability, we use:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Meaning:

- Favourable outcomes = what we want
- Total outcomes = All possible results

Example 1

A die has numbers: 1, 2, 3, 4, 5, 6

What is the probability of getting 3?

- Favourable outcomes = 1 (only number 3)
- Total outcomes = 6

$$P(3) = \frac{1}{6}$$



Application activity 10.12

1) True or False

1. A tree diagram shows all possible outcomes of an experiment.
2. The probability of an impossible event is 1.
3. When two events are independent, the outcome of one does not affect the other.
4. In a tree diagram, the probability of a path is found by **adding** probabilities along branches.
5. A coin tossed twice has 4 possible outcomes: HH, HT, TH, TT .

2) Choose the correct answer

1. A coin is tossed two times. What is the probability of getting exactly one head?

- a) $\frac{1}{4}$ b) $\frac{1}{2}$ c) $\frac{3}{4}$ d) 1

2. A bag contains 2 red and 1 blue ball. One ball is drawn, **then another without replacement**. What is $P(\text{red then red})$?

- a) $\frac{1}{3}$ b) $\frac{2}{3}$ c) $\frac{1}{2}$ d) $\frac{2}{9}$

3. Open Questions

1. A bag contains 3 red balls and 2 blue balls. Two balls are drawn **without replacement**.

- a) Draw a tree diagram showing all possible outcomes.
b) Find the probability of:

- i) Both balls being red
- ii) One red and one blue

2. Two dice are rolled.

- a) List all possible sums of the two dice.
b) Find the probability of getting a sum of 7.

3. A card is drawn from a standard deck of 52 cards.

- a) What is the probability of drawing a Queen?
b) What is the probability of drawing a red card or a face card?

10.13 Bayes theorem and its applications



Activity 10.13

Suppose that entire output of a factory is produced on three machines. Let B_1 denote the event that a randomly chosen item was made by machine 1, B_2 denote the event that a randomly chosen item was made by machine 2 and B_3 denote the event that a randomly chosen item was made by machine 3. Let A denote the event that a randomly chosen item is defective.

- 1) Use conditional probability formula and give the relation should be used to find the probability that the chosen item is defective, $p(A)$, given that it is made by machine 1 or machine 2 or machine 3.

If we need the probability that the chosen item is produced by machine 1 given that is found to be defective, i.e $P(B_1 | A)$, give the formula for this conditional probability. Recall that

$P(B_i \cap A)$ can be written as $P(A|B_i)P(B_i)$. Do the same if the item is produced by machine 2 and by machine 3. Give the general formula if the item is produced by machine i (i from 1 to 3).



Content Summary

Bayes' Theorem is used to find the probability of an event **based on new information**.

It allows us to determine the probability of a **cause** after observing a **result**

Let $B_1, B_2, B_3, \dots, B_n$ be incompatible and exhaustive events and let A be an arbitrary event.

$$\text{We have: } P(B_i | A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(A | B_i) \times P(B_i)}{\sum_{i=1}^n P(A | B_i) \times P(B_i)}$$

This formula is called **Bayes' formula**.

Remark

We also have the **Bayes' standard form**:

$$P(B | A) = \frac{P(A | B) \times P(B)}{P(A)}$$

Example 1:

Suppose that machines M_1, M_2, M_3 , produce, respectively, 500, 1000, and 1500 parts per day, of which 5%, 6%, and 7% are defective. A part produced by one of these machines is taken at random, at the end of a given workday, and it is found to be defective. What is the probability that it was produced by machine M_3 ?

Solution

Let A_i be the event "the part taken at random was produced by machine M_i " for $i = 1, 2, 3$; and let D be "the part taken at random is defective."

Using Bayes' formula, we seek

$$P(A_3 | D) = \frac{P(D | A_3)P(A_3)}{\sum_{i=1}^3 P(D | A_i)P(A_i)} = \frac{(0.07)\left(\frac{1500}{3000}\right)}{(0.05)\left(\frac{1}{6}\right) + (0.06)\left(\frac{1}{3}\right) + (0.07)\left(\frac{1}{2}\right)} = \frac{105}{190} = \frac{21}{38}$$

Example 2: Two machines A and B produce 60% and 40% respectively of total output of a factory. Of the parts produced by machine A, 3% are defective and of the parts produced by machine B, 5% are defective. A part is selected at random from a day's production and found to be defective. What is the probability that it came from machine A?

Solution: Let E be the event that the part came from machine A

C be the event that the part came from machine B and

D be the event that the part is defective.

Now, $P(E) \times P(D | E) = 0.6 \times 0.03 = 0.018$ and

$P(D) = P(E \cap D) + P(C \cap D) = 0.018 + 0.4 \times 0.05 = 0.038$

Therefore, the required probability is $\frac{0.018}{0.038} = \frac{9}{19}$



Application activity 10.13

1. True or False

1. Bayes' Theorem is used to find the probability of an event **given that another event has occurred**.
2. The sum of probabilities of all incompatible and exhaustive events is always 1.
3. Conditional probability can only be used when events are independent.
4. Bayes' Theorem can be applied to real-life problems such as defect rates in factories or alarms in factories.

2. Choose the correct answer

A. A factory has three machines producing 500, 1000, and 1500 parts/day, with defect rates 5%, 6%, and 7% respectively. A part is selected at random and found defective. The probability that it was produced by Machine 1 is approximately:

- a) 0.13 b) 0.20 c) 0.25 d) 0.30

B. Which of the following statements is **true** about Bayes' Theorem?

- a. It finds probability of an event **without any condition**.
- b. It updates probabilities **based on new information**.

- c. It only applies to one machine or one event.
- d. It is used only for independent events.

C. Two machines A and B produce 60% and 40% of total output. Defective rates are 3% and 5% respectively. A part is defective. Probability it came from machine A is:

- a) 0.36 b) 0.45 c) 0.60 d) 0.72



End of unit assessment

Assessing Knowledge and Understanding

A. True or False

1. Probability is always a number between 0 and 1.
2. The sample space is the set of all possible outcomes of an experiment.
3. Two events are independent if the occurrence of one affects the probability of the other.
4. For mutually exclusive events, the probability of A or B is the sum of their probabilities.
5. Conditional probability is used to find the probability of event B **given** event A has already occurred.

B. Multiple Choice Questions

1. A die is rolled once. What is the probability of getting a number greater than 4?

- a) $\frac{1}{2}$
- b) $\frac{1}{3}$
- c) $\frac{1}{6}$
- d) $\frac{2}{3}$

2. A bag contains 3 red and 2 blue balls. Two balls are drawn **without replacement**. What is P (Red then Blue)?

- a) $\frac{3}{10}$
- b) $\frac{1}{5}$
- c) $\frac{2}{5}$
- d) $\frac{3}{5}$

Assessing Skills

3. Open Questions

1. The probability that it is Friday and that a student is absent is 0.03. Since there are 5 school days in a week, the probability that it is Friday is 0.2. What is the probability that a student is absent given that today is Friday?
2. At Kennedy Middle School, the probability that a student takes Technology and Spanish is 0.087. The probability that a student takes Technology is 0.68. What is the probability that a student takes Spanish given that the student is taking Technology?
3. A car dealership is giving away a trip to Rome to one of their 120 best customers. In this group, 65 are women, 80 are married and 45 married women. If the winner is married, what is the probability that it is a woman?

Assessing Attitudes and Values

- 1) A game offers a prize if you pick a red ball from a bag containing 1 red ball and 9 blue balls.
 - a) Is this a fair game? (Yes/No)
 - b) Explain your reasoning using probability.
- 2) A doctor uses probability to explain the chance of a patient recovering from a disease.
 - a) Why is probability important in making such decisions?
 - b) Give another real-life situation where probability is useful.

BIBLIOGRAPHY

1. Rwanda Education Board (2015). Mathematics Syllabus for Ordinary Level Senior 1-3, Kigali.
2. Rwanda Basic Education Board (2025). Mathematics syllabus for ordinary level senior 1-3, Kigali.
3. Kenya Institute of Curriculum development (2024). Junior School Curriculum Design Mathematics Grade 7, Nairobi, Kenya.
4. Allen R (2004). Intermediate Algebra for College Students, Pearson Education, Inc, New Jersey.
5. Mugo K., Maina L., Ondera J (2003). Excel and Succeed Mathematics Form 1, Longhorn Publishers, Nairobi.
6. Mugo K., Maina L., Ondera J (2003). Excel and Succeed Mathematics Form 2, Longhorn Publishers, Nairobi.
7. Penguin Books Ltd (1998). The Penguin Dictionary of Mathematics, England.
8. A. J. Sadler, D. W. S. Thorning (1987). Understanding Pure Mathematics, Oxford University Press.

9. David Rayner (2000). Higher GCSE Mathematics, Oxford University Press
10. Direction des Programmes de l'Enseignement Secondaire. Géométrie de l'Espace 1er Fascule. Kigali, October 1988
11. Direction des Programmes de l'Enseignement Secondaire. Géométrie de l'Espace 2ème Fascule. Kigali, October 1988
12. George B. Thomas, Maurice D. Weir & Joel R. Hass, Thomas' (2010) Calculus Twelfth Edition, Pearson Education, Inc.
13. Geoff Mannall & Michael Kenwood (1995). Pure Mathematics 2, Heinemann Educational Publishers
14. H.K. DASS (2007). Engineering Mathematics. New Delhi, S. Chand & Company LTD, thirteenth revised edition.
15. Jean Paul Beltramonde, Vincent Brun, Claude Felloneau, LydiaMisset, Claude Talamoni (2005). Declic 1re S Mathématiques. Hachette-education, Paris.
16. JF Talber & HH Heing (1995), Additional Mathematics 6th Edition Pure & Applied, Pearson Education South Asia Pte Ltd
17. J.K. Backhouse, SPTHouldsworth B.E.D. Copper and P.J.F. Horril (1985). Pure Mathematics 2. Longman, third edition 1985, fifteenth impression 1998.
18. M.Nelkon, P. Parker. Advanced Level Physics (1995), Seventh Edition. Heinemann.
19. Peter Smythe: Mathematics HL & SL with HL options, Revised Edition (2005), Mathematics Publishing Pty. Limited.
20. Rwanda Basic Education Board (2015), Mathematics Syllabus for Advanced Level S4-S6, Ministry of Education: Rwanda.
21. Seymour Lipschutz. Schaum's outline of Theory and Problems of linear algebra. McGraw-Hill 1968.
22. Shampiyona Aimable: Mathématiques 6. Kigali, Juin 2005.
23. Tom Duncan, Advanced Physics (1998). Fourth edition. John Murray (Publishers) Ltd, London
24. Yves Noirot, Jean-Paul Parisot, Nathalie Brouillet. Cours de Physique Mathématiques pour la Physique. Paris, DUNOD, 19